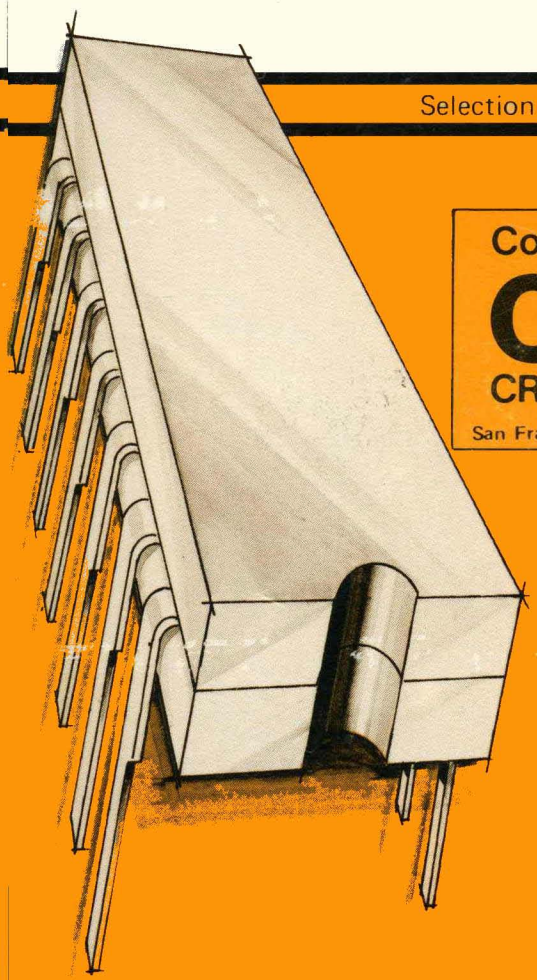
RCA

LINEAR Integrated Circuits

Selection Guide / Data



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RCA
LINEAR Integrated Circuits

This DATABOOK contains complete data on linear integrated circuits and DMOS (discrete MOS field-effect) devices presently available from RCA Solid State Division as standard products. Application notes on both linear IC's and DMOS devices are contained in a separate DATABOOK, SSD-202C.

For ease of type selection, product matrix charts are given on pages 10-18. Data sheets for linear IC's and DMOS devices are then included as nearly as possible in alphanumerical sequence of type numbers. Because some data sheets include more than one type number, however, some types may be out of sequence. If you don't find the type you're looking for where you expect it to be, please consult the Index to Devices on pages 6-7.

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The RCA Solid State DATABOOKS are supplemented throughout the year by a comprehensive data service system that keeps you aware of all new device announcements and lets you obtain as much or as little product information as you need — when you need it.

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Each newsletter issue contains a "bingo"-type fast-response form for your use in requesting information on new devices of interest to you. If you wish to receive all new product information published throughout the year, without having to use the newsletter response form, you may subscribe to a mailing service which will bring you all new data sheets and application notes in a package every other month. You can also obtain a binder for easy filing of all your supplementary material. Provisions for obtaining information on the update mailing service and the binder are included in the order form on page 4.

Because we are interested in your reaction to this approach to data service, we invite you to add your comments to the form when you return it, or to send your remarks to one of the addresses listed at the top of the form. We solicit your constructive criticism to help us improve our service to you.

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Please fill out just one copy of this form, and mail it to:

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RCA Solid State Division
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Somerville, N. J., U.S.A. 08876

(b) from Latin America and Far East:

RCA Solid State
International Sales
Somerville, N. J., U.S.A. 08876

(c) from United Kingdom, Europe, Middle East, and Africa:

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Middlesex TW16 7HW, England

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Liege, Belgium

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Address (Number) (Street, RFD, P.O. Box)

Home ☐
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(Country) (Zip or Pstl. Zone)

Function: (Check One)

A ☐ Executive/Administration
B ☐ Purchasing/Procurement
C ☐ Research/Development
D ☐ Design Engineer
E ☐ Application/Components Engineer
F ☐ Production/Manufacturing
G ☐ Documentation/Library
H ☐ Reliability/QA
I ☐ Education/Training
J ☐ Program/Project Management
K ☐ Marketing

Activity: (Check One)

A ☐ Broadcast
B ☐ Communication
C ☐ Instrumentation/Control
D ☐ Computer/Data Processing
E ☐ Computer, Peripheral
F ☐ Automotive
G ☐ Industrial
H ☐ Medical
I ☐ Research
J ☐ Transportation
K ☐ Consumer, Electronic
L ☐ Consumer, Appliance
M ☐ Space
N ☐ Ordnance
O ☐ Avionics
P ☐ Electronic Warfare

Product Interest: (Indicate order of interest if more than one is marked)

A ☐ Linear IC's
B ☐ Digital IC's, COS/MOS
C ☐ Digital IC's, Bipolar
D ☐ Thyristors/Rectifiers
E ☐ Liquid Crystals
F ☐ Semiconductor Diodes
G ☐ RF Power Semiconductors
H ☐ MOSFETS
I ☐ Power Transistors
J ☐ Power Hybrid Circuits

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*CA3022	190	low-power video amplifier	243
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*CA3095	470	super-beta transistor array	591
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Notes: For types available in beam-lead versions (File No. 515), refer to page 613.
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Linear IC New Products Program

The linear integrated circuits listed below are some of the devices scheduled for introduction during 1975. For further information concerning announcement schedules and availability, contact your RCA representative or supplier or watch for announcement in the RCA Solid State Announcement Newsletter, "What's New In Solid State", referred to on the inside front cover of this DATABOOK.

Consumer Types

Description	Similar Industry Type
CD-4 Quadraphonic Demodulator and Noise-Suppression Circuit	
TV Horizontal Processor, Positive Sync	MC1391
TV Horizontal Processor, Negative Sync	MC1394
FM Detector and Limiter With Voltage Regulator	ULN2136
High-Gain Dual-Gate MOS/FET for TV Mixer Applications	3N211
Two-Terminal Temperature Sensor	
7-Watt Audio Amplifier with Load Dump Protection (Positive Voltage Transients)	

Industrial Types

Type No.	Description	Similar Industry Type	Availability
CA7800	Voltage Regulator Series	μA7800	1 Q '75
CA1488	Quad Line Driver	MC1488	2 Q '75
CA1489, 89A	Quad Line Receivers	MC1489A	2 Q '75
CA75361A	Memory Driver	SN75361A	2 Q '75
CA521	Memory Sense Amplifier	NE521	2 Q '75

Linear IC New Products Program (cont'd)

Consumer Types and Industrial Types Custom Programs

RCA has considerable experience in the design and production of custom circuits for high-volume equipment in the applications listed below. Our engineering staff is available for the design of circuits through either specialized chip designs or the remetallization of one of our existing standard circuits.

Applications:

- Automotive
- Home Entertainment
- Appliances
- Computer Interface
- Telecommunications
- Controls

Linear IC's Direct-Replacement Guide

Analog Devices Type No.	RCA Direct Replacement
AD101AH	CA101AT
AD201H	CA201T
AD301AH	CA301AT
AD741	CA741T
AD741C	CA741CT

Advanced Micro Devices Type No.	RCA Direct Replacement
AM101T	CA101T
AM101AT	CA101AT
AM201T	CA201T
AM301AT	CA301AT
AM307T	CA307T
AM741T	CA741T
AM741CT	CA741CT
AM747T	CA747T
AM747CT	CA747CT
AM748T	CA748T
AM748CT	CA748CT

Fairchild Semiconductor Type No.	RCA Direct Replacement
LM101AH	CA101AT
LM101H	CA101T
LM107H	CA107T
LM108AH	CA108AT
LM108H	CA108T
LM111H	CA111T
LM201AH	CA201AT
LM201H	CA201T
LM207H	CA207T
LM208AH	CA208AT
LM208H	CA208T
LM301AH	CA301AT
LM301AN	CA301AE
LM307H	CA307T
LM307N	CA307E
LM308AH	CA308AT
LM308H	CA308T
LM311H	CA311T
μA723HC	CA723CT
μA723HM	CA723T
741DC	CA741CE
741DM	CA741E
741HM	CA741T
741HC	CA741CT
741TC	CA741E
747DC	CA747CE
747DM	CA747E
747HC	CA747CT
747HM	CA747T

Fairchild Semiconductor Type No.	RCA Direct Replacement
748HC	CA748CT
748HM	CA748T
748TC	CA748CE
μA758	CA758E
FPQ3724P	CA3724G
FPQ3725P	CA3725G
U5B7748312	CA748T
U5B7748393	CA748CT
U5R7723312	CA723T
U5R7723393	CA723CT
U6A7723393	CA723CE

Intersil Type No.	RCA Direct Replacement
ICL-101-TY	CA101T
ICL-101A-TY	CA101AT
ICL-108-TY	CA108T
ICL-108A-TY	CA108AT
ICL-201-TY	CA201T
ICL-201A-TY	CA201AT
ICL-208-TY	CA208T
ICL-208A-TY	CA208AT
ICL-301A-PA	CA301AE
ICL-301A-TY	CA301AT
ICL-308-TY	CA308T
ICL-308A-TY	CA308AT
ICL-723-TZ	CA723T
ICL-723C-TZ	CA723CT
ICL-741-TY	CA741T
ICL-741C-TY	CA741CT
ICL-741-LNC	CA741CE
ICL-748-TY	CA748T
ICL-748C-TY	CA748CT
ICL-8741-PA	CA741E
ICL-8741C-PA	CA741CE
ICL-8748-PA	CA748E
ICL-8748C-PA	CA748CE

Motorola Semiconductor Type No.	RCA Direct Replacement
MLM101AG	CA101AT
MLM107G	CA107T
MLM111G	CA111T
MLM201AG	CA201AT
MLM207G	CA207T
MLM211G	CA211T
MC1310P	CA1310E
MC1311P	CA758E
MC1352P	CA1352E
MC1357P	CA2111AE
MC1398P	CA1398E
MC1441L	CA1541D

Motorola Semiconductor Type No.	RCA Direct Replacement
MC1458G	CA1458T
MC1458CP1	CA1458E
MC1541L	CA1541D
MC1558G	CA1558T
MC1723CG	CA723CT
MC1723CH	CA723CH
MC1723G	CA723T
MC1741G	CA741T
MC1741CG	CA741CT
MC1741CP1	CA741CE
MC1741P2*	CA741E
MC1741CP2*	CA741CE
MC1748G	CA748T
MC1748CG	CA748CT
MC3302P	CA339AE
MC3401P	CA3401E
MPQ3724P	CA3724G
MPQ3725P	CA3725G

National Semiconductor Type No.	RCA Direct Replacement
LM101AH	CA101AT
LM101H	CA101T
LM107H	CA107T
LM108AH	CA108AT
LM108H	CA108T
LM111H	CA111T
LM201AH	CA201AT
LM201H	CA201T
LM207H	CA207T
LM208AH	CA208AT
LM208H	CA208T
LM211H	CA211T
LM301AH	CA301AT
LM301AN	CA301AE
LM307H	CA307T
LM307N	CA307E
LM308AH	CA308AT
LM308H	CA308T
LM311H	CA311T
LM324N	CA324E
LM339AN	CA339AE
LM339D	CA339E
LM723CH	CA723CT
LM723H	CA723T
LM741H	CA741T
LM741CH	CA741CT
LM741CN	CA741CE
LM747H	CA747T
LM747CH	CA747CT
LM747CN	CA747CE
LM748H	CA748T
LM748CH	CA748CT

Linear IC's Direct-Replacement Guide (cont'd)

National Semiconductor Type No.	RCA Direct Replacement
LM748CN	CA748CE
LM1310N	CA1310E
LM1458H	CA1458T
LM1458N	CA1458E
LM1558H	CA1558T
LM1800N	CA758E
LM2111	CA2111AE
DH3724CN	CA3724G
DH3725CN	CA3725G

Precision Monolithic Type No.	RCA Direct Replacement
SSS108J	CA108T
SSS108AJ	CA108AT
SSS208J	CA208T
SSS208AJ	CA208AT
SSS308J	CA308T
SSS308AJ	CA308AT

Raytheon Type No.	RCA Direct Replacement
LM101AH	CA101AT
LM101H	CA101T
LM107H	CA107T
LM108AH	CA108AT
LM108H	CA018T
LM111H	CA111T
LM201AH	CA201AT
LM201H	CA201T
LM207H	CA207T
LM211H	CA211T
LM301AH	CA301AT
LM301AN	CA301AE
LM307H	CA307T
LM307N	CA307E
LM308AH	CA308AT
LM308H	CA308T
LM311H	CA311T
RC723T	CA723CT
RM723T	CA723T
RC741DN	CA741CE
RC741TE	CA741CT
RM741TE	CA741T
RC748DN	CA748CE
RC748TE	CA748CT
RM748TE	CA748T
RC1458DN	CA1458E
RC1458T	CA1458T
SP3724QD	CA3724G
SP3725QD	CA3725G
RM4558TE	CA1558T

Signetics Type No.	RCA Direct Replacement
LM101AH	CA101AT
LM101H	CA101T
LM107H	CA107T
LM201AH	CA201AT
LM201H	CA201T
LM301AH	CA301AT
LM301AN	CA301AE
LM307H	CA307T
LM307N	CA307E
μA723CL	CA723CT
μA723L	CA723T
N5558T	CA1458T
N5558V	CA1458E
N5741T	CA741CT
N5741V	CA741CE
N5747A	CA747CE
N5748T	CA748CT
N5748V	CA748CE
S5558T	CA1558T
S5741T	CA741T
S5748T	CA748T

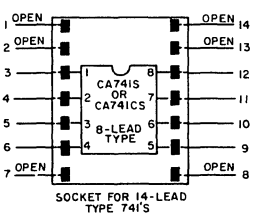
Silicon General Type No.	RCA Direct Replacement
SG101AT	CA101AT
SG101M	CA101S
SG101T	CA101T
SG107T	CA107T
SG108AT	CA108AT
SG108T	CA108T
SG111T	CA111T
SG201AT	CA201AT
SG201T	CA201T
SG207T	CA207T
SG208AT	CA208AT
SG208T	CA208T
SG211T	CA211T
SG301AM	CA301AE
SG301AT	CA301AT
SG307M	CA307E
SG307T	CA307T
SG308AT	CA308AT
SG308T	CA308T
SG311T	CA311T
SG339N	CA339E
SG339AN	CA339AE
SG723CT	CA723CT
SG723T	CA723T
SG741M	CA741E
SG741T	CA741T
SG741CM	CA741CE
SG741CT	CA741CT
SG747T	CA747T
SG747CT	CA747CT
SG748M	CA748E
SG748T	CA748T
SG748CM	CA748CE
SG748CT	CA748CT
SG1458M	CA1458E

Solitron Type No.	RCA Direct Replacement
UC4741	CA741T
UC7471C	CA741CT

Sprague Type No.	RCA Direct Replacement
ULN2111A	CA2111AE
ULN2125A	CA3120E
ULN2269A	CA3121E
ULN2741D	CA741T
ULN2747A	CA747CE
ULS2741D	CA741CT
ULX2210	CA1310E
ULX2244	CA758E

Texas Instruments Type No.	RCA Direct Replacement
SN52101AL	CA101AT
SN52101L	CA101T
SN52107L	CA107T
SN52108L	CA108T
SN52108AL	CA108AT
SN52558L	CA1558T
SN52558P	CA1558S
SN52741L	CA741T
SN52741P	CA741S
SN52747L	CA747T
SN52748L	CA748T
SN52748P	CA748S
SN72301AL	CA301AT
SN72301AN	CA301AE
SN72307L	CA307T
SN72307P	CA307E
SN72308L	CA308T
SN72308AL	CA308AT
SN72558L	CA1458T
SN72558P	CA1458E
SN72741L	CA741CT
SN72741P	CA741CE
SN72747	CA747CT
SN72748L	CA748CT
SN72748P	CA748CE
SN72301AL	CA301AT
SN72301AP	CA301AE
SN72307P	CA307E
SN72558P	CA1458E
SN76115N	CA1310E
SN76116N	CA758E
SN76650N	CA1352E

• Can be substituted for the corresponding 14-lead dual-in-line type by inserting device into 14-pin socket such that terminal No. 1 of the CA741 coincides with socket terminal No. 3 of the type to be replaced.



Note: RCA types in TO-5 packages are also supplied with dual-in-line formed leads ("DIL-CAN" package) and are designated with suffix letter (S). These types are both pin and electrical direct replacements for the corresponding 8-lead "Mini-Dip" dual-in-line types.

For prices and availability of RCA Linear Integrated Circuits, contact your RCA Distributor or your local RCA Sales Office.

Operational Amplifiers

		Micropower						High-Current							General Purpose																	
															Single Unit																	
															Low Noise																	COS/MOS Output, MOS/FET Input
		Single OTA ●		Triple OTA ●			Single OP-AMP																									
		CA3080	CA3080A	CA3060A	CA3060B	CA3060	CA3078A	CA3078	CA3033	CA3033A	CA3047	CA3047A	CA3094	CA3094A	CA3094B	CA6741	CA6078A ▲	CA101	CA101A	CA201	CA201A	CA301A	CA107	CA207	CA307	CA741C	CA741	CA748C	CA748	CA3130	CA3130A	CA3130B
	File No.	475		537			535		360				598			592		786					785			531				817		
APPLICATIONS	Sample and Hold																													■	■	■
	Switching	■	■	■	■	■							■	■	■																	
	Schmitt Trigger	■	■	■	■	■			■	■	■	■	■	■	■															■	■	■
	Multivibrator	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■
	Modulator	■	■	■	■	■	■	■					■	■	■		■															
	Mixer	■	■	■	■	■	■	■					■	■	■		■															
	Detector	■	■	■	■	■	■	■					■	■	■		■															
	Comparator	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■				■	■	■	■	■	■	■
	DC Amplifier	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■									■	■	■	■	■	■	■
	Timer												■	■	■			■	■	■	■	■	■	■	■					■	■	■
	Wideband Large Signal																															
FEATURES	Multiple Unit			■	■	■																										
	AGC Capability	■	■	■	■	■							■	■	■																	
	Balanced Input	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■
	Short-Circuit Protection	■	■	■	■	■	■	■								■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■
	Internal Frequency Compensation															■							■	■	■	■	■					
	Offset Adjustment						■	■	■	■	■	■					■	■	■	■	■	■						■	■	■	■	■
	Negative V_{ICR} near V^-	■	■	■	■	■	■	■					■	■	■		■														■	■
	Low Power Supply Current (<1 mA)	■	■	■	■	■	■	■					■	■	■		■															
	Ultra-Low I_{IB}	■	■	■	■	■	■	■					■	■	■		■														■	■
	Very Low V_{IO} & I_{IO}																															
PACKAGE		TYPE DESIGNATION SUFFIX LETTER (See Note 1)																														
	Flat Pack Ceramic																															
	Dual In-Line Ceramic (DIC)			D	D	D			■	■																						
	Dual In-Line Plastic (DIP)	E*				E					■	■	E*	E*									E*			E*	E*		E*			
	TO-5 Style Straight Lead	■	■				T	T					T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T
	TO-5 Style Dual-In-Line (DIL-CAN)	S	S				S	S					S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S
	Frit Seal Dual In-Line Ceramic	F	F				F	F																								
	Beam Lead																										L					
	Chip							H																		H		H				

Note 1: The indicated suffix letter identifies the package type for the device type number having a suffix letter; a black square is shown for a type number with no suffix letter.

● Operational Transconductance Amplifier ▲ Micropower Type *Available in 8-lead DIP (MINI-DIP)

Operational Amplifiers

		General Purpose					Wideband																Precision					
		Multiple Unit				Quad.																						
		CA747C	CA747	CA1458	CA1558	CA3401	CA3008	CA3008A	CA3010	CA3010A	CA3015	CA3015A	CA3016	CA3016A	CA3029	CA3029A	CA3030	CA3030A	CA3037	CA3037A	CA3038	CA3038A	CA3100	CA108	CA108A	CA208	CA208A	CA308	CA308A
	File No.	531				630	316	310	316	310	316	310	316	310	316	310	316	310	316	310	316	310	625	621					
APPLICATIONS	Switching																												
	Schmitt Trigger																												
	Multivibrator	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■		■	■	■	■	■	■
	Modulator																												
	Mixer																												
	Detector																												
	Comparator	■	■	■	■																		■	■	■	■	■	■	■
	DC Amplifier	■	■	■	■		■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■		■	■	■	■	■	■
	Timer																							■	■	■	■	■	■
	Wideband Large Signal																						■						
FEATURES	Multiple Unit	■	■	■	■	■																							
	AGC Capability																												
	Balanced Input	■	■	■	■		■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■		■	■	■	■	■	■
	Short-Circuit Protection	■	■	■	■																			■	■	■	■	■	■
	Internal Frequency Compensation	■	■	■	■																			■	■	■	■	■	■
	Offset Adjustment	■	■				■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■							
	Negative V_{ICR} near V^-					■																							
	Low Power Supply Current (< 1 mA)																						■	■	■	■	■	■	■
	Ultra-Low I_{IB}																						■	■	■	■	■	■	■
	Very Low V_{IO} & I_{IO}																							■		■		■	
PACKAGE	TYPE DESIGNATION SUFFIX LETTER (See Note 1)																												
	Flat Pack Ceramic						■	■					■	■															
	Dual In-Line Ceramic (DIC)																		■	■	■	■							
	Dual In-Line Plastic (DIP)	E		E*		E									■	■	■	■											
	TO-5 Style Straight Lead	T	T	T	T				■	■	■	■											T	T	T	T	T	T	T
	TO-5 Style Dual In-Line (DIL-CAN)			S	S																		S	S	S	S	S	S	S
	Frit Seal Dual In-Line Ceramic																						F						
	Beam Lead																												
	Chip	H				H																	H				H		

Note 1: The indicated suffix letter identifies the package type for the device type number having a suffix letter; a black square is shown for a type number with no suffix letter.

*Available in 8-lead DIP (MINI-DIP)

Arrays

		Diode Arrays		Transistor Arrays																
		Individual	Quad Plus Two	General-Purpose								2 Transistors, 2 Zener Diodes, 1 Diode	Dual Darlington Connected			Darlington Connected Pair Plus Two Individual			
				n-p-n					p-n-p		p-n-p & n-p-n								
		CA3019	CA3039	CA3081	CA3082	CA3083	CA3183A	CA3183	CA3084	CA3096	CA3096A	CA3093	CA3036	CA3050	CA3051	CA3018	CA3018A	CA3118A	CA3118
	File No.	236	343	480		481	532		482	595		533	275	361		338		532	
Applications	Comparator					■	■	■				■							
	Detector	■	■			■	■	■		■	■			■	■	■	■	■	■
	Differential Amplifier					■	■	■	■	■	■	■	■	■	■	■	■	■	■
	Limiter	■	■			■	■	■	■							■	■	■	■
	Mixer	■	■			■	■	■	■							■	■	■	■
	Modulator	■	■			■	■	■	■							■	■	■	■
	Multivibrator	■	■			■	■	■	■							■	■	■	■
	Oscillator			■	■	■	■	■	■							■	■	■	■
	Schmitt Trigger					■	■	■											
	Sense Amplifier					■	■	■		■	■								
	Switching	■	■	■	■	■	■	■	■										
	Thyristor & SCR Control			■	■	■	■	■		■	■	■				■	■	■	■
	Timer																		
	VHF																		
	Regulator											■							
Features	High Input Resistance																		
	Balanced Input					■	■	■	■	■	■		■	■	■	■	■	■	■
	Balanced Output					■	■	■	■				■	■	■	■	■	■	■
	Low Noise									■	■		■						
	AGC Capability															■	■	■	■
	Multiple Unit												■	■	■	■	■	■	■
	Wide Band															■	■	■	■
		TYPE DESIGNATION SUFFIX LETTER (SEE NOTE 1)																	
Package	Flat Pack Ceramic																		
	Dual In-Line Ceramic													■					
	Dual In-Line Plastic			■	■	■	E	E	■	E	E	E			■				
	TO-5 Style Straight Lead	■	■													■	■	T	T
	TO-5 Style Formed Lead																		
	Frit Seal Dual-In-Line Ceramic			F	F	F													
	Chip	H	H	H	H	H		H	H	H		H				H			H
	Beam-Lead					L			L							L			

NOTE 1: The indicated suffix letter identifies the package type for the device type number having a suffix letter; a black square is shown for a type number with no suffix letter.

Arrays

		Transistor Arrays									Amplifier Arrays					
		Differentially Connected Pair Plus Three Individual					Super β Diff. Amp. Plus 3 n-p-n Trans.	1 n-p-n & 1 p-n-p/n-p-n transistors, 1 zener diode, 1 PUT* 1 SCR▲ (Thyristor)	COS/MOS Array 3 n-channel & 3 p-channel transistors	High-Freq. n-p-n	Dual Independent (Differential)				Three Ampl.	Four Ampl.
		CA3045	CA3046	CA3086	CA3146A	CA3146	CA3095	CA3097	CA3600	CA3127	CA3026	CA3049	CA3102	CA3054	CA3035	CA3048
	File No.	341		483	532		591	633	619	662	388	611		388	274	377
Applications	Comparator							■	■							
	Detector	■	■	■	■	■	■			■	■	■	■	■		
	Differential Amplifier	■	■	■			■		■	■	■	■	■			
	Limiter			■					■							
	Mixer			■			■		■	■		■	■			■
	Modulator			■						■		■	■			
	Multivibrator			■			■		■	■		■	■			■
	Oscillator			■			■	■	■	■		■	■			
	Schmitt Trigger			■						■						
	Sense Amplifier			■						■		■	■			
	Switching			■				■	■	■		■	■			
	Thyristor & SCR Control			■				■	■							
	Timer							■	■							
	VHF							■		■						
	Regulator							■								
Features	High Input Resistance								■							
	Balanced Input	■	■	■	■	■	■			■		■	■			
	Balanced Output	■	■	■	■	■	■			■		■	■			
	Low Noise						■		■						■	■
	AGC Capability						■			■		■	■			
	Multiple Unit	■	■	■	■	■	■		■	■		■	■	■	■	■
	Wide Band	■	■	■	■	■	■			■		■	■		■	
Package	TYPE DESIGNATION SUFFIX LETTER (SEE NOTE 1)															
	Flat Pack Ceramic															
	Dual In-Line Ceramic	■	■	■												
	Dual In-Line Plastic				E	E	E	E	E	E			E	■		■
	TO-5 Style Straight Lead										■	■			■	
	TO-5 Style Formed Lead														VI	
	Frit Seal Dual-In-Line Ceramic	F														
	Chip	H				H	H	H			H	H	H	H	H	H
	Beam-Lead	L										L		L		

* Programmable Unijunction Transistor ▲Silicon Controlled Rectifier

Broadband (Video) and Differential Amplifiers, and AM/FM Communications Circuits

		Broadband (Video) Amplifiers								Differential Amplifiers														AM/FM Communications Circuits										
		CA3002	CA1352	CA3020	CA3020A	CA3021	CA3022	CA3023	CA3040 †	CA3000	CA3001 •	CA3004	CA3005	CA3006	CA3007	CA3026	CA3028A	CA3028B	CA3049	CA3050	CA3051	CA3053	CA3054	CA3102E	CA3011	CA3012	CA3013	CA3014	CA3043	CA3075	CA3076	CA3088	CA3089	CA2111A	CA3123
	File No.	123	Prel.	339			243		363	121	122	124	125	126	388	382	611	361	382	388	611	128	129	331	424	430	560	561	612	631				
Applications	Voltage Regulator																																	
	Comparator									■	■	■	■	■		■	■	■	■					■											
	Comparator – High Current Output																																	
	Control – Relays, Heaters, LED's Lamps, etc.																																	
	Detector	■										■	■	■		■			■	■	■		■	■			■	■	■		■	■	■		
	Differential Amplifier	■	■	■	■				■	■	■	■	■	■		■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■		■	■	
	Limiter			■	■	■	■	■				■	■	■		■	■	■				■		■	■	■	■	■	■	■			■	■	
	Mixer	■		■	■				■	■	■	■	■	■		■			■	■	■		■	■										■
	Modulator	■							■	■	■	■	■	■		■	■	■	■	■	■	■	■	■											
	Multivibrator									■	■					■				■	■		■												
	Oscillator			■	■					■							■	■	■			■		■											■
	Schmitt Trigger	■							■	■	■					■	■	■	■	■	■	■	■	■											
	Sense Amplifier	■							■	■	■					■	■	■	■	■	■	■	■	■	■	■									
	Switching			■	■											■	■	■	■	■	■	■		■											
	Thyristor & SCR Control																																	
	Freq. Doubler, Mult., Divide, Sq. Root, Squarer																																	
	Display Decoder-Driver																																	
	Timer																																	
Features	Balanced Input	■		■	■				■	■	■	■	■	■	■	■	■	■	■	■		■	■	■	■	■	■	■	■	■		■	■	
	Balanced Output			■	■				■	■	■	■	■	■	■	■	■	■	■	■		■	■		■	■								
	Low Noise (1/f)																																	
	Regulated Power Supply			■	■																							■	■	■	■	■	■	■
	Class B Output			■	■																													
	AGC Capability	■	■			■	■	■	■	■	■	■	■	■			■	■				■								■	■		■
	Multiple Unit															■			■	■	■			■	■									
	Wide Band	■	■	■				■	■		■	■	■	■		■	■	■	■					■	■	■	■	■	■	■		■	■	
	Micropower					■																												
	Decimal Pt. Output																																	
	Ripple Blanking																																	
	Type Designation Suffix Letter ■ = No Suffix Letter																																	
Package	Flat Pack (FP)																																	
	Dual-In-Line Ceramic (DIC)																			■														
	Dual-In-Line Plastic (DIP)		E																		■		■	E						■‡		E	E	E‡	E
	TO-5	■		■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	T			■			■	■	■	■			■			
	Chip	H		H				H		H	H		H			H	H		H				H	H		H			H	H	H			
	Beam Lead																L		L				L											
	Frit Seal															F	F					F								F				
	TO-5 Style Dual-In-Line (DIL-CAN)															S	S																	

NOTE 1: The indicated suffix letter identifies the package type for the device type number having a suffix letter; a black square is shown for a type number with no suffix letter.

- • CA3001 is also useful as a Broadband (Video) Amplifier.
- † CA3040 is also useful as a Differential Amplifier.
- ‡ In quad-in-line package (QUIP)

Power-Control and Voltage-Regulator Circuits, Analog Multiplier, and Computer-Interface Circuits

		Power Control										Voltage Regulators			Analog Multiplier	Computer Interface Circuits				
		Thyristor Control				Power Control Switch/ Ampl.			Programmable Comparator	Timers	Photo Det.					Sense Ampl.	Decoder Drivers			
		CA3058	CA3059	CA3079	CA3097‡	CA3094A	CA3094B	CA3094	CA3099	CA555, C	CA3062	CA3085	CA3085A	CA3085B	CA3091	CA1541	CD2500E	CD2501E	CD2502E	CD2503E
	File No.	490			633	598			620	834	421	491			534	536	392			
Applications	Voltage Regulator					■	■	■				■	■	■		Sense Amplifier				
	Comparator	■	■	■	■	■	■	■	■											
	Comparator – High Current Output	■	■	■		■	■	■	■			■	■	■						
	Control – Relays, Heaters, LED's, Lamps, Etc.	■	■	■	■	■	■	■	■											
	Detector	■	■	■							■				■					
	Differential Amplifier	■	■	■		◘	◘	◘	◘											
	Limiter																			
	Mixer					■	■	■							■					
	Modulator					■	■	■	■	■					■					
	Multivibrator				■	■	■	◘	■		■									
	Oscillator	■	■	■	■	■	■	■	■	■										
	Schmitt Trigger				■	■	■	■	■											
	Sense Amplifier																			
	Switching	■	■	■		■	■	■	■			■								
	Thyristor & SCR Control	■	■	■	■	■	■	■	■			■	■	■						
	Freq. Doubler, Mult , Divide, Sq. Root, Squarer														■					
	Display Decoder-Driver																■	■	■	■
	Timer	■	■	■	■	■	■	■	■	■										
Features	Balanced Input					■	■	■	■											
	Balanced Output										■									
	Low Noise (1/f)					■	■	■				■	■	■						
	Regulated Power Supply	■	■	■					■											
	Class B Output																			
	AGC Capability					■	■	■												
	Multiple Unit																			
	Wide Band																			
	Micropower					■	■	■	■											
	Decimal Pt. Output																■		■	
	Ripple Blanking																	■		■
		Type Designation Suffix Letter ■ = No Suffix Letter																		
Package	Flat Pack (FP)																			
	Dual-In-Line Ceramic (DIC)	■													D	D				
	Dual-In-Line Plastic (DIP)		■	■	E			E	E	E		E					E	E	E	E
	TO-5					T	T	T		T	■	■	■	■						
	Chip		H		H			H	H			H			H	H				
	Beam Lead											L								
	Frit Seal																			
	TO-5 Style Dual-In-Line (DIL-CAN)					S	S	S		S		S	S	S						

NOTE 1: The indicated suffix letter identifies the package type for the device type number having a suffix letter; a black square is shown for a type number with no suffix letter.

CA555, CA555C, CA3085, CA3085A available in 8-lead DIP (MINI-DIP) package.

Consumer Circuits

		Audio Circuits						Multiplex Decoders			AM Rcvr. Ckts.		FM Receiver Circuits						
		Pre-Amp.		Drivers		Power Amplifiers							FM IF Subsystems				FM IF Gain Blocks		
		CA3036	CA3052	CA3094	CA3094A, B	CA810	CA3131, CA3132	CA758	CA1310	CA3090A	CA3088	CA3123	CA2111A	CA3089	CA3075	CA3043	CA3013, CA3014	CA3011	CA3012	CA3076
	File No.	387		598		Prel.	Prel.	760	761	684	560	631	612	561	424	331	129	128	430
Circuit Functions	Audio Driver			■	■			Stereo Multiplex Decoder											
	Audio Preamplifier	■	■	■	■						■			■	■	■	■		
	ACC																		
	AFC/AFT													■					
	AFPC																		
	AGC			■	■						■			■					
	Chroma Amplifier																		
	Chroma Demodulator																		
	Chroma Signal Processor																		
	Converter										■	■							
	Detector										■		■	■	■	■	■		
	Video Amplifier																	■	■ ■
	Sync Processor																		
	IF Amplifier										■	■	■	■	■	■	■	■	■ ■
	Limiter												■	■	■	■	■	■	■ ■
	Oscillator											■							
	Audio Power Amplifier					■	■												
	Tint Control																		
Package		(TYPE DESIGNATION SUFFIX LETTER (See Note 1)																	
	Dual-In-Line Plastic		■				EM	E	E		E	E	E	E					
	Quad-In-Line Plastic					Q,QM				Q			Q		■				
	TO-5 Standard Lead	■		T	T											■	■	■	■ ■
	TO-5 Formed Lead																		

Note 1: Where a code letter is shown (E, EM, Q, T, V1), add the code letter as a suffix to the type number to identify the package (and lead configuration) option. A black square indicates no suffix code is added to the type number for that package option.

Consumer Circuits

TV Receiver Circuits

		Remote Control	Automatic Fine-Tuning (AFT)		IF Systems: Sound						IF Systems: Pix		Chroma Systems: 2 Package								Chroma Systems: 3 Package			"Jungle" Circuit
		CA3035	CA3044	CA3064	CA3134	CA3041	CA3042	CA3065	CA2111A	CA3068	CA1352	CA3066	CA3067	CA3070	CA3121	CA3067	CA3126	CA1398	CA3125	CA3128	CA3070	CA3071	CA3072	CA3120
	File No.	274	340	396	Prel.	318	319	412	612	467	Prel.	466		468	688	466	860	686	685	Prel.		468		691
Circuit Functions	Audio Driver					■	■	■																
	Audio Preamplifier	■				■	■	■	■															
	ACC											■		■	■		■	■		■	■	■		
	AFC/AFT		■	■																				
	AFPC													■			■			■	■			
	AGC									■	■													■
	Chroma Amplifier											■			■		■	■		■		■		
	Chroma Demodulator												■		■	■			■				■	
	Chroma Processor; PAL Systems											■		■	■		■	■			■	■		
	Converter																			■				
	Detector		■	■	■	■	■	■	■	■				■							■			
	Video Amplifier									■									■			■		■
	Sync Processor																							■
	IF Amplifier		■	■	■	■	■	■	■	■	■													
	Limiter		■		■	■	■	■	■															
	Oscillator											■		■		■	■	■		■	■			
	Audio Power Amplifier				■																			
	Tint Control												■	■				■			■			
Package	(TYPE DESIGNATION SUFFIX LETTER (See Note 1)																							
	Dual-In-Line Plastic			E	E				E		E			■	E			E	E		■	■	■	E
	Quad-In-Line Plastic					■	■	■	Q	■		■	■			■	Q			Q				
	TO-5 Standard Lead	■	■																					
	TO-5 Formed Lead	V1	V1	■																				

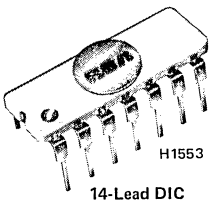
Note 1 Where a code letter is shown (E, EM, Q, T, V1), add the code letter as a suffix to the type number to identify the package (and lead configuration) option. A black square indicates no suffix code is added to the type number for that package option.

DMOS (Discrete MOS Field-Effect) Devices

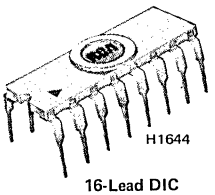
		Industrial Types													Consumer Types												
		Single-Gate							Dual-Gate			Dual-Gate Protected			Single-Gate			Dual-Gate				Dual-Gate Protected					
		3N128	3N138	3N139	3N142	3N143	3N152	3N153	3N140	3N141	3N159	3N187	3N200	40819	40467A	40468A	40559A	40600	40601	40602	40603	40604	40673	40820	40821	40822	40823	40841
	File No.	309	283	284	286	309	314	320	335	285	285	326	436	437	463	324	323	323	333	333	333	334	334	381	464	464	465	465	489
Applications	RF Amplifier, Mixer	■			■	■	■		■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■
	Chopper		■					■																					■
	General-Purpose Amplifier			■	■															■									■
	Oscillator	■		■	■	■	■		■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■
Features	Low-Noise						■				■	■	■																
	Low-Leakage		■					■	■	■	■							■	■	■	■	■							
	High-Gain						■		■		■	■	■																
	Gain-Controlled								■	■	■	■	■	■				■	■	■	■	■	■	■	■	■	■	■	■
	Premium-Performance						■				■	■	■																
All MOS/FET devices are supplied in the JEDEC TO-72 package																													

RCA LINEAR IC PACKAGES AND LEAD FORMS

D-Suffix: Dual-In-Line Welded-Seal Ceramic Package (DIC)

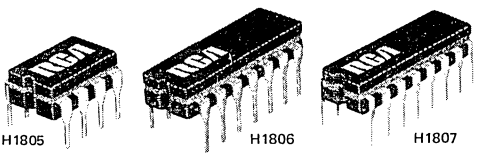


14-Lead DIC



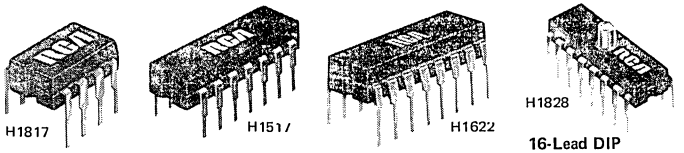
16-Lead DIC

F-Suffix: Dual-In-Line Frit-Seal Ceramic Package



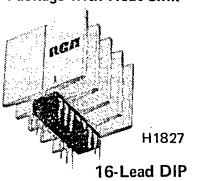
8-Lead Frit Seal | 14-Lead Frit Seal | 16-Lead Frit Seal

E-Suffix: Dual-In-Line Plastic Package (DIP)



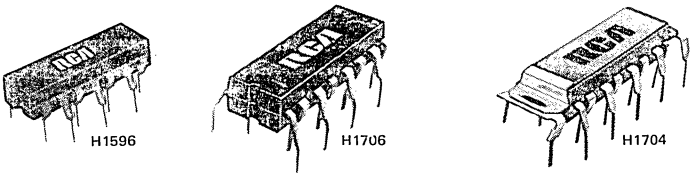
8-Lead MINI-DIP | 14-Lead DIP | 16-Lead DIP | 16-Lead DIP with "Power Stud"

EM-Suffix: Modified Dual-In-Line Plastic Package with Heat Sink



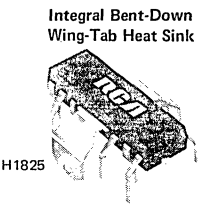
16-Lead DIP

Q-Suffix: Quad-In-Line Plastic Package (QUIP)



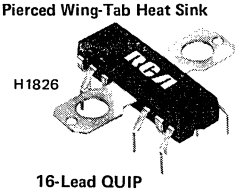
14-Lead QUIP | 16-Lead QUIP | 20-Lead QUIP

Integral Bent-Down Wing-Tab Heat Sink



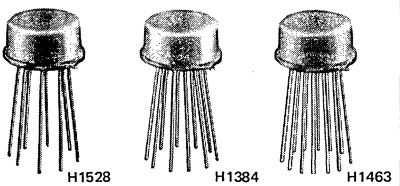
16-Lead Modified QUIP

QM-Suffix: Modified Quad-In-Line Plastic Package (QUIP) with Integral Flat and Pierced Wing-Tab Heat Sink



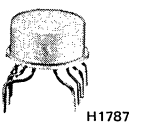
16-Lead QUIP

T-Suffix: TO-5 Style Package with Straight Leads



8-Lead TO-5 | 10-Lead TO-5 | 12-Lead TO-5

S-Suffix: TO-5 Style Package with Dual-In-Line Formed Leads (DIL-CAN)



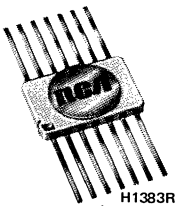
8-Lead TO-5 DIL-CAN

V1-Suffix: TO-5 Style Package with Radial Formed Leads



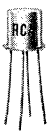
10-Formed-Lead TO-5

K-Suffix: Ceramic Flat Package (FP)

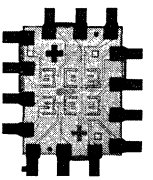


14-Lead FP

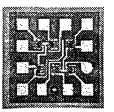
JEDEC TO-72: MOS/FET Devices



L-Suffix: IC Beam Lead



H-Suffix: IC Chip



RCA
Solid State
Division

Operating Considerations for RCA Solid State Devices

Solid state devices are being designed into an increasing variety of electronic equipment because of their high standards of reliability and performance. However, it is essential that equipment designers be mindful of good engineering practices in the use of these devices to achieve the desired performance.

This Note summarizes important operating recommendations and precautions which should be followed in the interest of maintaining the high standards of performance of solid state devices.

The ratings included in RCA Solid State Devices data bulletins are based on the Absolute Maximum Rating System, which is defined by the following Industry Standard (JEDEC) statement:

Absolute-Maximum Ratings are limiting values of operating and environmental conditions applicable to any electron device of a specified type as defined by its published data, and should not be exceeded under the worst probable conditions.

The device manufacturer chooses these values to provide acceptable serviceability of the device, taking no responsibility for equipment variations, environmental variations, and the effects of changes in operating conditions due to variations in device characteristics.

The equipment manufacturer should design so that initially and throughout life no absolute-maximum value for the intended service is exceeded with any device under the worst probable operating conditions with respect to supply-voltage variation, equipment component variation, equipment control adjustment, load variation, signal variation, environmental conditions, and variations in device characteristics.

It is recommended that equipment manufacturers consult RCA whenever device applications involve unusual electrical, mechanical or environmental operating conditions.

GENERAL CONSIDERATIONS

The design flexibility provided by these devices makes possible their use in a broad range of applications and under many different operating conditions. When incorporating these devices in equipment, therefore, designers should anticipate the rare possibility of device failure and make certain that no safety hazard would result from such an occurrence.

The small size of most solid state products provides obvious advantages to the designers of electronic equipment. However, it should be recognized that these compact devices usually provide only relatively small insulation area between adjacent leads and the metal envelope. When these devices are used in moist or contaminated atmospheres, therefore, supplemental protection must be provided to prevent the development of electrical conductive paths across the relatively small insulating surfaces. For specific information on voltage creepage, the user should consult references such as the JEDEC Standard No. 7 "Suggested Standard on Thyristors," and JEDEC Standard RS282 "Standards for Silicon Rectifier Diodes and Stacks".

The metal shells of some solid state devices operate at the collector voltage and for some rectifiers and thyristors at the anode voltage. Therefore, consideration should be given to the possibility of shock hazard if the shells are to operate at voltages appreciably above or below ground potential. In general, in any application in which devices are operated at voltages which may be dangerous to personnel, suitable precautionary measures should be taken to prevent direct contact with these devices.

Devices should not be connected into or disconnected from circuits with the power on because high transient voltages may cause permanent damage to the devices.

TESTING PRECAUTIONS

In common with many electronic components, solid-state devices should be operated and tested in circuits which have reasonable values of current limiting resistance, or other forms of effective current overload protection. Failure to observe these precautions can cause excessive internal heating of the device resulting in destruction and/or possible shattering of the enclosure.

TRANSISTORS AND THYRISTORS WITH FLEXIBLE LEADS

Flexible leads are usually soldered to the circuit elements. It is desirable in all soldering operatings to provide some slack or an expansion elbow in each lead to prevent excessive tension on the leads. It is important during the soldering operation to avoid excessive heat in order to prevent possible damage to the devices. Some of the heat can be absorbed if the flexible lead of the device is grasped between the case and the soldering point with a pair of pliers.

TRANSISTORS AND THYRISTORS WITH MOUNTING FLANGES

The mounting flanges of JEDEC-type packages such as the TO-3 or TO-66 often serve as the collector or anode terminal. In such cases, it is essential that the mounting flange be securely fastened to the heat sink, which may be the equipment chassis. Under no circumstances, however, should the mounting flange of a transistor be soldered directly to the heat sink or chassis because the heat of the soldering operation could permanently damage the device. Soldering is the preferred method for mounting thyristors; see "Rectifiers and Thyristors," below. Devices which cannot be soldered can be installed in commercially available sockets. Electrical connections may also be made by soldering directly to the terminal pins. Such connections may be soldered to the pins close to the pin seals provided care is taken to conduct excessive heat away from the seals; otherwise the heat of the soldering operation could crack the pin seals and damage the device.

During operation, the mounting-flange temperature is higher than the ambient temperature by an amount which depends on the heat sink used. The heat sink must have sufficient thermal capacity to assure that the heat dissipated in the heat sink itself does not raise the device mounting-flange temperature above the rated value. The heat sink or chassis may be connected to either the positive or negative supply.

In many applications the chassis is connected to the voltage-supply terminal. If the recommended mounting hardware shown in the data bulletin for the specific solid-state device is not available, it is necessary to use either an anodized aluminum insulator having high thermal conductivity or a mica insulator between the mounting-flange and the chassis. If an insulating aluminum washer is required, it should be drilled or punched to provide the two mounting holes for the terminal pins. The burrs should then be removed from the washer and the washer anodized. To insure that the anodized insulating layer is not destroyed during mounting, it is necessary to remove the burrs from the holes in the chassis.

It is also important that an insulating bushing, such as glass-filled nylon, be used between each mounting bolt and the chassis to prevent a short circuit. However, the insulating bushing should not exhibit shrinkage or softening under the operating temperatures encountered. Otherwise the thermal resistance at the interface between device and heat sink may increase as a result of decreasing pressure.

PLASTIC POWER TRANSISTORS AND THYRISTORS

RCA power transistors and thyristors (SCR's and triacs) in molded-silicone-plastic packages are available in a wide range of power-dissipation ratings and a variety of package configurations. The following paragraphs provide guidelines for handling and mounting of these plastic-package devices, recommend forming of leads to meet specific mounting requirements, and describe various mounting arrangements, thermal considerations, and cleaning methods. This information is intended to augment the data on electrical characteristics, safe operating area, and performance capabilities in the technical bulletin for each type of plastic-package transistor or thyristor.

Lead-Forming Techniques

The leads of the RCA VERSAWATT in-line plastic packages can be formed to a custom shape, provided they are not indiscriminately twisted or bent. Although these leads can be formed, they are not flexible in the general sense, nor are they sufficiently rigid for unrestrained wire wrapping

Before an attempt is made to form the leads of an in-line package to meet the requirements of a specific application, the desired lead configuration should be determined, and a lead-bending fixture should be designed and constructed. The use of a properly designed fixture for this operation eliminates the need for repeated lead bending. When the use of a special bending fixture is not practical, a pair of long-nosed pliers may be used. The pliers should hold the lead firmly between the bending point and the case, but should not touch the case.

When the leads of an in-line plastic package are to be formed, whether by use of long-nosed pliers or a special bending fixture, the following precautions must be observed to avoid internal damage to the device:

1. Restrain the lead between the bending point and the plastic case to prevent relative movement between the lead and the case.
2. When the bend is made in the plane of the lead (spreading), bend only the narrow part of the lead.
3. When the bend is made in the plane perpendicular to that of the leads, make the bend at least 1/8 inch from the plastic case.
4. Do not use a lead-bend radius of less than 1/16 inch.
5. Avoid repeated bending of leads.

The leads of the TO-220AB VERSAWATT in-line package are not designed to withstand excessive axial pull. Force in this direction greater than 4 pounds may result in permanent damage to the device. If the mounting arrangement tends to impose axial stress on the leads, some method of strain relief should be devised.

Wire wrapping of the leads is permissible, provided that the lead is restrained between the plastic case and the point of the wrapping. Soldering to the leads is also allowed. The maximum soldering temperature, however, must not exceed 275°C and must be applied for not more than 5 seconds at a distance not less than 1/8 inch from the plastic case. When

wires are used for connections, care should be exercised to assure that movement of the wire does not cause movement of the lead at the lead-to-plastic junctions.

The leads of RCA molded-plastic high-power packages are not designed to be reshaped. However, simple bending of the leads is permitted to change them from a standard vertical to a standard horizontal configuration, or conversely. Bending of the leads in this manner is restricted to three 90-degree bends; repeated bendings should be avoided.

Mounting

Recommended mounting arrangements and suggested hardward for the VERSAWATT package are given in the data bulletins for specific devices and in RCA Application Note AN-4142. When the package is fastened to a heat sink, a rectangular washer (RCA Part No. NR231A) is recommended to minimize distortion of the mounting flange. Excessive distortion of the flange could cause damage to the package. The washer is particularly important when the size of the mounting hole exceeds 0.140 inch (6-32 clearance). Larger holes are needed to accommodate insulating bushings; however, the holes should not be larger than necessary to provide hardware clearance and, in any case, should not exceed a diameter of 0.250 inch.

Flange distortion is also possible if excessive torque is used during mounting. A maximum torque of 8 inch-pounds is specified. Care should be exercised to assure that the tool used to drive the mounting screw never comes in contact with the plastic body during the driving operation. Such contact can result in damage to the plastic body and internal device connections. An excellent method of avoiding this problem is to use a spacer or combination spacer-isolating bushing which raises the screw head or nut above the top surface of the plastic body. The material used for such a spacer or spacer-isolating bushing should, of course, be carefully selected to avoid "cold flow" and consequent reduction in mounting force. Suggested materials for these bushings are diallphtalate, fiberglass-filled nylon, or fiberglass-filled polycarbonate. Unfilled nylon should be avoided.

Modification of the flange can also result in flange distortion and should not be attempted. The package should not be soldered to the heat sink by use of lead-tin solder because the heat required with this type of solder will cause the junction temperature of the device to become excessively high.

The TO-220AA plastic package can be mounted in commercially available TO-66 sockets, such as UID Electronics Corp. Socket No. PTS-4 or equivalent. For testing purposes, the TO-220AB in-line package can be mounted in a Jetron Socket No. DC74-104 or equivalent. Regardless of the mounting method, the following precautions should be taken:

1. Use appropriate hardware.
2. Always fasten the package to the heat sink before the leads are soldered to fixed terminals.
3. Never allow the mounting tool to come in contact with the plastic case.
4. Never exceed a torque of 8 inch-pounds.
5. Avoid oversize mounting holes.
6. Provide strain relief if there is any probability that axial stress will be applied to the leads.
7. Use insulating bushings to prevent hot-creep problems. Such bushings should be made of diallphthalate, fiberglass-filled nylon, or fiberglass-filled polycarbonate.

The maximum allowable power dissipation in a solid state device is limited by the junction temperature. An important factor in assuring that the junction temperature remains below the specified maximum value is the ability of the associated thermal circuit to conduct heat away from the device.

When a solid state device is operated in free air, without a heat sink, the steady-state thermal circuit is defined by the junction-to-free-air thermal resistance given in the published data for the device. Thermal considerations require that a free flow of air around the device is always present and that the power dissipation be maintained below the level which would cause the junction temperature to rise above the maximum rating. However, when the device is mounted on a heat sink, care must be taken to assure that all portions of the thermal circuit are considered.

To assure efficient heat transfer from case to heat sink when mounting RCA molded-plastic solid state power devices, the following special precautions should be observed:

1. Mounting torque should be between 4 and 8 inch-pounds.
2. The mounting holes should be kept as small as possible.
3. Holes should be drilled or punched clean with no burrs or ridges, and chamfered to a maximum radius of 0.010 inch.
4. The mounting surface should be flat within 0.002 inch/inch.
5. Thermal grease (Dow Corning 340 or equivalent) should always be used on both sides of the insulating washer if one is employed.
6. Thin insulating washers should be used. (Thickness of factory-supplied mica washers range from 2 to 4 mils).
7. A lock washer or torque washer, made of material having sufficient creep strength, should be used to prevent degradation of heat sink efficiency during life.

A wide variety of solvents is available for degreasing and flux removal. The usual practice is to submerge components in a solvent bath for a specified time. However, from a reliability stand point it is extremely important that the solvent, together with other chemicals in the solder-cleaning system (such as flux and solder covers), do not adversely affect the life of the component. This consideration applies to all non-hermetic and molded-plastic components.

It is, of course, impractical to evaluate the effect on long-term device life of all cleaning solvents, which are marketed with numerous additives under a variety of brand names. These solvents can, however, be classified with

respect to their component parts as either acceptable or unacceptable. Chlorinated solvents tend to dissolve the outer package and, therefore, make operation in a humid atmosphere unreliable. Gasoline and other hydrocarbons cause the inner encapsulant to swell and damage the transistor. Alcohol is an acceptable solvent. Examples of specific, acceptable alchols are isopropanol, methanol, and special denatured alcohols, such as SDA1, SDA30, SDA34, and SDA44.

Care must also be used in the selection of fluxes for lead soldering. Rosin or activated rosin fluxes are recommended, while organic or acid fluxes are not. Examples of acceptable fluxes are:

1. Alpha Reliaros No. 320-33
2. Alpha Reliaros No. 346
3. Alpha Reliaros No. 711
4. Alpha Reliafoam No. 807
5. Alpha Reliafoam No. 809
6. Alpha Reliafoam No. 811-13
7. Alpha Reliafoam No. 815-35
8. Kester No. 44

If the completed assembly is to be encapsulated, the effect on the molded-plastic transistor must be studied from both a chemical and a physical standpoint.

RECTIFIERS AND THYRISTORS

A surge-limiting impedance should always be used in series with silicon rectifiers and thyristors. The impedance value must be sufficient to limit the surge current to the value specified under the maximum ratings. This impedance may be provided by the power transformer winding, or by an external resistor or choke.

A very efficient method for mounting thyristors utilizing the "modified TO-5" package is to provide intimate contact between the heat sink and at least one half of the base of the device opposite the leads. This package can be mounted to the heat sink mechanically with glue or an expoxy adhesive, or by soldering, the most efficient method.

The use of a "self-jigging" arrangement and a solder preform is recommended. If each unit is soldered individually, the heat source should be held on the heat sink and the solder on the unit. Heat should be applied only long enough to permit solder to flow freely. For more detailed thyristor mounting considerations, refer to Application Note AN3822, "Thermal Considerations in Mounting of RCA Thyristors".

MOS FIELD-EFFECT TRANSISTORS

Insulated-Gate Metal Oxide-Semiconductor Field-Effect Transistors (MOS FETs), like bipolar high-frequency transistors, are susceptible to gate insulation damage by the electrostatic discharge of energy through the devices. Electrostatic discharges can occur in an MOS FET if a type with an unprotected gate is picked up and the static charge, built in the handler's body capacitance, is discharged through the device. With proper handling and applications procedures, however, MOS transistors are currently being extensively used in production by numerous equipment manufacturers in military, industrial, and consumer applications, with virtually no problems of damage due to electrostatic discharge.

In some MOS FETs, diodes are electrically connected between each insulated gate and the transistor's source. These diodes offer protection against static discharge and in-circuit transients without the need for external shorting mechanisms. MOS FETs which do not include gate-protection diodes can be handled safely if the following basic precautions are taken:

1. Prior to assembly into a circuit, all leads should be kept shorted together either by the use of metal shorting springs attached to the device by the vendor, or by the insertion into conductive material such as "ECCOSORB* LD26" or equivalent.
   (NOTE: Polystyrene *insulating* "SNOW" is not sufficiently conductive and should not be used.)
2. When devices are removed by hand from their carriers, the hand being used should be grounded by any suitable means, for example, with a metallic wristband.
3. Tips of soldering irons should be grounded.
4. Devices should never be inserted into or removed from circuits with power on.

RF POWER TRANSISTORS

Mounting and Handling

Stripline rf devices should be mounted so that the leads are not bent or pulled away from the stud (heat sink) side of the device. When leads are formed, they should be supported to avoid transmitting the bending or cutting stress to the ceramic portion of the device. Excessive stresses may destroy the hermeticity of the package without displaying visible damage.

Devices employing silver leads are susceptible to tarnishing; these parts should not be removed from the original tarnish-preventive containers and wrappings until ready for use. Lead solderability is retarded by the presence of silver tarnish; the tarnish can be removed with a silver cleaning solution, such as thiourea.

The ceramic bodies of many rf devices contain beryllium oxide as a major ingredient. These portions of the transistors should not be crushed, ground, or abraded in any way because the dust created could be hazardous if inhaled.

Operating

Forward-Biased Operation. For Class A or AB operation, the allowable quiescent bias point is determined by reference to the infrared safe-area curve in the appropriate data bulletin. This curve depicts the safe current/voltage combinations for extended continuous operation.

Load VSWR. Excessive collector load or tuning mismatch can cause device destruction by over-dissipation or secondary breakdown. Mismatch capability is generally included on the data bulletins for the more recent rf transistors.

See RCA RF Power Transitor Manual, Technical Series RMF-430, pp 39-41, for additional information concerning the handling and mounting of rf power transistors.

*Trade Mark: Emerson and Cumming, Inc.

INTEGRATED CIRCUITS

Handing

All COS/MOS gate inputs have a resistor/diode gate protection network. All transmission gate inputs and all outputs have diode protection provided by inherent p-n junction diodes. These diode networks at input and output interfaces protect COS/MOS devices from gate-oxide failure in handling environments where static discharge is not excessive. In low-temperature, low-humidity environments, improper handling may result in device damage. See ICAN-6000, "Handling and Operating Considerations for MOS Integrated Circuits", for proper handling procedures.

Mounting

Integrated circuits are normally supplied with lead-tin plated leads to facilitate soldering into circuit boards. In those relatively few applications requiring welding of the device leads, rather than soldering, the devices may be obtained with gold or nickel plated Kovar leads.* It should be recognized that this type of plating will not provide complete protection against lead corrosion in the presence of high humidity and mechanical stress. The aluminum-foil-lined cardboard "sandwich pack" employed for static protection of the flat-pack also provides some additional protection against lead corrosion, and it is recommended that the devices be stored in this package until used.

When integrated circuits are welded onto printed circuit boards or equipment, the presence of moisture between the closely spaced terminals can result in conductive paths that may impair device performance in high-impedance applications. It is therefore recommended that conformal coatings or potting be provided as an added measure of protection against moisture penetration.

In any method of mounting integrated circuits which involves bending or forming of the device leads, it is extremely important that the lead be supported and clamped between the bend and the package seal, and that bending be done with care to avoid damage to lead plating. In no case should the radius of the bend be less than the diameter of the lead, or in the case of rectangular leads, such as those used in RCA 14-lead and 16-lead flat-packages, less than the lead thickness. It is also extremely important that the ends of the bent leads be straight to assure proper insertion through the holes in the printed-circuit board.

Operating

Unused Inputs

All unused input leads must be connected to either V_{SS} or V_{DD}, whichever is appropriate for the logic circuit involved. A floating input on a high-current type, such as the CD4049 or CD4050, not only can result in faulty logic operation, but can cause the maximum power dissipation of 200 milliwatts to be exceeded and may result in damage to the device. Inputs to these types, which are mounted on printed-circuit boards that may temporarily become unterminated, should have a pull-up resistor to V_{SS} or V_{DD}. A useful range of values for such resistors is from 10 kilohms to 1 megohm.

Input Signals

Signals shall not be applied to the inputs while the device power supply is off unless the input current is limited to a steady state value of less than 10 milliamperes. Input currents of less than 10 milliamperes prevent device damage; however, proper operation may be impaired as a result of current flow through structural diode junctions.

Output Short Circuits

Shorting of outputs to V_{SS} or V_{DD} can damage many of the higher-output-current COS/MOS types, such as the CD4007, CD4041, CD4049, and CD4050. In general, these types can all be safely shorted for supplies up to 5 volts, but will be damaged (depending on type) at higher power-supply voltages. For cases in which a short-circuit load, such as the base of a p-n-p or an n-p-n bipolar transistor, is directly driven, the device output characteristics given in the published data should be consulted to determine the requirements for a safe operation below 200 milliwatts.

For detailed COS/MOS IC operating and handling considerations, refer to Application Note ICAN-6000 "Handling and Operating Considerations for MOS Integrated Circuits".

SOLID STATE CHIPS

Solid state chips, unlike packaged devices, are non-hermetic devices, normally fragile and small in physical size, and therefore, require special handling considerations as follows:

1. Chips must be stored under proper conditions to insure that they are not subjected to a moist and/or contaminated atmosphere that could alter their electrical, physical, or mechanical characteristics. After the shipping container is opened, the chip must be stored under the following conditions:
   A. Storage temperature, 40°C max.
   B. Relative humidity, 50% max.
   C. Clean, dust-free environment.
2. The user must exercise proper care when handling chips to prevent even the slightest physical damage to the chip.
3. During mounting and lead bonding of chips the user must use proper assembly techniques to obtain proper electrical, thermal, and mechanical performance.
4. After the chip has been mounted and bonded, any necessary procedure must be followed by the user to insure that these non-hermetic chips are not subjected to moist or contaminated atmosphere which might cause the development of electrical conductive paths across the relatively small insulating surfaces. In addition, proper consideration must be given to the protection of these devices from other harmful environments which could conceivably adversely affect their proper performance.

*Mil-M-38510A, paragraph 3.5.6.1 (a), lead material.

Technical Data-Linear IC's

RCA
Solid State
Division

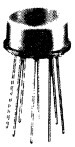
Linear Integrated Circuits

Monolithic Silicon

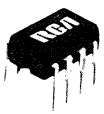
CA101, CA101A, CA201, CA201A, CA301A Types



S-Suffix Types
8-LEAD TO-5
with Dual-In-Line
Formed Leads
(DIL-CAN)
H-1787



T-Suffix Types
8-LEAD TO-5
H-1528



CA301AE
8-LEAD DUAL-
IN-LINE PLASTIC
PACKAGE
(MINI-DIP) H-1817

Operational Amplifiers

For Military, Industrial, and Commercial Applications

Features:

- **Short-circuit protection and latch-free operation**
- **Unity-gain phase compensation with a single 30-pF capacitor**

		CA101	CA201	CA101A	CA201A	CA301A	
Max. V_{IO}	T_A = 25°C	5	7.5	2	2	7.5	mV
Max. I_{IO}		200	500	10	10	50	nA
Min. A_{OL}		50	20	50	50	25	V/mV
T_A Range (Operating)		−55 to +125	0 to +70	−55 to +125	−25 to +85	0 to +70	°C
Slew Rate (Summing ampl.)		–	–	10	10	10	V/μs

RCA-CA101, CA101A, CA201, CA201A, and CA301A are general-purpose, high-gain operational amplifiers for use in military, industrial, and commercial applications.

These types, which are externally phase compensated, permit a choice of operation for optimum high-frequency performance at a selected gain; unity-gain compensation can be obtained with a single 30-pF capacitor.

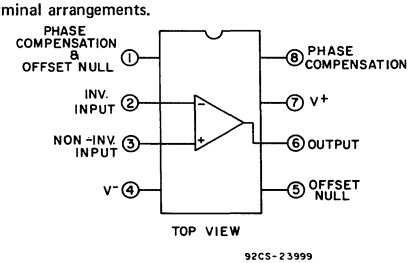
Types CA101A and CA201A have all the desirable features and characteristics of the CA101 and CA201, respectively, plus superior input-offset characteristics, and improved noise performance.

All types are available in 8-lead TO-5 style packages with standard leads (T suffix) and with dual-in-line formed leads "DIL-CAN" (S suffix). Type CA301A is also available in an 8-lead dual-in-line plastic package "MINI-DIP" (E suffix), and in chip form (H suffix).

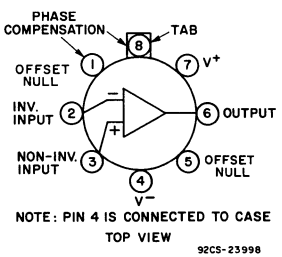
Types CA101T, S; CA101AT, S; CA201T, S; CA201AT, S; and CA301AT, S, E are direct replacements for industry types 101, 101A, 201, 201A, 301A in packages with similar terminal arrangements.

Applications:

- **Long-interval integrator**
- **Timers**
- **Sample and hold circuits**
- **Summing amplifiers**
- **Multivibrators**
- **Comparators**
- **Instrumentation**
- **AC/DC converters**
- **Inverting amplifiers**
- **Sine- & square-wave generators**
- **Capacitance multipliers & simulated inductors**



b — Plastic package for CA301A



a — TO-5 style package for all types

Fig. 1 — Functional diagrams.

Maximum Ratings, *Absolute-Maximum Values at $T_A = 25^oC$*

DC SUPPLY VOLTAGE (between V^+ and V^- terminals):		
CA101, CA101A, CA201, CA201A	44	V
CA301A	36	V
DC INPUT VOLTAGE	±15	V
(For supply voltage less than ±15 V, the Input Voltage rating is equal to the DC Supply Voltage)		
DIFFERENTIAL INPUT VOLTAGE	±30	V
OUTPUT SHORT-CIRCUIT DURATION	Indefinite*	
DEVICE DISSIPATION:		
Up to $T_A = 75^oC$	500	mW
Above $T_A = 75^oC$	derate linearly at 6.67	mW/°C
AMBIENT TEMPERATURE RANGE:		
Operating —		
CA101, CA101A	−55 to +125	°C
CA201A	−25 to +85	°C
CA201, CA301A	0 to +70	°C
Storage (All types)	−65 to +150	°C
LEAD TEMPERATURE (During Soldering):		
At a distance 1/16" ± 1/32" (1.59 ± 0.79 mm) from case for 10 seconds max.	+265	°C

* At $T_A \leq 70^oC$ and $T_C \leq 125^oC$ (CA101);
$T_A \leq 75^oC$ and $T_C \leq 125^oC$ (CA101A, CA201A);
$T_A \leq 55^oC$ and $T_C \leq 70^oC$ (CA201, CA301A).

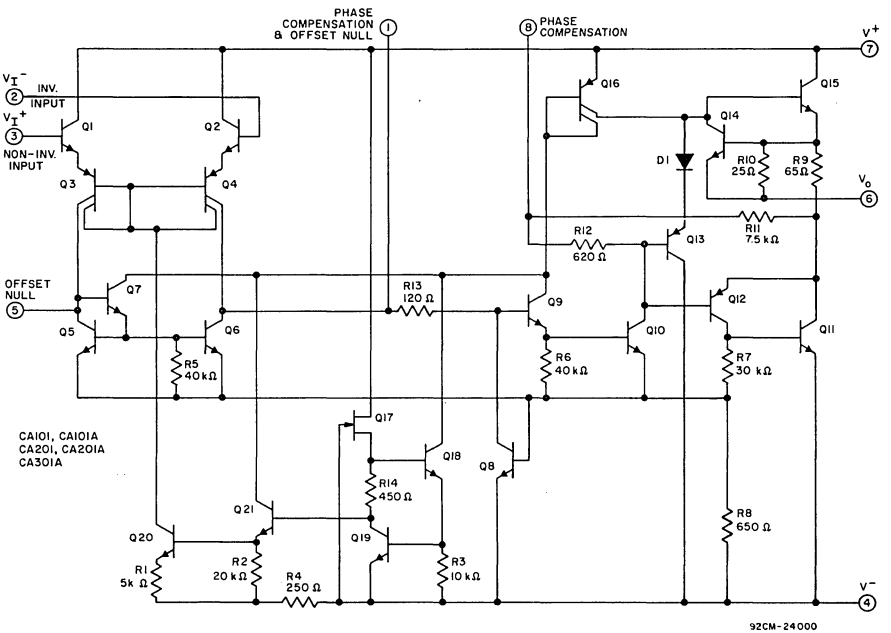


Fig.2 — Schematic diagram.

ELECTRICAL CHARACTERISTICS

CHARACTERISTICS		TEST CONDITIONS▲ Supply Voltage ($V^{\pm}$) = 5 to 15 V		LIMITS												UNITS
				CA101			CA101A CA201A			CA201			CA301A			
				Min.	Typ.	Max.	Min.	Typ.	Max.	Min.	Typ.	Max.	Min.	Typ.	Max.	
Input Offset Voltage	V_{IO}	T_A=25°C	$R_S \leq 10k\Omega$	–	1	5	–	–	–	–	2	7.5	–	–	–	mV
			$R_S \leq 50k\Omega$	–	–	–	–	0.7	2	–	–	–	–	2	7.5	
			$R_S \leq 10k\Omega$	–	–	6	–	–	–	–	–	10	–	–	–	
			$R_S \leq 50k\Omega$	–	–	–	–	–	3	–	–	–	–	–	10	
Average Temperature Coefficient of Input Offset Voltage	αV_{IO}		$R_S \leq 10k\Omega$	–	6	–	–	–	–	–	10	–	–	–	–	µV/°C
			$R_S \leq 50\Omega$	–	3	–	–	–	–	–	6	–	–	–	–	
				–	–	–	–	3	15	–	–	–	–	6	30	
Average Temperature Coefficient of Input Offset Current	αI_{IO}	−55°C to +25°C		–	–	–	–	0.02	0.2	–	–	–	–	–	–	nA/°C
		0°C to +25°C		–	–	–	–	–	–	–	–	–	–	0.02	0.6	
		+25°C to +70°C		–	–	–	–	–	–	–	–	–	–	0.01	0.3	
		+25°C to +125°C		–	–	–	–	0.01	0.1	–	–	–	–	–	–	
Input Offset Current	I_{IO}	T_A=0°C		–	–	–	–	–	–	–	150	750	–	–	–	nA
		T_A=25°C		–	40	200	–	1.5	10	–	100	500	–	3	50	
		T_A=70°C		–	–	–	–	–	–	–	50	400	–	–	–	
		T_A=125°C		–	10	200	–	–	–	–	–	–	–	–	–	
				–	–	–	–	–	20	–	–	–	–	–	70	
		T_A=−55°C		–	100	500	–	–	–	–	–	–	–	–	–	
Input Bias Current	I_{IB}	T_A=−55°C		–	0.28	1.5	–	–	–	–	–	–	–	–	–	µA
		T_A=0°C		–	–	–	–	–	–	–	0.32	2	–	–	–	
		T_A=25°C		–	0.12	0.5	–	0.03	0.075	–	0.25	1.5	–	0.07	0.25	
				–	–	–	–	–	0.1	–	–	–	–	–	0.3	
Supply Current	$I^{\pm}$	T_A=25°C	$V^{\pm}$=15V	–	–	–	–	–	–	–	–	–	–	1.8	3	mA
			$V^{\pm}$=20V	–	1.8	3	–	1.8	3	–	1.8	3	–	–	–	
		T_A=125°C	$V^{\pm}$=20V	–	1.2	2.5	–	1.2	2.5	–	–	–	–	–	–	
Open-Loop Differential Voltage Gain	A_{OL}	T_A=25°C V_O=±10V	$V^{\pm}$=15V $R_L \geq 2k\Omega$	50	160	–	50	160	–	20	150	–	25	160	–	V/mV
		V_O=±10V	$V^{\pm}$=15V $R_L \geq 2k\Omega$	25	–	–	25	–	–	15	–	–	15	–	–	
Input Resistance	R_I	T_A=25°C		0.3	0.8	–	1.5	4	–	0.1	0.4	–	0.5	2	–	MΩ
Output Voltage Swing	V_{OPP}	$V^{\pm}$=15V	R_L=10kΩ	±12	±14	–	±12	±14	–	±12	±14	–	±12	±14	–	V
		$V^{\pm}$=15V	R_L=2kΩ	±10	±13	–	±10	±13	–	±10	±13	–	±10	±13	–	
Common-Mode Input-Voltage Range	V_{ICR}	$V^{\pm}$=15V		±12	–	–	–	–	–	±12	–	–	±12	–	–	V
		$V^{\pm}$=20V		–	–	–	±15	–	–	–	–	–	–	–	–	
Common-Mode Rejection Ratio	CMRR		$R_S \leq 10k\Omega$	70	90	–	–	–	–	65	90	–	–	–	–	dB
			$R_S \leq 50k\Omega$	–	–	–	80	96	–	–	–	–	70	90	–	
Supply-Voltage Rejection Ratio	PSRR		$R_S \leq 10k\Omega$	70	90	–	–	–	–	70	90	–	–	–	–	dB
			$R_S \leq 50k\Omega$	–	–	–	80	96	–	–	–	–	70	90	–	

▲ Characteristics applicable over operating temperature range (T_A) as shown below, unless otherwise specified:
CA101, C101A: −55 to +125°C CA201A: −25 to +85°C CA201, CA301A: 0 to 70°C

TYPICAL STATIC CHARACTERISTICS

Type CA101

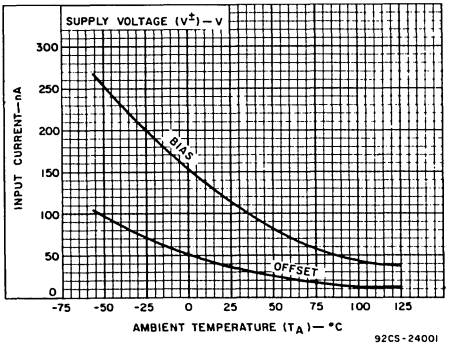


Fig.3 — Input current (I_{IO}, I_{IB}) vs. temperature.

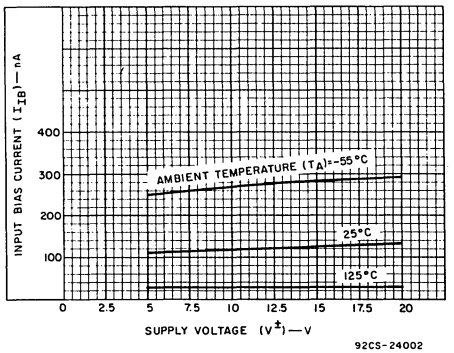


Fig.4 — Input bias current vs. supply voltage.

Types CA101A and CA201A

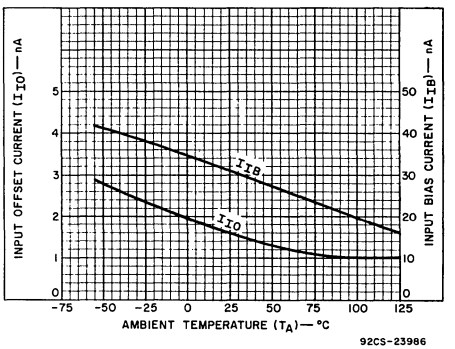


Fig. 5 — Input current (I_{IO}, I_{IB}) vs. temperature.

Types CA101, CA101A, and CA201A

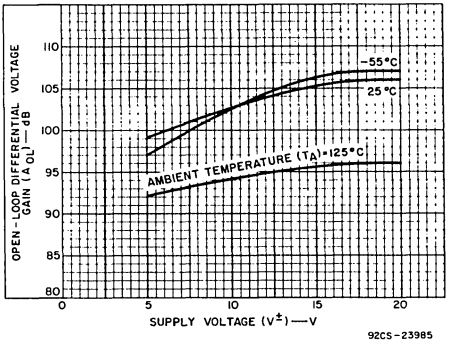


Fig.6 — Voltage gain vs. supply voltage.

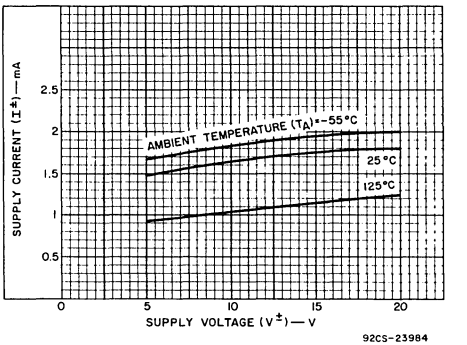


Fig.7 — Supply characteristics.

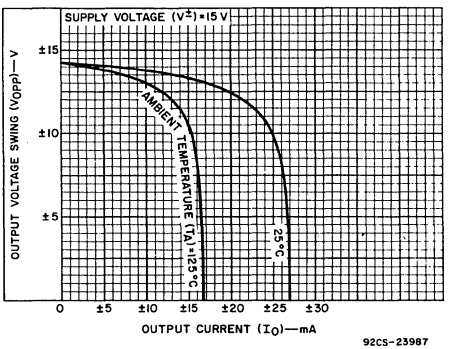


Fig.8 — Output characteristics.

TYPICAL STATIC CHARACTERISTICS (Cont'd)

Type CA201

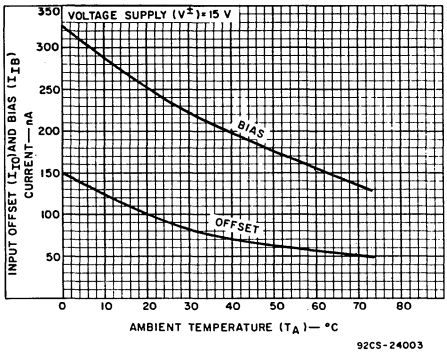


Fig.9 — Input current (I_{IO}, I_{IB}) vs. temperature.

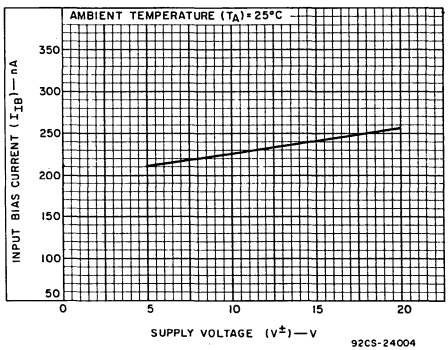


Fig.10 — Input bias current (I_{IB}) vs. supply voltage.

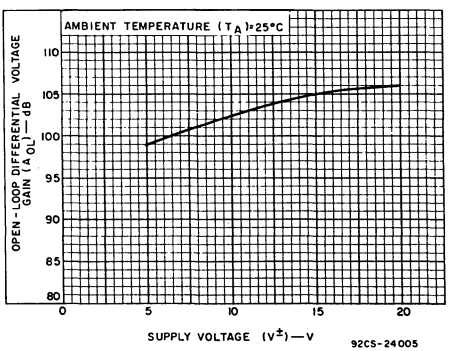


Fig.11 — Voltage gain vs. supply voltage.

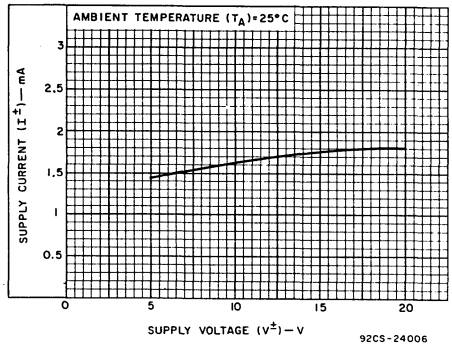


Fig.12 — Supply characteristics.

Type CA301A

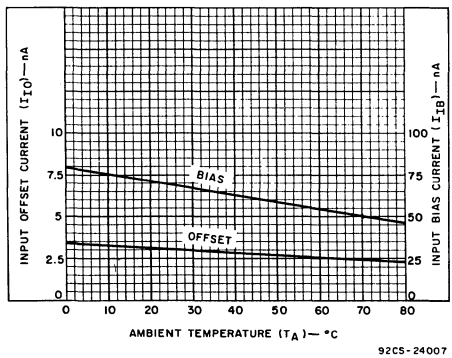


Fig.13 — Input current (I_{IO}, I_{IB}) vs. temperature.

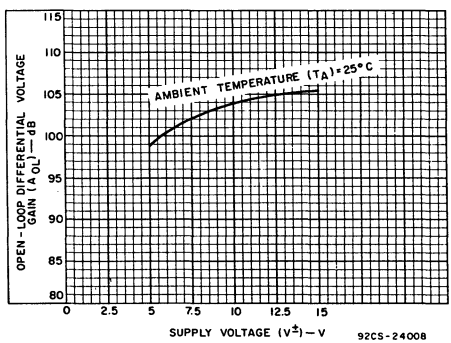


Fig.14 — Voltage gain vs. supply voltage.

TYPICAL STATIC CHARACTERISTICS (Cont'd)
Type CA301A

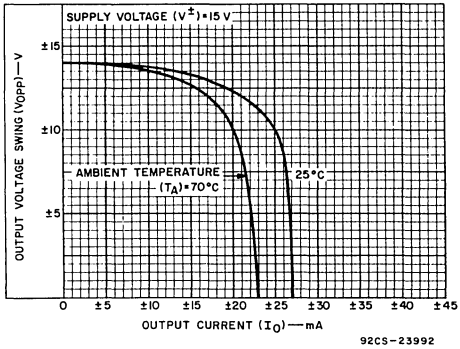


Fig.15 – Output characteristics.

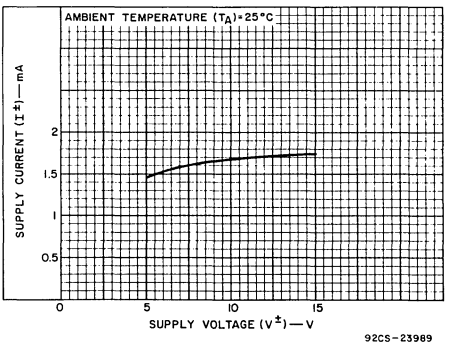


Fig.16 – Supply characteristics.

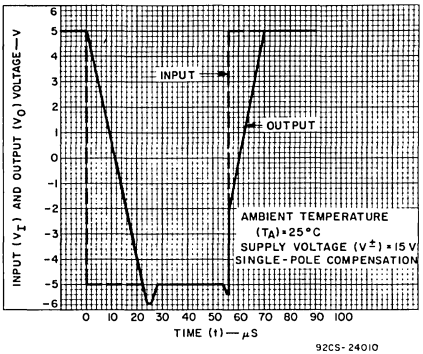


Fig.18 – Voltage follower (V_I, V_O) pulse response.

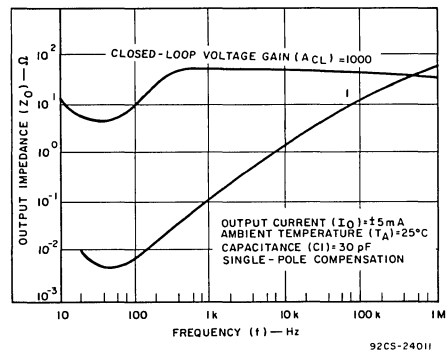


Fig.19 – Closed-loop output impedance vs. frequency.

TYPICAL DYNAMIC CHARACTERISTICS AND TEST CIRCUITS FOR TYPES CA101A AND CA201A
Single-Pole Compensation

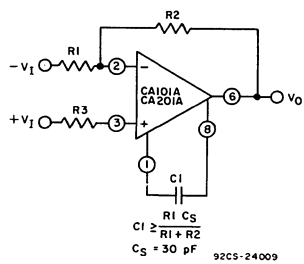


Fig.17 – Test circuit employing single-pole compensation.

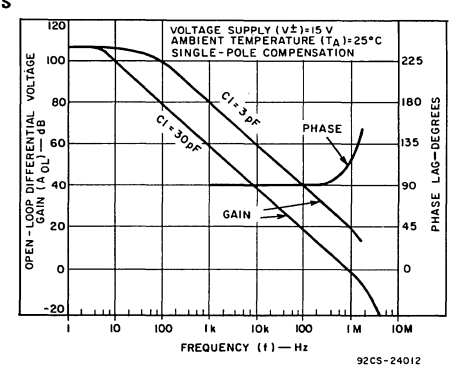


Fig.20 – Voltage gain and phase lag vs. frequency.

TYPICAL DYNAMIC CHARACTERISTICS AND TEST CIRCUITS FOR TYPES CA101A AND CA201A

Single-Pole Compensation (Cont'd)

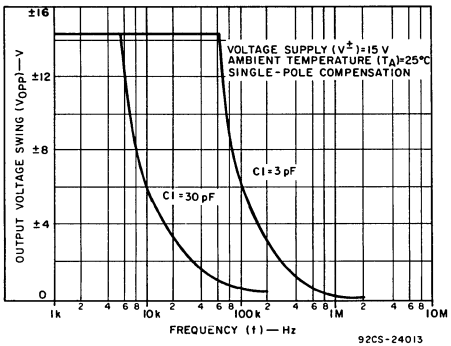


Fig.21 — Output voltage swing vs. frequency.

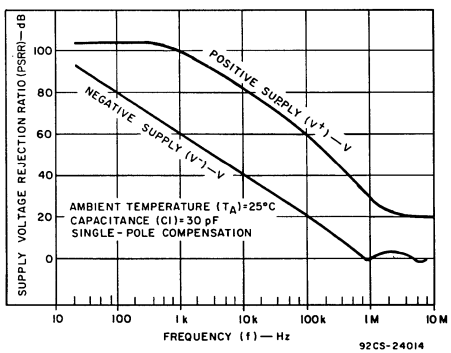


Fig.22 — Supply voltage rejection ratio vs. frequency.

FOR TYPES CA101A AND CA201A

Two-Pole Compensation

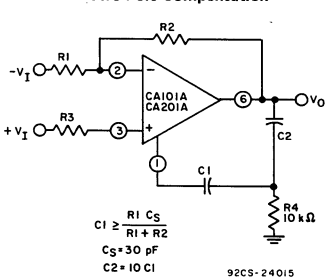


Fig.23 — Test circuit employing two-pole compensation.

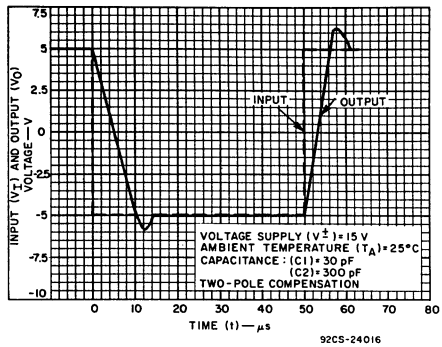


Fig.24 — Voltage follower pulse response.

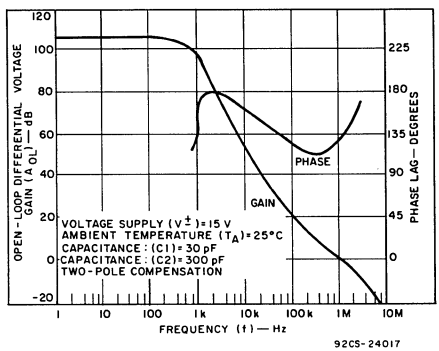


Fig.25 — Voltage gain and phase lag vs. frequency.

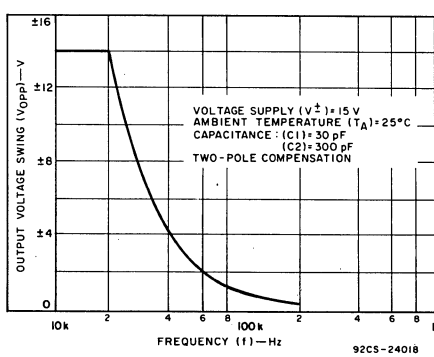


Fig.26 — Output voltage swing vs. frequency.

TYPICAL DYNAMIC CHARACTERISTICS AND TEST CIRCUITS (Cont'd) FOR TYPES CA101A AND CA201A

Feed-Forward Compensation

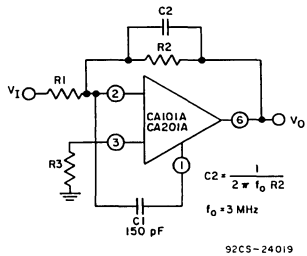


Fig.27 – Test circuit employing feedforward compensation.

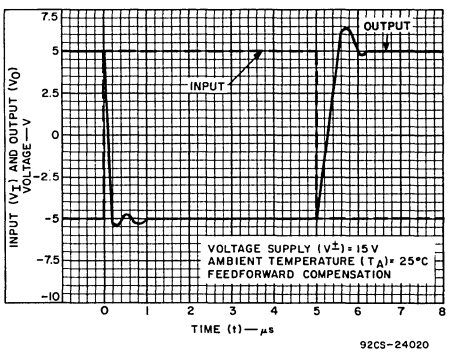


Fig.28 – Inverter pulse response.

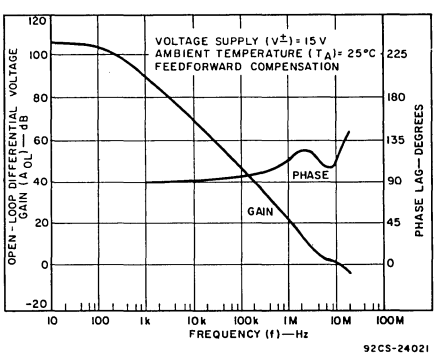


Fig.29 – Voltage gain and phase lag vs. frequency.

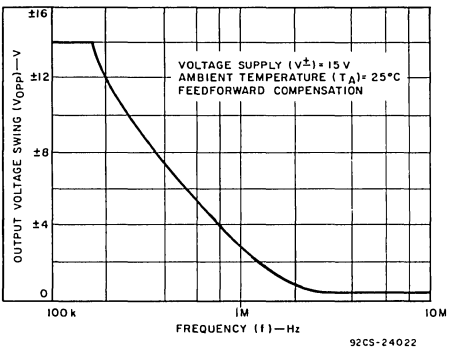


Fig.30 – Output voltage swing vs. frequency.

TYPICAL DYNAMIC CHARACTERISTICS FOR TYPES CA101A AND CA201A

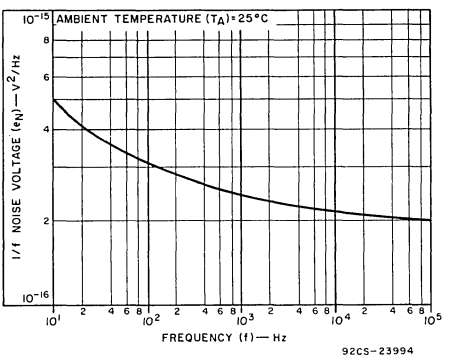


Fig.31 – 1/f noise voltage vs. frequency.

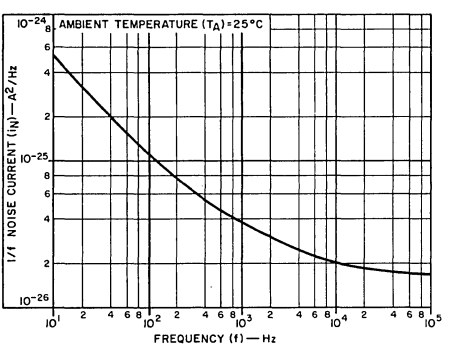


Fig.32 – 1/f noise current vs. frequency.

TYPICAL DYNAMIC CHARACTERISTICS (Cont'd)

FOR TYPES CA101A AND CA201A

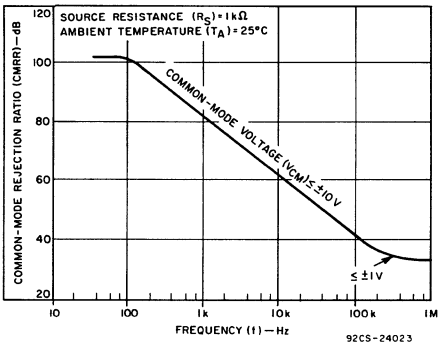


Fig.33 – Common-mode rejection ratio vs. frequency.

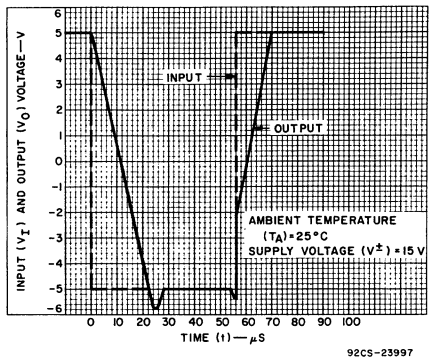


Fig.36 – Voltage follower pulse response.

Types CA101, CA201, CA301A

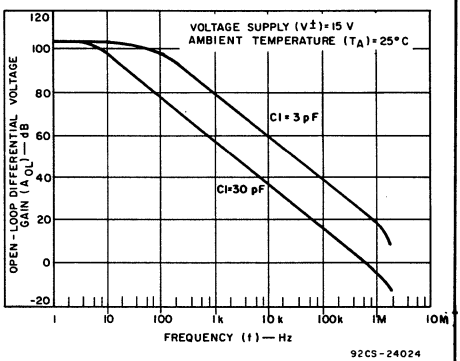


Fig.34 – Voltage gain vs. frequency.

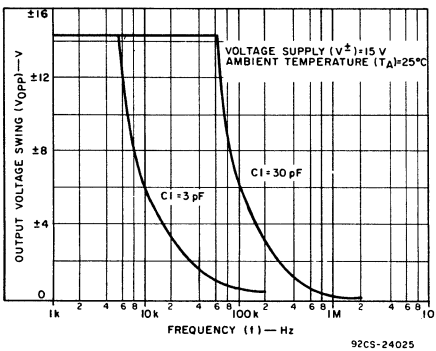


Fig.35 – Output voltage swing vs. frequency.

Type CA301A

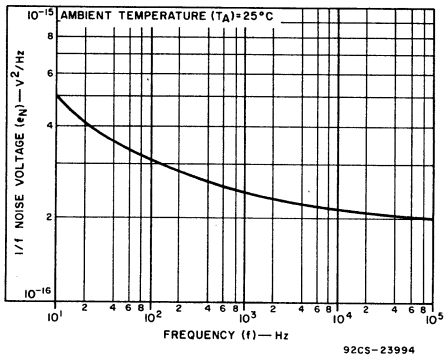


Fig.37 – I/f noise voltage vs. frequency.

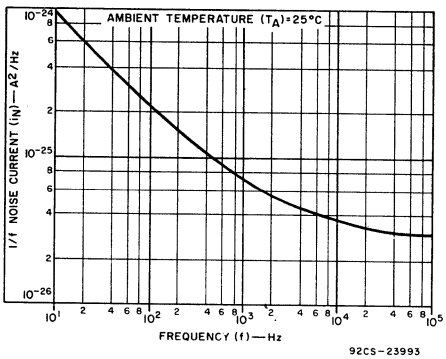


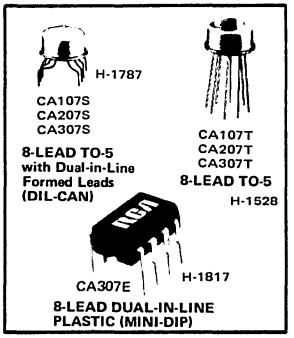
Fig.38 – I/f noise current vs. frequency.

RCA
Solid State
Division

Linear Integrated Circuits

Monolithic Silicon

CA107, CA207, CA307 Types



Operational Amplifiers

For Military, Industrial, and Commercial Applications

Type \ Feature	Max. V_{IO} (mV)	Max. I_{IO} (nA)	Max. I_{IB} (nA)	Temp. Range (T_A) °C
CA107	3	20	100	−55 to +125
CA207	3	20	100	−25 to +85
CA307	10	70	300	0 to +70

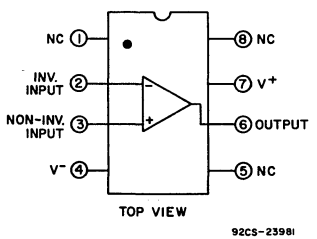
RCA-CA107, CA207, CA307 are general-purpose operational amplifiers intended for use in military, industrial, and commercial applications. A 30-pF on-chip capacitor provides internal frequency compensation. Low input current over temperature range (100 nA max.) for the CA107 and CA207 make these types especially well suited for applications such as long interval timers and sample-and-hold circuits.

The CA107, CA207, and CA307 are supplied in the standard 8-lead TO-5 style package ("T" suffix), the 8-lead TO-5 style with dual-in-line formed leads ("S" suffix), and in chip form ("H" suffix). The CA307 is also available in the 8-lead dual-in plastic "MINI-DIP" package ("E" suffix).

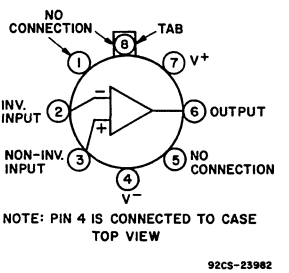
Applications:

- **Long-interval integrators**
- **Timers**
- **Sample-and-hold circuits**
- **Summing amplifiers**
- **Multivibrators**

The CA107T,S, CA207T,S, and CA307T,S,E are direct replacements for industry types 107, 207, and 307 in packages with similar terminal arrangements.



FUNCTIONAL DIAGRAM FOR PLASTIC PACKAGE.



FUNCTIONAL DIAGRAM FOR TO-5 STYLE PACKAGES.

Maximum Ratings, *Absolute-Maximum Values at T_A = 25°C:*

DC SUPPLY VOLTAGE (Between V^+ and V^- Terminals):	
CA107, CA207	44 V
CA307	36 V
DC INPUT VOLTAGE	±15 V
(For supply voltages less than ±15 V, the absolute maximum input voltage is equal to the supply voltage)	
DIFFERENTIAL INPUT VOLTAGE	±30 V
OUTPUT SHORT-CIRCUIT DURATION*	Indefinite
DEVICE DISSIPATION UP TO T_A = 70°C	500 mW
Above T_A = 70°C Derate linearly at	6.67 mW/°C
AMBIENT TEMPERATURE RANGE:	
Operating — CA107	−55°C to +125°C
CA207	−25°C to +85°C
CA307	0°C to +70°C
Storage — All Types	−65°C to +150°C
LEAD TEMPERATURE (During Soldering):	
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max.	+265°C

* For CA307 continuous short circuit is allowed for Case Temperature to +70°C and ambient temperature to +55°C.

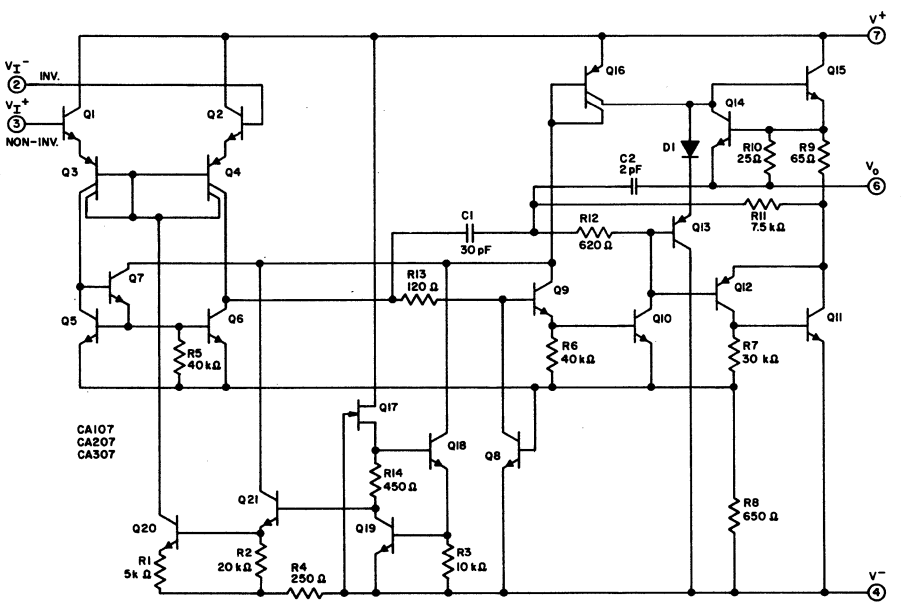


Fig. 1 — Schematic diagram of CA107, CA207, and CA307.

ELECTRICAL CHARACTERISTICS

CHARACTERISTIC	SYMBOL	TEST CONDITIONS▲ Supply Voltage ($V^{\pm}$) = 5 V to 15 V	LIMITS						UNITS
			CA107 CA207			CA307			
			Min.	Typ.	Max.	Min.	Typ.	Max.	
Input Offset Voltage	V_{IO}	$T_A = 25°C$, $R_S \leq 50\ k\Omega$	–	0.7	2	–	2	7.5	mV
		$R_S \leq 50\ k\Omega$	–	–	3	–	–	10	
Average Temperature Coefficient of Input Offset Voltage	αV_{IO}		–	3	15	–	6	30	$\mu V/°C$
Input Offset Current	I_{IO}		–	–	20	–	–	70	nA
		$T_A = 25°C$	–	1.5	10	–	3	50	
Average Temperature Coefficient of Input Offset Current	αI_{IO}	See Note 1	–	0.01	0.1	–	0.01	0.3	nA/°C
		See Note 2	–	0.02	0.2	–	0.02	0.6	
Input Bias Current	I_{IB}		–	–	100	–	–	300	nA
		$T_A = 25°C$	–	30	75	–	70	250	
Supply Current	$I^{\pm}$	$T_A = +125°C$, $V^{\pm} = 20\ V$	–	1.2	2.5	–	–	–	mA
		$T_A = 25°C$, $V^{\pm} = 20\ V$, (CA307 $V^{\pm} = 15\ V$)	–	1.8	3	–	1.8	3	
Open-Loop Differential Voltage Gain	A_{OL}	$V^{\pm} = 15\ V$, $V_O = \pm 10\ V$, $R_L \geq 2\ k\Omega$	25	–	–	15	–	–	V/mV
		$V^{\pm} = 15\ V$, $V_O = \pm 10V$ $R_L \geq 2\ k\Omega$, $T_A = 25°C$	50	160	–	25	160	–	
Input Resistance	R_I	$T_A = 25°C$	1.5	4	–	0.5	2	–	MΩ
Output Voltage Swing	V_{OPP}	$V^{\pm} = 15\ V$, $R_L = 10\ k\Omega$	±12	±14	–	±12	±14	–	V
		$V^{\pm} = 15\ V$, $R_L = 2\ k\Omega$	±10	±13	–	±10	±13	–	
Input Voltage Range	V_{ICR}	$V^{\pm} = 20\ V$, (CA307 $V^{\pm} = 15\ V$)	±15	–	–	±12	–	–	V
Common-Mode Rejection Ratio	CMRR	$R_S \leq 50\ k\Omega$	80	96	–	70	90	–	dB
Supply-Voltage Rejection Ratio	PSRR	$R_S \leq 50\ k\Omega$	80	96	–	70	96	–	dB

Note 1: For CA107, +25 to +125°C; For CA207, +25 to +85°C: For CA307, +25 to 70°C.

Note 2: For CA107, −55 to +25°C; For CA207, −25 to +25°C: For CA307, 0 to +25°C.

▲ Characteristics applicable over operating temperature range as shown below unless otherwise specified.

CA107 – T_A = −55 to +125°C

CA207 – T_A = −25 to +85°C

CA307 – T_A = 0 to 70°C

TYPICAL CHARACTERISTICS FOR CA107, CA207

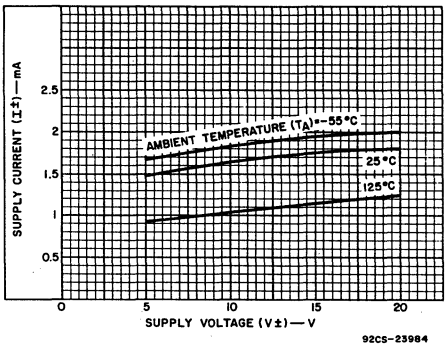


Fig. 2 – Supply current vs. supply voltage.

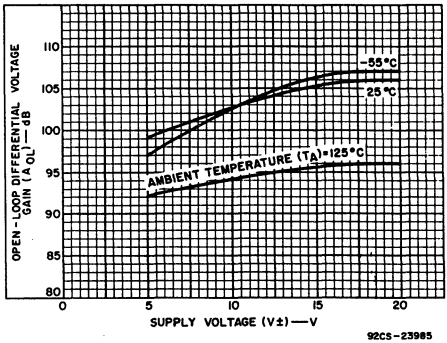


Fig. 3 – Open-loop differential voltage gain vs. supply voltage.

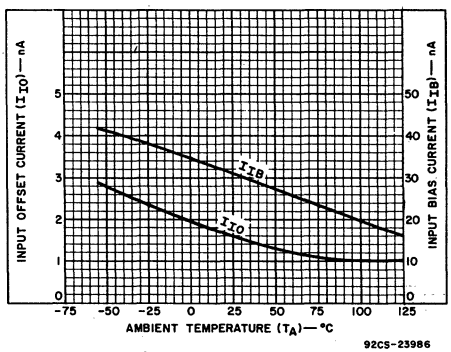


Fig. 4 – Input offset and input bias current vs. ambient temperature.

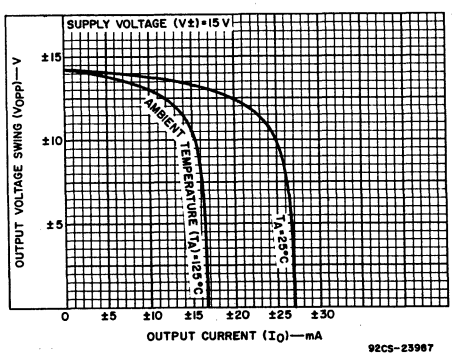


Fig. 5 – Output voltage swing vs. output current.

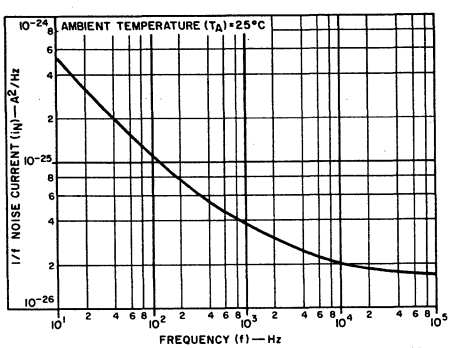


Fig. 6 – 1/f noise current vs. frequency.

TYPICAL CHARACTERISTICS FOR CA307

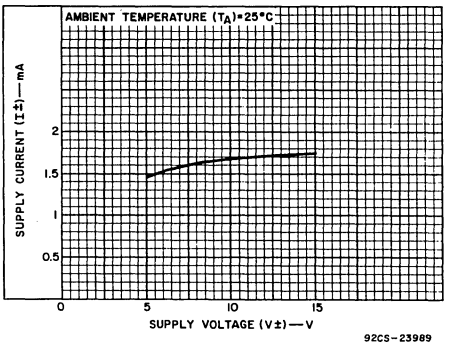


Fig. 7 – Supply current vs. supply voltage.

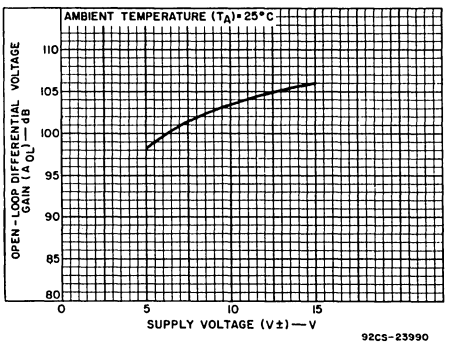


Fig. 8 – Open-loop differential voltage gain vs. supply voltage.

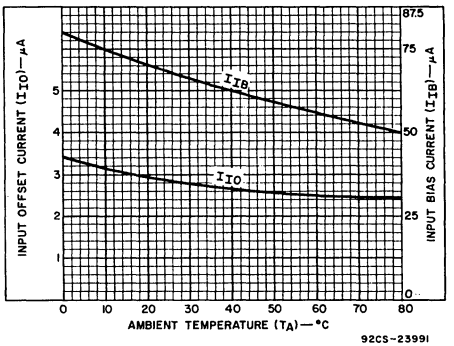


Fig. 9 – Input offset and input bias current vs. ambient temperature.

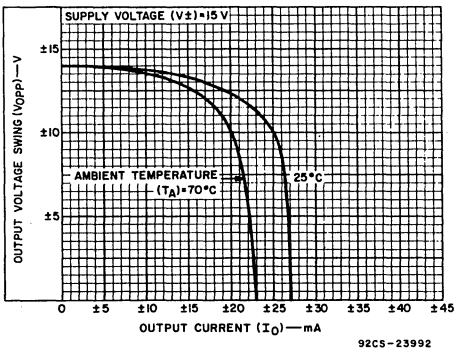


Fig. 10 – Output voltage swing vs. output current.

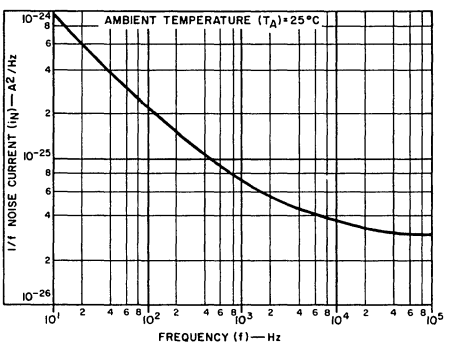


Fig. 11 – 1/f noise current vs. frequency.

TYPICAL CHARACTERISTICS FOR ALL TYPES

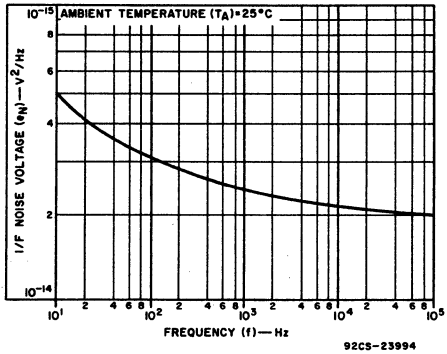


Fig. 12 – 1/f noise voltage vs. frequency.

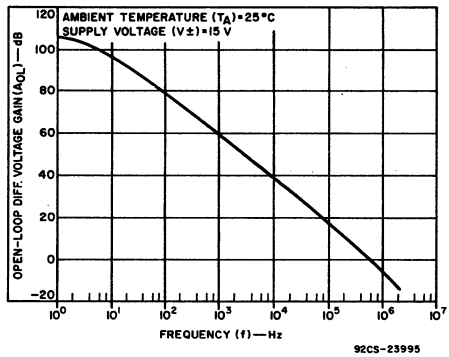


Fig. 13 – Open-loop differential voltage gain vs. frequency.

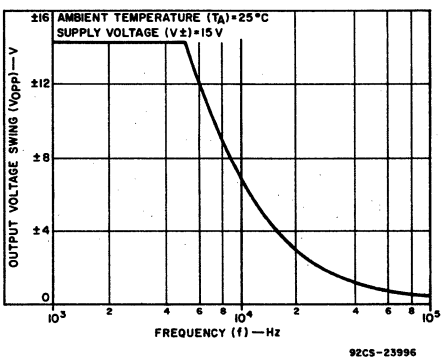


Fig. 14 – Output voltage swing vs. frequency.

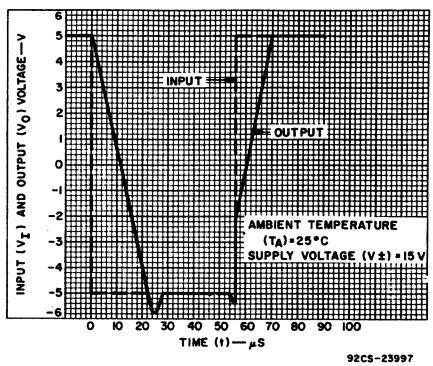


Fig. 15 – Input and output voltage vs. time.

RCA
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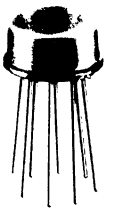
CA108T	CA108S	CA108AT	CA108AS
CA208T	CA208S	CA208AT	CA208AS
CA308T	CA308S	CA308AT	CA308AS



CA108S, AS
CA208S, AS
CA308S, AS

8-LEAD TO-5 with Dual-In-Line Formed Leads

H-1787



CA108T, AT
CA208T, AT
CA308T, AT

8-LEAD TO-5

H-1528

Precision Operational Amplifiers

For Military, Industrial, and Commercial Applications

Features:

- Maximum input bias current — 2 nA for CA108 & CA208 series
  7 nA for CA308 series
- Maximum input offset current — 0.2 nA for CA108 & CA208 series
  1 nA for CA308 series
- Supply current of only 300 μA, even in saturation
- Maximum input offset voltage of 0.5 mV for "A" suffix types

RCA-CA108T, CA108AT, CA108S, CA108AS, CA208T, CA208AT, CA208S, CA208AS, CA308T, CA308AT, CA308S, and CA308AS are uncompensated precision operational amplifiers using super-beta transistors and feature very low offset parameters, high input impedance, and defined drift rates with temperature change.

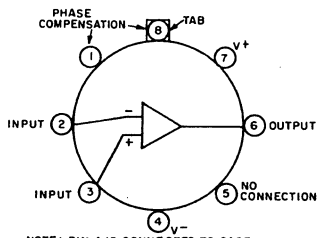
In addition to low drift, these super-beta op-amps have input currents sufficiently low to insure low drift, even when using high source resistances, e.g., 10 megohms.

These devices have sufficient supply rejection to operate from unregulated power supplies within a range of ±2 V to ±20 V, and the input bias current is specifically controlled for use in sample-and-hold applications.

The "A" versions have all the desirable features and characteristics of their prototypes plus exceptionally low input offset voltage characteristics. The CA108, CA108A, CA208, CA208A, CA308, and CA308A are direct replacements for industry types 108,108A,208,208A,308,308A, and they are supplied in either standard 8-lead TO-5 packages or in 8-lead TO-5 packages with dual-in-line formed leads ("DIL-CAN").

Applications:

- Instrumentation
- Summing amplifier
- Comparator
- Multivibrators
- Band-pass filters
- Sample and hold

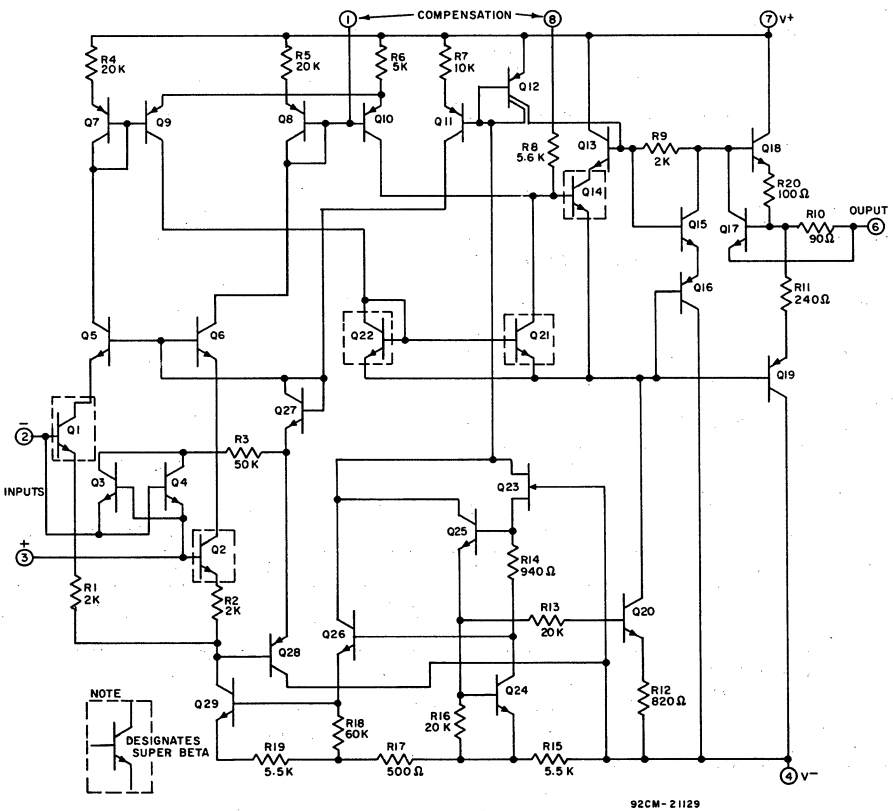


FUNCTIONAL DIAGRAM

ELECTRICAL CHARACTERISTICS, MAXIMUM VALUES AT T_A = 25°C	CA108T CA108S	CA108AT CA108AS	CA208T CA208S	CA208AT CA208AS	CA308T CA308S	CA308AT CA308AS
Input Offset Voltage (V_{IO})	2 mV	0.5 mV	2 mV	0.5 mV	7.5 mV	0.5 mV
Input Offset Current (I_{IO})	0.2 nA				1 nA	
Input Bias Current (I_{IB})	2 nA			7 nA		
Average Temperature Coefficient of Input Offset Voltage ($\Delta V_{IO}/\Delta T$)	15 μV/°C	5 μV/°C	15 μV/°C	5 μV/°C	30 μV/°C	5 μV/°C
Ambient Operating-Temperature Range	−55 to +125°C		−25 to +85°C		0 to +70°C	

Maximum Ratings, *Absolute-Maximum Values at T_A = 25°C:*

DC Supply Voltage (Between V^+ and V^- Terminals):	
CA108, CA108A, CA208, CA208A	40 V
CA308, CA308A	36 V
DC Input Voltage	±15 V
(For supply voltages less than ± 15 V, the absolute maximum input voltage is equal to the supply voltage)	
Differential Input Current	±10 mA
Output Short-Circuit Duration	Indefinite
Device Dissipation	500 mW
Ambient Temperature Range:	
Operating – CA108, CA108A	−55°C to +125°C
CA208, CA208A	−25°C to +85°C
CA308, CA308A	0°C to +70°C
Storage – All Types	−65°C to +150°C
Lead Temperature (During Soldering):	
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max.	+300 °C



ELECTRICAL CHARACTERISTICS

CHARACTERISTICS	SYMBOL	TEST CONDITIONS FIG. No.	TEST CONDITIONS Supply Voltage (V) = ±5 V to ±15 V	LIMITS CA108 CA208 Min.	CA108 CA208 Typ.	CA108 CA208 Max.	CA108A CA208A Min.	CA108A CA208A Typ.	CA108A CA208A Max.	CA308 Min.	CA308 Typ.	CA308 Max.	CA308A Min.	CA308A Typ.	CA308A Max.	UNITS
Input Offset Voltage	V_{IO}	6,7	T_A = 25°C	–	0.7	2	–	0.3	0.5	–	2	7.5	–	0.3	0.5	mV
			Note 1	–	–	3	–	–	1	–	–	10	–	–	0.73	
Average Temperature Coefficient of Input Offset Voltage	$\frac{\Delta V_{IO}}{\Delta T}$		Note 1	–	3	15	–	1	5	–	6	30	–	1	5	µV/°C
Input Offset Current	I_{IO}	8,9	Note 1	–	–	0.4	–	–	0.4	–	–	1.5	–	–	1.5	nA
			T_A = 25°C	–	0.05	0.2	–	0.05	0.2	–	0.2	1	–	0.2	1	
Average Temperature Coefficient of Input Offset Current	$\frac{\Delta I_{IO}}{\Delta T}$		Note 1	–	0.5	2.5	–	0.5	2.5	–	2	10	–	2	10	pA/°C
Input Bias Current	I_{IB}	10, 11	Note 1	–	–	3	–	–	3	–	–	10	–	–	10	nA
			T_A = 25°C	–	0.8	2	–	0.8	2	–	1.5	7	–	1.5	7	
Supply Current	I_O	12, 13	T_A = +125°C	–	0.15	0.4	–	0.15	0.4	–	–	–	–	–	–	mA
			T_A = 25°C	–	0.3	0.6	–	0.3	0.6	–	0.3	0.8	–	0.3	0.8	
Large-Signal Voltage Gain	A_V	2,14 15	V = ±15 V, T_A = 25°C V_O = ±10 V, $R_L \geq 10$ kΩ	50	300	–	80	300	–	25	300	–	80	300	–	V/mV
			V = ±15 V, V_O = ±10 V $R_L \geq 10$ kΩ, Note 1	25	–	–	40	–	–	15	–	–	60	–	–	
Input Resistance	R_I		T_A = 25°C	30	70	–	30	70	–	10	40	–	10	40	–	MΩ
Output Voltage	V_O	16, 17	V = ±15 V, R_L = 10 kΩ, Note 1	±13	±14	–	±13	±14	–	±13	±14	–	±13	±14	–	V
Input Voltage Range	V_I		V = ±15 V, Note 1	±13.5	–	–	±13.5	–	–	±14	–	–	±14	–	–	
Common-Mode Rejection Ratio	CMRR		Note 1	85	100	–	96	110	–	80	100	–	96	110	–	dB
Supply-Voltage Rejection Ratio	V_{RR}		Note 1	80	96	–	96	110	–	80	96	–	96	110	–	

Note 1: Ambient Temperature (T_A) over applicable operating temperature range as shown below unless otherwise specified.

CA108 CA108A	CA208 CA208A	CA308 CA308A
−55 to +125°C	−25 to +85°C	0 to +70°C

TEST CIRCUITS

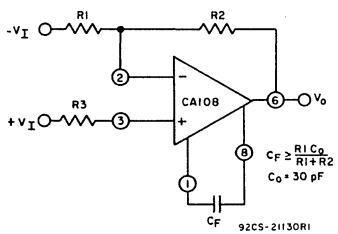


Fig. 2—Standard frequency-compensation.

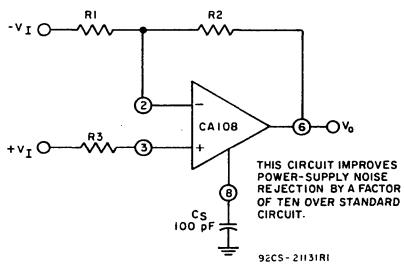


Fig. 3—Alternate frequency-compensation.

TYPICAL APPLICATIONS

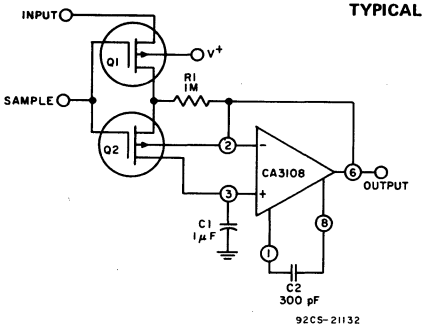


Fig. 4 – Sample-and-hold circuit.

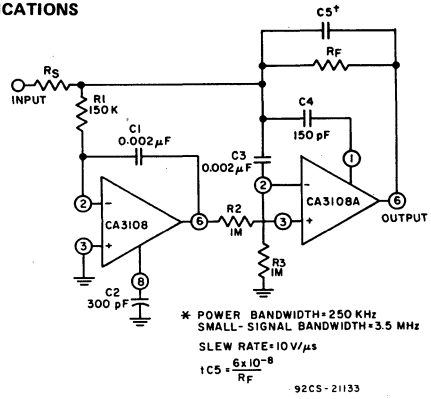


Fig. 5 – Fast summing amplifier circuit.*

CHARACTERISTIC CURVES

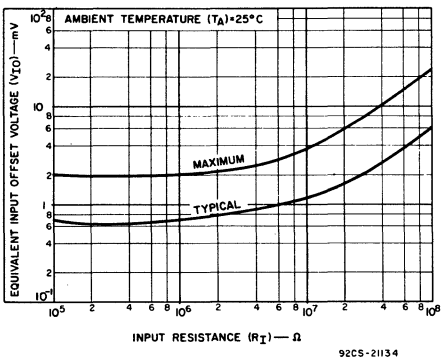


Fig. 6 – Input offset error for CA108, CA108A, CA208, and CA208A.

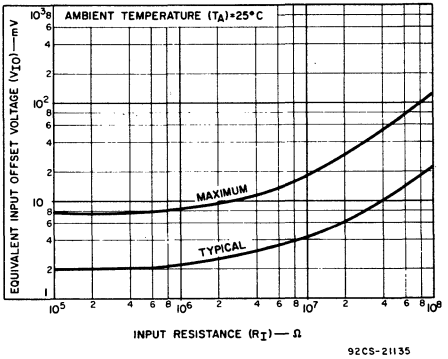


Fig. 7 – Input offset error for CA308 and CA308A.

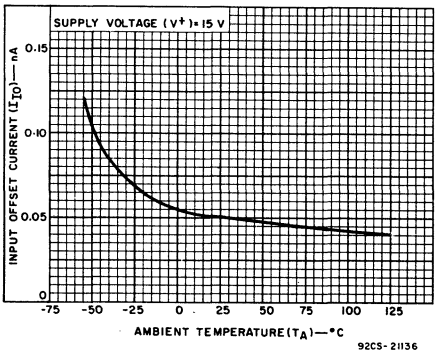


Fig. 8 – Input offset current vs. temperature for CA108, CA108A, CA208, and CA208A.

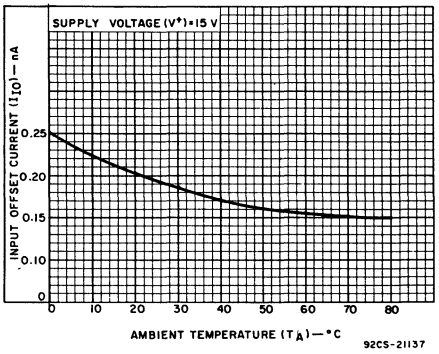


Fig. 9 – Input offset current vs. temperature for CA308 and CA308A.

CHARACTERISTIC CURVES (Cont'd)

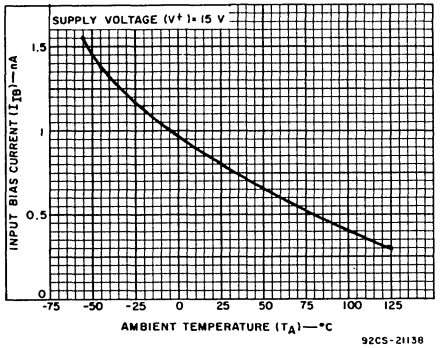


Fig. 10 — Input bias current vs. temperature for CA108, CA108A, CA208, and CA208A.

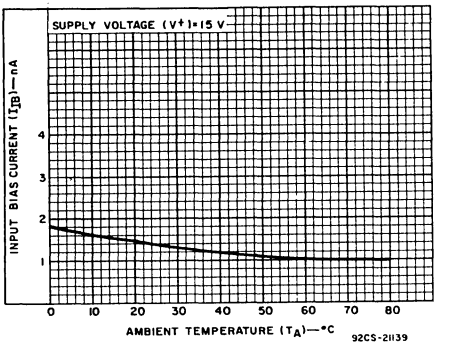


Fig. 11 — Input bias current vs. temperature for CA308 and CA308A.

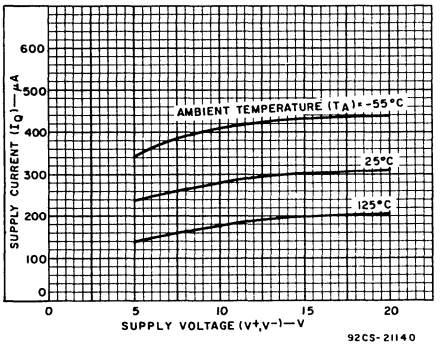


Fig. 12 — Supply current vs. supply voltage for CA108, CA108A, CA208, and CA208A.

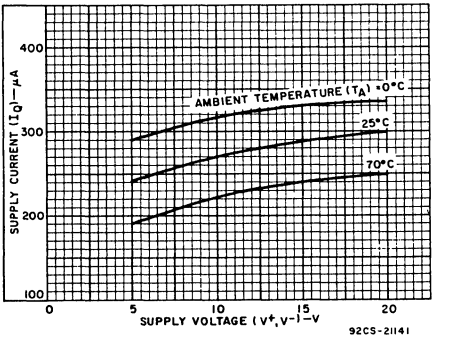


Fig. 13 — Supply current vs. supply voltage for CA308 CA308A.

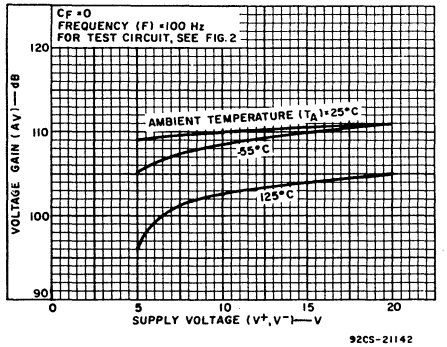


Fig. 14 — Voltage gain vs. supply voltage for CA108, CA108A, CA208, and CA208A.

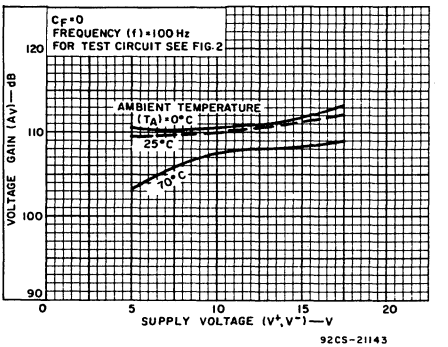


Fig. 15 — Voltage gain vs. supply voltage for CA308 and CA308A.

CHARACTERISTIC CURVES (Cont'd)

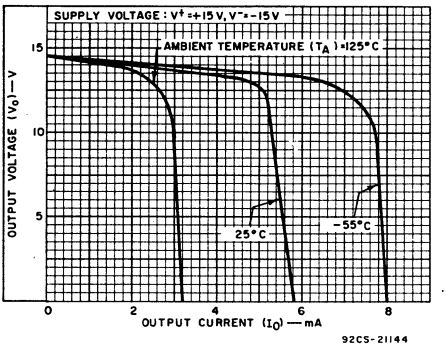


Fig. 16 – Output voltage vs. output current for CA108, CA108A, CA208, and CA208A.

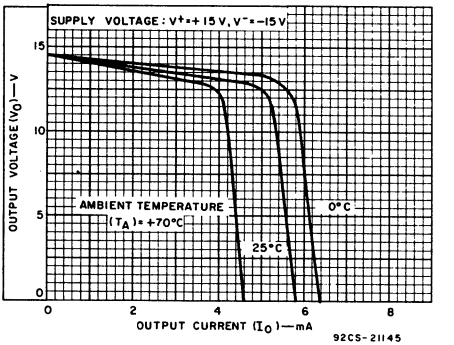


Fig. 17 – Output voltage vs. output current for CA308 and CA308A.

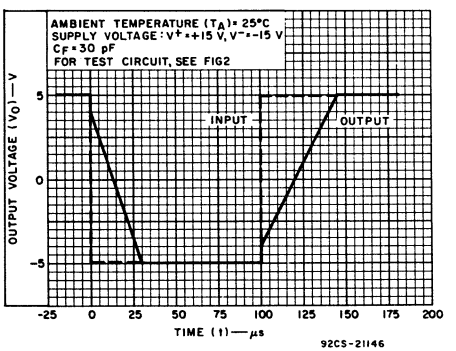


Fig. 18 – Voltage-follower pulse response for all types.

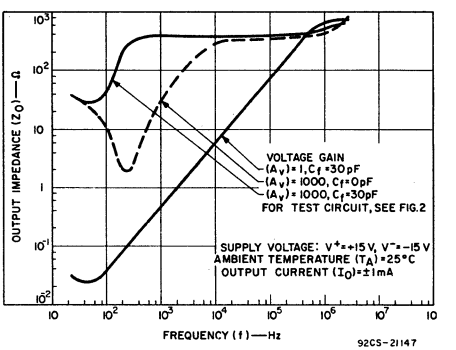


Fig. 19 – Closed-loop output impedance for all types.

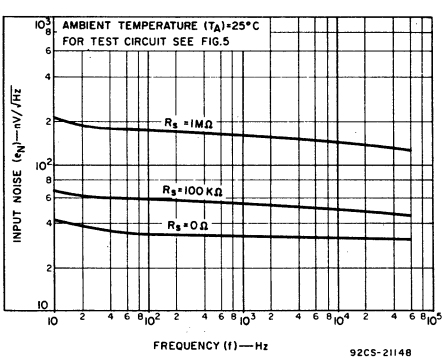


Fig. 20 – Input noise voltage for all types.

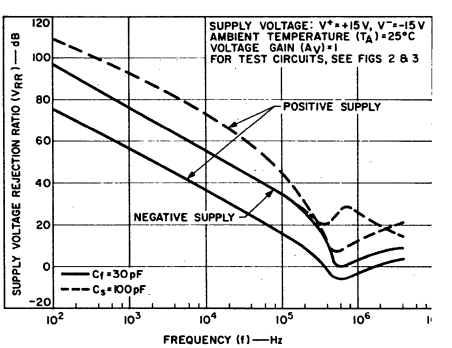


Fig. 21 – Power-supply rejection for all types.

CHARACTERISTIC CURVES (Cont'd)

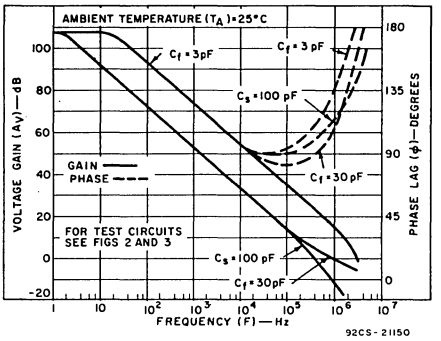


Fig. 22 – Open-loop frequency response for all types.

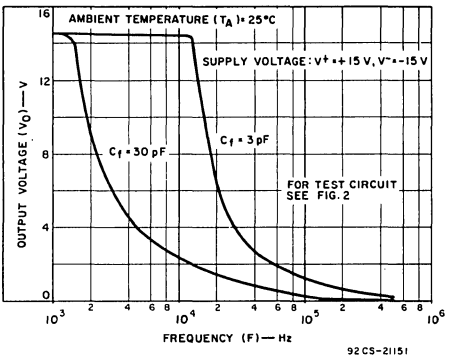


Fig. 23 – Large-signal frequency response for all types.

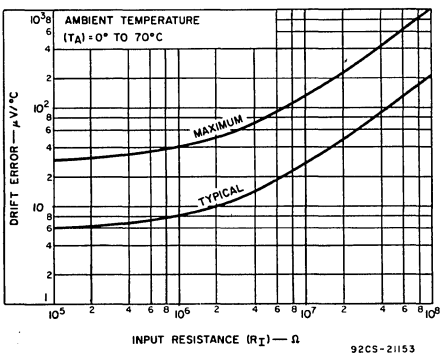


Fig. 24 – Drift error vs. input resistance for CA108, CA108A, CA208, and CA208A.

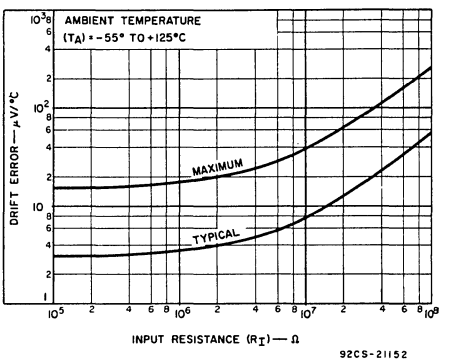


Fig. 25 – Drift error vs. input resistance for CA308 and CA308A.

RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA111, CA211, CA311 Types



("S" SUFFIX)

8-LEAD TO-5 with Dual-In-Line Formed Leads "DIL-CAN"

H-1787



("T" SUFFIX)

8-LEAD TO-5

H-1528

Voltage Comparators

For Commercial and Industrial Applications

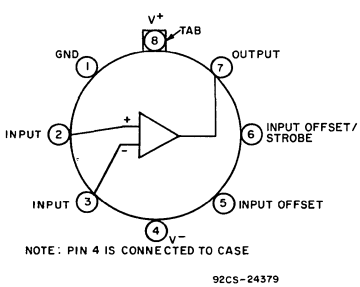
Features

- **Single- or dual-supply operation**
- **Power consumption — 135 mW at ±15 V**
- **Strobe capability**
- **Low input-offset current:**
  CA111, CA211 — 4 nA (typ.)
  CA311 — 6 nA (typ.)
- **Differential input-voltage range — ±30 V**
- **Directly interchangeable with National Semiconductor types LM111H, LM211H, and LM311**

The RCA-CA111, CA211, and CA311 are monolithic voltage comparators that operate from dual supplies up to ±15 V, or from single supplies down to 5 V. This single-supply capability makes the outputs of these devices compatible with RTL, DTL, TTL, and MOS circuits. In addition, they can drive lamps or relays, and switch voltages up to 50 V (CA311, 40 V) at currents as high as 50 mA.

The inputs and the outputs of the CA111, CA211, and CA311 can be isolated from system ground, allowing the output to drive loads referred to ground, V^+, or V^-.

The CA111, CA211, and CA311 are supplied in the 8-lead TO-5 style package ("T" suffix), and the "DIL-CAN", an 8-lead TO-5 style package with dual-in-line formed leads ("S" suffix). The CA311 is also supplied in chip form ("H" suffix). These types are direct replacements for industry types 111, 211, and 311 in similar packages.



FUNCTIONAL DIAGRAM

Applications

- **Multivibrators**
- **Positive and negative peak detectors**
- **Crystal oscillators**
- **Zero-crossing detectors**
- **Solenoid, relay, and lamp drivers**

MAXIMUM RATINGS, *Absolute Maximum Values at $T_A = 25^\circ C$*

DC SUPPLY VOLTAGE (between V^+ and V^- terminals)	36 V
DC INPUT VOLTAGE*	±15 V
DIFFERENTIAL INPUT VOLTAGE	±30 V
OUTPUT TO NEGATIVE SUPPLY VOLTAGE (V_{7-4}):	
CA111, CA211	50 V
CA311	40 V
GROUND TO NEGATIVE SUPPLY VOLTAGE (V_{1-4})	30 V
OUTPUT SHORT-CIRCUIT DURATION	10 s
DEVICE DISSIPATION:	
Up to $T_A = 25^\circ C$	500 mW
Above $T_A = 25^\circ C$	derate linearly at 6.67 mW/°C
AMBIENT TEMPERATURE RANGE:	
Operating:	
CA111	−55 to +125°C
CA211	−25 to +85°C
CA311	0 to +70°C
Storage, all types	−65 to +150°C
LEAD TEMPERATURE (DURING SOLDERING):	
At distance 1/16 ± 1/32 in. (1.59 ± 0.79 mm) from case for 10 seconds max.	+265°C

*This rating applies for ±15 V supplies. The positive input-voltage limit is 30 V above the negative supply. The negative input-voltage limit is equal to the negative supply voltage or 30 V below the positive supply. The negative input-voltage limit is equal to the negative supply voltage or 30 V below the positive supply, whichever is less.

ELECTRICAL CHARACTERISTICS

CHARACTERISTICS	SYMBOL	TEST CONDITIONS FIG. NO.	SUPPLY VOLTAGE ($V^{\pm}$) = 15 V UNLESS OTHERWISE SPECIFIED		LIMITS CA111 CA211 TYP.	CA111 CA211 MAX.	CA311 TYP.	CA311 MAX.	UNITS
Input Offset Voltage	V_{IO}	16,24	$R_s \leq 5$ kΩ, Note 2	T_A=25°C	0.7	3	2	7.5	mV
				Note 1	–	4	–	10	
Saturation Voltage		13,22.	$V_I = -5$ mV, $I_o = 50$ mA (For CA311, $V_I \leq -10$ mV)	T_A=25°C	0.75	1.5	–	–	V
			$V^+ \geq 4.5$ V, $V^- = 0$, $V_I \leq -6$ mV, $I_{SINK} \leq 8$ mA (For CA311, $V_I \leq -10$ mV)	Note 1	0.23	0.4	–	–	
Input Voltage Range	V_{IPP}			Note 1	±14	–	±14	–	V
Input Offset Current	I_{IO}	9,18	Note 2	T_A=25°C	4	10	6	50	nA
				Note 1	–	20	–	70	
Input Bias Current	I_{IB}	8,17	Note 2	T_A=25°C	60	100	100	250	nA
				Note 1	–	150	–	300	
Positive Supply Current	I^+			T_A=25°C	5.1	6	5.1	7.5	mA
Negative Supply Current	I^-			T_A=25°C	4.1	5	4.1	5	mA
Output Leakage Current		15,24	$V_I \geq 5$ mV, $V_o = 35$ V (For CA311, $V_I \geq -10$ mV)	T_A=25°C	0.2	10	–	–	nA
				Note 1	0.1	0.5	–	–	μA
Strobe On Current				T_A=25°C	3	–	3	–	mA
Voltage Gain	A			T_A=25°C	200	–	200	–	V/mV
Response Time		2,3, 4,5	100 mV Input Step with 5 mV overdrive voltage	T_A=25°C	200	–	200	–	ns

Note 1: Ambient temperature (T_A) over applicable operating temperature range as shown below.

CA111	CA211	CA311
−55 to +125°C	−25 to +85°C	0 to +70°C

Note 2: The input offset characteristics given are the values required to drive the output to within 1 V of either supply with a 1-mA load. These characteristics define an error band which takes into account the worst-case effects of voltage gain and input impedance. The input offset voltage, input offset current, and input bias current specifications apply for any supply voltage from a 5 V single supply up to a ±15 V dual supply.

TYPICAL CHARACTERISTICS – ALL TYPES

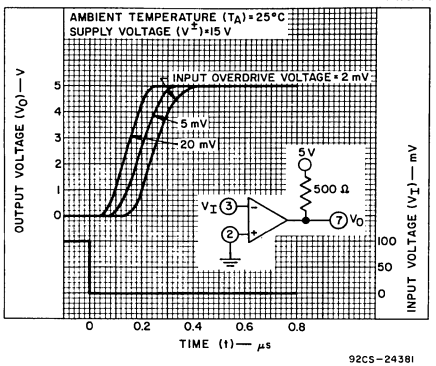


Fig. 1—Response time for various input overdrive voltages—positive input.

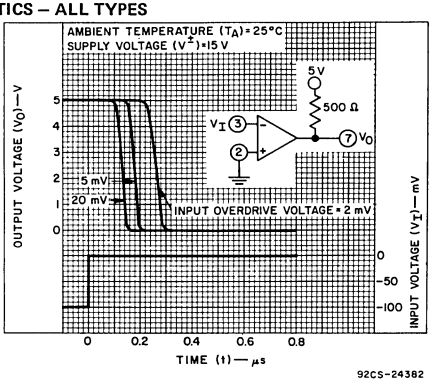


Fig. 2—Response time for various input overdrive voltages—negative input.

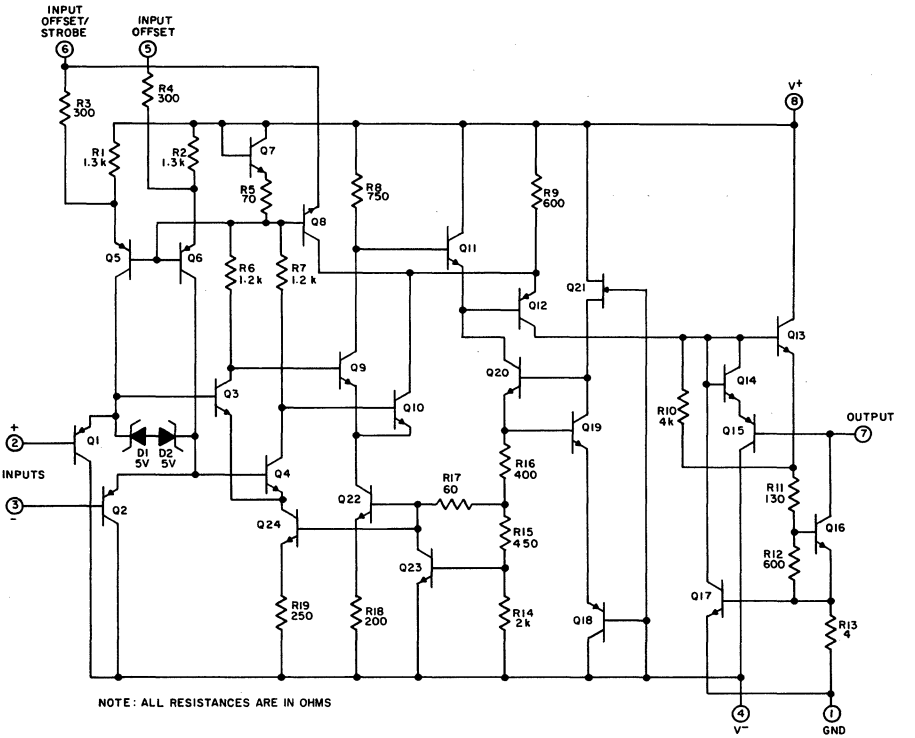


Fig. 3—Schematic diagram for CA111, CA211, and CA311. 92CM-24380

TYPICAL CHARACTERISTICS – ALL TYPES (CONT'D)

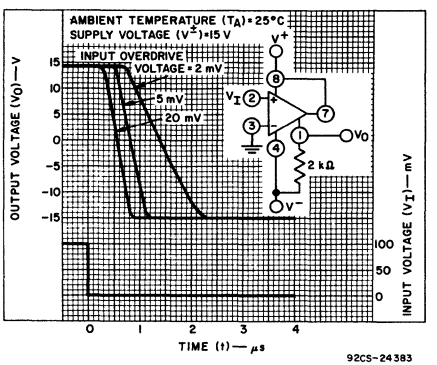


Fig. 4—Response time for various input overdrive voltages—positive input.

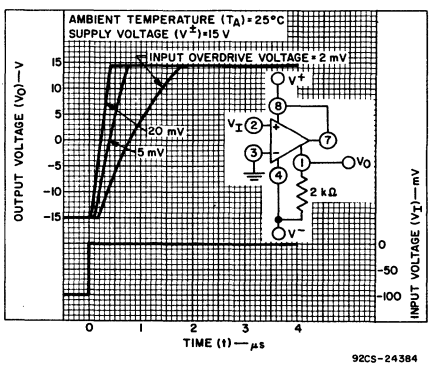


Fig. 5—Response time for various input overdrive voltages—negative input.

TYPICAL CHARACTERISTICS – ALL TYPES (CONT'D)

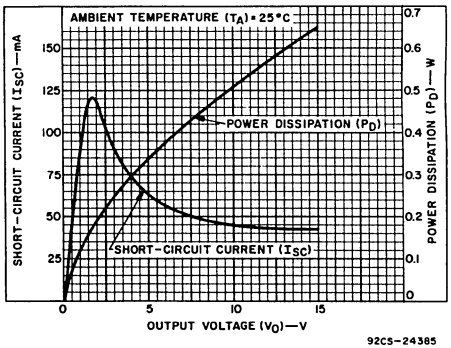


Fig. 6—Ouptut limiting characteristics.

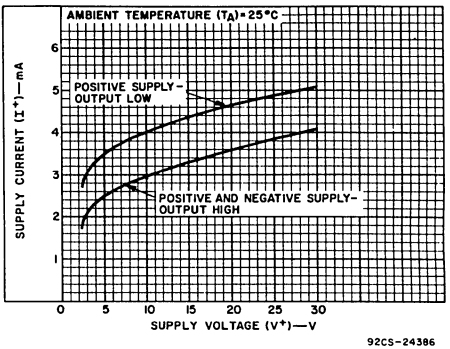


Fig. 7—Supply current vs. supply voltage.

TYPICAL CHARACTERISTICS – CA111, CA211

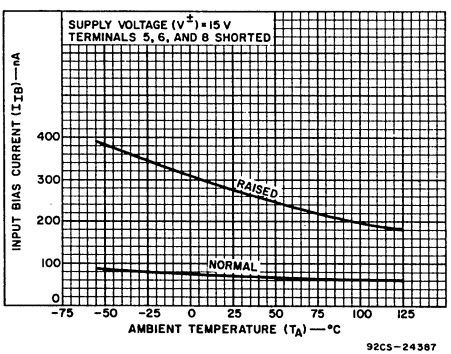


Fig. 8—Input bias current vs. ambient temperature.

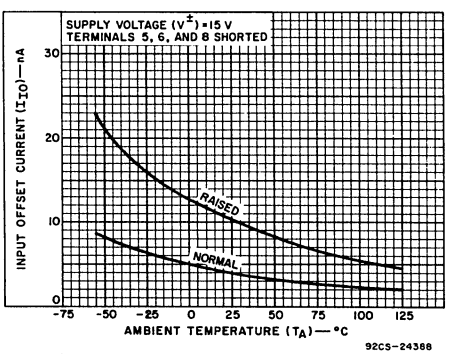


Fig. 9—Input offset current vs. ambient temperature.

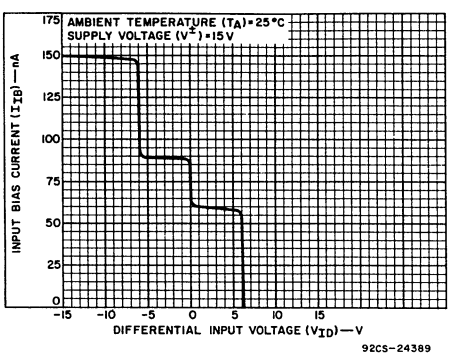


Fig. 10—Input characteristics.

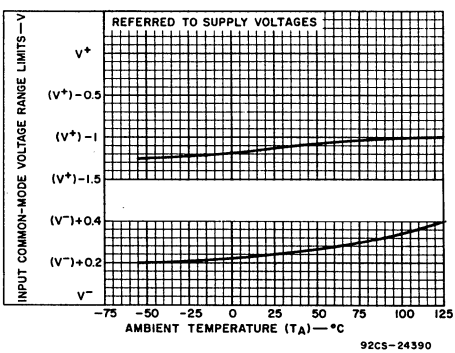


Fig. 11—Common-mode voltage range limits vs. ambient temperature.

TYPICAL CHARACTERISTICS – CA111, CA211 (CONT'D)

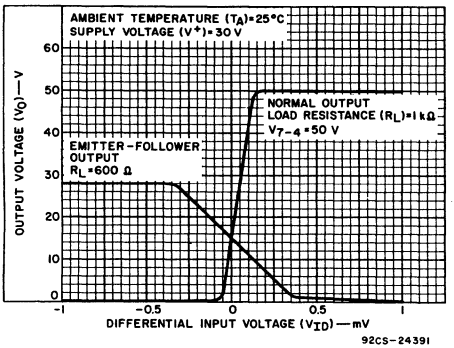


Fig. 12—Transfer function.

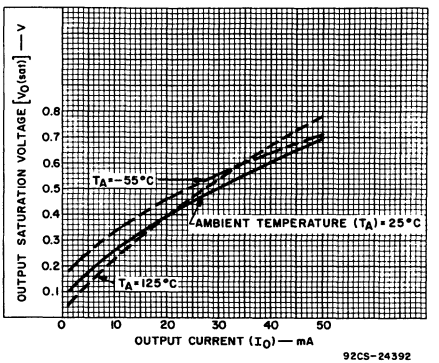


Fig. 13—Output saturation voltage vs. output current.

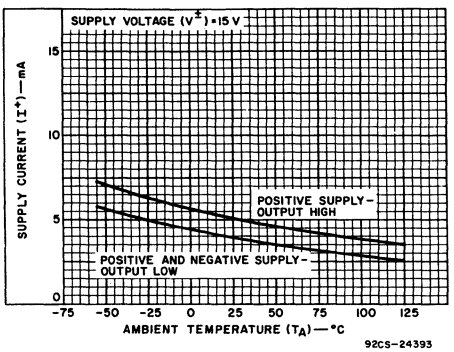


Fig. 14—Supply current vs. ambient temperature.

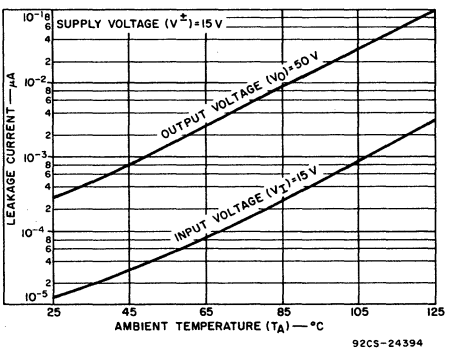


Fig. 15—Input and output leakage current vs. ambient temperature.

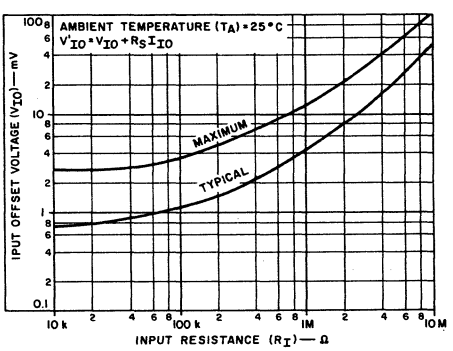


Fig. 16—Offset error.

TYPICAL CHARACTERISTICS – CA311

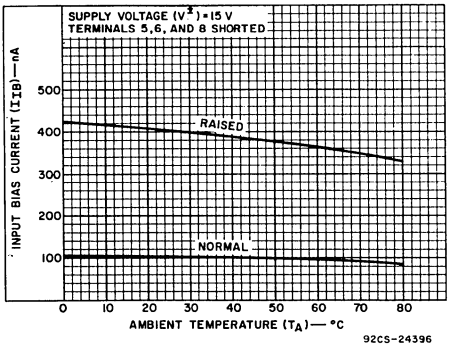


Fig. 17—Input bias current vs. ambient temperature.

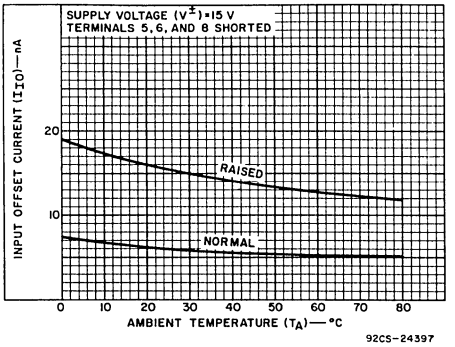


Fig. 18—Input offset current vs. ambient temperature.

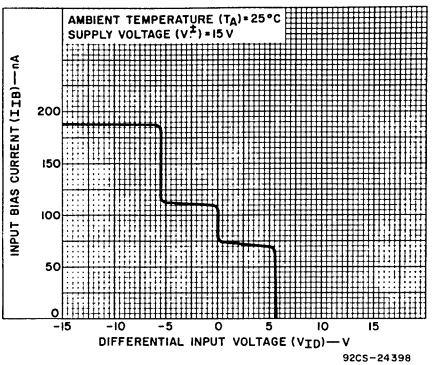


Fig. 19—Input characteristics.

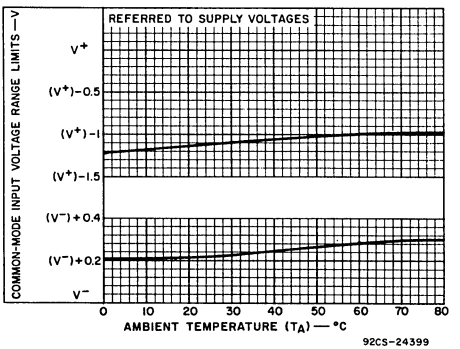


Fig. 20—Common-mode voltage range limits vs. ambient temperature.

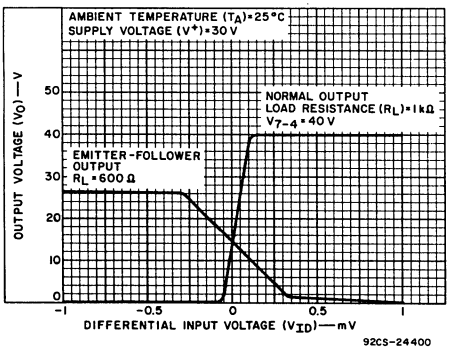


Fig. 21—Transfer function.

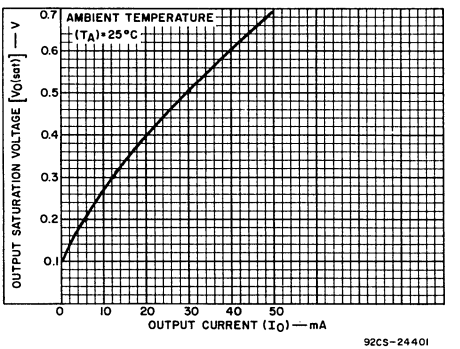


Fig. 22—Output saturation voltage vs. output current.

TYPICAL CHARACTERISTICS – CA311 (CONT'D)

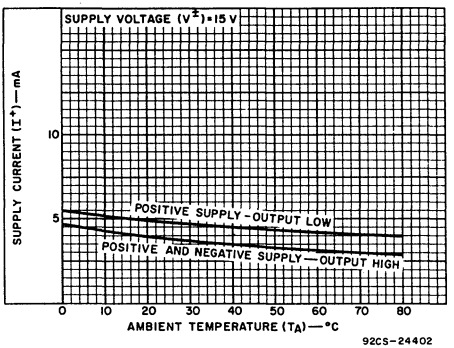


Fig. 23–Supply current vs. ambient temperature.

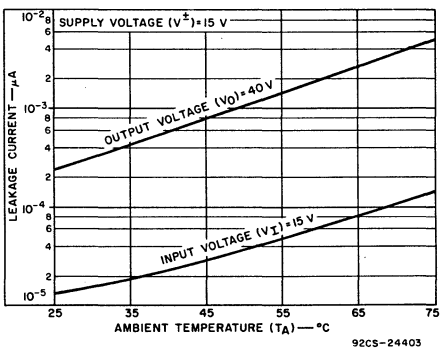


Fig. 24–Input and output leakage current vs. ambient temperature.

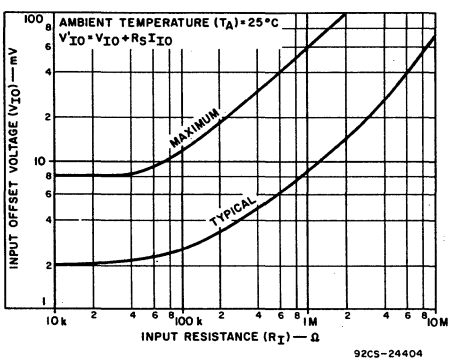


Fig. 25–Offset error.

TYPICAL APPLICATIONS

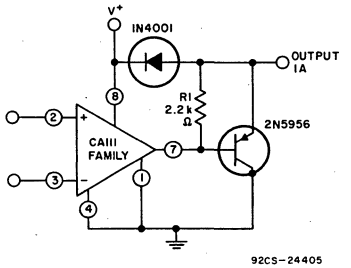


Fig. 26–Comparator and solenoid driver.

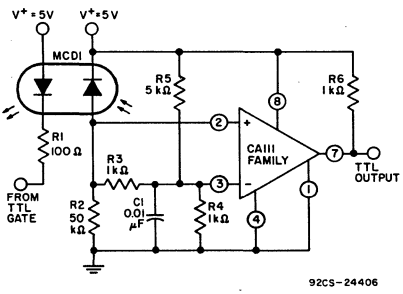


Fig. 27–Digital transmission isolator.

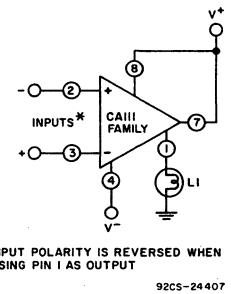


Fig. 28–Driving a ground-referred load.

TYPICAL APPLICATIONS (CONT'D)

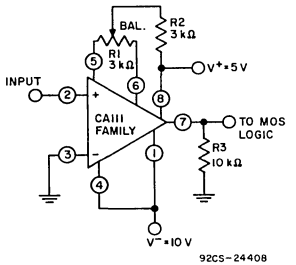


Fig. 29—Zero-crossing detector driving MOS logic.

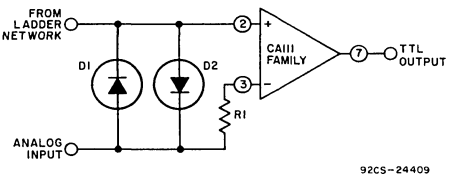


Fig. 30—Using clamp diodes to improve response.

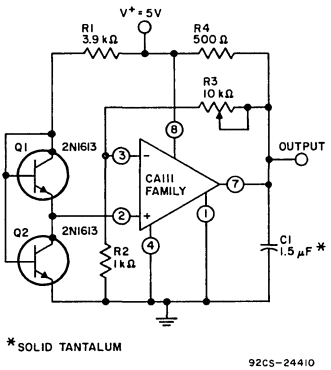


Fig. 31—Low-voltage adjustable-reference supply.

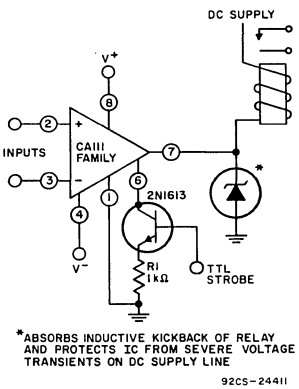


Fig. 32—Relay driver with strobe.

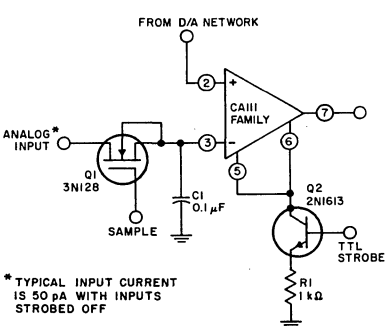


Fig. 33—Strobing off both input and output stages.

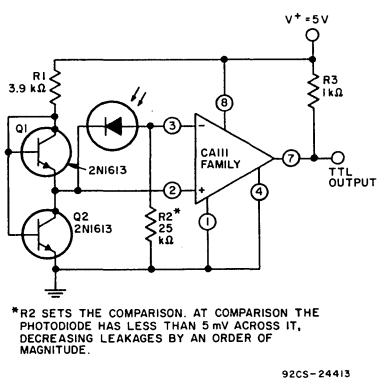


Fig. 34—Precision photodiode comparator.

TYPICAL APPLICATIONS (CONT'D)

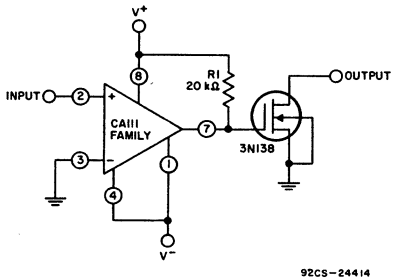


Fig. 35—Zero-crossing detector driving an MOS switch.

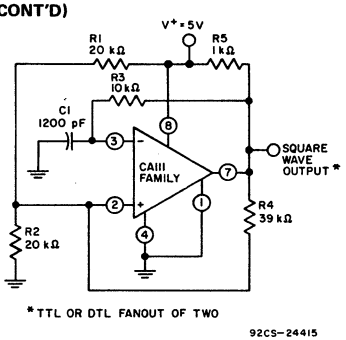


Fig. 36—100-kHz free-running multivibrator.

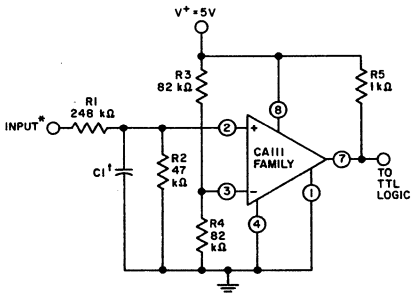


Fig. 37—TTL interface with high-level logic.

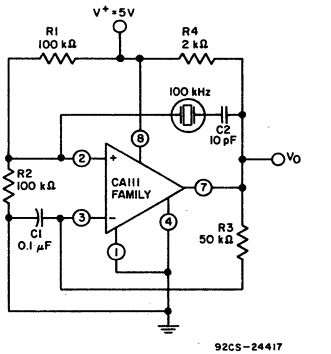


Fig. 38—Crystal oscillator.

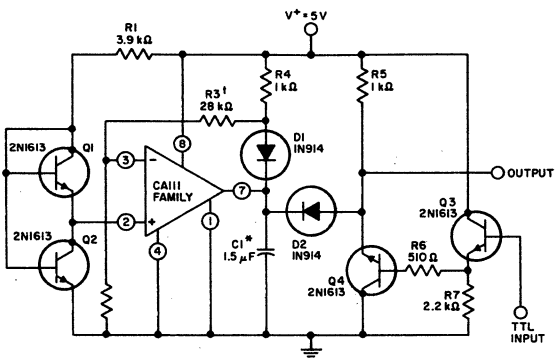


Fig. 39—Precision squarer.

TYPICAL APPLICATIONS (CONT'D)

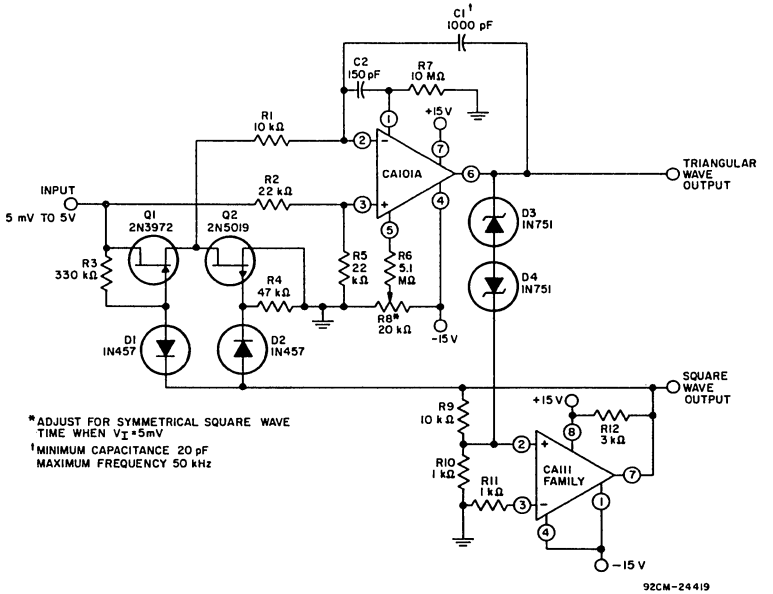


Fig. 40—10-Hz to 10-kHz voltage controlled oscillator.

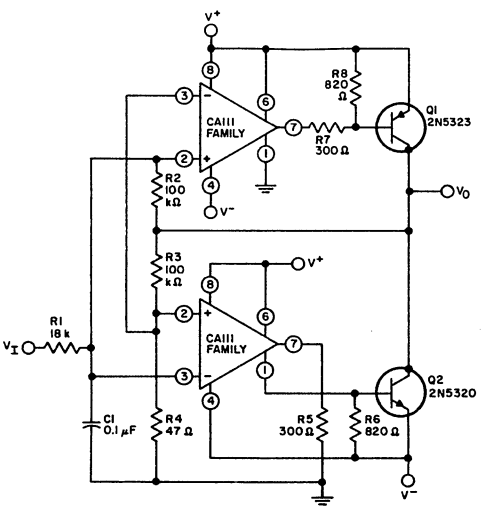


Fig. 41—Switching power amplifier.

TYPICAL APPLICATIONS (CONT'D)

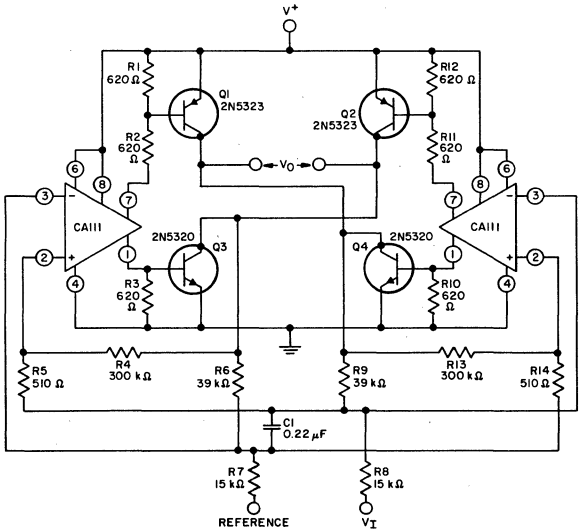


Fig. 42–Switching power amplifier.

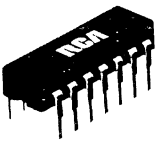
92CS-24421

RCA
Solid State
Division

Linear Integrated Circuits

Monolithic Silicon

CA339E
CA339AE



14-Lead Dual-in-Line
Plastic Package

H-1517

Quad Voltage Comparators

For Industrial and Commercial Applications

Features:

- Operation from single or dual supplies
- Common-mode input-voltage range to ground
- Output voltage compatible with TTL, DTL, ECL, MOS, and CMOS
- Differential input-voltage range equal to the supply voltage
- Maximum input offset voltage:
  CA339A -- 2 mV
  CA339 -- 5 mV

The RCA-CA339E and CA339AE types consist of four independent single- or dual-supply voltage comparators on a single monolithic substrate. The common-mode input-voltage range includes ground even when operated from a single supply, and the low power-supply current drain makes these comparators suitable for battery operation. These types are designed to interface directly with TTL and CMOS.

The CA339AE has all the features and characteristics of the CA339E plus an even lower input-offset-voltage characteristic. They are supplied in the 14-lead dual-in-line plastic package ("E" suffix) and are direct replacements for industry types 339 and 339A in similar packages. The CA339 is also available in chip form (CA339H).

Applications:

- Square-wave generators
- Time-delay generators
- Pulse generators
- Multivibrators
- High-voltage digital logic gates
- A/D converters
- MOS clock timers

MAXIMUM RATINGS, *Absolute-Maximum Values at* $T_A = 25^oC$*:*

DC SUPPLY VOLTAGE	36 V or ± 18 V
DC DIFFERENTIAL INPUT VOLTAGE	±36 V
INPUT VOLTAGE	–0.3 V to + 36 V
INPUT CURRENT (V_I < –0.3 V)*	50 mA
OUTPUT SHORT CIRCUIT TO GROUND▲	
(Single Supply)	Continuous
DEVICE DISSIPATION:	
Up to $T_A = 25^oC$	570 mW
Above $T_A = 25^oC$	derate linearly at 6.67 mW/°C
AMBIENT TEMPERATURE RANGE:	
Operating	–55°C to +125°C
Storage	–65 to +150 °C
LEAD TEMPERATURE (DURING SOLDERING):	
At distance 1/16 ± 1/32 in. (1.59 ± 0.79 mm) from case for 10 seconds max.	+265 °C

* Inputs must not go more negative than –0.3 V.

▲ Short circuits from the output to V^+ can cause excessive heating and eventual destruction. The maximum output current independent of V^+ is approximately 20 mA.

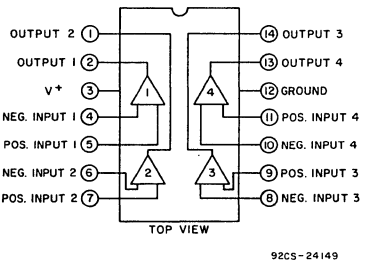


Fig. 1 — Functional diagram.

ELECTRICAL CHARACTERISTICS

CHARACTERISTIC	SYMBOL	TEST CONDITIONS $V^+ = 5$ V, Unless otherwise indicated		LIMITS CA339 / CA339A MIN.	TYP.	MAX.	UNITS
Input Offset Voltage: CA339	V_{IO}	At Output Switch Point V ≅ 1.4 V, V_{REF} = 1.4 V, R_S = 0	25°C	–	2	5	mV
			0 to 70 °C	–	–	9	
CA339A			25°C	–	1	2	
			0 to 70 °C	–	–	4	
Differential Input Voltage	V_{ID}	Keep all inputs ⩾ 0 V for V^- (If Used), TA = 0 to 70°C, NOTE 1		–	–	36	V
Saturation Voltage	V_{SAT}	V_I^- = 1 V, V_I^+ = 0 V, I_{SINK} ⩽ 4 mA	25°C	–	250	500	mV
			0 to 70 °C	–	–	700	
Common-Mode Input Voltage Range	V_{ICR}	NOTE 2	25°C	0	–	$(V^+) - 1.5$	V
			0 to 70 °C	0	–	$(V^+) - 2$	
Input Offset Current	I_{IO}	$I_I^+ - I_I^-$	25°C	–	5	50	nA
			0 to 70 °C	–	–	150	
Input Bias Current	I_{IB}	I_I^+ or I_I^- with Output in Linear Range	25°C	–	25	250	nA
			0 to 70 °C	–	–	400	
Supply Current	I^+	$R_L = \infty$ on all comparators, T_A = 25°C		–	0.8	2	mA
Output Leakage Current		$V_I^+ \geqslant 1$ V, $V_I^- = 0$, V_O = 5 V	25°C	–	0.1	–	nA
		$V_I^+ \geqslant 1$ V, $V_I^- = 0$, V_O = 30 V	0 to 70 °C	–	–	1	μA
Output Sink Current		$V_I^- \geqslant 1$ V, $V_I^+ = 0$, $V_O \leqslant +1.5$ V, T_A = 25°C		6	16	–	mA
Voltage Gain CA339E	A	$R_L \geqslant 15$ kΩ, T_A = 25°C		–	200	–	V/mV
CA339AE		$R_L \geqslant 15$ kΩ, T_A = 25°C		50	200	–	V/mV
Large Signal Response Time		V_I = TTL Logic Swing, V_{REF} = +1.4 V, V_{RL} = 50 V, R_L = 5.1 kΩ, T_A = 25°C		–	300	–	ns
Response Time		V_{RL} = 5 V, R_L = 5.1 kΩ, T_A = 25°C, See Figs. 5 & 6		–	1.3	–	μs

NOTE 1: The comparator will provide a proper output state even if the positive swing of the inputs exceeds the power supply voltage level, if the other input remains within the common-mode voltage range. The low input voltage state must not be less than −0.3 V (or 0.3 V below the magnitude of the negative power supply, if used).

NOTE 2: The upper end of the common-mode voltage range is (V^+) −1.5 V, but either or both inputs can go to +30 V without damage.

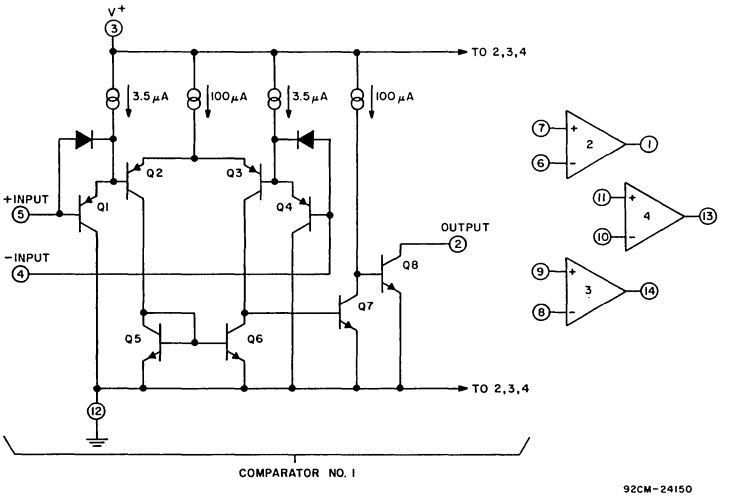


Fig.2 – Schematic diagram.

TYPICAL CHARACTERISTICS

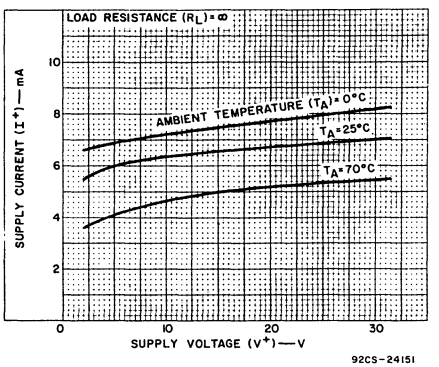


Fig.3 – Supply current vs. supply voltage.

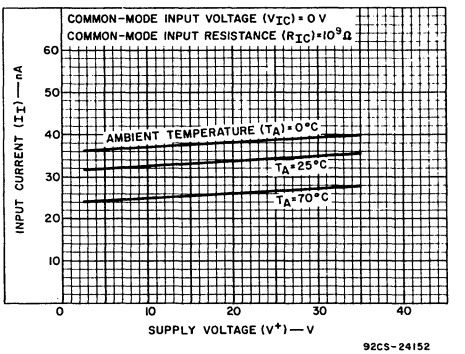


Fig.4 – Input current vs. supply voltage.

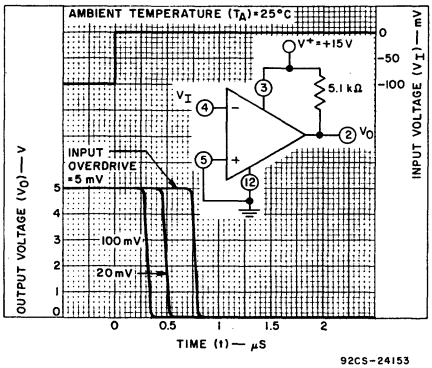


Fig.5 — Response time for various input overdrives — negative transition.

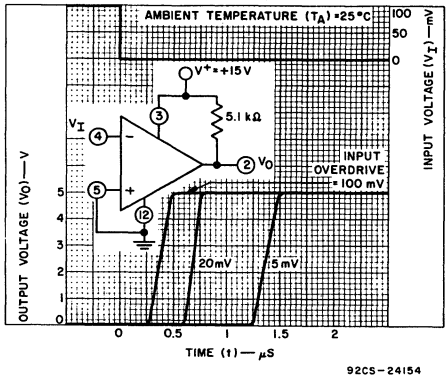


Fig.6 — Response time for various input overdrives — positive transition.

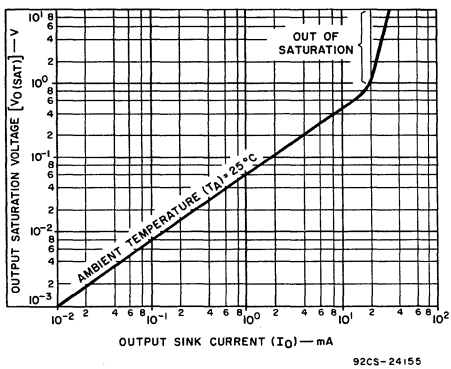


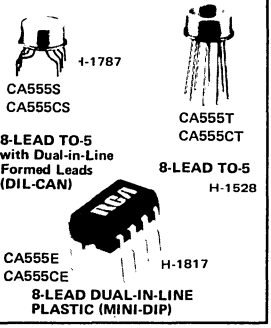
Fig.7 — Output saturation voltage vs. output sink current.

RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA555, CA555C



Timers

For Timing Delays &
Oscillator Application

Features:

- **Accurate timing from microseconds through hours**
- **Astable and monostable operation**
- **Adjustable duty cycle**
- **Output capable of sourcing or sinking up to 200 mA**
- **Output capable of driving TTL devices**
- **Normally ON and OFF outputs**
- **High-temperature stability — 0.05%/°C**
- **Directly interchangeable with SE555, NE555, MC1555, and MC1455**

The RCA-CA555 and CA555C are highly stable timers for use in precision timing and oscillator applications. As timers, these monolithic integrated circuits are capable of producing accurate time delays for periods ranging from microseconds through hours. These devices are also useful for astable oscillator operation and can maintain an accurately controlled free-running frequency and duty cycle with only two external resistors and one capacitor.

The circuits of the CA555 and CA555C may be triggered by the falling edge of the waveform signal, and the output of these circuits can source or sink up to a 200-milliampere current or drive TTL circuits.

The CA555 and CA555C are supplied in standard 8-lead TO-5 style packages (T suffix), 8-lead TO-5 style packages with dual-in-line formed leads ("DIL-CAN", S suffix), 8-lead dual-in-line plastic packages ("MINI-DIP", E suffix), and in chip form (H suffix). These types are direct replacements for industry types in packages with similar terminal arrangements (e.g. SE555 and NE555, MC1555 and MC1455, respectively).

Applications:

- **Precision timing**
- **Sequential timing**
- **Time-delay generation**
- **Pulse generation**
- **Pulse-width and -position modulation**
- **Pulse detector**

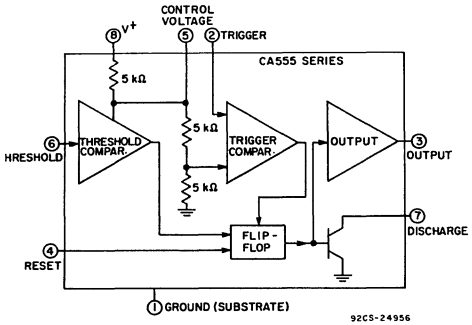


Fig. 1 — Functional diagram of the CA555 series.

MAXIMUM RATINGS, *Absolute-Maximum Values:*

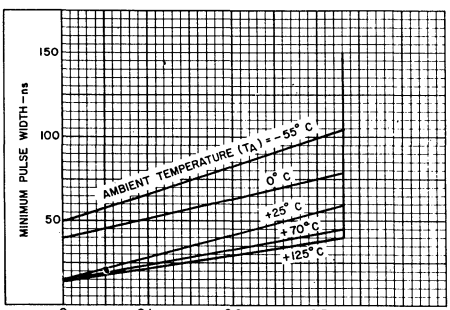
DC SUPPLY VOLTAGE	18	V
DEVICE DISSIPATION:		
Up to T_A = 55°C	600	mW
Above T_A = 55°C .....Derate linearly	5	mW/°C
AMBIENT TEMPERATURE RANGE:		
Operating	−55 to +125	°C
Storage	−65 to +150	°C
LEAD TEMPERATURE (During Soldering):		
At distance 1/16" ±1/32" (1.59 ±0.79) from case for 10 seconds max.	+265	°C

ELECTRICAL CHARACTERISTICS, *At T_A = 25°C, V^+ = 5 to 15 V unless otherwise specified*

CHARACTERISTIC	SYMBOL	TEST CONDITIONS		LIMITS						UNITS
				CA555			CA555C			
				Min.	Typ.	Max.	Min.	Typ.	Max.	
DC Supply Voltage	V^+			4.5	–	18	4.5	–	16	V
DC Supply Current (Low State)*	I^+	V^+ = 5 V, R_L = ∞		–	3	5	–	3	6	mA
		V^+ = 15 V, R_L = ∞		–	10	12	–	10	15	
Threshold Voltage	V_{TH}			–	(2/3)V^+	–	–	(2/3)V^+	–	V
Trigger Voltage		V^+ = 5 V		1.45	1.67	1.9	–	1.67	–	V
		V^+ = 15 V		4.8	5	5.2	–	5	–	
Trigger Current				–	0.5	–	–	0.5	–	μA
Threshold Current▲	I_{TH}			–	0.1	0.25	–	0.1	0.25	μA
Reset Voltage				0.4	0.7	1.0	0.4	0.7	1.0	V
Reset Current				–	0.1	–	–	0.1	–	mA
Control Voltage Level		V^+ = 5 V		2.9	3.33	3.8	2.6	3.33	4	V
		V^+ = 15 V		9.6	10	10.4	9	10	11	
Output Voltage Drop: Low State	V_{OL}	V^+ = 5 V	I_{SINK} = 5 mA	–	–	–	–	0.25	0.35	V
			I_{SINK} = 8 mA	–	0.1	0.25	–	–	–	
		V^+ = 15 V	I_{SINK} = 10 mA	–	0.1	0.15	–	0.1	0.25	V
			I_{SINK} = 50 mA	–	0.4	0.5	–	0.4	0.75	
			I_{SINK} = 100 mA	–	2.0	2.2	–	2.0	2.5	
			I_{SINK} = 200 mA	–	2.5	–	–	2.5	–	
High State	V_{OH}	V^+ = 5 V	I_{SOURCE} = 100 mA	3.0	3.3	–	2.75	3.3	–	V
		V^+ = 15 V	I_{SOURCE} = 100 mA	13.0	13.3	–	12.75	13.3	–	
			I_{SOURCE} = 200 mA	–	12.5	–	–	12.5	–	
Timing Error (Monostable): Initial Accuracy		R_A, R_B = 1 to 100 kΩ		–	0.5	2	–	1	–	%
Frequency Drift with Temperature		C = 0.1 μF		–	30	100	–	50	–	p/m/°C
Drift with Supply Voltage		Tested at V^+ = 5 V and V^+ = 15 V		–	0.05	0.2	–	0.1	–	%/V
Output Rise Time	t_r			–	100	–	–	100	–	ns
Output Fall Time	t_f			–	100	–	–	100	–	ns

* When the output is in a high state, the dc supply current is typically 1 mA less than the low-state value.

▲ The threshold current will determine the sum of the values of R_A and R_B to be used in Fig. 6 (astable operation); the maximum total $R_A + R_B$ = 20 MΩ.



*WHERE x IS THE DECIMAL MULTIPLIER OF THE SUPPLY VOLTAGE

Fig. 2 – Minimum pulse width vs. minimum trigger voltage.

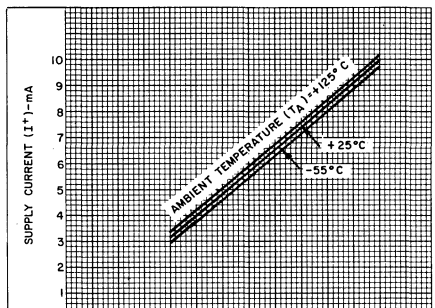


Fig. 3 – Supply current vs. supply voltage.

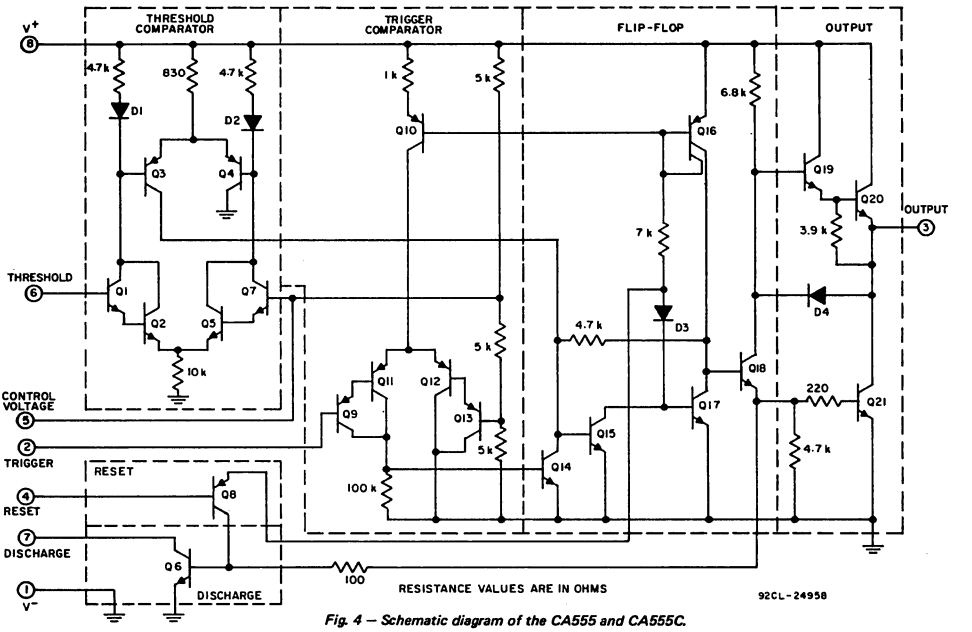


Fig. 4 – Schematic diagram of the CA555 and CA555C.

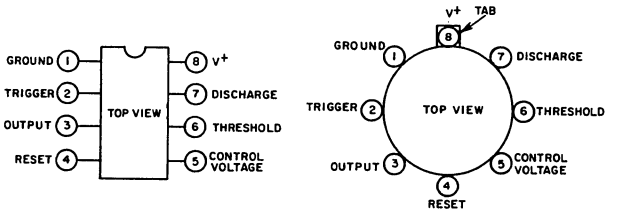


Fig. 5 – Terminal assignment diagrams.

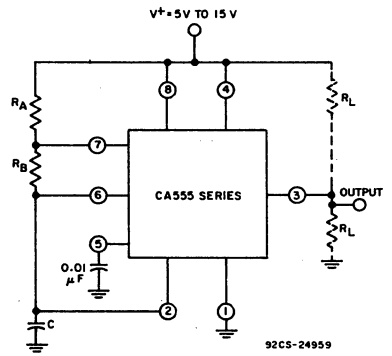


Fig. 6 – Astable oscillator circuit.

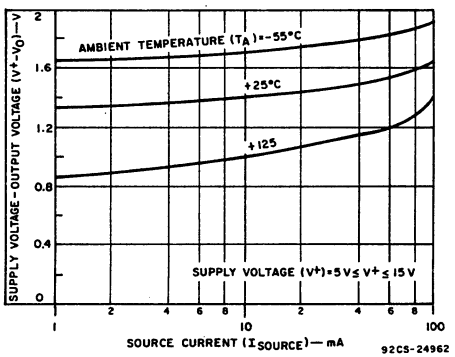


Fig. 7 – Output voltage drop (high state) vs. source current.

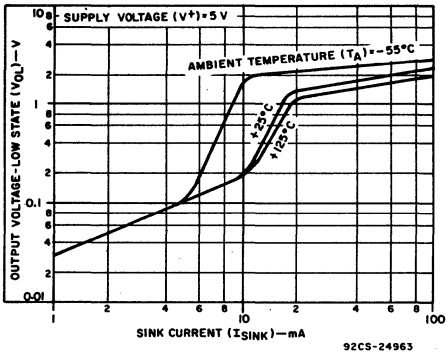


Fig. 8 — Output voltage-low state vs. sink current at $V^+ = 5$ V.

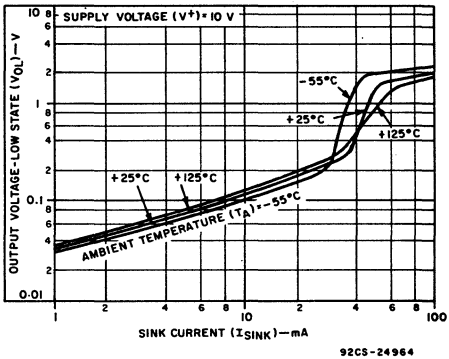


Fig. 9 — Output voltage-low state vs. current at $V^+ = 10$ V.

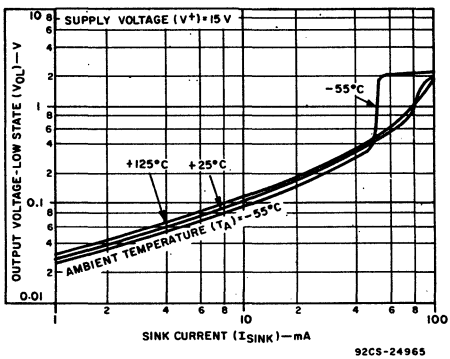


Fig. 10 — Output voltage-low state vs. sink current at $V^+ = 15$ V.

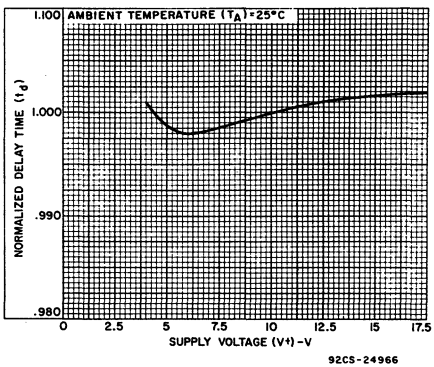


Fig. 11 — Delay time vs. supply voltage.

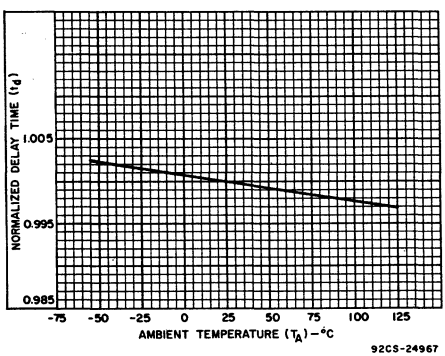


Fig. 12 — Delay time vs. temperature.

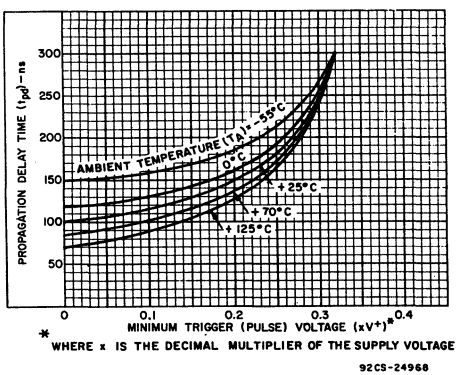


Fig. 13 — Propagation delay time vs. trigger voltage.

RCA
Solid State
Division

Linear Integrated Circuits

Monolithic Silicon

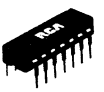
CA723T, CA723CT
CA723E, CA723CE



"T" Suffix

10-Lead TO-5

H-1384



"E" Suffix

14-Lead Dual-In-Line
Plastic Package

H-1517

Voltage Regulators

For Regulated Output Voltages Adjustable from 2 V to 37 V at Currents up to 150 mA Without External Pass Transistors

Features:

- **Up to 150 mA output current**
- **Positive and negative voltage regulation**
- **Regulation in excess of 10 A with suitable pass transistors**
- **Input and output short-circuit protection**
- **Load and line regulation: 0.03%**
- **Direct replacement for 723 and 723C industry types**
- **Adjustable output voltage: 2 to 37 V**

RCA-CA723 and CA723C are silicon monolithic integrated circuits designed for service as voltage regulators at output voltages ranging from 2 to 37 volts at currents up to 150 milliamperes.

Each type includes a temperature-compensated reference amplifier, an error amplifier, a power series pass transistor, and a current-limiting circuit. They also provide independently accessible inputs for adjustable current limiting and remote shutdown and, in addition, feature low standby current drain, low temperature drift, and high ripple rejection.

The CA723 and CA723C may be used with positive and negative power supplies in a wide variety of series, shunt, switching, and floating regulator applications. They can provide regulation at load currents greater than 150 milliamperes and in excess of 10 amperes with the use of suitable n-p-n or p-n-p external pass transistors.

The CA723 and CA723C are supplied in the 10-lead TO-5-style ceramic package (T suffix), and the 14-lead dual-in-line plastic package (E suffix), and are direct replacements for industry types 723, 723C, μA723, and μA723C in packages with similar terminal arrangements. They are also available in chip form ("H" suffix).

All CA723-series circuits can be operated over the full military-temperature range of −55°C to +125°C. Refer to the chart of Electrical Characteristics for characteristics limits over several temperature ranges.

Applications

- **Series and shunt voltage regulator**
- **Floating regulator**
- **Switching voltage regulator**
- **High-current voltage regulator**
- **Temperature controller**

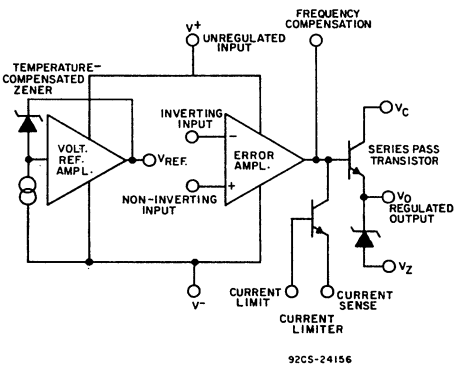


Fig. 1 — Functional diagram of the CA723 and CA723C.

MAXIMUM RATINGS, *Absolute-Maximum Values:*

DC SUPPLY VOLTAGE (Between V^+ and V^- Terminals)	40	V
PULSE VOLTAGE FOR 50-ms PULSE WIDTH (Between V^+ and V^- Terminals)	50	V
DIFFERENTIAL INPUT-OUTPUT VOLTAGE	40	V
DIFFERENTIAL INPUT VOLTAGE:		
Between Inverting and Non-Inverting Inputs	±5	V
Between Non-Inverting Input and V^-	8	V
CURRENT FROM ZENER DIODE TERMINAL (V_Z)	25	mA
CURRENT FROM VOLTAGE REFERENCE TERMINAL (V_{REF})	15	mA

DEVICE DISSIPATION:

Up to T_A = 25°C —	
CA723T, CA723CT	800 mW
CA723E, CA723CE	1000 mW
Above T_A = 25°C —	
CA723T, CA723CT	Derate linearly 6.3 mW/°C
CA723E, CA723CE	Derate linearly 8.3 mW/°C
AMBIENT TEMPERATURE RANGE:	
Operating	−55 to +125 °C
Storage	−65 to +150 °C
LEAD TEMPERATURE (During Soldering):	
At a distance 1/16" ± 1/32" (1.59 ± 0.79 mm) from case for 10 seconds max.	+265 °C

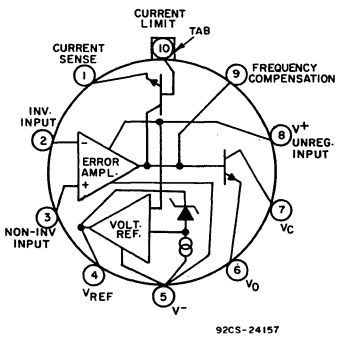


Fig.2 — Terminal arrangement of the CA723T and CA723CT in the TO-5 style package.

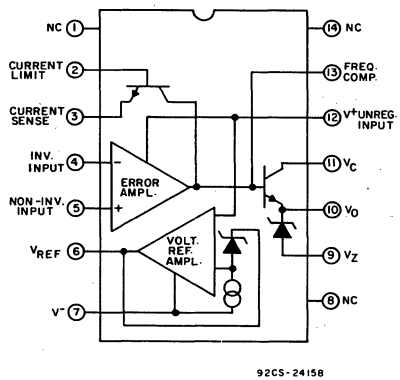


Fig.3 — Terminal arrangement of the CA723E and CA723CE in the dual-in-line plastic package.

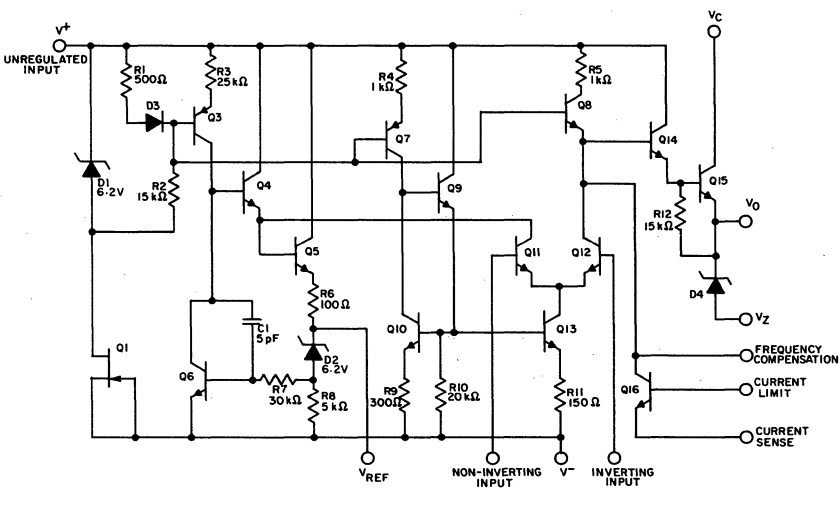


Fig.4 — Equivalent schematic diagram of the CA723 and CA723C.

ELECTRICAL CHARACTERISTICS:

CHARACTERISTIC	SYMBOL	TEST CONDITIONS (See Note)	CA723			CA723C			UNIT
		T_A=25°C, V_I=V^+=V_C=12V, V^-=0, V_O=5V, I_L=1mA, C_I=100 pF, $Z_{DIVIDER}$≤ 10 kΩ (into error amplifier as shown in Fig.23) unless otherwise indicated	Min.	Typ.	Max.	Min.	Typ.	Max.	
Quiescent Regulator Current	I_Q	I_L=0, V_I=30 V	–	2.3	3.5	–	2.3	4.0	mA
Input Voltage Range	V_I		9.5	–	40	9.5	–	40	V
Output Voltage Range	V_O		2.0	–	37	2.0	–	37	V
Differential Input-Output Voltage	V_I–V_O		3.0	–	38	3.0	–	38	V
Reference Voltage	V_{REF}		6.95	7.15	7.35	6.80	7.15	7.50	V
Line Regulation		V_I=12 to 40 V	–	0.02	0.2	–	0.1	0.5	%V_O
		V_I=12 to 15 V	–	0.01	0.1	–	0.01	0.1	
		V_I=12 to 15 V, T_A=–55 to +125°C	–	–	0.3	–	–	–	
		V_I=12 to 15 V, T_A=0 to 70°C	–	–	–	–	–	0.3	
Load Regulation		I_L=1 to 50 mA	–	0.03	0.15	–	0.03	0.2	%V_O
		I_L=1 to 50 mA, T_A=–55 to +125°C	–	–	0.6	–	–	–	
		I_L=1 to 50 mA, T_A=0 to 70°C	–	–	–	–	–	0.6	
Output-Voltage Temperature Coefficient	ΔV_O	T_A=–55 to +125°C	–	0.002	0.015	–	–	–	%/°C
		T_A=0 to 70°C	–	–	–	–	0.003	0.015	
Ripple Rejection		f=50 Hz to 10 kHz	–	74	–	–	74	–	dB
		f=50 Hz to 10 kHz, C_{REF}=5 μF	–	86	–	–	86	–	
Short-Circuit Limiting Current	I_{LIM}	R_{SCP}=10 Ω, V_O=0	–	65	–	–	65	–	mA
Equivalent Noise Output Voltage	V_{NOISE}	BW=100 to 10 kHz, C_{REF}=0	–	20	–	–	20	–	μV_{RMS}
		BW=100 to 10 kHz, C_{REF}=5 μF	–	2.5	–	–	2.5	–	

Note: Line and load regulation specifications are given for condition of a constant chip temperature; for high dissipation conditions, temperature drifts must be separately taken into account.

TYPICAL CHARACTERISTICS CURVES FOR TYPE CA723

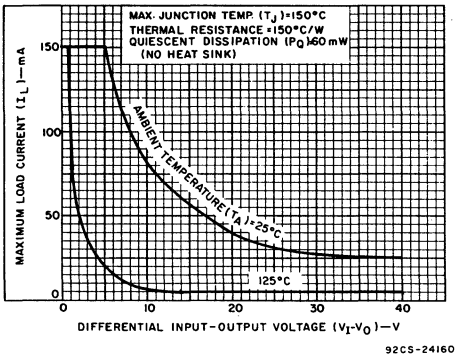


Fig.5 — Max. load current vs differential input-output voltage.

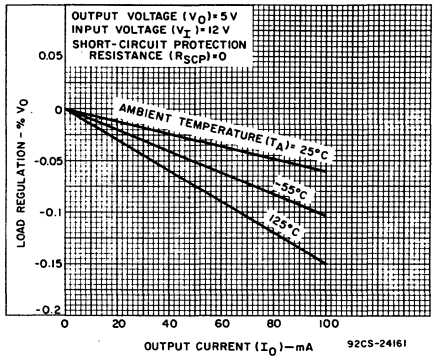


Fig.6 — Load regulation without current limiting.

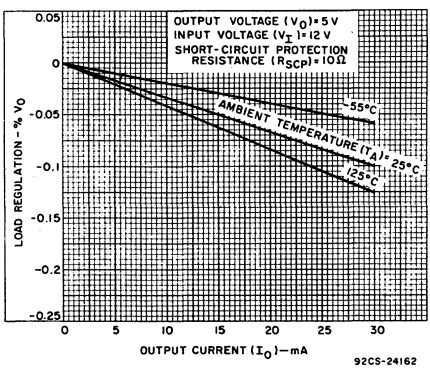


Fig.7 — Load regulation with current limiting.

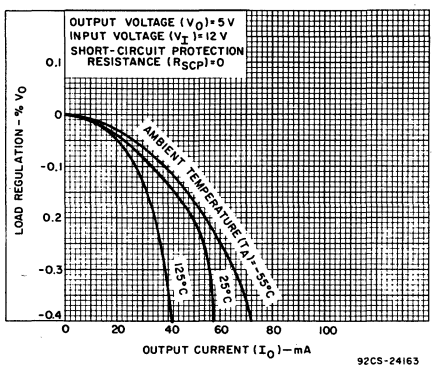


Fig.8 — Load regulation with current limiting.

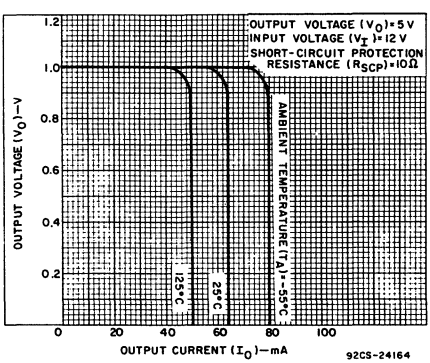


Fig.9 — Current limiting characteristics.

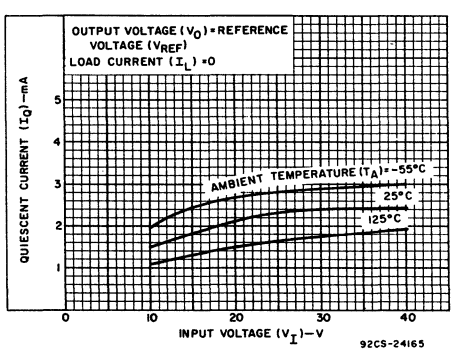


Fig.10 — Quiescent current vs. input voltage.

TYPICAL CHARACTERISTICS CURVES FOR TYPE CA723C

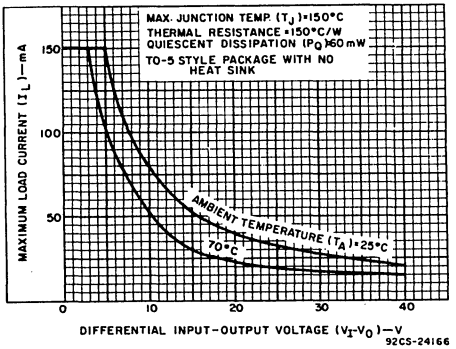


Fig.11 – Max. load current vs differential input-output voltage for CA723CT.

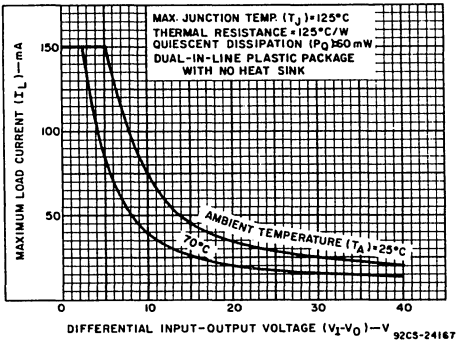


Fig.12 – Max. load current vs differential input-output voltage for CA723CE.

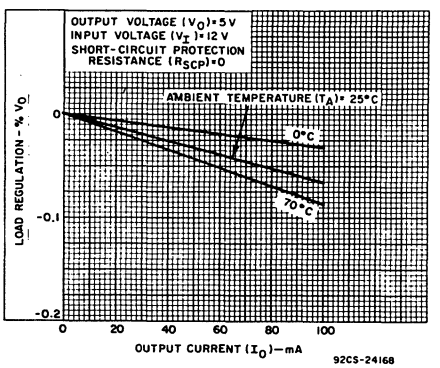


Fig.13 – Load regulation without current limiting.

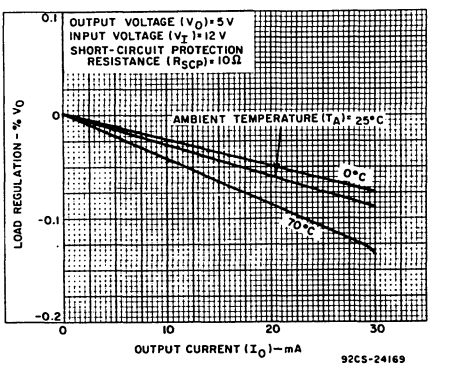


Fig.14 – Load regulation with current limiting.

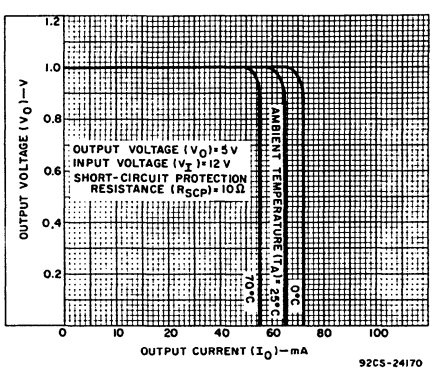


Fig.15 – Current limiting characteristics.

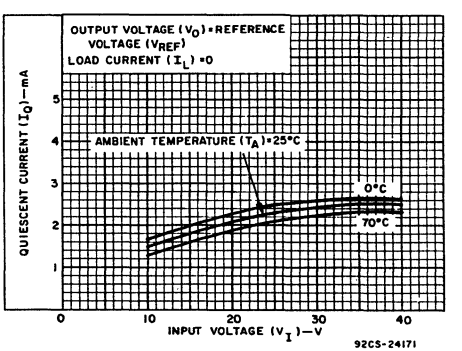


Fig.16 – Quiescent current vs. input voltage.

TYPICAL CHARACTERISTICS CURVES FOR TYPES CA723 AND CA723C

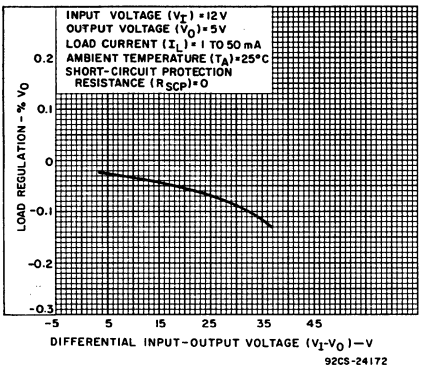


Fig.17 – Load regulation vs. differential input-output voltage.

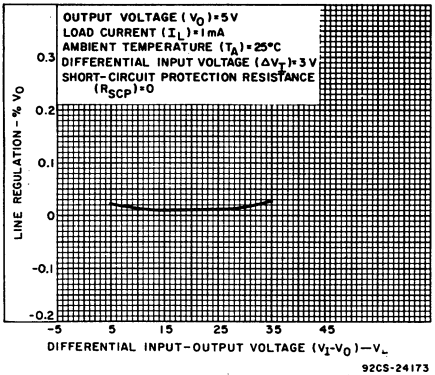


Fig.18 – Line regulation vs. differential input-output voltage.

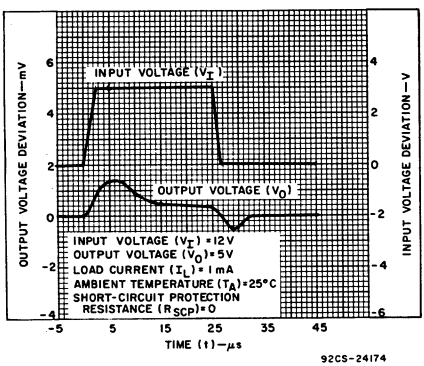


Fig.19 – Line transient response.

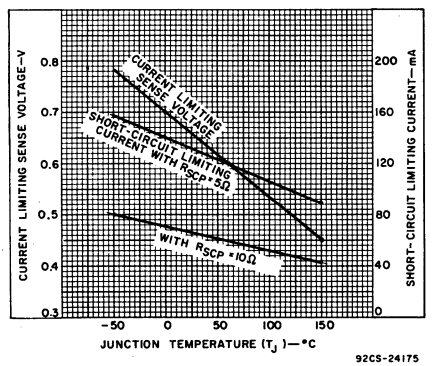


Fig.20 – Current limiting characteristics vs. junction temperature.

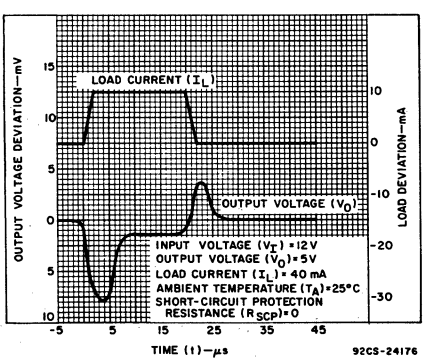


Fig.21 – Load transient response.

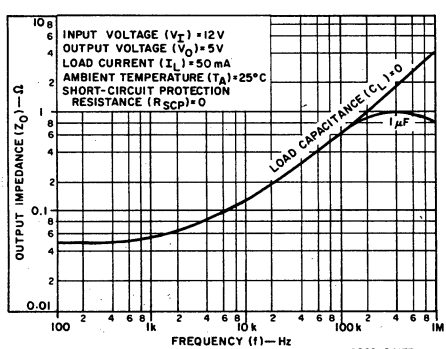
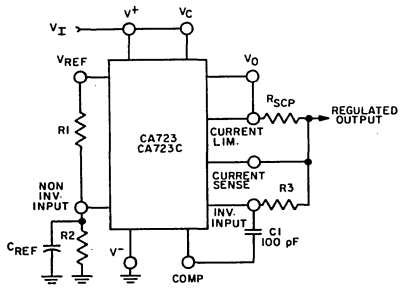


Fig.22 – Output impedance vs. frequency.

TYPICAL APPLICATION CIRCUITS

Voltage-Resistance Chart and Equations for the following Circuits are shown on pages 9 and 10.



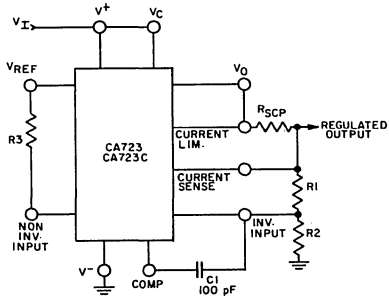
CIRCUIT PERFORMANCE DATA:

REGULATED OUTPUT VOLTAGE	5 V
LINE REGULATION (ΔV_I = 3 V)	0.5 mV
LOAD REGULATION (ΔI_L = 50 mA)	1.5 mV

Note: $R3 = \frac{R1\ R2}{R1+R2}$ for minimum temperature drift

92CS-24178

Fig.23 – Low-voltage regulator circuit (V_O = 2 to 7 volts).



CIRCUIT PERFORMANCE DATA:

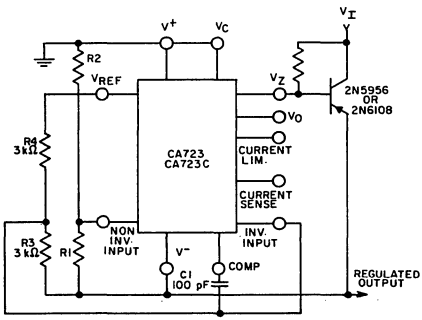
REGULATED OUTPUT VOLTAGE	15 V
LINE REGULATION (ΔV_I = 3 V)	1.5 mV
LOAD REGULATION (ΔI_L = 50 mA)	4.5 mV

Note: $R3 = \frac{R1\ R2}{R1+R2}$ for minimum temperature drift

R3 may be eliminated for minimum component count.

92CS-24179

Fig.24 – High-voltage regulator circuit (V_O = 7 to 37 volts).



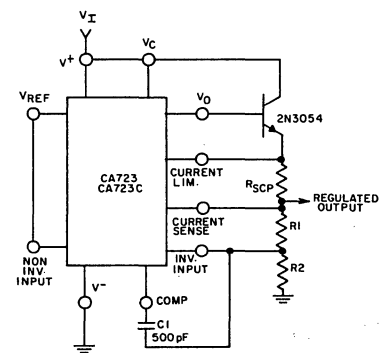
CIRCUIT PERFORMANCE DATA:

REGULATED OUTPUT VOLTAGE	−15 V
LINE REGULATION (ΔV_I = 3 V)	1 mV
LOAD REGULATION (ΔI_L = 100 mA)	2 mV

Note: For applications employing the TO-5 style package and where V_Z is required, an external 6.2-volt zener diode should be connected in series with V_O (Terminal 6).

92CS-24180

Fig.25 – Negative-voltage-regulator circuit.



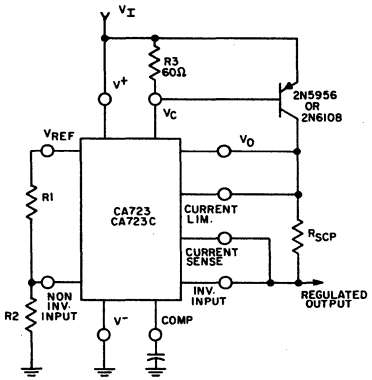
CIRCUIT PERFORMANCE DATA:

REGULATED OUTPUT VOLTAGE	15 V
LINE REGULATION (ΔV_I = 3 V)	1.5 mV
LOAD REGULATION (ΔI_L = 1 A)	15 mV

92CS-24181

Fig.26 – Positive-voltage-regulator circuit (with external n-p-n pass transistor).

TYPICAL APPLICATION CIRCUITS (Cont'd)

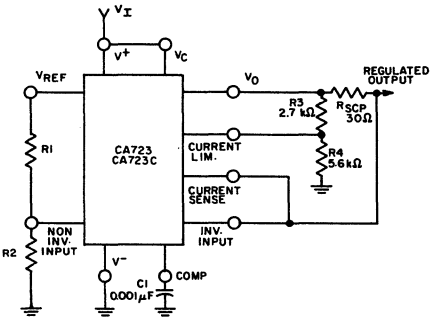


CIRCUIT PERFORMANCE DATA:

REGULATED OUTPUT VOLTAGE	5 V
LINE REGULATION (ΔV_I = 3 V)	0.5 mV
LOAD REGULATION (ΔI_L = 1 A)	5 mV

92CS-24182

Fig.27 — Positive voltage-regulator circuit (with external p-n-p pass transistor).

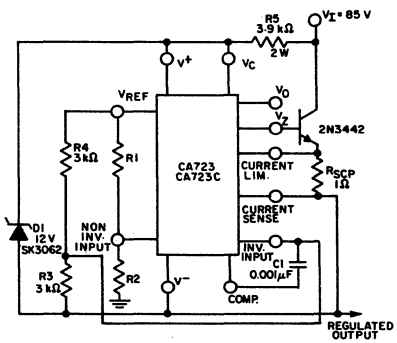


CIRCUIT PERFORMANCE DATA:

REGULATED OUTPUT VOLTAGE	5 V
LINE REGULATION (ΔV_I = 3 V)	0.5 mV
LOAD REGULATION (ΔI_L = 10 mA)	1 mV
SHORT-CIRCUIT CURRENT	20 mA

92CS-24183

Fig.28 — Foldback current-limiting circuit.



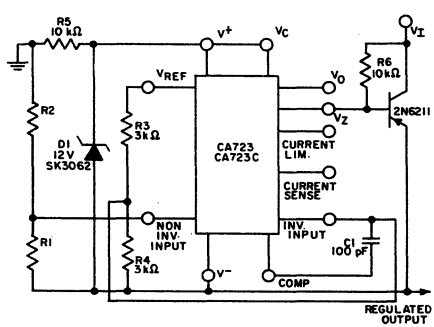
Note: For applications employing the TO-5 style package and where V_Z is required, an external 6.2-volt zener diode should be connected in series with V_O (Terminal 6).

CIRCUIT PERFORMANCE DATA:

REGULATED OUTPUT VOLTAGE	50 V
LINE REGULATION (ΔV_I = 20 V)	15 mV
LOAD REGULATION (ΔI_L = 50 mA)	20 mV

92CS-24184

Fig.29 — Positive-floating regulator circuit.



CIRCUIT PERFORMANCE DATA:

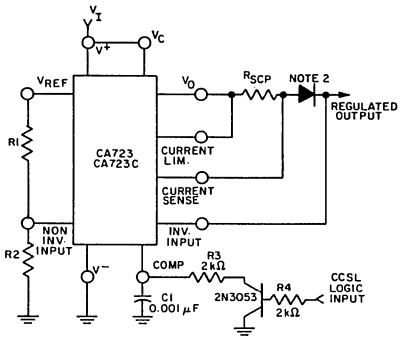
REGULATED OUTPUT VOLTAGE	−100 V
LINE REGULATION (ΔV_I = 20 V)	30 mV
LOAD REGULATION (ΔI_L = 100 mA)	20 mV

Note: For applications employing the TO-5 style package and where V_Z is required, an external 6.2-volt zener diode should be connected in series with V_O (Terminal 6).

92CS-24185

Fig.30 — Negative-floating regulator circuit.

TYPICAL APPLICATION CIRCUITS (Cont'd)



CIRCUIT PERFORMANCE DATA:

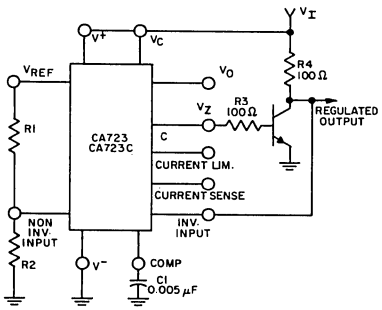
REGULATED OUTPUT VOLTAGE . . .	5 V
LINE REGULATION (ΔV_I = 3 V)	0.5 mV
LOAD REGULATION (ΔI_L = 50 mA). . .	1.5 mV

Note 1: A current limiting transistor may be used for shutdown if current limiting is not required.

Note 2: Add a diode if $V_O > 10$ V.

92CS-24186

Fig.31 – Remote shutdown regulator circuit with current limiting.



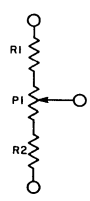
CIRCUIT PERFORMANCE DATA:

REGULATED OUTPUT VOLTAGE . . .	5 V
LINE REGULATION (ΔV_I = 10 V) . . .	0.5 mV
LOAD REGULATION (ΔI_L = 100 mA) . .	1.5 mV

Note: For applications employing the TO-5 style package and where V_Z is required, an external 6.2-volt zener diode should be connected in series with V_O (Terminal 6).

92CS-24187

Fig.32 – Shunt regulator circuit.



92CS-24196

Fig.33 – Resistance network for adjustable output voltage.

EQUATIONS FOR DERIVING INTERMEDIATE OUTPUT VOLTAGES AND CURRENT LIMITING

1. Output Voltages from 2 to 7 Volts
   Use circuits with Fig. Nos. 21, 25, 26, 30, (24 – See Note 1)

$$V_O = V_{REF} \times \frac{R2}{R1+R2}$$

2. Output Voltages from 7 to 37 Volts
   Use circuits with Fig. Nos. 22, 24, (25, 26, 30 – See Note 1)

$$V_O = V_{REF} \times \frac{R1+R2}{R2}$$

3. Output Voltages from 4 to 250 Volts
   Use circuit with Fig. No. 27

$$V_O = \frac{V_{REF}}{2} \times \frac{R2-R1}{R1} \; ; \; R3 = R4$$

4. Output Voltages from −6 to −250 Volts
   Use circuits with Fig. Nos. 23 and 28

$$V_O = \frac{V_{REF}}{2} \times \frac{R1+R2}{R1} \; ; \; R3 = R4$$

5. Current Limiting

$$I_{LIMIT} = \frac{V_{SENSE}}{R_{SCP}}$$

6. Foldback Current Limiting

$$I_{KNEE} = \frac{V_O R3}{R_{SCP} R4} + \frac{V_{SENSE}}{R_{SCP} R4} \frac{(R3+R4)}{}$$

$$I_{SHORT\ CKT} = \frac{V_{SENSE}}{R_{SCP}} \times \frac{R3+R4}{R4}$$

RESISTANCE VALUES (Resistors R1, R2, and Potentiometer P1) FOR STANDARD OUTPUT VOLTAGES

Positive Output Voltage	Applicable Circuit Fig. No. (Note 1)	Fixed Output Resistance ±5%		Adjustable Output Resistance ±10% (Note 2)			Negative Output Voltage	Applicable Circuit Fig. No. (Note 1)	Fixed Output Resistance ±5%		5% Adjustable Output Resistance ±10%		
+V		R1 kΩ	R2 kΩ	R1 kΩ	R2 kΩ	P1 kΩ	−V		R1 kΩ	R2 kΩ	R1 kΩ	R2 kΩ	P1 kΩ
3.0	23, 27, 28 32, (26)	4.12	3.01	1.8	1.2	0.5							
3.6	23, 27, 28 32, (26)	3.57	3.65	1.5	1.5	0.5							
5.0	23, 27, 28, 32, (26)	2.15	4.99	0.75	2.2	0.5							
6.0	23, 27, 28, 32, (26)	1.15	6.04	0.5	2.7	0.5	−6.0 (Note 3)	25	3.57	2.43	1.2	0.75	0.5
9.0	24, 26 (27, 28, 32)	1.87	7.15	0.75	2.7	1.0	−9.0	25	3.48	5.36	1.2	2.0	0.5
12	24, 26, (27, 28, 32)	4.87	7.15	2.0	3.0	1.0	−12	25	3.57	8.45	1.2	3.3	0.5
15	24, 26, (27, 28, 32)	7.87	7.15	3.3	3.0	1.0	−15	25	3.65	11.5	1.2	4.3	0.5
28	24, 26, (27, 28, 32)	21.0	7.15	5.6	2.0	1.0	−28	25	3.57	24.3	1.2	10	0.5
45	29	3.57	48.7	2.2	39	10	−45	30	3.57	41.2	2.2	33	10
75	29	3.57	78.7	2.2	68	10							
100	29	3.57	102	2.2	91	10	−100	30	3.57	97.6	2.2	91	10
250	29	3.57	255	2.2	240	10	−250	30	3.57	249	2.2	240	10

Note 1 – Circuits whose Fig. Nos. are shown in parentheses may be used if divider network R1 – R2 is placed on opposite side of Error Amplifier.

Note 2 – Use the divider network shown in Fig. 31 in place of the R1 – R2 network.

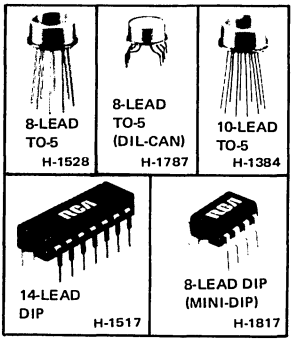
Note 3 – A supply voltage of 3 V or greater must be applied to V^+.

RCA
Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA741, CA741C, CA747, CA747C, CA748, CA748C, CA1458, CA1558 Types



Operational Amplifiers

High-Gain Single and Dual Operational Amplifiers For Military, Industrial and Commercial Applications

Applications:

- Comparator
- DC amplifier
- Integrator or differentiator
- Multivibrator
- Narrow-band or band-pass filter
- Summing amplifier

Features:

- Input bias current (all types): 500 nA max.
- Input offset current (all types): 200 nA max.

RCA-CA1458, CA1558 (dual types); CA741C, CA741 (single-types); CA747C, CA747 (dual types); and CA748C, CA748 (single types) are general-purpose, high-gain operational amplifiers for use in military, industrial, and commercial applications.

These monolithic silicon integrated-circuit devices provide output short-circuit protection and latch-free operation. These types also feature wide common-mode and differential-mode signal ranges and have low-offset voltage nulling capability when used with an appropriately valued potentiometer. A 5-megohm potentiometer is used for offset nulling types. CA748C, CA748 (See Fig. 9); a 10-kilohm potentiometer is used for offset nulling types CA741C, CA741, CA747CE, CA747E (See Fig. 8); and types CA1458, CA1558, CA747CT, CA747T have no specific terminals for offset nulling. Each type consists of a differential-input amplifier that effectively drives a gain and level-shifting stage having a complementary emitter-follower output.

This operational amplifier line also offers the circuit designer the option of operation with internal or external phase compensation. Types CA748C and CA748, which are externally phase compensated (terminals 1 and 8) permit a choice of operation for improved bandwidth and slew-rate capabilities. Unity gain with externa. phase compensation can be obtained with a single 30-pF capacitor. All the other types are internally phase-compensated.

RCA's manufacturing process makes it possible to produce IC operational amplifiers with low-burst ("popcorn") noise characteristics. Type CA6741, a low-noise version of the CA741, gives limit specifications for burst noise in the data bulletin, File No. 530. Contact your RCA Sales Representative for information pertinent to other operational amplifier types that meet low-burst noise specifications.

ORDERING INFORMATION

When ordering any of these types, it is important that the appropriate suffix letter for the package required be affixed to the type number. For example: If a CA1458 in a TO-5 package is desired, order CA1458T.

RCA Type No.	No. of Ampl.	Phase Comp.	Offset Voltage Null	Min. A_{OL}	Max. V_{IO} (mV)	Operating-Temperature Range (°C)	Package Type and Suffix Letter						
							TO-5			Plastic		Chip	Beam-Lead
							8L	10L	DIL-CAN	8L	14L		
CA1458	dual	int.	no	20k	6	0 to +70	T		S	E		H	
CA1558	dual	int.	no	50k	5	−55 to +125	T		S				
CA741C	single	int.	yes	20k	6	0 to +70	T		S	E		H	
CA741	single	int.	yes	50k	5	−55 to +125	T		S				L
CA747C	dual	int.	yes*	20k	6	0 to +70		T			E	H	
CA747	dual	int.	yes*	50k	5	−55 to +125		T			E		
CA748C	single	ext.	yes	20k	6	0 to +70	T		S	E		H	
CA748	single	ext.	yes	50k	5	−55 to +125	T		S				

*In the 14-lead dual-in-line plastic package only.

ELECTRICAL CHARACTERISTICS

For Equipment Design

Characteristics	Symbols	Test Conditions	Ambient Temperature (T_A)	LIMITS						Units
		Supply Volts: $V^+ = 15$, $V^- = -15$		CA741C, CA747C*, CA748C*, CA1458			CA741, CA747*, CA748*, CA1558			
				Min.	Typ.	Max.	Min.	Typ.	Max.	
Input Offset Voltage	V_{IO}	$R_S \leq 10$ kΩ	25°C	–	2	6	–	1	5	mV
			0 to 70°C	–	–	7.5	–	–	–	
			−55 to +125°C	–	–	–	–	1	6	
Input Offset Current	I_{IO}		25°C	–	20	200	–	20	200	nA
			−55°C	–	–	–	–	85	500	
			+125°C	–	–	–	–	7	200	
			0 to 70°C	–	–	300	–	–	–	
Input Bias Current	I_{IB}		25°C	–	80	500	–	80	500	nA
			−55°C	–	–	–	–	300	1500	
			+125°C	–	–	–	–	30	500	
			0 to 70°C	–	–	800	–	–	–	
Input Resistance	R_I			0.3	2	–	0.3	2	–	MΩ
Open-Loop Differential Voltage Gain	A_{OL}	$R_L \geq 2$ kΩ, $V_O = \pm 10$ V	25°C	20,000	200,000	–	50,000	200,000	–	
			0 to 70°C	15,000	–	–	–	–	–	
			−55 to +125°C	–	–	–	25,000	–	–	
Common-Mode Input Voltage Range	V_{ICR}		25°C	±12	±13	–	–	–	–	V
			−55 to +125°C	–	–	–	±12	±13	–	
Common-Mode Rejection Ratio	CMRR	$R_S \leq 10$ kΩ	25°C	70	90	–	–	–	–	dB
			−55 to +125°C	–	–	–	70	90	–	
Supply Voltage Rejection Ratio	V_{RR}	$R_S \leq 10$ kΩ	25°C	–	30	150	–	–	–	μV/V
			−55 to +125°C	–	–	–	–	30	150	
Output Voltage Swing	$V_{O(P-P)}$	$R_L \geq 10$ kΩ	25°C	±12	±14	–	–	–	–	V
			−55 to +125°C	–	–	–	±12	±14	–	
		$R_L \geq 2$ kΩ	25°C	±10	±13	–	–	–	–	
			0 to 70°C	±10	±13	–	–	–	–	
			−55 to +125°C	–	–	–	±10	±13		
Supply Current			25°C	–	1.7	2.8	–	1.7	2.8	mA
			−55°C	–	–	–	–	2	3.3	
			+125°C	–	–	–	–	1.5	2.5	
Device Dissipation	P_D		25°C	–	50	85	–	50	85	mW
			−55°C	–	–	–	–	60	100	
			+125°C	–	–	–	–	45	75	

*Values apply for each section of the dual amplifiers.

ELECTRICAL CHARACTERISTICS

Typical Values Intended Only for Design Guidance

Characteristics	Symbols	Test Conditions (Supply Volts: $V^+ = 15$, $V^- = -15$)	LIMITS: CA741C, CA747C*, CA748C*, CA1458	LIMITS: CA741, CA747*, CA748*, CA1558	Units
Input Capacitance	C_I		1.4	1.4	pF
Offset Voltage Adjustment Range			±15	±15	mV
Output Resistance	R_O		75	75	Ω
Output Short-Circuit Current			25	25	mA
Transient Response Risetime	t_r	Unity Gain, V_I = 20 mV, R_L = 2 kΩ, $C_L \leq$ 100 pF	0.3	0.3	μs
Overshoot			5.0	5.0	%
Slew Rate: Closed Loop	SR	$R_L \geq$ 2 kΩ	0.5	0.5	V/μs
Open Loop▲			40	40	

▲Open-loop slew rate applies only for types CA748C and CA748.

*Values apply for each section of the dual amplifiers.

MAXIMUM RATINGS, Absolute-Maximum Values at T_A = 25°C

DC Supply Voltage (between V^+ and V^- terminals):
- CA741C, CA747C▲, CA748C, CA1458▲ . . . . . 36 V
- CA741, CA747▲, CA748, CA1558▲ . . . . . 44 V

Differential Input Voltage . . . . . ±30 V

DC Input Voltage* . . . . . ±15 V

Output Short-Circuit Duration . . . . . Indefinite

Device Dissipation:
- Up to 70°C (CA741C, CA748C) . . . . . 500 mW
- Up to 75°C (CA741, CA748) . . . . . 500 mW
- Up to 30°C (CA747) . . . . . 800 mW
- Up to 25°C (CA747C) . . . . . 800 mW
- Up to 30°C (CA1558) . . . . . 680 mW
- Up to 25°C (CA1458) . . . . . 680 mW

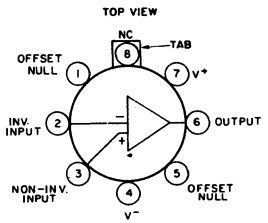
For Temperatures Indicated Above . . . . . Derate linearly 6.67 mW/°C

Voltage between Offset Null and V^- (CA741C, CA741, CA747CE) . . . . . ±0.5 V

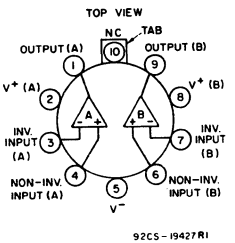
Ambient Temperature Range:
- Operating – CA741, CA747E, CA748, CA1558 . . . . . −55 to +125 °C
- CA741C, CA747C, CA748C, CA1458 . . . . . 0 to +70 °C
- Storage . . . . . −65 to +150 °C

Lead Temperature (During Soldering):
- At distance 1/16±1/32 inch (1.59±0.79 mm) from case for 10 seconds max. . . . . 265°C

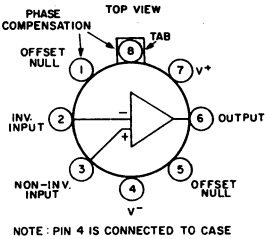
▲Voltage values apply for each of the dual operational amplifiers.



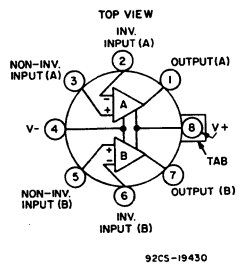
1a. – CA741CS, CA741CT, CA741S, & CA741T with internal phase compensation.



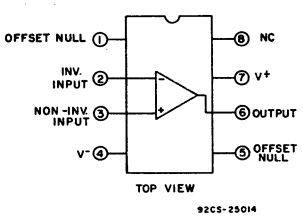
1b. – CA747CT and CA747T with internal phase compensation.



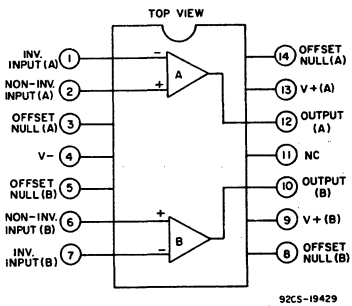
1c. – CA748CS, CA748CT, CA748S, and CA748T with external phase compensation.



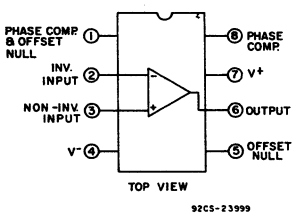
1d. – CA1458S, CA1458T, CA1558S, and CA1558T with internal phase compensation.



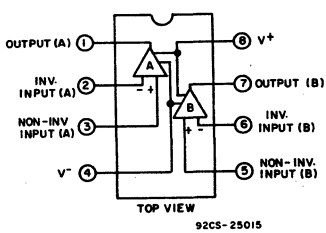
1e. – CA741CE with internal phase compensation.



1f. – CA747CE and CA747E with internal phase compensation.



1g. – CA748CE with external phase compensation.



1h. – CA1458E with internal phase compensation.

Fig. 1 – Functional diagrams.

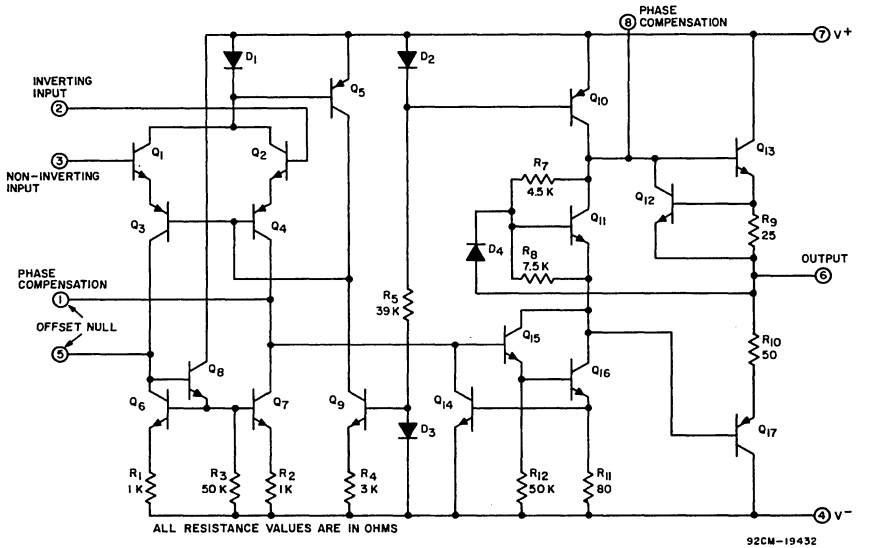


Fig.2–Schematic diagram of operational amplifier with external phase compensation for CA748C and CA748.

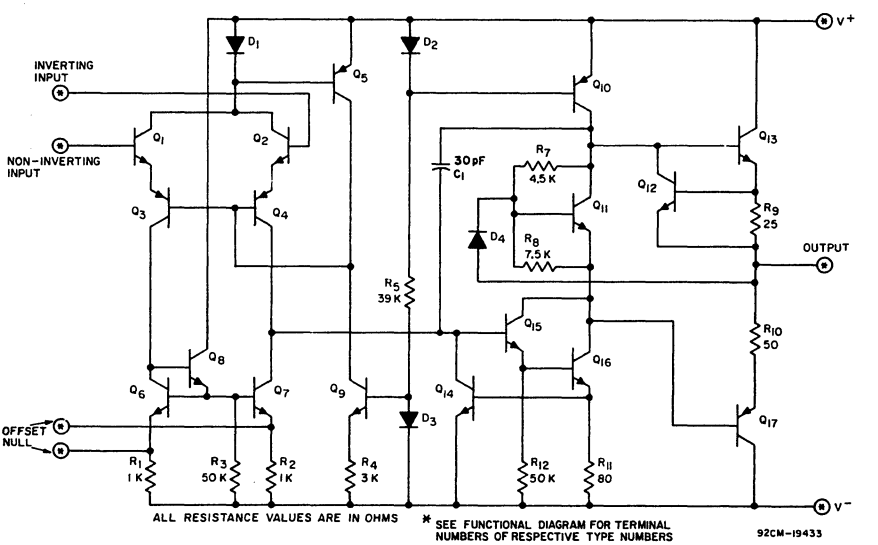


Fig.3–Schematic diagram of operational amplifiers with internal phase compensation for CA741C and CA741 and for each amplifier of the CA747C, CA747, CA1458, and CA1558.

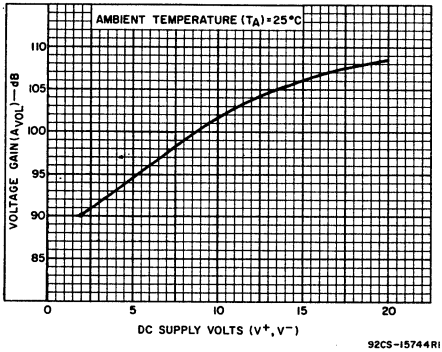


Fig. 4 — Open-loop voltage gain vs. supply voltage for all types except CA748 and CA748C.

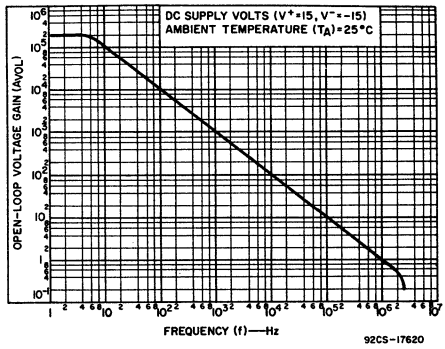


Fig. 5 — Open-loop voltage gain vs. frequency for all types except CA748 and CA748C.

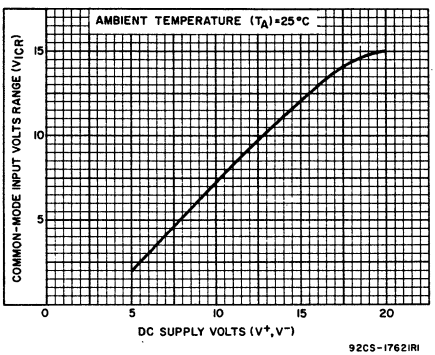


Fig. 6 — Common-mode input voltage range vs. supply voltage for all types.

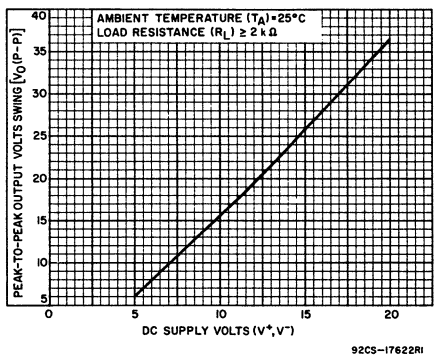


Fig. 7 — Peak-to-peak output voltage vs. supply voltage for all types except CA748 and CA748C.

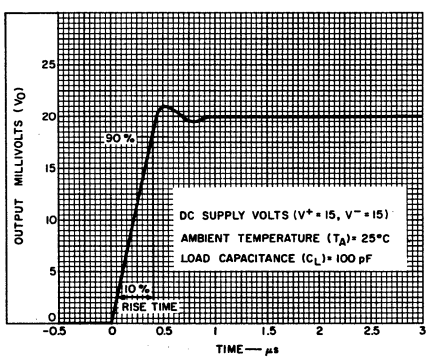


Fig. 8 — Output voltage vs. transient response time for CA741C and CA741.

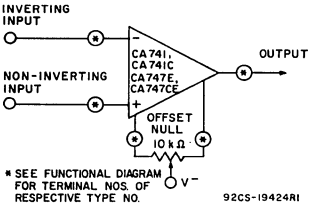


Fig. 9 – Voltage-offset null circuit for CA741C, CA741, CA747CE and CA747E.

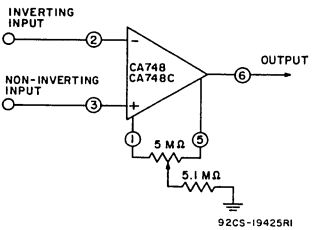


Fig. 10 – Voltage-offset null circuit for CA748C and CA748.

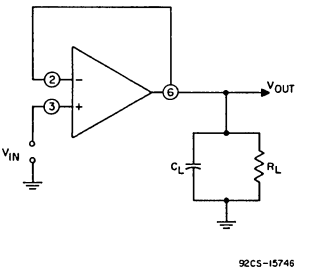


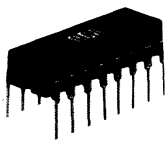
Fig. 11 – Transient response test circuit for all types.

RCA
Solid State
Division

Linear Integrated Circuits

Monolithic Silicon

CA758E



16-Lead Dual-in-Line
Plastic Package

H-1622

RC Phase-Lock-Loop Stereo Decoder

For FM Multiplex Systems

Features:

- **Low distortion (THD): 0.4% (typ.)**
- **Excellent SCA rejection: 70 dB typ.**
- **RC oscillator**
- **High-audio-channel separation: 45 dB**
- **Power supply range: 10 to 16 V dc**
- **Requires only one adjustment for complete alignment**
- **Low-impedance outputs**
- **Stereo indicator lamp drive: 150 mA typ.**

RCA-CA758E is a monolithic silicon integrated circuit RC phase-lock loop stereo decoder intended for FM solid-state stereo multiplex systems.

The CA758E is pin compatible and electrically equivalent to industry types μA758, MC1311P, LM1800, and ULX2244.

The CA758E decodes the multiplexed stereo input signal into left and right channel audio output signals. The decoder also suppresses SCA (storecast) transmissions when present in the composite stereo signal.

The decoder uses a minimum of external components, and requires one adjustment (oscillator frequency) for complete alignment. In addition, the CA758E provides automatic mono-stereo mode switching and energizes a stereo indicator lamp.

The CA758E is supplied in a 16-lead dual-in-line plastic package and operates over an ambient temperature range of −40 to +85°C.

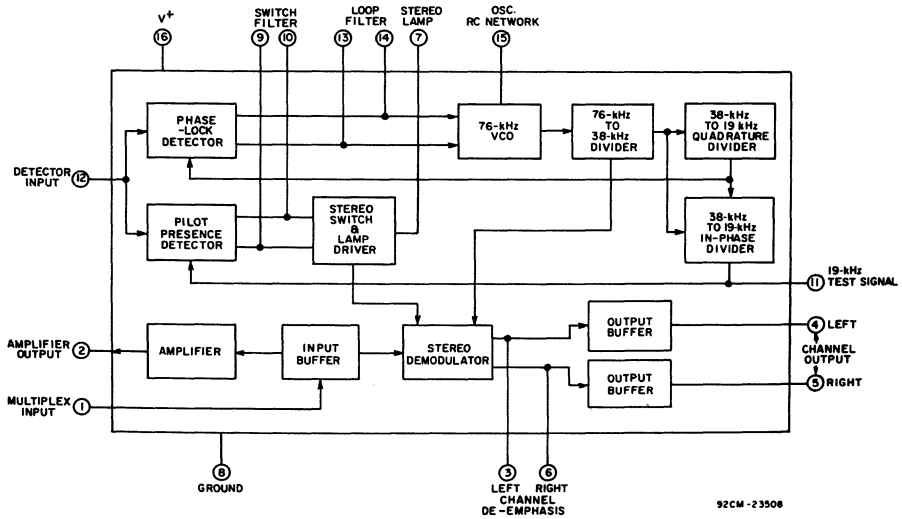


Fig. 1 – Functional block diagram of the CA758E.

MAXIMUM RATINGS, *Absolute-Maximum Values at $T_A = 25^oC$*

DC Supply Voltage	+18 V
DC Supply Voltage (for ≤a 15-second period)	+22 V
DC Voltage at Term. 7 (Lamp Driver Circuit with Lamp "OFF")	+22 V
Device Dissipation:	
Up to $T_A = 70^oC$	730 mW
Above $T_A = 70^oC$ derate linearly	9.1 mW/^{o}C
Ambient Temperature Range:	
Operating	−40 to +85°C
Storage	−65 to +150°C
Lead Temperature (During soldering):	
At a distance not less than 1/32" (0.79 mm) from case for 10 s max.	+265°C

ELECTRICAL CHARACTERISTICS

CHARACTERISTIC	TEST CONDITIONS (Referenced to Fig. 3 unless otherwise specified) $V^+ = 12$ V, $T_A = 25^oC$ Multiplex Input Signal (L=R, pilot "OFF") = 300 mV RMS 19-kHz Pilot Level = 30 mV RMS f (modulation) = 400 Hz or 1 kHz	LIMITS			UNITS
		Min.	Typ.	Max.	
Static Characteristics					
Total Current	Lamp "OFF"	–	26	35	mA
Maximum Available Lamp Current		75	150	–	mA
DC Voltage at Term. 7 (Lamp Driver)	I (Lamp) = 50 mA	–	I.3	1.8	V
DC Voltage Shift at either Term. 4 or 5 (Output)	Stereo-to-Mono Operation	–	30	150	mV
Dynamic Characteristics					
Power Supply Ripple Rejection	For a 200-Hz, 200-mV RMS Signal	35	45	–	dB
Input Resistance		20	35	–	kΩ
Output Resistance		0.9	1.3	2.0	kΩ
Channel Separation (Stereo)	At f = 100 Hz	–	40	–	dB
	f = 400 Hz	30	45	–	dB
	f = 10 kHz	–	45	–	dB
Channel Balance (Monaural)		–	0.3	1.5	dB
Voltage Gain	At f = 1 kHz	0.5	0.9	1.4	V/V
Pilot Input Level:					
19-kHz Input	Lamp "ON"	–	15	20	mV RMS
19-kHz Input	Lamp "OFF"	2.0	7.0	–	mV RMS
Hysteresis	Lamp "OFF"	3.0	7.0	–	dB
Capture Range (Deviation from 76-kHz Center Frequency)		±2.0	±4.0	±6.0	%
Total Harmonic Distortion	Multiplex Input Signal = 600 mV RMS (Pilot "OFF")	–	0.4	1.0	%
19-kHz Rejection		25	35	–	dB
38-kHz Rejection		25	45	–	dB
SCA (Storecast) Rejection	Measured Composite Signal: 80% Stereo, 10% Pilot, 10% SCA	–	70	–	dB
Voltage-Controlled Oscillator (VCO) Tuning Resistance	Total Resistance (Term. 15 to 8) required to set f_{REF} = 19 kHz ± 10 Hz (Term. 11)	21.0	23.3	25.5	kΩ
Voltage-Controlled Oscillator	$0^o \leq T_A \leq 25^oC$	–	+0.1	±2	%
Frequency Drift	$25^o \leq T_A \leq 70^oC$	–	−0.4	±2	%

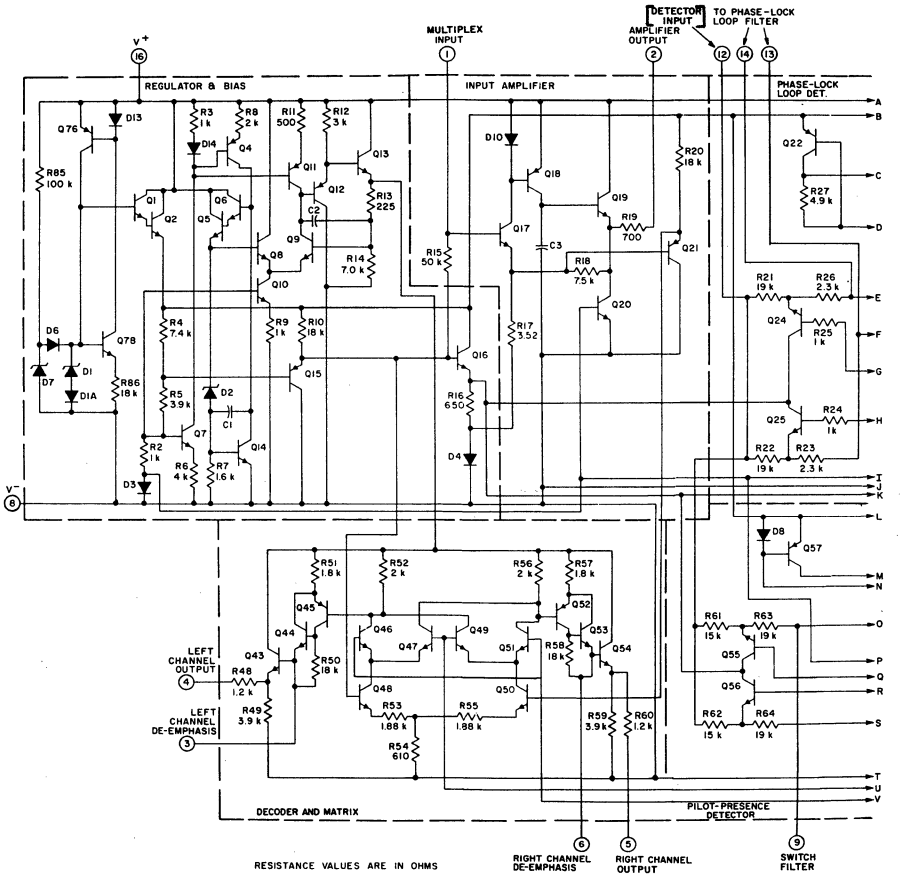


Fig. 2 – Schematic diagram of the CA758E.

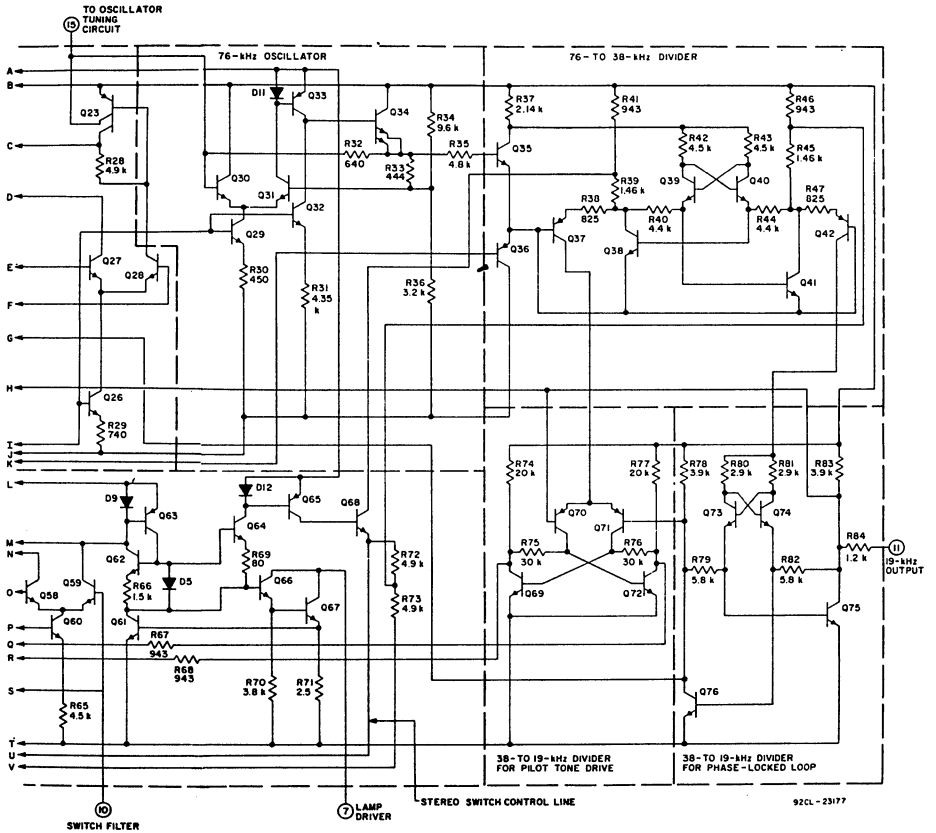


Fig.2 – Schematic diagram of the CA758E (Cont'd).

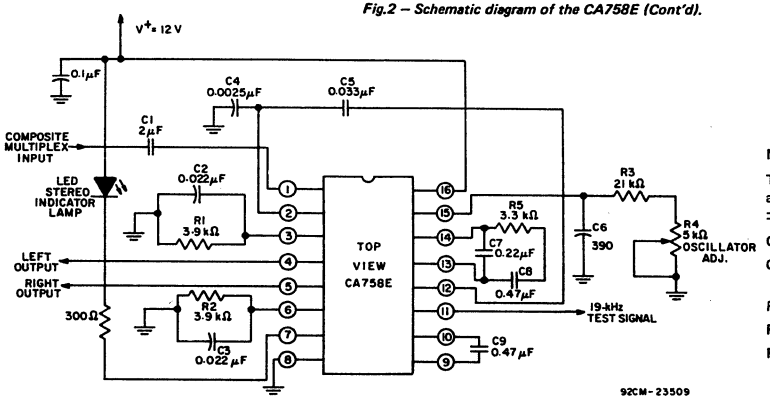


Fig. 3 – Test circuit for measurement of dynamic characteristics.

NOTES:

Tolerance on resistors is ±5% and tolerance on capacitors is ±20% unless otherwise specified.

C_1 = +100%, –20%

C_6 = ±1% in test circuit and ±5% in typical application.

R_3 = ±1%

R_4 = ±10%

R_1 and R_2 = ±1% in test circuit and ±5% in typical application.

TYPICAL PERFORMANCE CHARACTERISTICS (Referenced to Fig. 3)

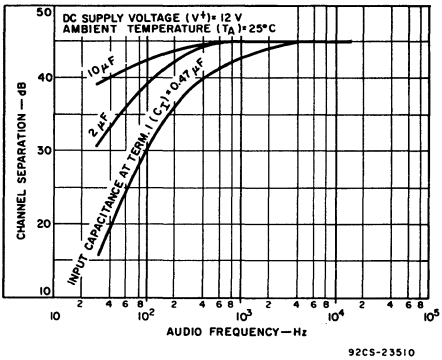


Fig. 4 — Channel separation vs. audio frequency.

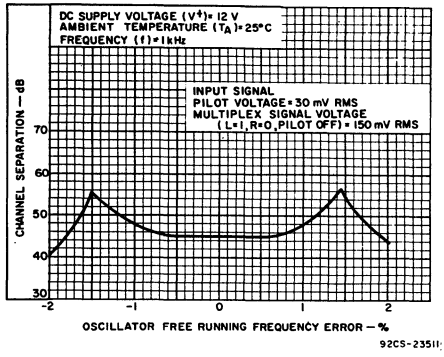


Fig. 5 — Channel separation vs. oscillator free running frequency error.

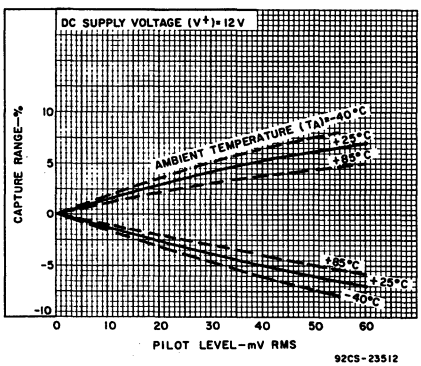


Fig. 6 — Capture range vs. pilot level.

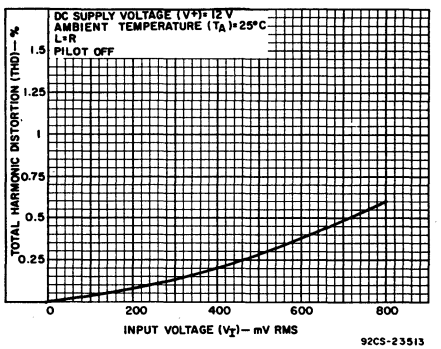


Fig. 7 — Total harmonic distortion vs. input level.

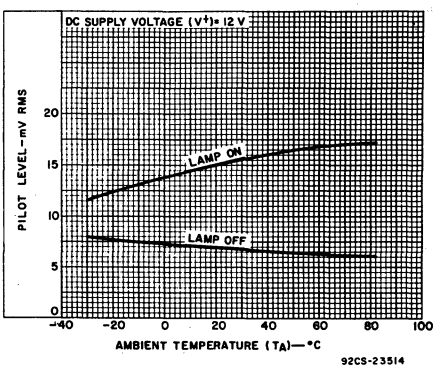


Fig. 8 — Lamp turn-on and turn-off sensitivity vs. ambient temperature.

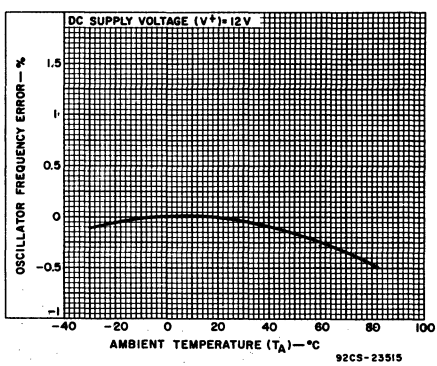


Fig. 9 — Oscillator free running frequency error vs. ambient temperature.

RCA Solid State Division

Linear Integrated Circuits

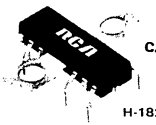
Monolithic Silicon

CA810Q
CA810QM

Preliminary Data



H-1825



H-1826

Modified 16-lead quad-in-line plastic packages with wing tabs

7-Watt Audio Power Amplifier With Thermal Shut-Down

Features

- Power output – 7 W with 4Ω load
- Supply voltage range – 4 to 20 V
- Peak output current – 2.5 A (max.)
- Very low harmonic and cross-over distortion

The RCA-CA810Q and CA810QM are monolithic audio amplifiers intended for class B operation. They are specifically designed for mobile equipment operating from 12-V battery supplies. They operate over a wide range of supply voltages (4 to 20 V) with very low harmonic and crossover distortion. The maximum repetitive peak output current is 2.5 A, and an integral thermal limiting circuit shuts the device down in case of output overload or excessive package temperature.

The CA810Q and CA810QM are supplied in modified 16-lead quad-in-line plastic packages ("Q" suffix) with integral wing-tab heat sinks. The tabs on the CA810Q are bent down for p.c. board insertion, and on the CA810QM they are flat and pierced for easy attachment to an external heat sink.

The CA810Q and CA810QM are electrically and mechanically equivalent to types TBA810S and TBA810AS, respectively. It should be noted that pin-numbering conventions for these devices may differ from manufacturer to manufacturer, however the devices are pin compatible and interchangeability is not affected.

MAXIMUM RATINGS, *Absolute Maximum Values at $T_A = 25°C$*

SUPPLY VOLTAGE			20 V
PEAK OUTPUT CURRENT (non-repetitive)			3.5 A
PEAK OUTPUT CURRENT (repetitive)			2.5 A
DEVICE DISSIPATION:			
At $T_A = 70°C$			1 W
At $T_{tab} = 100°C$			5 W
AMBIENT TEMPERATURE RANGE:			
Operating			−40°C to (Refer to Fig. 7 for typical high-temperature limit)
Storage			−40 to +150°C
THERMAL RESISTANCE:	CA810Q	CA810QM	
Junction to tab	12	10	°C/W
Junction to ambient	70*	80	°C/W

*Value obtained with tabs soldered to printed-circuit board

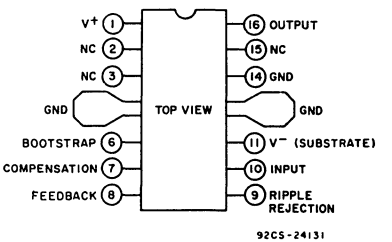


Fig. 1 – Terminal diagram of CA810Q and CA810QM. The wing tabs on the CA810Q are bent down, and on the CA810QM they are flat and pierced.

ELECTRICAL CHARACTERISTICS, *at* $T_A = 25°C$

CHARACTERISTIC	SYMBOL	TEST CONDITIONS		LIMITS			UNIT
		Supply Voltage (V^+) = 14.4 V Unless Otherwise Specified		CA810Q CA810QM			
				MIN.	TYP.	MAX.	
Supply Voltage	V^+			4	–	20	V
Input Voltage	V_I			–	–	220	mV
Input Sensitivity	e_I	P_O=6 W, R_L=4 Ω, R1 = 56 Ω, f = 1 kHz		–	80	–	mV
Quiescent Output Voltage	V_O			6.4	7.2	8	V
Quiescent Current Drain	I_O			–	12	20	mA
Input Noise Voltage	e_N	R_g=0, BW (–3 dB) =20 to 20,000 Hz		–	2	–	μV
Bias Current	I_{IB}			–	0.4	–	μA
Output Power	P_O	f=1 kHz, R_L=4Ω, THD = 10%	V^+= 14.4 V	–	6	–	W
			V^+= 6 V	–	1	–	
Input Resistance	R_I			–	5	–	MΩ
Total Harmonic Distortion	THD	P_O=50 mW to 3W, R_L 4 Ω, f = 1 kHz		–	0.3	–	%
Open-Loop Voltage Gain	A_{OL}	R_L = 4 Ω, f = 1 kHz		–	80	–	dB
Closed-Loop Voltage Gain	A	R_L = 4 Ω, f = 1 kHz, R1 = 56 Ω		34	37	40	dB
Efficiency	η	P_O = 5 W, R_L = 4 Ω, f = 1 kHz		–	70	–	%

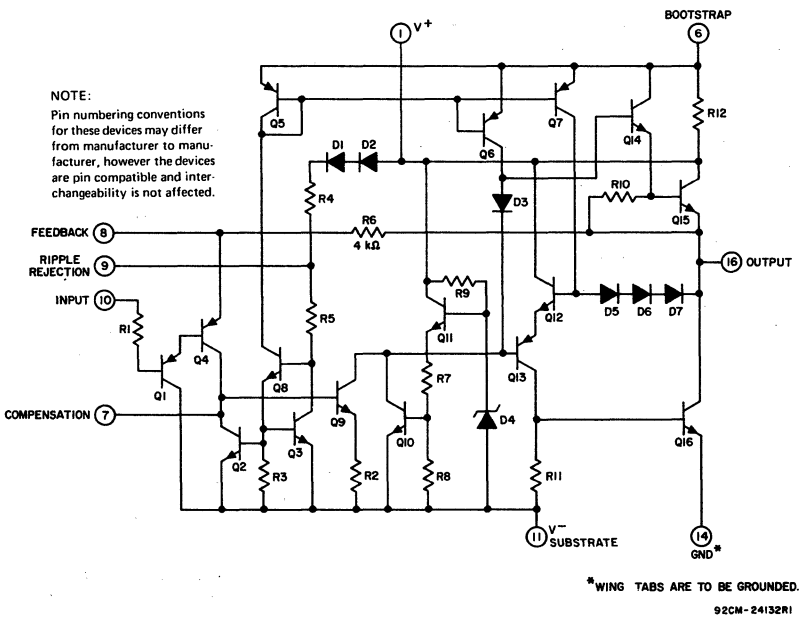


Fig. 2 – Schematic diagram of CA810Q, CA810QM.

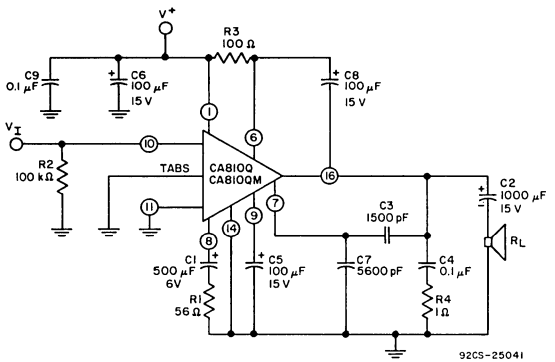
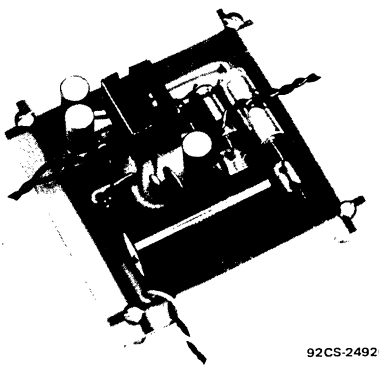


Fig. 3 – Test and circuit application for the CA810Q and CA810QM.

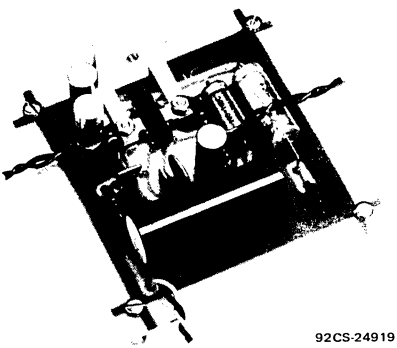


Fig. 4 – Bottom view of printed-circuit boards shown in Figs. 5 and 6.



Circuit heat is dissipated by a combination of free air and printed-circuit board foil.

Fig. 5 – Component view of printed-circuit board for CA810Q.



Circuit arrangement for use with chassis having a thermal resistance of $\leq 5^{o}C/W$. Vertical bracket should make good thermal contact to chassis.

Fig. 6 – Component view of printed-circuit board for CA810QM.

The thermal-limiting network incorporated in the CA810 Series circuits provides protection against damage due to excessive semiconductor temperatures that may result from high ambient temperatures and/or excessive dissipation, e.g., as encountered in sustained overloads. As indicated in Fig. 7, the thermal-limiting feature automatically reduces the supply current (and output power) at the higher temperatures.

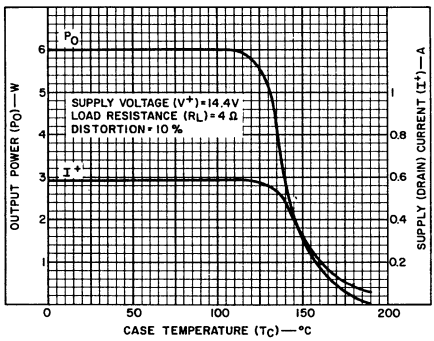


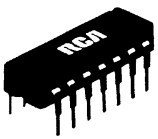
Fig. 7 – Typical output power and drain current vs. case temperature.

RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA1310E



14-Lead Dual-in-Line Plastic Package H-1517

RC Phase-Lock-Loop Stereo Decoder

For FM Multiplex Systems

Features:

- Low distortion (THD): 0.3% typ.
- Excellent SCA (storecast) rejection: 75 dB typ.
- RC oscillator
- High audio channel separation: 40 dB
- Operates from a wide range of power supplies: 8 to 14 V dc
- Requires only one adjustment for complete alignment
- Drives a stereo indicator lamp up to 75 mA — surge current limiting

RCA-CA1310E is a monolithic silicon integrated circuit RC phase-lock-loop stereo decoder intended for FM solid-state stereo multiplex systems.

The CA1310E is a direct replacement for industry types MC1310P, LM1310, and SN76115N.

This decoder uses a minimum of external components. In addition the stereo decoder requires only one adjustment (oscillator frequency) for complete alignment.

The CA1310E is supplied in a 14-lead dual-in-line plastic package and operates over an ambient temperature range of –40 to +85°C.

MAXIMUM RATINGS, *Absolute-Maximum Values at $T_A = 25°C$*

DC Supply Voltage	14 V
Current (Lamp) at Term. 6	75 mA
Device Dissipation:	
Up to $T_A = 25°C$	625 mW
Above $T_A = 25°C$ derate linearly	5 mW/°C
Ambient Temperature Range:	
Operating	–40 to +85°C
Storage	–65 to +150°C
Lead Temperature (During soldering):	
At distance not less than 1/32" (0.79 mm) from case for 10 s max.	+265°C

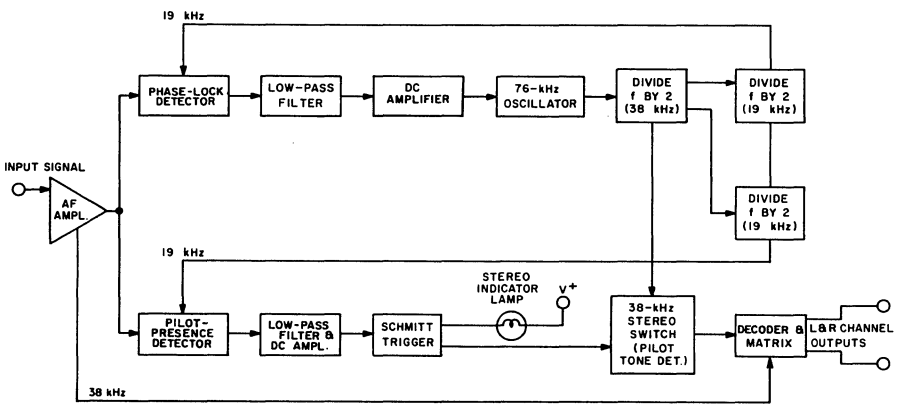


Fig. 1 — Functional block diagram system using the CA1310E. 92CS-23500

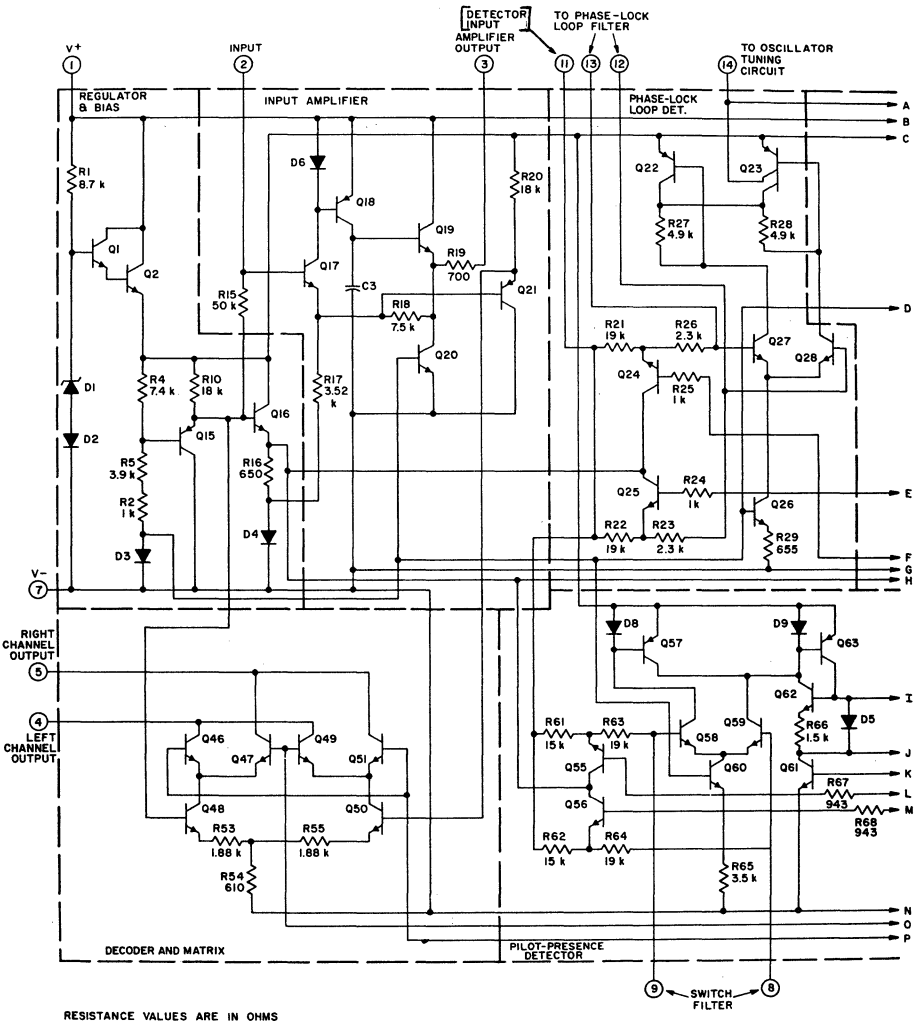


Fig. 2 – Schematic diagram of the CA1310E.

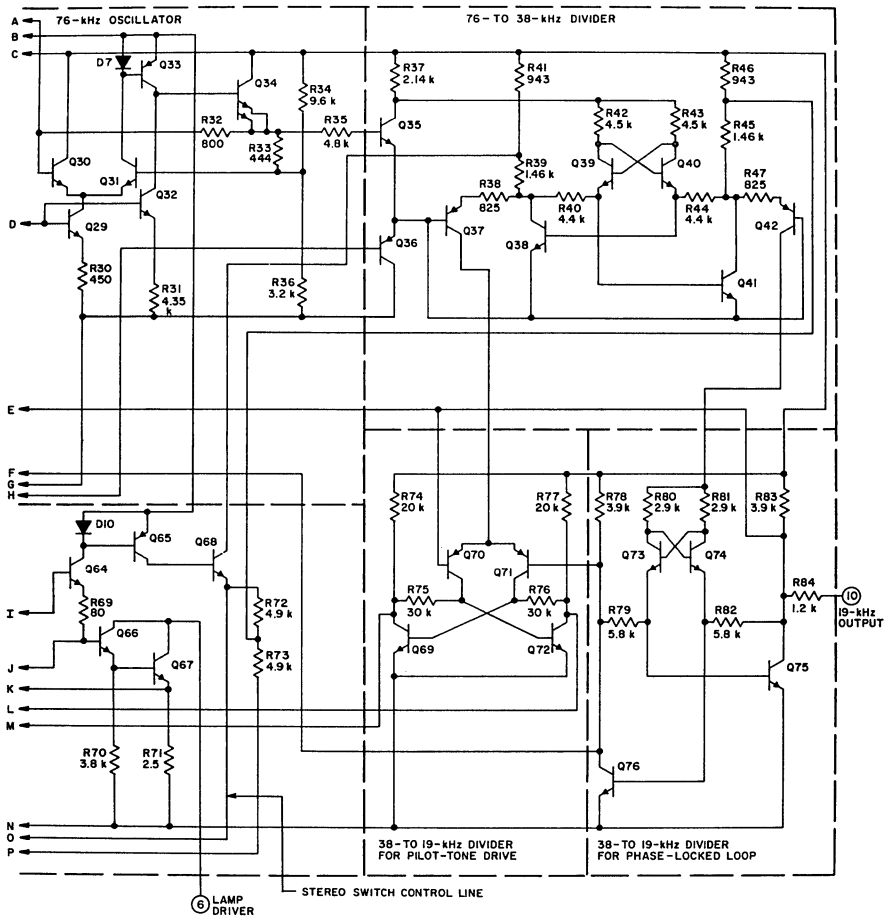
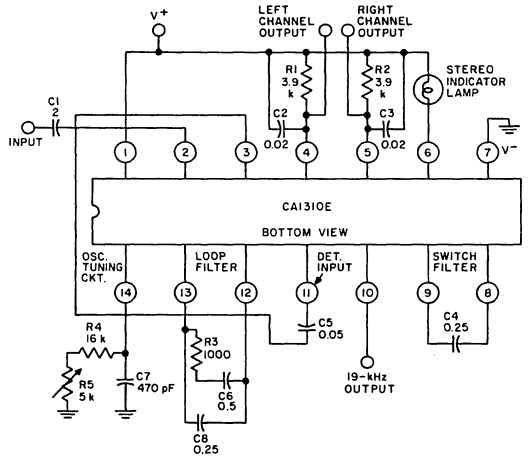


Fig. 2 – Schematic diagram of the CA1310E (Cont'd).

ELECTRICAL CHARACTERISTICS

CHARACTERISTIC	TEST CONDITIONS (Referenced to Fig. 3) $V^+ = 12$ V $T_A = 25^oC$ Composite Multiplex Input Signal = 560 mV RMS (2.8 V p-p) Only L or R Channel modulated; and with 100-mV RMS (10%) Pilot Level	LIMITS			UNITS
		Min.	Typ.	Max.	
Static Characteristics					
DC Supply Voltage	For 8-V operation, reduce load to 2.7 kΩ	8	–	14	V
Total Current	Lamp "OFF"	–	13	–	mA
Dynamic Characteristics					
Input Impedance		20	50	–	kΩ
Channel Separation (Stereo)	50 Hz – 15 kHz	30	40	–	dB
Audio Output Voltage (For any one channel)		–	485	–	mV RMS
Channel Balance (Monaural)	Pilot Tone "OFF"	–	–	1.5	dB
Capture Range (Permissible tuning error of internal oscillator)		–	±3.5	–	%
Total Harmonic Distortion		–	0.3	–	%
Ultrasonic Frequency Rejection:					
19 kHz		–	34.4	–	dB
38 kHz		–	45	–	dB
SCA (Storecast) Rejection	f = 67 kHz, 9-kHz beat note measured with 1-kHz modulation "OFF"	–	75	–	dB
Stereo Switch Level:					
19-kHz Input Level (For lamp on)		–	–	20	mV RMS
19-kHz Input Level (For lamp off)		5	–	–	mV RMS
Maximum Composite (Stereo) Input	0.5% THD	2.8	–	–	V p-p
Maximum Monaural Input	1% THD	2.8	–	–	V p-p



RESISTANCE VALUES ARE IN OHMS.
CAPACITANCE VALUES ARE IN MICROFARADS.

92CS-23501

NOTES

A buffered 3-volt positive-going square wave is available at Term. 10. The alignment of the free-running oscillator frequency may be checked at this point with a frequency counter.

C1: A lower value input coupling capacitor may be used in place of the 2-μF value if reduced separation at low frequencies is acceptable.

C4: The time constant for the stereo switch level detector circuit is calculated by C4 x 53,000 ohms ±30% with a maximum dc voltage drop across C4 of 0.25 volt (Term. 8 positive) and a pilot level voltage of 100 mV RMS. Signal voltage across C4 is negligible.

C5: The recommended 0.05-μF capacitor provides a 1.75° phase lead at 19 kHz.

R1, R2: Load resistance values are related to supply voltage as follows:

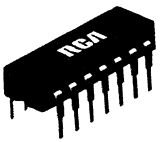
Minimum Supply Voltage	8	10	12	V
Maximum Load Resistance	2.7	4.3	6.2	kΩ

R3, C6, C8: C8 may be omitted, R3 = 100 ohms and C6 = 0.25 μF, if relaxed circuit performance is acceptable.

R4, R5, C7: If a capture range greater than ±3% typ. is required, reduce value of C7 and increase values of R4, R5 proportionally. However, beat-note distortion is increased at high signal levels because of oscillator-phase jitter. R4, C7 = ±1% in test circuit and ±5% in typical application.

Fig. 3 – Test circuit for measurement of dynamic characteristics.

Preliminary Data



14-Lead Dual-In-Line Plastic Package

H-1517

TV Video Amplifier

With AGC and Keyer Circuit

Features

- High 45-MHz gain – 53 dB (typ.)
- High-gain gated AGC system – with either positive- or negative-going sync.
- Adjustable rf AGC delay to tuner
- AGC gain reduction – 68 dB (typ.)

The RCA-CA1352E is a monolithic integrated circuit designed for use as an if amplifier in monochrome or color TV receivers. It features a high-gain gated AGC system with a 68-dB range (typ.). A delayed forward AGC output is adjustable by means of a potentiometer. Either positive- or negative-going sync may be used for this system.

MAXIMUM RATINGS, *Absolute-Maximum Values*
At T_A = 25

SUPPLY VOLTAGE:
- Between terminals 4 and 11 ..... 18 V
- Between terminals 7 or 8 and 4 ..... 18 V

INPUT VOLTAGE (terminal 1 or 2) ..... 10 V p-p
AGC INPUT VOLTAGE (terminal 6 or 10) ..... 6 V
DEVICE DISSIPATION:
- Up to T_A = 55°C ..... 750 mW
- Above T_A = 55°C derate linearly at ..... 7.9 mW/°C

AMBIENT TEMPERATURE RANGE:
- Operating ..... −40 to +85°C
- Storage ..... −65 to +150°C

LEAD TEMPERATURE (During Soldering):
- At distance 1/16 ± 1/32 in. (1.59 ± 0.79 mm) from case for 10 seconds max. ..... +265°C

The CA1352E is supplied in the 14-lead dual-in-line plastic package, and is directly interchangeable with the industry type 1352 in similar packages.

TYPICAL STATIC CHARACTERISTICS
at T_A = 25°C, V^+ = 12 V

Total Current ($I_7 + I_8 + I_{11}$) ..... 27 mA
Output Stage Current ($I_7 + I_8$) ..... 5.7 mA

TYPICAL DYNAMIC CHARACTERISTICS
at T_A = 25°C, V^+ = 12 V

AGC Range ..... 68 dB
Power Gain ..... 53 dB
Minimum rf AGC Range (term. 12) ..... 0.2 V
Maximum rf AGC Range (term. 12) ..... 7 V

SYNC POLARITY	* VOLTAGE AT TERMINAL 6	* VOLTAGE AT TERMINAL 10	* VALUE OF RI−Ω
NEGATIVE	5.5 V, −2 V, 0 V	1 TO 4 V NOM=2 V	0
POSITIVE	1 TO 8 V NOM=4.5	4.5 V, 0 V	3.9k

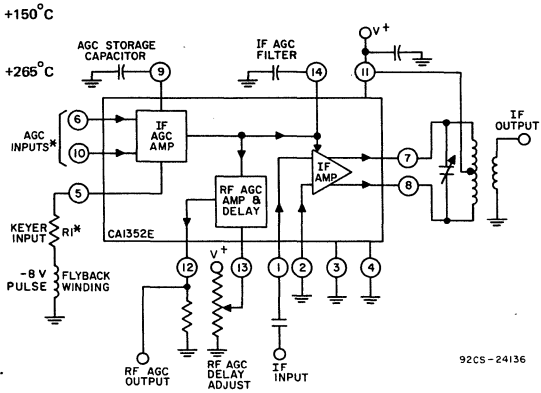


Fig. 1 – CA1352E block diagram and typical AGC test set-up.

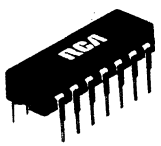
RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA1398E

Television Chroma Processor



14-Lead Dual-In-Line Plastic Package H-1517

Features

- Minimum number of external components required
- Injection-lock oscillator with internal feedback
- DC chroma gain control and hue control circuits
- Low-impedance internal voltage regulation

RCA-CA1398E is a monolithic silicon integrated-circuit chroma processor containing chroma-amplifier and gain-control, color-killer, color subcarrier oscillator, hue control, and ACC circuitry. It has been designed for interchangeability with other "1398"-type chroma-processor devices. It functions compatibly with the RCA-CA3125E Chroma Demodulator as well as other commercially available chroma demodulators in color-TV receivers. Fig. 2 shows a functional block diagram of a 2-package TV chroma system incorporating the CA1398E and CA3125E. The CA1398E is supplied in a 14-lead dual-in-line plastic package.

Maximum Ratings, *Absolute-Maximum Values at* $T_A = 25^\circ C$

Peak Horizontal-Pulse Input Current	250 μA
Supply Current (Terminal 14)	35 mA
Ambient Temperature Range:	
Operating	−40 to +85°C
Storage	−65 to +150°C
Lead Temperature (During Soldering):	
At distance 1/16" ± 1/32" (1.59 ± 0.79 mm) from case for 10 s max.	265°C

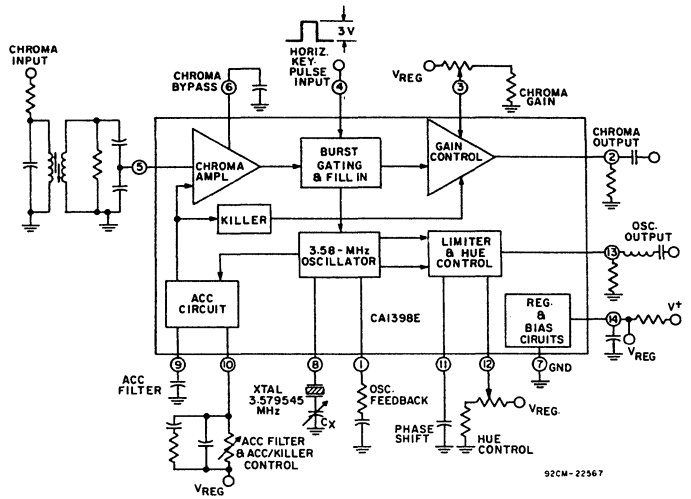


Fig. 1 – Functional block diagram of the CA1398E.

ELECTRICAL CHARACTERISTICS at T_A = 25°C and Referenced to Test Circuit (Fig. 4)

CHARACTERISTIC	TERMINAL MEASURED AND SYMBOL	TEST CONDITIONS						LIMITS			UNITS
		SWITCH POSITION (S1)	CONTROL SETTING CHROMA	HUE	KILLER	V_{BURST} mV p-p	V_{CHROMA} mV p-p	MIN.	TYP.	MAX.	
Static Characteristics											
Regulated Supply Voltage	V_{14}	2	max.	max.	max.	0	0	8.9	9.5	11.5	V
Chroma Output Bias	V_{14} to V_2	2	max.	max.	max.	6	0	1.2	2.4	3.6	V
Regulator Impedance	See Note 1	2	max.	max.	max.	0	0	–	12	25	Ω
Dynamic Characteristics (Refer to Test Set-Up Procedure for Oscillator)											
Max. Chroma Gain	V_2	1	max.	max.	See Note 2	6	5	310	425	–	mV p-p
Min. Chroma Gain	V_2	1	min.	max.		6	5	–	–	7	mV p-p
ACC Action	V_2 (dB up from gain test)	1	max.	max.		50	50	2	7	11	dB
Killer Function:											
Kill	V_2	2	max.	max.		0	5	–	–	7	mV p-p
Unkill	V_2	1	max.	max.		15	5	100	–	–	mV p-p
Oscillator Lock-Up:											
Voltage	V_{13}	1	max.	max.		6	0	250	340	390	mV p-p
Phase (Referenced to burst)	ϕ_{13}	1	max.	max.		6	0	–20	0	+20	degrees
Hue Control Range:											
Voltage	V_{13}	1	max.	min.		6	0	250	340	390	mV p-p
Phase (Referenced to burst)	ϕ_{13}	1	max.	min.		6	0	95	110	140	degrees

Note 1 – Measure V_{14} at I_{SUPPLY} = 38 mA and 18 mA. Calculate the regulator impedance:
$Z_{reg.} = [V_{14}\text{ (at 38 mA)} - V_{14}\text{ (at 18 mA)}]/0.02$

Note 2 – Increase the killer potentiometer resistance from minimum until the circuit unkills. This condition is evidenced by a shift in bias voltage at Term. 2 Maintain this potentiometer setting for all the dynamic tests.

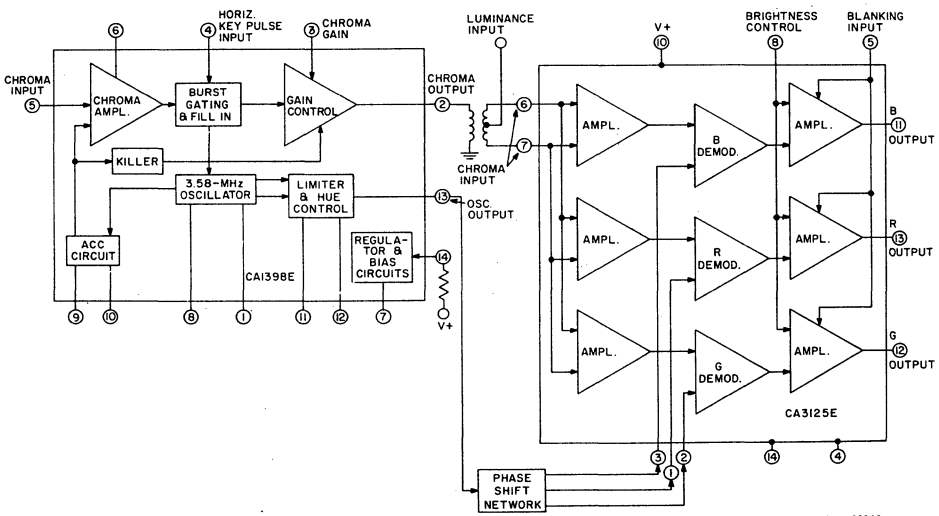


Fig. 2 – TV chroma system functional block diagram.

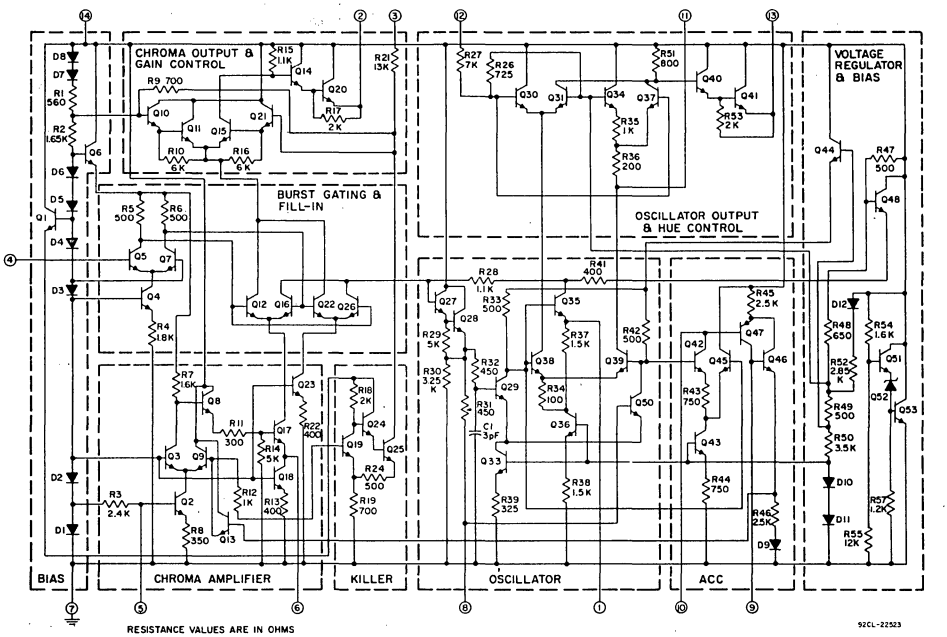


Fig. 3 – Schematic diagram of the CA1398E.

TEST SET-UP PROCEDURE FOR OSCILLATOR

Remove the horizontal keying and chroma inputs and adjust C_X to obtain a free-running oscillator frequency of 3.579545 MHz ±10 Hz. Under the same Test Conditions described in the Electrical Characteristics Chart for Oscillator Lock-Up, vary L1 (approx. 20 μH) and/or C1 (approx. 1000 pF) to obtain the initial conditions for amplitude and phase oscillator lock-up.

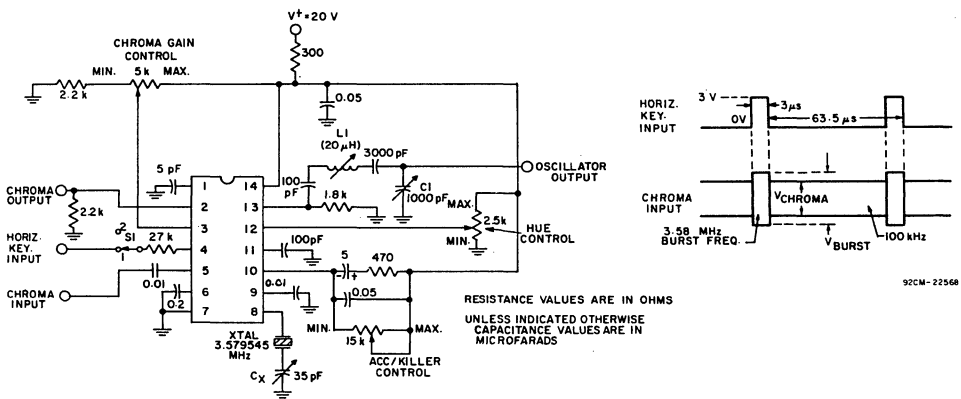


Fig. 4 – Typical static and dynamic characteristics test circuit for the CA1398E.

RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA1541D



14-Lead Dual-In-Line Ceramic Package H-1553

Dual-Input Memory Sense Amplifier

Features

- Complete dual input core memory sense amplifier
- Two available outputs: —Saturated logic output
  —Linear output (positive output for either polarity input)
- Nominal threshold voltage: 17 mV
- Adjustable threshold: 10 to 35 mV
- Low threshold uncertainty range: ±3 mV
- Fast overload recovery time: —Differential-Mode: 15 ns typ.
  —Common-Mode: 30 ns typ.
- Independent channel gate and strobe terminals compatible with saturated logic levels
- Suitable for core memories having cycle times ≥ 0.4 μs
- Input offset voltage: 6 mV max.

RCA-CA1541D*, a monolithic silicon integrated circuit, is a dual-input memory sense amplifier intended for core memory applications.

The sense amplifier, consisting of two differential input amplifiers, a common second stage amplifier, and an output logic gate (See Fig. 1), converts low-level core-memory "1" pulses to saturated logic-level output pulses. Either one of the input amplifiers may be gated ON with a saturated logic signal so that an incoming "1" pulse of positive or negative polarity can be detected from either of two sense lines.

The CA1541D features an external switching threshold adjustment, plus its gate and strobe inputs are compatible with saturated logic levels. The sense amplifier is suitable for operation with core memories having cycle times equal to or greater than 0.4 μs and is unilaterally interchangeable with industry types 1541L and 1441.

The CA1541D is supplied in 14-lead dual-in-line ceramic package and is rated for operation over the full military temperature range of −55°C to +125°C.

*Formerly Developmental Type TA5820.

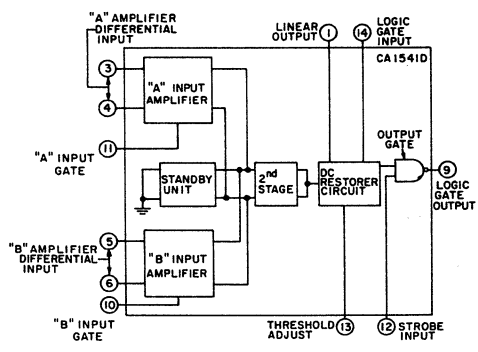


Fig. 1 — Functional block diagram of the CA1541D.

MAXIMUM RATINGS, Absolute Maximum Values, at T_A = 25°C

Except for Differential Input Voltage, all voltages are measured with respect to ground (Term. 8).

DC Supply Voltage:	
V^+ (Term. 2)	+10 V
V^- (Term. 7)	−10 V
Differential Input Voltage	±5 V
Common-Mode Input Voltage	±5 V
"A" or "B"-Gate Input Voltage*	V^- to V^+
Strobe Terminal Voltage	V^- to +6V
Output Terminal Load Current	±25 mA
Device Dissipation:	
Up to T_A = 75°C	750 mW
Above T_A = 75°C	Derate Linearly 8 mW/°C
Ambient Temperature Range:	
Operating	−55 to +125 °C
Storage	−65 to +150 °C
Lead Temperature (during soldering):	
At distance not less than 1/32 inch (0.79 mm) from case for 10 seconds max.	+265°C

*Note: The "A" or "B"-Gate Input Voltage is also referred to, as the Channel-Gate Input Voltage.

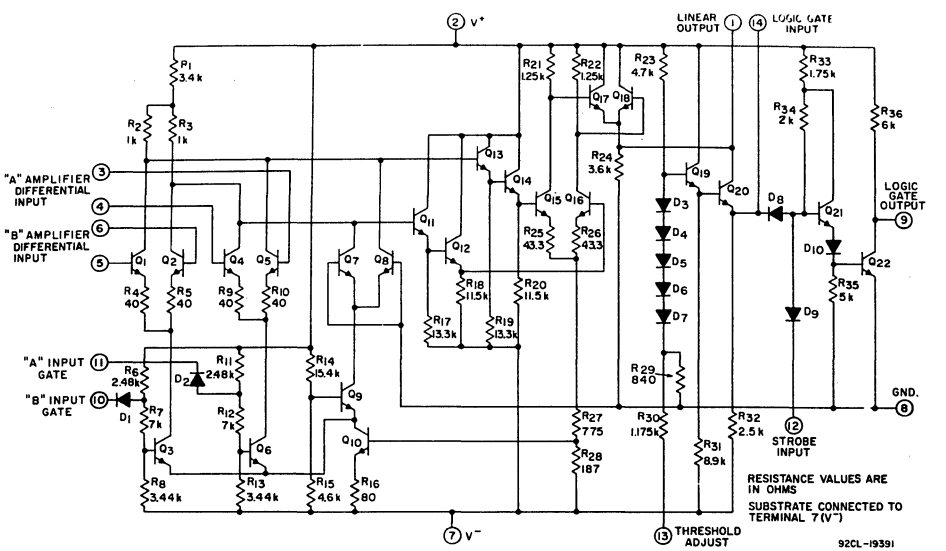


Fig. 2 – Schematic diagram of the CA1541D.

ELECTRICAL CHARACTERISTICS

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS (V^+ = 5V, V^- = −5V; V_{TH} ADJ. = −5V ± 1%, (Term. 13); C_{EXT} = 0.01 μF; T_A = 25°C (unless indicated otherwise))		LIMITS MIN.	LIMITS TYP.	LIMITS MAX.	UNITS
Static (DC) Characteristics							
Power Dissipation	P_D			–	140	180	mW
Input Offset Current	I_{IO}			–	1	2	μA
Input Bias Current:	I_{IB}		V_5 = V_6 = V_3 = V_4 = 0				μA
T_A = 25°C				–	5	25	
T_A = −55°C				–	–	50	
Output Voltage:							
High	V_{OH}	I_{OM} = 200 μA		3	–	–	V
Low –	V_{OL}	V_{14} = 5 V, I_9 = 10 mA					mV
T_A = 25°C				–	–	350	
T_A = 125°C				–	–	400	
Strobe Load Current	I_S	V_{12} = 0		–	–	1.5	mA
Strobe Reverse Current:	I_{SR}	V_{12} = 5V					μA
T_A = 25°C				–	–	2	
T_A = 125°C				–	–	25	
Input Gate Load Current	I_G	V_{10} = V_{11} = 0		–	–	2.5	mA
Input Gate Reverse Current:	I_{GR}	V_{10} = V_{11} = 5V					μA
T_A = 25°C				–	–	2	
T_A = 125°C				–	–	25	
Switching Characteristics							
Input Threshold Voltage:	V_{TH}						mV
T_A = 25°C				14	17	20	
T_A = −55 to 125°C				12	17	22	
Input Offset Voltage	V_{IO}			–	1	6	mV
Input Gate Voltage:		V_3 = V_5 = 25 mV, V_4 = V_6 = 0					V
High	V_{GH}			–	1.6	—	
Low	V_{GL}			–	0.7	–	
Common-Mode Range:	V_{CM}						V
Input Gate High				–	±1.5	–	
Input Gate Low				–	±1.5	–	
Differential-Mode Range:	V_{DH}			–	±600	–	mV
Input Gate High							
Input Gate Low	V_{DL}			–	±1.5	–	V
Propagation Delay:							ns
Input to Amplifier Output	t_{IA}	V_3 = 25 mV (pulsed), V_{12} = 2V		–	10	15	
Input to Output	t_{IO}			–	20	30	
Strobe to Output	t_{SO}	V_3 = V_4 = V_5 = V_6 = 0, V_{12} = 2V (pulsed)		–	15	20	
Gate Input to Amplifier Output	t_{GA}	V_{11} = 2V (pulsed)		–	10	15	
Gate Input to Amplifier Input	t_{GI}	V_3 = 25 mV		–	30	35	
Common-Mode Recovery Time:	t_{CMR}	V_3 = V_5 = 1.5 V					ns
Input Gate High				–	15	30	
Input Gate Low				–	15	30	
Differential-Mode Recovery Time:	t_{DR}	V_3 = V_5 = 400 mV					ns
Input Gate High				–	30	–	
Input Gate Low				–	0	–	

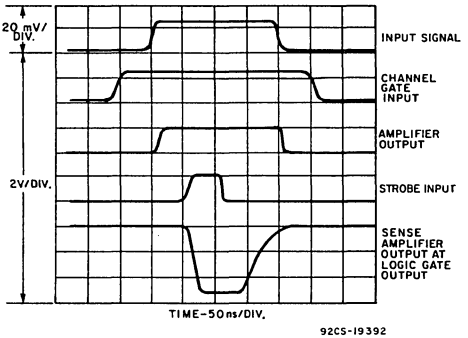


Fig. 3 – Typical operational wave forms.

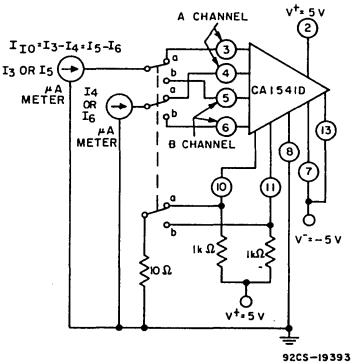


Fig. 4 – Input bias (I_{IB}) and input-offset current (I_{IO}) test circuit.

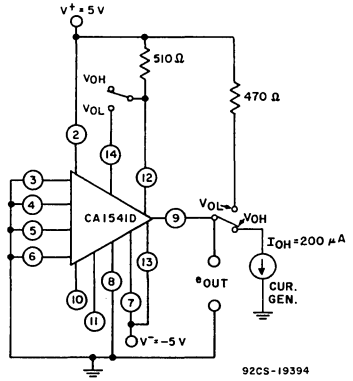


Fig. 5 – Test circuit for measurement of low (V_{OL}) and high (V_{OH}) output voltage levels.

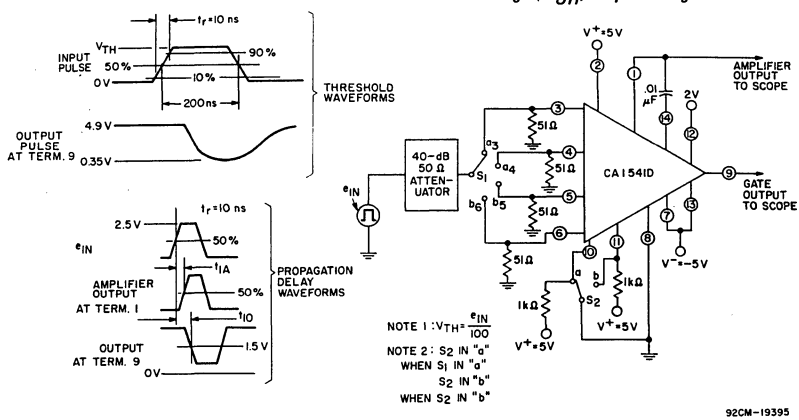


Fig. 6 – Threshold propagation delay, gate and input-offset test circuit with associated pulse wave forms.

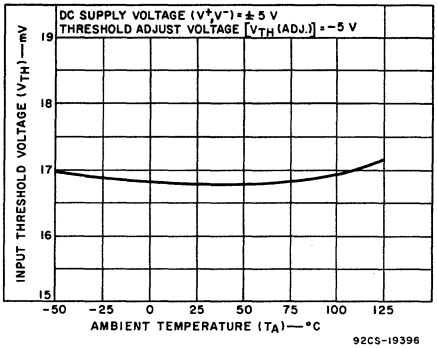


Fig. 7a – Input V_{TH} vs. T_A.

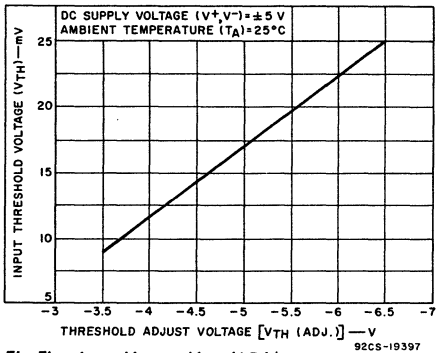


Fig. 7b – Input V_{TH} vs. V_{TH} (ADJ.).

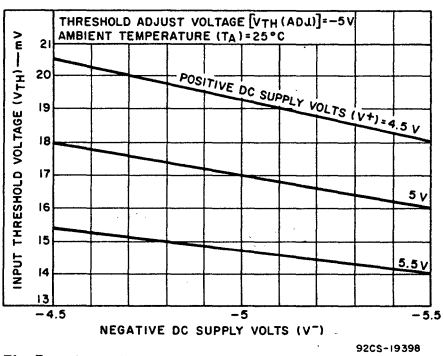


Fig. 7c – Input V_{TH} vs. V^-.

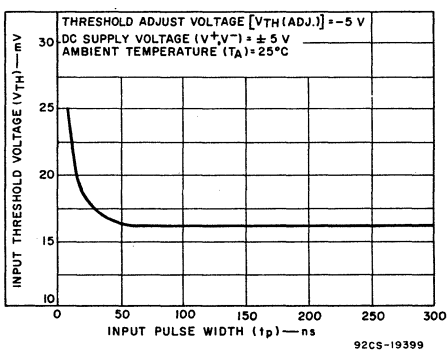


Fig. 7d – Input V_{TH} vs. input pulse width.

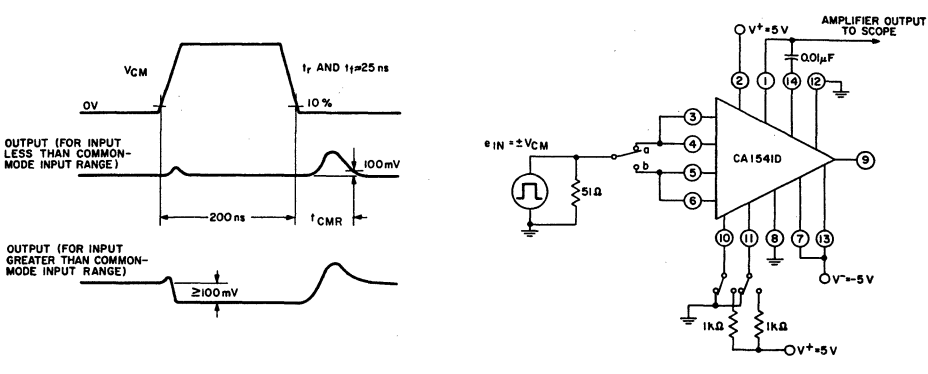


Fig. 8 – Common-mode input range test circuit with associated pulse wave forms.

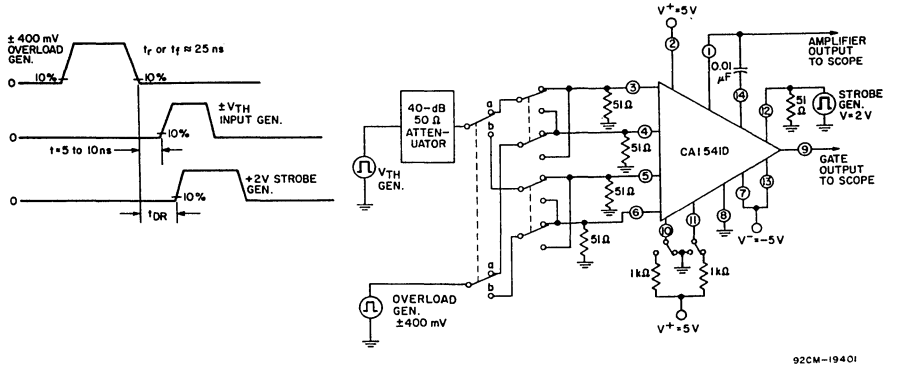


Fig. 9 – Differential-mode input range and recovery test circuit with associated pulse wave forms.

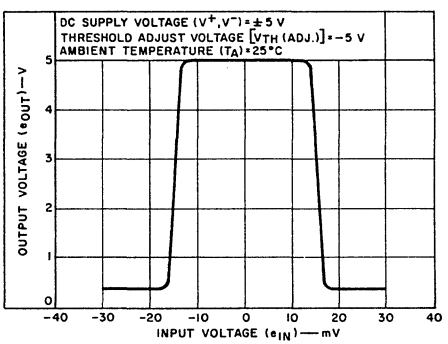


Fig. 10 – Input-output transfer characteristics.

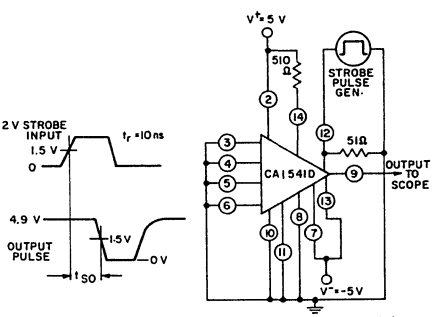


Fig. 11 – Strobe to output test circuit with associated pulse wave-forms.

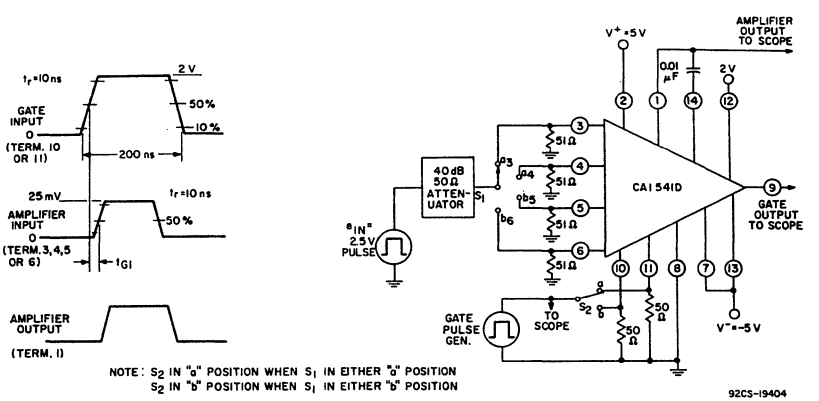


Fig. 12 – Gate input to amplifier input (t_{GI}) test circuit with associated pulse wave forms.

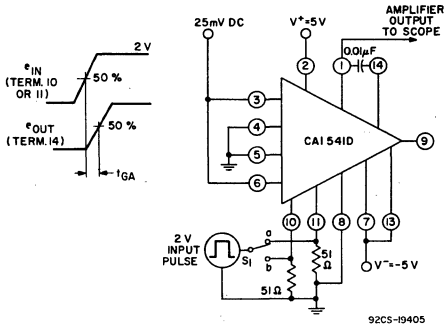


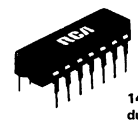
Fig. 13 – Gate input to amplifier output (t_{GA}) with associated pulse wave forms.

RCA
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Linear Integrated Circuits

Monolithic Silicon

CA2111AE
CA2111AQ



H-1517

14-Lead dual-in-line plastic package



H-1596

14-Lead quad-in-line plastic package

FM IF Amplifier-Limiter and Quadrature Detector

For FM IF and TV Sound IF Applications

Features:

- **Direct replacement for ULN2111A and MC1357**
- **Good sensitivity: Input limiting voltage (knee) (400 μV typ. at 10.7 MHz; 250 μV typ. at 4.5 MHz and 5.5 MHz)**
- **Excellent AM rejection (45 dB typ. at 10.7 MHz)**
- **Provision for output from 3-stage IF amplifier section**
- **Low harmonic distortion**
- **Quadrature detection permits simplified single-coil tuning**
- **Extremely low AFC voltage drift over full operating-temperature range**
- **Minimum number of external parts required**

The CA2111A, on a single monolithic chip, provides a multi-stage wideband amplifier-limiter, a quadrature detector, and an emitter-follower output stage. This device is designed for use in FM receivers and in the sound IF sections of TV receivers. In addition, an output terminal is provided which allows the use of the amplifier-limiter as a straight 60-dB wideband amplifier.

The amplifier-limiter features the excellent limiting characteristics of 3 cascaded differential amplifiers.

The quadrature detector requires only one coil in the associated outboard circuit and therefore, tuning is a simple procedure.

A unique feature of the CA2111A is its exceptionally low AFC voltage drift over the full operating-temperature range.

This device can be supplied in either dual-in-line or quad-in-line 14-lead plastic packages (CA2111AE and CA2111AQ, respectively).

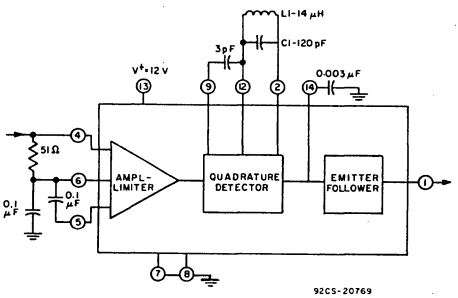


Fig. 1—Block diagram of CA2111A and associated outboard components.

MAXIMUM RATINGS, *Absolute-Maximum Values at T_A=25°C*

DC Supply Voltage [between terminals 13 (V^+) and 7 (V^-)]	16	V
Device Dissipation:		
Up to T_A = 60°C	600	mW
Above T_A = 60°C	derate linearly 6.7 mW/°C	
Ambient Temperature Range:		
Operating	−55 to +125	°C
Storage	−65 to +150	°C
Lead Temperature (During Soldering):		
At distance 1/16 ± 1/32 in. (1.59 ± 0.79 mm) from case for10s max.	+265	°C

ELECTRICAL CHARACTERISTICS at $T_A = 25°C$

CHARACTERISTIC	SYMBOL	TEST CONDITIONS	LIMITS			UNITS
			MIN.	TYP.	MAX.	
DC Voltage:		$V^+ = 12V$	–	5.4	–	V
At Terminal 1	V_1	$= 8V$	–	3.7	–	
At Terminals 4, 5, 6, 10	$V_{4,5,6,10}$	$V^+ = 8V$	–	1.35	–	
At Terminals 2, 12	$V_{2,12}$		–	3.5	–	
DC Current (into Terminal 13)						mA
At $V^+ = 8V$			–	14	–	
At $V^+ = 12V$	I_{13}		–	16	–	
Amplifier Input Resistance	R_4	$f_o = 10.7$ MHz	–	7	–	kΩ
Amplifier Input Capacitance	C_4		–	11	–	pF
Detector Input Resistance	R_{12}		–	70	–	kΩ
Detector Input Capacitance	C_{12}		–	2.7	–	pF
Amplifier Output Resistance	R_{10}		–	60	–	Ω
Detector Output Resistance	R_1		–	200	–	Ω
De-Emphasis Resistance	R_{14}		–	8.8	–	kΩ

DYNAMIC ELECTRICAL CHARACTERISTICS at $T_A = 25°C$
FM Modulation Frequency = 400 Hz, Source Resistance = 50Ω

CHARACTERISTIC	SYMBOL	TEST CONDITIONS								UNITS	TEST CIRCUIT OR CHARACTERISTIC CURVES FIG. NO.
		$f_o = 10.7$ MHz, $\Delta f = \pm 75$ KHz				$f_o = 4.5$ MHz, $\Delta f = \pm 25$ KHz		$f_o = 5.5$ MHz, $\Delta f = \pm 50$ KHz			
		$V^+ = 12V$		$V^+ = 8V$		$V^+ = 12V$		$V^+ = 12V$			
		LIMITS									
		TYP.	MAX.	TYP.	MAX.	TYP.	MAX.	TYP.	MAX.		
AMPL-LIMITER											
Input Limiting Threshold Voltage	V_i(lim) (4)	400	600	400	600	250	400	250	400	V (RMS)	3, 7, 8, 9
AM Rejection‡ *	AMR(1)	45	–	37	–	36	–	40	–	dB	3, 4, 5, 6
Ampl. Voltage Gain▲	A_V(10)	55	–	55	–	60	–	60	–	dB	3
DETECTOR											
Recovered Audio‡ Output Voltage	V_o(AF) (1)	0.48	–	0.3	–	0.72	–	1.2	–	V (RMS)	3, 7, 8, 9
Total Harmonic‡ Distortion	THD(1)	1	–	1	–	1.5	–	3	–	%	3

‡V_i = 10 mV (RMS) ▲$V_i \leq 50\ \mu V$ (rms) *100% FM, 30% AM

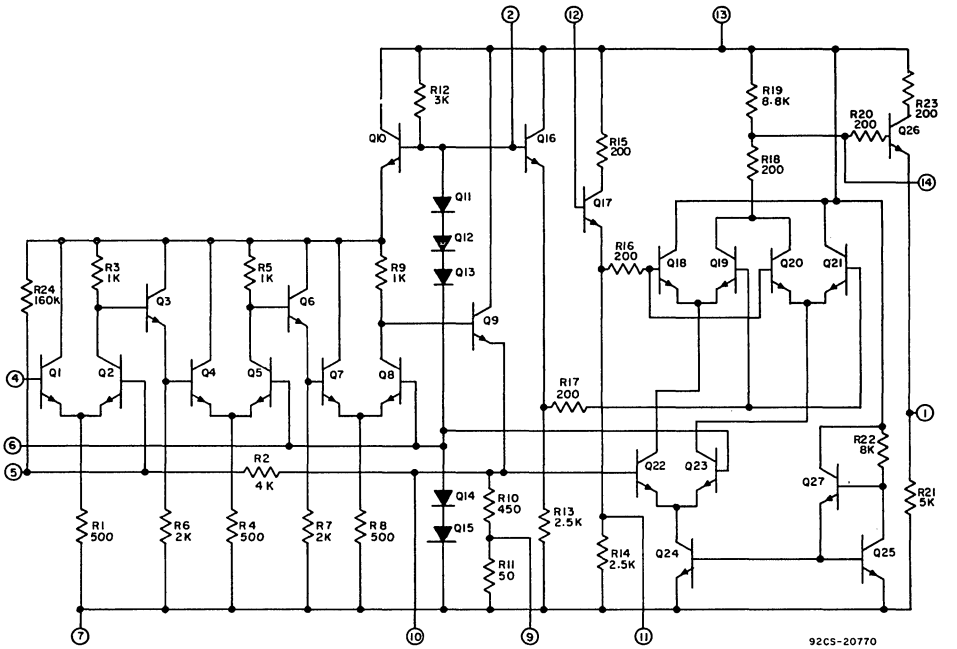
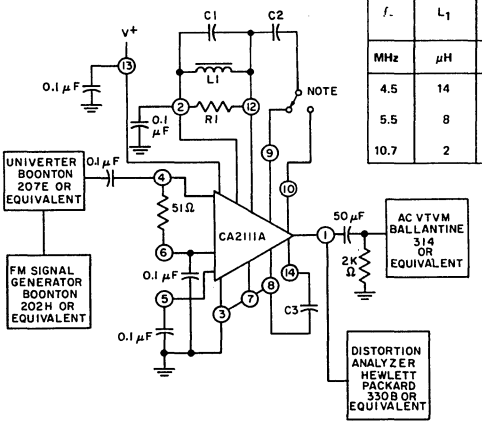


Fig. 2—Circuit schematic—CA2111A

COMPONENT VALUES							DETECTOR TRANSFER CHARACTERISTICS	
f_o	L_1	C_1	R_1	Q	C_2	C_3	UPPER PEAK	LOWER PEAK
MHz	μH	pF	KΩ	–	pF	μF	MHz	MHz
4.5	14	120	20	30	3	0.003	4.58	4.42
5.5	8	100	20	30	3	0.003	5.63	5.37
10.7	2	120	3.9	20	4.7	0.01	10.9	10.5



NOTE:

Input to the quadrature coil can be from either terminal 9 or terminal 10. Terminal 9 is normally used because it lessens the possibility of overloads during tuning. The use of terminal 10 increases the limiting sensitivity significantly and has been used successfully in these tests.

Fig. 3—Test circuit.

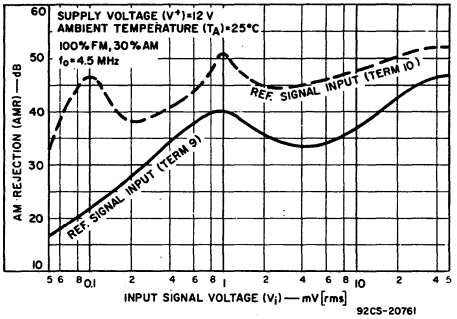


Fig. 4—AM rejection vs input voltage (4.5 MHz).

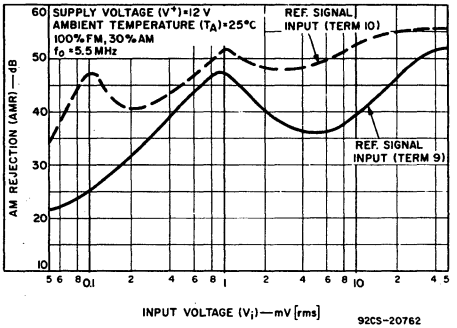


Fig. 5—AM rejection vs input voltage (5.5 MHz).

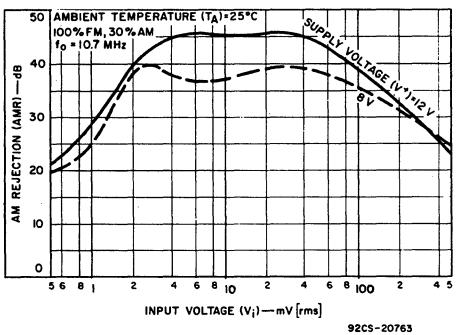


Fig. 6—AM rejection vs input voltage (10.7 MHz).

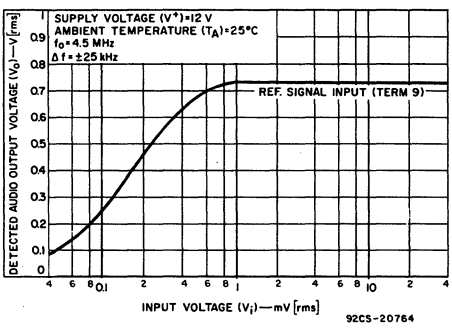


Fig. 7—Detected audio output vs input voltage (4.5 MHz).

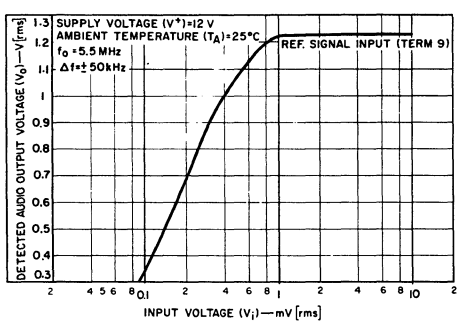


Fig. 8—Detected audio output vs input voltage (5.5 MHz).

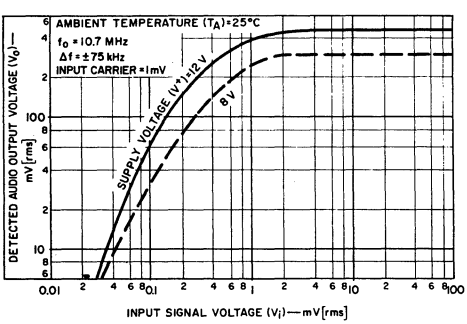


Fig. 9—Detected audio output voltage vs input voltage (10.7 MHz).

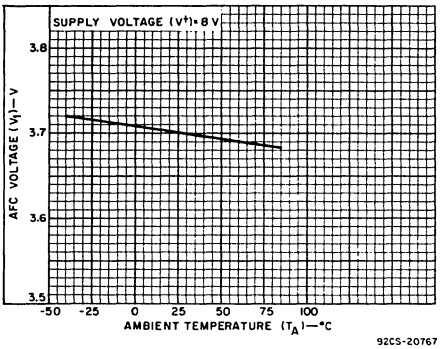


Fig. 10—AFC voltage vs ambient temp.

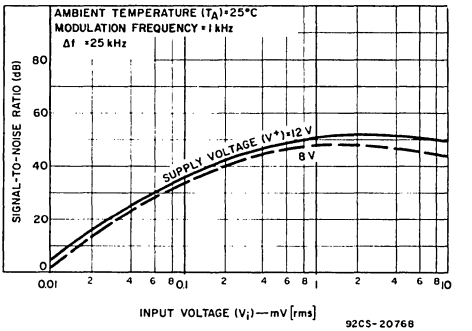


Fig. 11—Signal-to-noise ratio vs input voltage.

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CA3000

DC Amplifier

Monolithic Silicon



- Designed for use in Communication, Telemetry, Instrumentation, and Data-Processing Equipment
- Balanced differential-amplifier configuration with controlled constant-current source to provide outstanding versatility
- Built-in temperature stability for operation from -55°C to +125°C
- Companion Application Note, ICAN 5030 "Applications of RCA CA3000 Integrated Circuit DC Amplifier" covers characteristics of different operating modes, frequency considerations, 10 MHz narrow band tuned amplifier design, crystal oscillator design, and many other application aids

HIGHLIGHTS

- Input Impedance . . . . . 195 KΩ typ.
- Voltage Gain. . . . . 30 dB typ.
- Common-Mode Rejection Ratio . . . 98 dB typ.
- Input Offset Voltage. . . . . 1.4 mV typ.
- Push-Pull Input and Output
- Frequency Capability
  DC to 30 MHz (with external C and R)
- Wide AGC Range. . . . . 90 dB typ.

APPLICATIONS

- Schmitt Trigger
- RC-Coupled Feedback Amplifier
- Mixer
- Comparator
- Modulator
- Crystal Oscillator
- Sense Amplifier

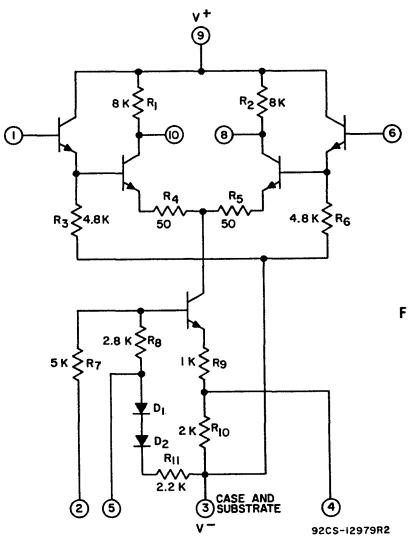


Fig. 1 SCHEMATIC DIAGRAM

Resistance values are in ohms

ABSOLUTE-MAXIMUM VOLTAGE LIMITS, at T_{FA} = 25°C

Indicated voltage limits for each terminal can be used under specified voltage conditions for other terminals

All voltages are with respect to ground (common terminal of Positive and Negative DC Supplies)

TERMINAL	VOLTAGE LIMITS		CONDITIONS	
	NEGATIVE	POSITIVE	TERMINAL	VOLTAGE
1	-2	+2	2	0
			3	-6
			6	0
			9	+6
2	-8	0	1	0
			3	-8
			6	0
			9	+6
3	-10	0	1	0
			2	0
			6	0
			9	+6
4	-8	0	1	0
			2	0
			6	0
			9	+6
5	-6	0	1	0
			2	0
			3	-6
			6	0
			9	+6
6	-2	+2	1	0
			2	0
			3	-6
			9	+6
7	NO CONNECTION			
8	0	+6	1	0
			2	0
			3	-6
			6	0
9	0	+10	1	0
			2	0
			3	-6
			6	0
10	0	+6	1	0
			2	0
			3	-6
			6	0
CASE	Internally Connected to Terminal No. 3 (Substrate) DO NOT GROUND			

OPERATING-TEMPERATURE RANGE . . −55°C to +125°C

STORAGE-TEMPERATURE RANGE . . . −65°C to +150°C

LEAD-TEMPERATURE (During Soldering):

At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max. . . . . +265°C

MAXIMUM SINGLE-ENDED INPUT-SIGNAL VOLTAGE . . ±4 V

MAXIMUM COMMON-MODE INPUT-SIGNAL VOLTAGE . . ±2 V

MAXIMUM DEVICE DISSIPATION:

From −55°C to 85°C. . . . . 450 mW

Above 85°C . . . . . Derate 5 mW/°C

STATIC CHARACTERISTICS AND TEST CIRCUITS FOR TYPE CA3000

INPUT OFFSET VOLTAGE AND CURRENT vs TEMPERATURE

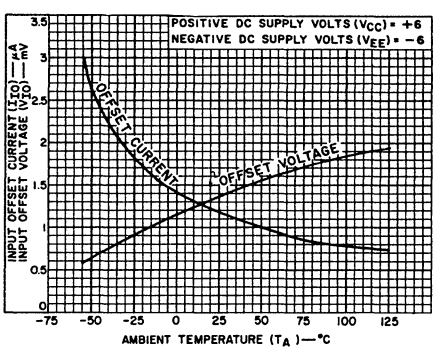


Fig.2

INPUT BIAS CURRENT vs TEMPERATURE

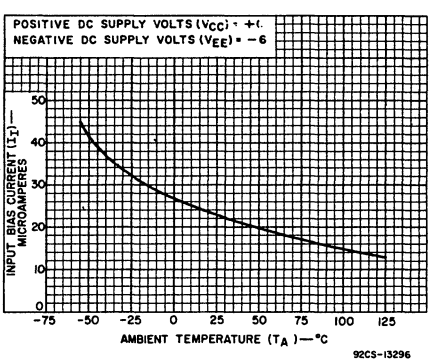


Fig.3

ELECTRICAL CHARACTERISTICS, at T_{FA} = 25°C, V_{CC} = +6V, V_{EE} = -6V, unless otherwise specified

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS Terminals No.4 & No.5 Not Connected Unless Specified		TEST CIRCUITS	LIMITS TYPE CA3000				TYPICAL CHARACTERISTICS CURVES
				Fig.	Min.	Typ.	Max.	Units	Fig.
STATIC CHARACTERISTICS									
Input Offset Voltage	V_{IO}				-	1.4	5	mV	2
Input Offset Current	I_{IO}				-	1.2	10	μA	2
Input Bias Current	I_{IB}				-	23	36	μA	3
Quiescent Operating Voltage	V_8 or V_{IO}	TERMINALS							
		4	5						
		NC	NC		-	2.6	-	V	4
		NC	VEE		-	4.2	-	V	4
		VEE	NC		-	-1.5	-	V	4
		VEE	VEE		-	0.6	-	V	4
Device Dissipation	P_D	NC	NC		-	30	-	mW	NONE
DYNAMIC CHARACTERISTICS									
Differential Voltage Gain Single-Ended Input	A_{DIFF}	Single-Ended Output f = 1 kHz		9	28	32	-	dB	5
		Double-Ended Output f = 1 kHz		9	-	38	-	dB	5
Bandwidth at -3 dB Point	BW	V_I = 10 mV, R_s = 1 kΩ			-	650	-	kHz	7
Maximum Output Voltage Swing	$V_{OUT(P-P)}$	f = 1 kHz		9	-	6.4	-	V(P-P)	NONE
Common-Mode Rejection Ratio	CMRR	f = 1 kHz		13	70	98	-	dB	8
Single-Ended Input Impedance	Z_{IN}	f = 1 kHz		15	70K	195K	-	Ω	10
Single-Ended Output Impedance	Z_{OUT}	f = 1 kHz		17	5.5K	8K	10.5K	Ω	12
Total Harmonic Distortion	THD	R_S=1kΩ f = 1 kHz V_O=42V_{p-p}			-	0.2	5	%	14
AGC Range (Maximum Voltage Gain to Complete Cutoff)	AGC	f = 1 kHz		20	80	90	-	dB	NONE

QUIESCENT OPERATING VOLTAGE vs TEMPERATURE

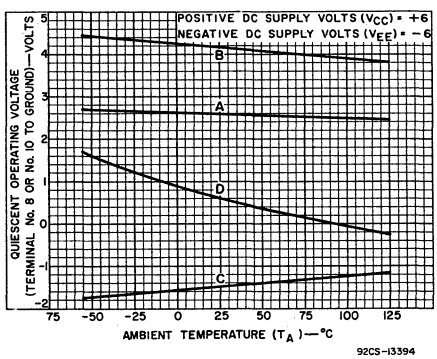


Fig.4

DYNAMIC CHARACTERISTICS AND TEST CIRCUIT FOR TYPE CA3000

DIFFERENTIAL VOLTAGE GAIN vs TEMPERATURE

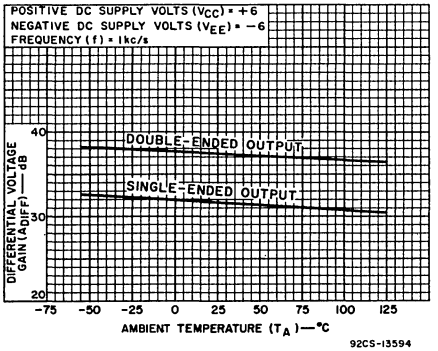


Fig.5

DIFFERENTIAL VOLTAGE GAIN AND MAXIMUM OUTPUT VOLTAGE SWING TEST CIRCUIT

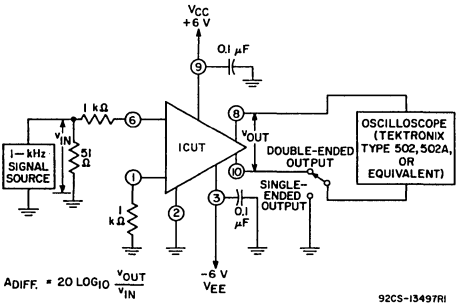


Fig.6

BANDWIDTH AT -3 dB POINT vs TEMPERATURE

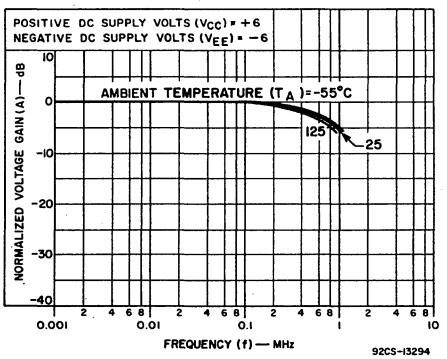


Fig.7

COMMON-MODE REJECTION RATIO vs TEMPERATURE

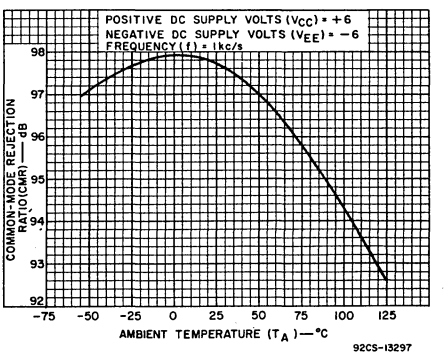


Fig.8

COMMON-MODE REJECTION RATIO TEST CIRCUIT

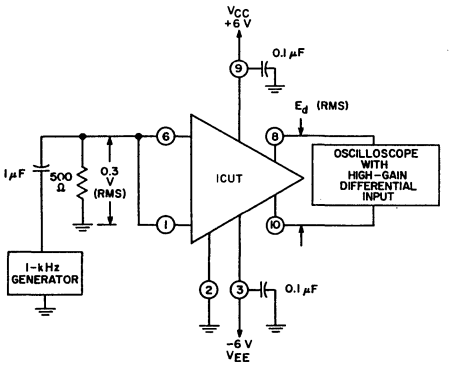


Fig.9

SINGLE-ENDED INPUT IMPEDANCE vs TEMPERATURE

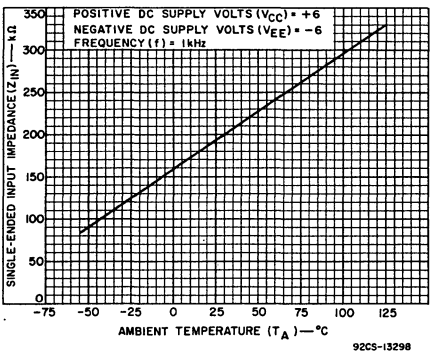


Fig.10

DYNAMIC CHARACTERISTICS AND TEST CIRCUITS FOR TYPE CA3000

SINGLE-ENDED INPUT IMPEDANCE TEST CIRCUIT

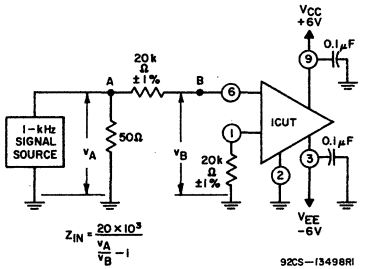


Fig. 11

SINGLE-ENDED OUTPUT IMPEDANCE vs TEMPERATURE

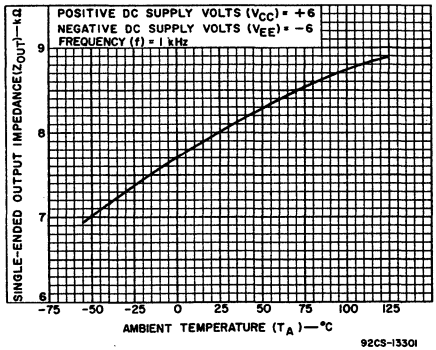


Fig. 12

SINGLE-ENDED OUTPUT IMPEDANCE TEST CIRCUIT

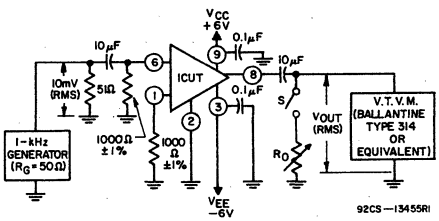


Fig. 13

TOTAL HARMONIC DISTORTION vs TEMPERATURE

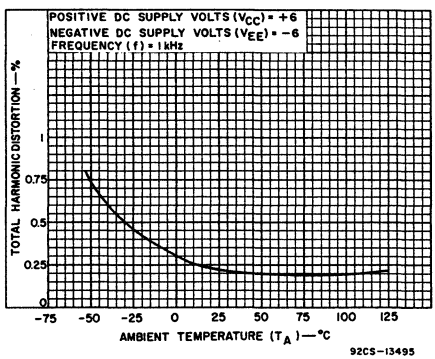


Fig. 14

AGC RANGE TEST CIRCUIT

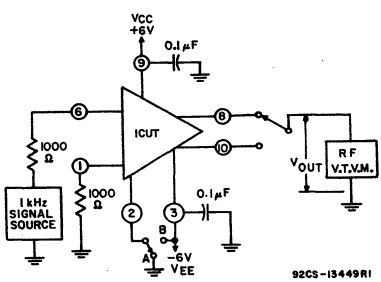


Fig. 15

RCA
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CA3001

Video and Wide-band Amplifier

Monolithic Silicon



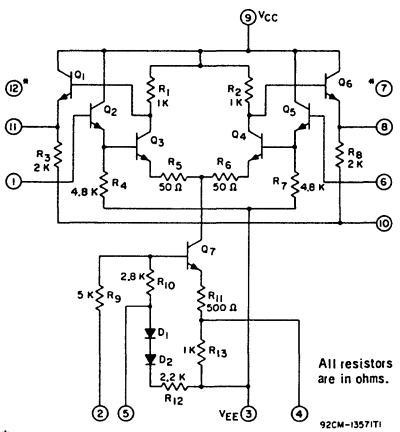
- Designed for use in Video Systems and Communication Equipment
- Balanced differential amplifier configuration with controlled constant-current source provides outstanding versatility
- Built-in temperature stability for operation from -55°C to +125°C
- Emitter follower input & output
- Companion Application Note ICAN5038 "Application of the RCA-CA3001 Integrated-Circuit Video Amplifier", covers different operating modes, gain control, distortion, swing capability, 3 stage amplifier design, and a Schmitt trigger study.

APPLICATIONS

- Schmitt Trigger
- Mixer
- Modulator
- DC, IF, & Video Amplifier

HIGHLIGHTS

- Push-Pull Input & Output
- AGC Range . . . . . 60 dB typ.
- Bandwidth. . . . . 29 MHz
- Input Resistance . . . . . 150 kΩ typ.
- Output Resistance . . . . . 45 Ω typ.
- Voltage Gain . . . . . 19 dB typ.
- Input Offset Voltage . . . . 1.5 mV typ.



* Internal Connection – DO NOT USE

Fig.1 - Schematic Diagram.

ABSOLUTE-MAXIMUM VOLTAGE AND CURRENT LIMITS at T_A = 25°C

Indicated voltage or current limits for each terminal can be applied under the specified conditions for other terminals. All Voltages are with respect to ground (common terminal of Positive and Negative DC Supplies).

TERMINAL	VOLTAGE OR CURRENT LIMITS		CONDITIONS	
	NEGATIVE	POSITIVE	TERMINAL	VOLTAGE
1	-2.5	+2.5	2, 6 3, 10 9	0 -6 +6
2	-8.5	0	1, 6 3, 10 9	0 -8.5 +6
3	-10	0	1, 2, 6 9 10	0 +6 -6
4	-8.5	0	1, 2, 6 9 10	0 +6 -6
5	-6	0	1, 2, 6 3, 10 9	0 -6 +6
6	-2.5	+2.5	1, 2 3, 10 9	0 -6 +6
7	INTERNAL CONNECTION DO NOT USE			

TERMINAL	VOLTAGE OR CURRENT LIMITS		CONDITIONS	
	NEGATIVE	POSITIVE	TERMINAL	VOLTAGE
8	25 mA		1, 2, 6, 10 3 9 200-Ω RESISTOR CONNECTED BETWEEN TERMINALS No.8 & No.10	0 -6 +6
9	0	+10	1, 2, 6, 10 3	0 -6
10	-10	0	1, 2, 6 3 9	0 -6 +6
11	25 mA		1, 2, 6, 10 3 9 200-Ω RESISTOR CONNECTED BETWEEN TERMINALS No.10 & No.11	0 -6 +6
12	INTERNAL CONNECTION DO NOT USE			
CASE	INTERNALLY CONNECTED TO TERMINAL No.3 (SUBSTRATE) DO NOT GROUND			

OPERATING TEMPERATURE RANGE	-55°C to +125°C
STORAGE TEMPERATURE RANGE	-65°C to +150°C
LEAD TEMPERATURE (During Soldering):	
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max.	+265°C
MAXIMUM SINGLE-ENDED INPUT-SIGNAL VOLTAGE	± 4 V
MAXIMUM COMMON-MODE INPUT-SIGNAL VOLTAGE	± 2.5 V
MAXIMUM DEVICE DISSIPATION:	
-55 to 85°C	450 mW
Above 85°C	Derate linearly 5 mW/°C

ELECTRICAL CHARACTERISTICS, AT T_A = 25°C, V_{CC} = +6 V, V_{EE} = –6 V

CHARACTERISTICS (See Page 2 for Definitions of Terms)	SYMBOLS	SPECIAL TEST CONDITIONS Terminals No.4 and No.5 Not Connected Unless Specified	TEST CIRCUITS	LIMITS TYPE CA3001				TYPICAL CHARACTERISTICS CURVES
			Fig.	Min.	Typ.	Max.	Units	Fig.
STATIC CHARACTERISTICS:								
Input Offset Voltage	V_{IO}		4	-	1.5	-	mV	2
Input Offset Current	I_{IO}		5	-	1	10	μA	2
Input Bias Current	I_I		5	-	16	36	μA	3
Output Offset Voltage	V_{OO}	R_S = 1 kΩ		-	54	300	mV	6
Quiescent Operating Voltage	V_8 OR V_{11}	TERMINALS MODE 4 5						
		A NC NC		3.8	4.4	5	V	7
		B NC V_{EE}		-	4.8	-	V	7
		C V_{EE} NC		-	2.7	-	V	7
		D V_{EE} V_{EE}		-	4	-	V	7
Device Dissipation	P_D	A NC NC		60	78	120	mW	8
		B NC V_{EE}		-	71	-	mW	8
		C V_{EE} NC		-	110	-	mW	8
		D V_{EE} V_{EE}		-	86	-	mW	8
DYNAMIC CHARACTERISTICS:								
Differential Voltage Gain (Single-ended input and output)	A_{DIFF}	f = 1.75 MHz f = 20 MHz		16 10	19 14	- -	dB dB	9 A, 9 B 9 B
Bandwidth at -3 dB Point	BW	R_S = 50Ω		16	29	-	MHz	NONE
Maximum Output Voltage Swing	$V_{OUT(P-P)}$	R_S = 50Ω f = 1.75 MHz		-	5	-	V_{P-P}	NONE
Noise Figure	NF_1	f = 1.75 MHz, R_S = 1 KΩ	14	-	5	8	dB	10
		f = 11.7 MHz, R_S = 1 KΩ	14	-	7.7	-	dB	10
Common-Mode Rejection Ratio	CMRR	f = 1 KHz	16	70	88	-	dB	12
Input Impedance Components:								
Parallel Input Resistance	R_{IN}	f = 1.75 MHz		50	140	-	KΩ	14
Parallel Input Capacitance	C_{IN}	f = 1.75 MHz		-	3.4	7	pF	14
Output Resistance	R_{OUT}	f = 1.75 MHz		-	45	70	Ω	NONE
AGC Range (Maximum voltage gain to complete cutoff)	AGC	f = 1.75 MHz	19	55	60	-	dB	NONE

TYPICAL STATIC CHARACTERISTICS

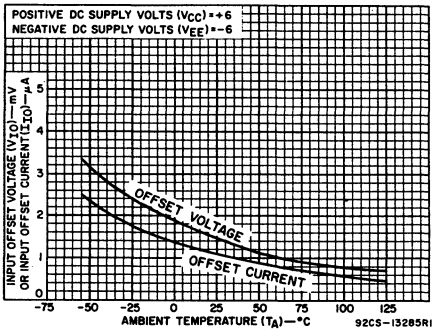


Fig.2 - Input offset voltage and current vs. temperature.

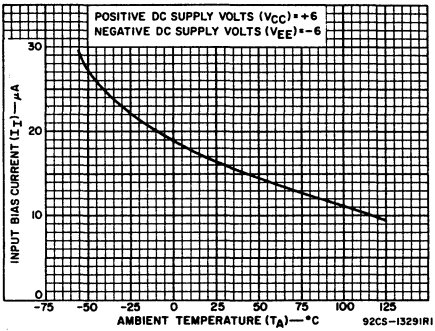
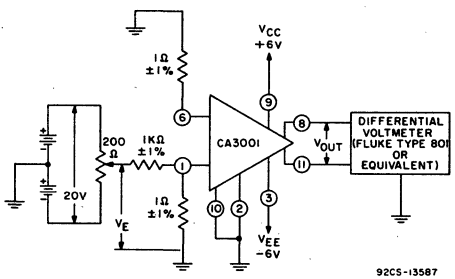


Fig.3 - Input bias current vs. temperature.

TYPICAL STATIC CHARACTERISTICS AND TEST CIRCUITS



1. Adjust V_E for $V_{OUT}(DC) = 0 \pm 0.1$ V 2. Measure V_E and record input offset voltage (V_{IO}) in mV as $V_{IO} = \frac{V_E}{1000}$

Fig.4 - Input offset voltage test circuit.

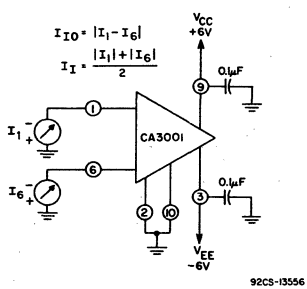


Fig.5 - Input offset current and input bias current test circuit.

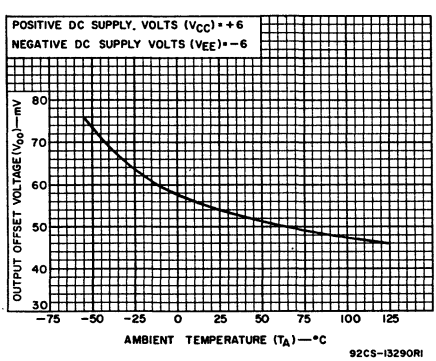


Fig.6 - Output offset voltage vs. temperature.

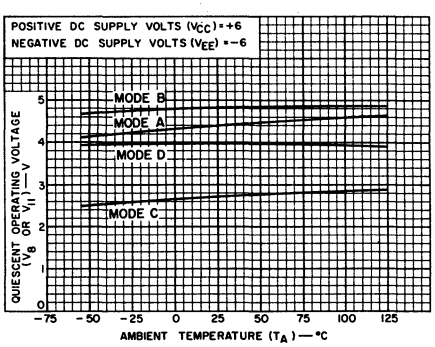


Fig.7 - Quiescent operating voltage vs. temperature.

TYPICAL STATIC CHARACTERISTICS

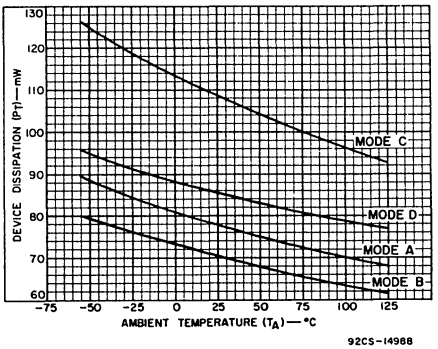


Fig.8 - Device dissipation vs. temperature.

TYPICAL DYNAMIC CHARACTERISTICS AND TEST CIRCUITS

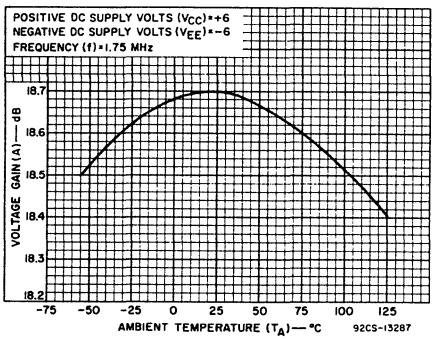


Fig.9a - Differential voltage gain vs. temperature.

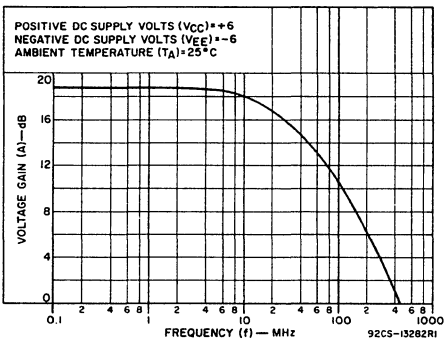


Fig.9b - Differential voltage gain vs. frequency.

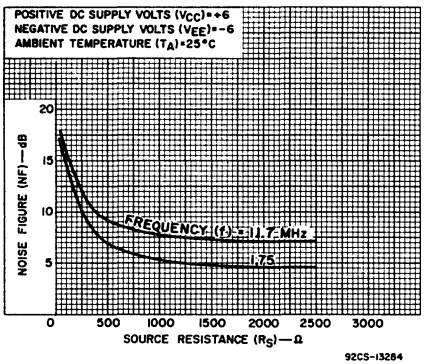
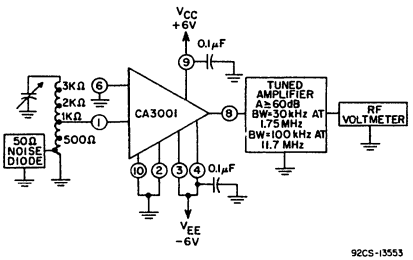


Fig.10 - Noise figure vs. source resistance and frequency.



* Separate tuned input circuits are used for 1.75 MHz and 11.7 MHz. Source-resistance matching taps adjusted with circuit tuned to resonance and with 50-ohm resistor connected to simulate noise diode.

Fig.11 - Noise figure test circuit.

TYPICAL DYNAMIC CHARACTERISTICS AND TEST CIRCUIT

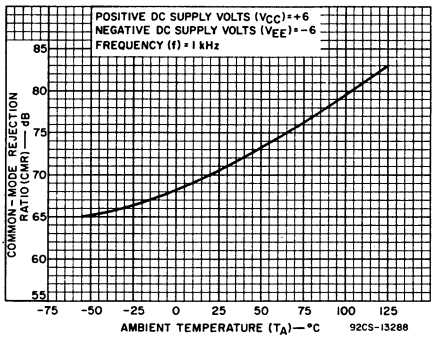


Fig.12 - Common-mode rejection ratio vs. temperature.

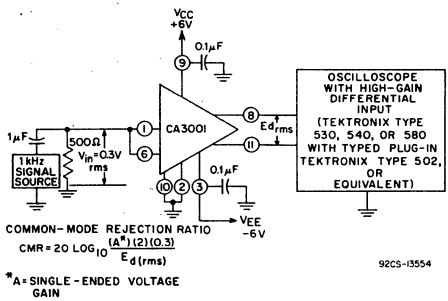


Fig.13 - Common-mode rejection ratio test circuit.

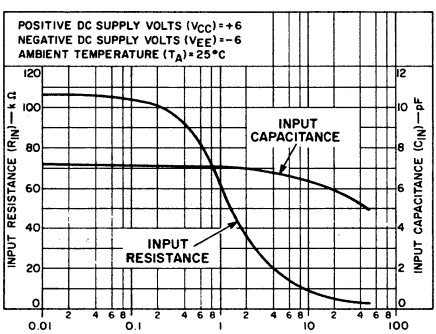


Fig.14 - Input impedance components vs. frequency.

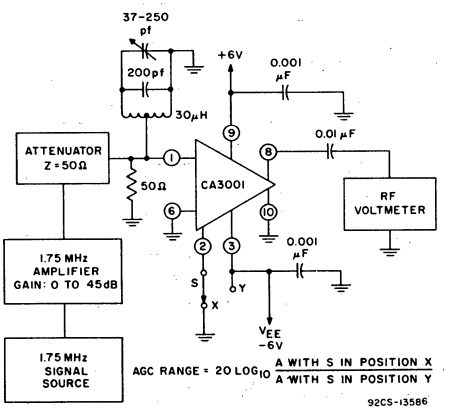


Fig.15 - AGC range test circuit.

RCA
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CA3002

IF Amplifier



- Designed for use in Communication Equipment
- Balanced differential amplifier configuration with controlled constant-current source provides outstanding versatility
- Built-in temperature stability for operation from -55°C to +125°C
- Companion Application Note ICAN-5036 "Application of the RCA-3002 Integrated-Circuit IF Amplifier" covers different operating modes, cross modulation, gain control, 4-stage amplifier design, and an envelope and product detector analysis.

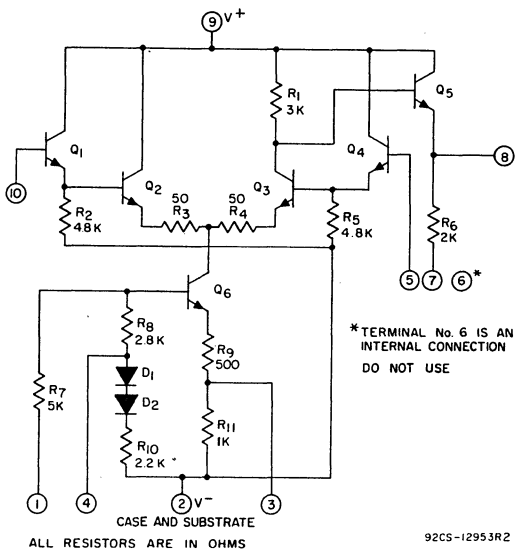
APPLICATIONS

- Product Detector
- IF & Video Amplifier
- AM Detector
- Schmitt Trigger

HIGHLIGHTS

- Input Resistance . . . . . 100 kΩ typ.
- Output Resistance . . . . . 70 Ω typ.
- Voltage Gain . . 24 dB typ. @ 1.75 MHz
- Push-Pull Input, Single-Ended Output
- -3 dB Bandwidth . . . . . 11 MHz typ.
- AGC Range . . . . . 80 dB typ.
- Useful Frequency Range DC to . . 15 MHz

SCHEMATIC DIAGRAM



ABSOLUTE-MAXIMUM VOLTAGE AND CURRENT LIMITS, at $T_A = 25^oC$

Indicated voltage or current limits for each terminal can be applied under the specified operating conditions for other terminals.

All voltages are with respect to ground ($-V_{CC}$, $+V_{EE}$) or common terminal of Positive and Negative DC supplies).

TERMINAL	VOLTAGE OR CURRENT LIMITS		CONDITIONS	
	NEGATIVE	POSITIVE	TERMINAL	VOLTAGE
1	-8 V	0 V	2, 7 5, 10 9	-8 0 +6
2	-10 V	0 V	1, 5, 10 9	0 +6
3	-8.5 V	0 V	1, 5, 10 7 9	0 -6 +6
4	-8 V	0 V	1, 5, 10 2, 7 9	0 -8 +6
5	-3.5 V	+3.5 V	1, 10 2, 7 9	0 -6 +6
CASE	INTERNALLY CONNECTED TO TERMINAL No.2 (SUBSTRATE) DO NOT GROUND			

TERMINAL	VOLTAGE OR CURRENT LIMITS		CONDITIONS	
	NEGATIVE	POSITIVE	TERMINAL	VOLTAGE
6	INTERNAL CONNECTION DO NOT USE			
7	-12 V	0 V	1, 5, 10 2 9	0 -6 +6
8	20 mA		1, 5, 7, 10 2 9 200 Ω Resistor Between Terminals 7 & 8	0 -6 +6
9	0 V	+10 V	1, 5, 10 2, 3, 7	0 -6
10	-3.5 V	+3.5 V	1, 5 2, 7 9	0 -6 +6

OPERATING-TEMPERATURE RANGE ..... -55°C to +125°C

STORAGE-TEMPERATURE RANGE ..... -65°C to +150°C

LEAD TEMPERATURE (During Soldering):

At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm)

from case for 10 seconds max. ..... +265°C

MAXIMUM INPUT-SIGNAL VOLTAGE ..... ± 4 V

MAXIMUM DEVICE DISSIPATION:

-55 to 85°C ..... 450 mW

Above 85°C ..... Derate linearly 5 mW/°C

STATIC CHARACTERISTICS AND TEST CIRCUITS

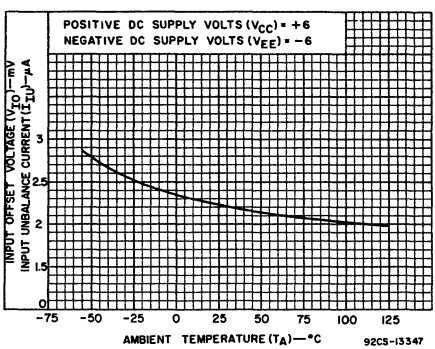


Fig.2 - Input unbalance voltage & current vs temperature.

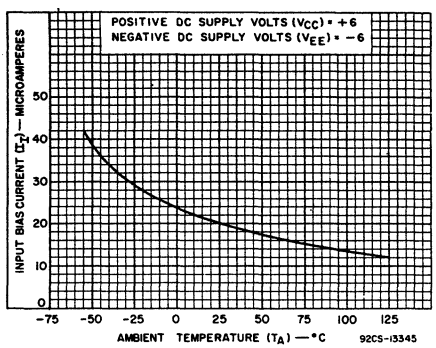


Fig.3 - Input bias current vs temperature.

ELECTRICAL CHARACTERISTICS, at T_A = 25°C, V_{CC} = +6 V, V_{EE} = -6 V

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS TERMINALS No.3 & No.4 NOT CONNECTED UNLESS OTHERWISE NOTED	TEST CIRCUITS	LIMITS CA3002				TYPICAL CHARACTERISTICS CURVES
			Fig.	Min.	Typ.	Max.	Units	Fig.
STATIC CHARACTERISTICS:								
Input Offset Voltage	V_{IO}		4	-	2.2	-	mV	2
Input Unbalance Current	I_{IU}			-	2.2	10	μA	2
Input Bias Current	I_I			-	20	36	μA	3
Quiescent Operating Voltage		MODE / TERMINAL 2 / TERMINAL 4						
		A / V_{EE} / NC		-	2.8	-	V	4
		B / V_{EE} / V_{EE}		-	3.9	-	V	4
Device Dissipation	P_T			-	55	-	mW	None
DYNAMIC CHARACTERISTICS:								
Differential Voltage Gain (Single-Ended Input and Output)	A_{DIFF}	V_{IN} = 10 mV, f = 1.75 MHz, R_S = 50Ω		19	24	-	dB	5 & 5
Bandwidth at -3 dB Point	BW	R_S = 50Ω, V_{IN} = 10 mV		-	11	-	MHz	6
Maximum Output Voltage Swing	$V_{OUT(P-P)}$	-		-	5.5	-	V_{P-P}	None
Noise Figure	NF	f = 1.75 MHz R_S = 1 kΩ	12	-	4	8	dB	7
Input Impedance Components: Parallel Input Resistance	R_{IN}	f = 1.75 MHz	None	-	100k	-	Ω	None
Parallel Input Capacitance	C_{IN}	f = 1.75 MHz	None	-	4	-	pF	None
Output Resistance	R_{OUT}	f = 1.75 MHz	14	-	70	-	Ω	9a & 9b
AGC Range (Maximum Voltage Gain to Complete Cutoff	AGC	f = 1.75 MHz	18	60	80	-	dB	12

STATIC CHARACTERISTICS AND TEST CIRCUITS

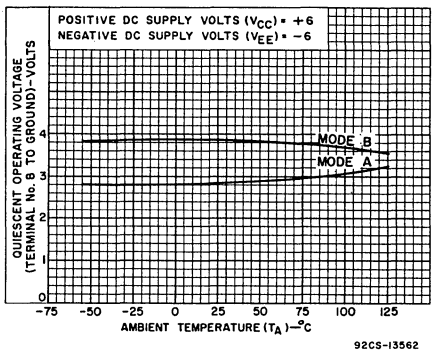


Fig.4 - Quiescent operating voltage vs temperature.

DYNAMIC CHARACTERISTICS

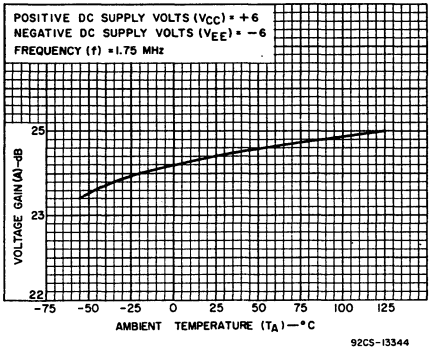


Fig. 5a - Differential voltage gain vs temperature.

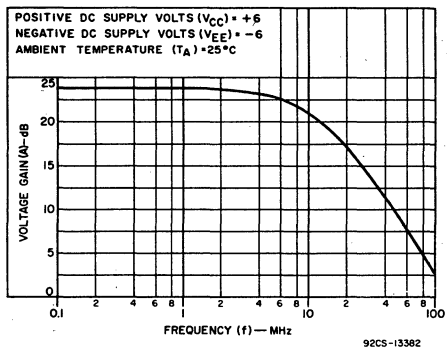


Fig. 5b - Differential voltage gain vs frequency.

DYNAMIC CHARACTERISTICS AND TEST CIRCUITS

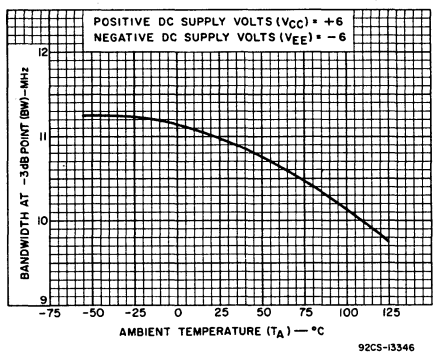


Fig. 6 - Bandwidth at -3 dB point vs temperature.

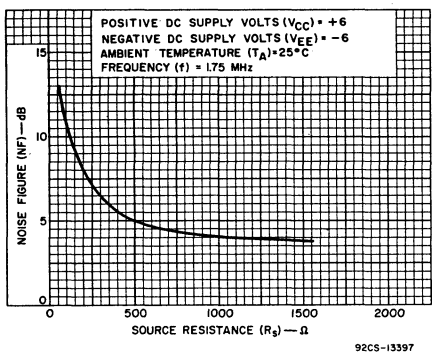
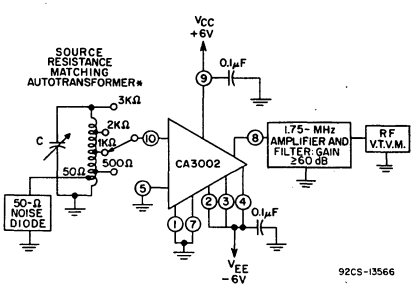


Fig. 7 - Noise figure vs source resistance.



*Taps are adjusted to provide indicated equivalent values of R_S with tank tuned to resonance at 1.75 MHz, and a 50-Ω resistor connected to simulate the noise diode.

Fig. 8 - Noise figure.

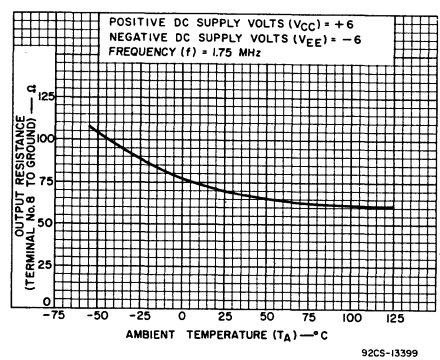


Fig. 9a - Output resistance vs temperature.

DYNAMIC CHARACTERISTIC AND TEST CIRCUIT

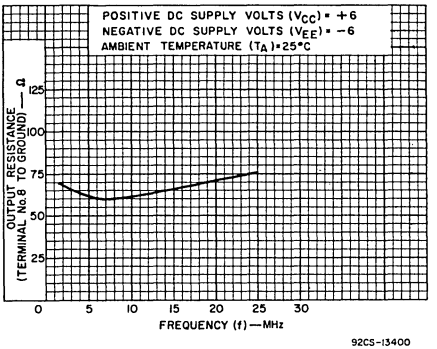


Fig. 9b - Output resistance vs frequency.

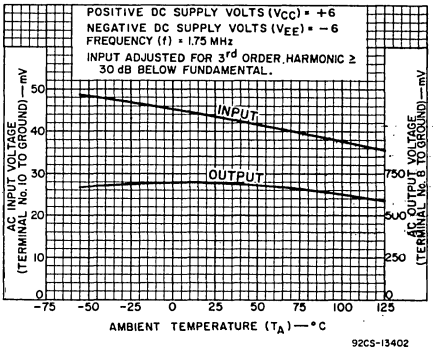
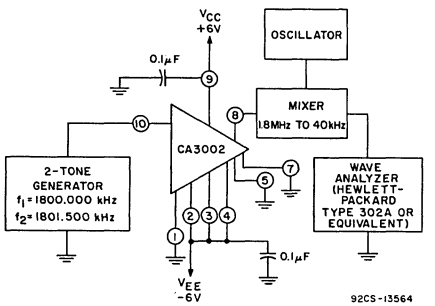


Fig. 10 - Input level for − 30 dB intermodulation vs. temperature



1) **Increase both input-signal tones until the $2f_2$-f1 and $2f_1$-f2 output-signal voltages are 30 dB below the f_1 and f_2 output-signal voltages.**
2) **Measure rms values of the input and output signal voltages.**
3) **The measured input signal voltage is that value when the 3rd-harmonic intermodulation products are 30 dB below the fundamental outputs.**

Fig. 11 - Intermodulation Test Circuit.

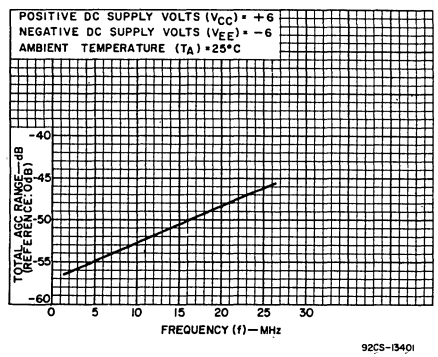


Fig. 12 - AGC range vs frequency.

1) Set attenuator at 80 dB attenuation.
2) Set variable dc supply voltage at 0 V.
3) Increase signal input voltage until RF V.T.V.M. indicates 5 mV output.
4) Set variable dc supply voltage at -6 V.
5) Adjust attenuator until RF V.T.V.M. again indicates 5 mV output.
6) Change in attenuator setting in dB is total AGC Range.

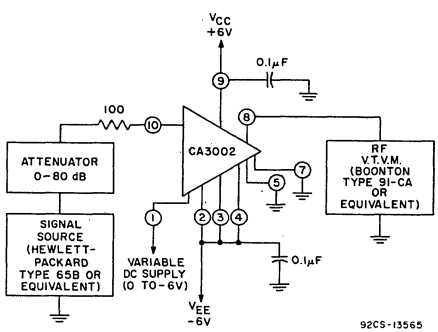


Fig. 13 - AGC range.

RCA
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Linear Integrated Circuits

CA3004

RF Amplifier

Monolithic Silicon

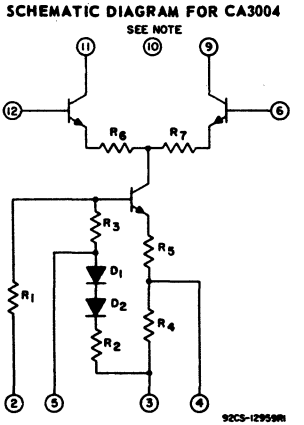
- Designed for use in Communications Equipment
- Balanced Differential-Amplifier Configuration with Controlled Constant-Current Source Provides Unexcelled Versatility



- Push-Pull Input and Output
- Wide and Narrow-Band Amplifier
- AGC
- Detector
- Operation from DC to 100 Mc/s
- Mixer
- Limiter
- Modulator
- RF, IF, and Video Frequency Capability

- Built-in Temperature Stability for Operation from -55° C to +125° C
- Similar to RCA CA3005 and CA3006, plus Emitter-Degeneration Resistors to Provide More Linear Transfer Characteristic and Increased Input-Signal Handling Capability
- Companion Application Note ICAN 5022 "Application of RCA CA3004, CA3005, and CA3006 Integrated Circuit RF Amplifiers", covers characteristics of different operating modes, noise performance, cross-modulation, mixer, AGC, limiter, detector, and amplifier design considerations.

SCHEMATIC DIAGRAM FOR CA3004



NOTE: Connect Terminal No. 10 to most positive dc supply voltage used for circuit.

Fig. 1

ABSOLUTE-MAXIMUM VOLTAGE LIMITS, at T_{FA} = 25°C

Voltage limits shown for each terminal can be applied under the indicated circuit conditions for other terminals. All voltages are with respect to GROUND (common terminal of Positive and Negative DC Supplies)

TERMINAL	VOLTAGE LIMITS		CONDITIONS	
	NEGATIVE	POSITIVE	TERMINAL	VOLTAGE
1	NO CONNECTION			
2	-9.5	0	6	0
			12	0
			3	-9.5
			9	+6
			10	+6
			11	+6
3	-12	0	2	0
			6	0
			9	+6
			10	+6
			11	+6
			12	0
4	-12	0	2	0
			6	0
			9	+6
			10	+6
			11	+6
			12	0
5	-6	0	2,6,12	0
			3	-6
			9	+6
			10	+6
			11	+6
6	-3.5	+3.5	2	0
			3	-6
			9	+6
			10	+6
			11	+6
			12	0

TERMINAL	VOLTAGE LIMITS		CONDITIONS	
	NEGATIVE	POSITIVE	TERMINAL	VOLTAGE
7	NO CONNECTION			
8	NO CONNECTION			
9	0	+12	2	0
			3	-6
			6	0
			10	+6
			11	+6
			12	0
10	0	+12	2	0
			3	-6
			6	0
			9	+6
			11	+6
			12	0
11	0	+12	2	0
			3	-6
			6	0
			10	+6
			1)	+6
			12	0
12	-3.5	+3.5	2	0
			3	-6
			6	0
			9	+6
			10	+6
			11	+6
CASE	INTERNALLY CONNECTED TO TERMINAL NO.3 (SUBSTRATE) DO NOT GROUND			

OPERATING-TEMPERATURE RANGE . . . . . -55°C to +125°C

STORAGE-TEMPERATURE RANGE . . . . . -65°C to +150°C

LEAD TEMPERATURE (During Soldering)
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm)
from case for 10 seconds max. . . . . +265°C

MAXIMUM SINGLE-ENDED INPUT-
SIGNAL VOLTAGE . . . . . ±3.5 V

MAXIMUM COMMON-MODE INPUT-
SIGNAL VOLTAGE . . . . . -2.5 V, +3.5 V

MAXIMUM DEVICE DISSIPATION . . . . . 300 mW

ELECTRICAL CHARACTERISTICS, at T_{FA} = 25° C, V_{CC} = +6V, V_{EE} = -6 V unless otherwise specified

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS Terminals No.4 and No.5 Open Unless Otherwise Specified		TEST CIRCUIT	LIMITS TYPE CA3004				TYPICAL CHARACTERISTICS CURVES
				Fig.	Min.	Typ.	Max.	Units	Fig.
STATIC CHARACTERISTICS									
Input Offset Voltage	V_{IO}			Fig.4	–	1.7	5	mV	Fig.2
Input Offset Current	I_{IO}			Fig.5	–	0.125	5	μA	Fig.2
Input Bias Current	I_I			Fig.5	–	21	40	μA	Fig.3
		TERMINALS							
		4	5						
Quiescent Operating Current	I_9 or I_{11}	NC	NC	Fig.8	–	1	–	mA	Fig.6
		V_{EE}	NC	Fig.8	–	2.7	–	mA	Fig.6
		NC	V_{EE}	Fig.8	–	0.45	–	mA	Fig.6
		V_{EE}	V_{EE}	Fig.8	–	1.25	–	mA	Fig.6
Quiescent Operating Current Ratio	I_9/I_{11}			Fig.8	–	1.1	–	–	Fig.7
Device Dissipation	P_T			Fig.8	–	26	–	mW	NONE
DYNAMIC CHARACTERISTICS									
Power Gain	G_P	f = 100 Mc/s		Fig.11	10	12	–	dB	Fig.9
Noise Figure	NF	f = 100 Mc/s		Fig.11	–	6.3	9	dB	Fig.10
Common Mode Rejection Ratio	CMR	f = 1 Kc/s		Fig.13	–	98	–	dB	Fig.12
AGC Range (Max. Voltage Gain to Complete Cutoff)	AGC	f = 1.75 Mc/s		Fig.14	-60	–	–	dB	NONE

DEFINITIONS OF TERMS

Input Offset Voltage

The difference in the dc voltages which must be applied to the input terminals to obtain equal quiescent operating voltages (zero output offset voltage) at the output terminals.

Input Offset Current

The difference in the currents at the two input terminals when the quiescent operating voltages at the two output terminals are equal.

Input Bias Current

The average value (one-half the sum) of the currents at the two input terminals when the quiescent operating voltages at the two output terminals are equal.

Quiescent Operating Current

The average (dc) value of the current in either output terminal.

Quiescent Operating Current Ratio

The ratio of the Quiescent operating currents in the two output terminals.

Device Dissipation

The total power drain of the device with no signal applied and no external load current.

Power Gain

The ratio of the signal power developed at the output of the device to the signal power applied to the input, expressed in dB.

Noise Figure

The ratio of the total noise power of the device and a resistive signal source to the noise power of the signal source alone, the signal source representing a generator of zero impedance in series with the source resistance.

Common-Mode Rejection Ratio

The ratio of the full differential voltage gain to the common-mode voltage gain.

Common-Mode Voltage Gain

The ratio of the signal voltages developed between the two output terminals to the signal voltage applied to the two input terminals connected in parallel for ac.

Differential Voltage Gain

The ratio of the change in output voltage at either output terminal with respect to ground, to a change in input voltage at either input terminal with respect to ground, with the other input terminal at ac ground.

AGC Range

The total change in voltage gain (from maximum gain to complete cutoff) which may be achieved by application of the specified range of dc voltage to the AGC input terminal of the device.

TYPICAL STATIC CHARACTERISTICS AND TEST CIRCUITS FOR TYPE CA3004

INPUT OFFSET VOLTAGE AND CURRENT VS TEMPERATURE

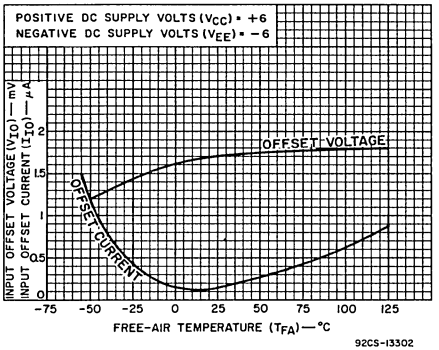


Fig.2

INPUT BIAS CURRENT VS TEMPERATURE

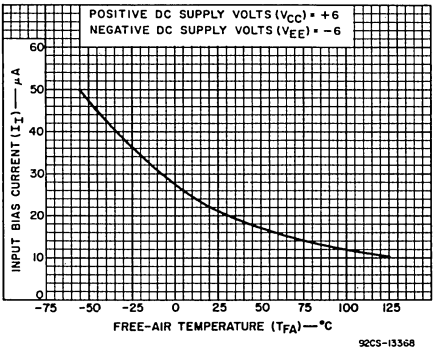


Fig.3

INPUT OFFSET VOLTAGE TEST CIRCUIT

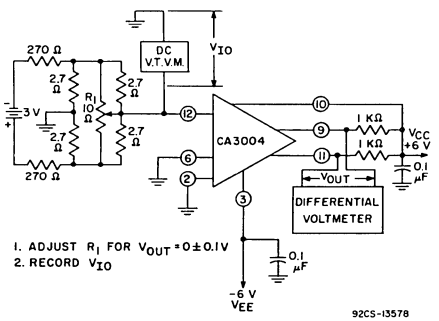


Fig.4

INPUT OFFSET CURRENT AND BIAS CURRENT TEST CIRCUIT

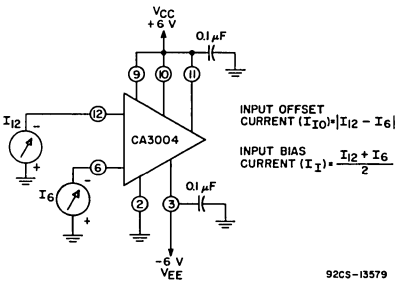


Fig.5

QUIESCENT OPERATING CURRENT VS TEMPERATURE

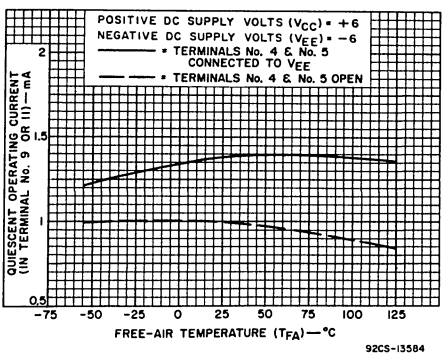


Fig.6

QUIESCENT OPERATING CURRENT RATIO VS TEMPERATURE

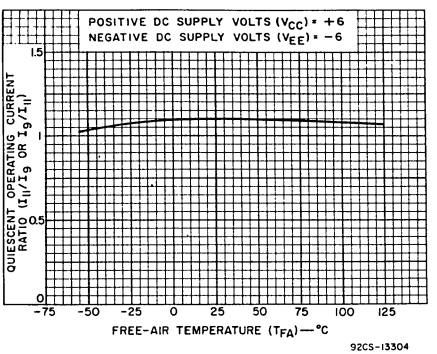
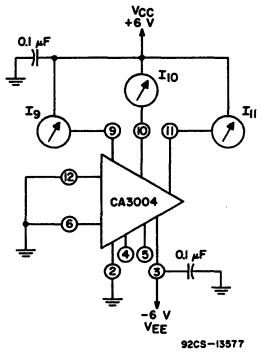


Fig.7

TEST CIRCUIT FOR TYPE CA3004

QUIESCENT OPERATING CURRENT, QUIESCENT OPERATING CURRENT RATIO, AND DEVICE DISSIPATION TEST CIRCUIT



$$P_T = V_{CC}(I_g + I_{10} + I_{11}) + V_{EE} I_3$$

Fig.8

TYPICAL DYNAMIC CHARACTERISTICS FOR TYPE CA3004

POWER GAIN VS FREQUENCY

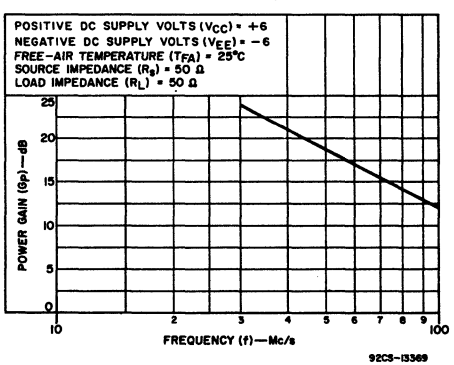


Fig.9

NOISE FIGURE VS FREQUENCY

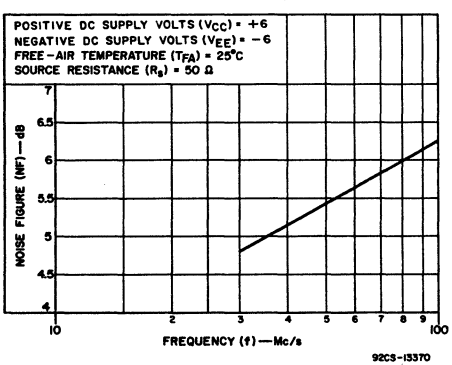


Fig.10

TYPICAL DYNAMIC CHARACTERISTICS AND TEST CIRCUITS FOR TYPE CA3004

100 Mc/s POWER GAIN AND NOISE FIGURE TEST CIRCUIT

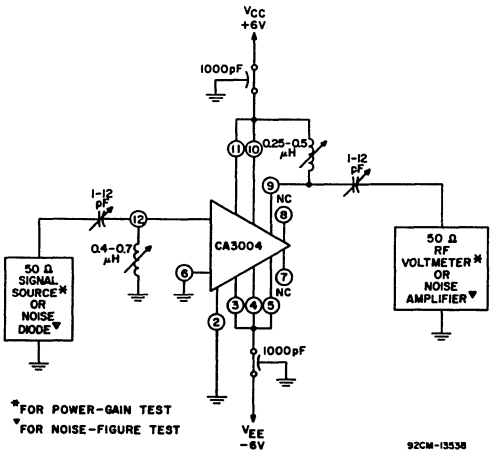


Fig. 11

COMMON-MODE REJECTION RATIO VS TEMPERATURE

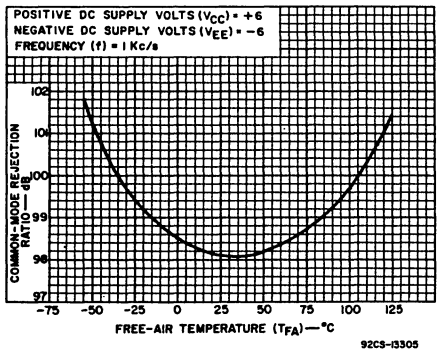


Fig. 12

COMMON-MODE REJECTION RATIO TEST CIRCUIT

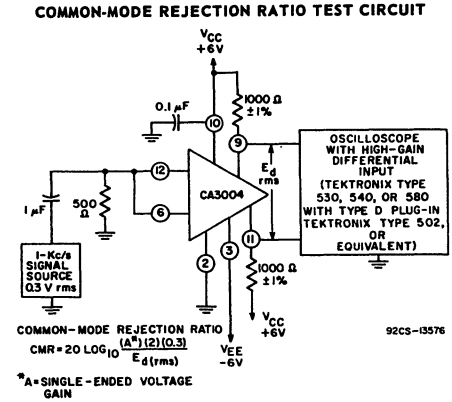


Fig. 13

AGC RANGE TEST CIRCUIT

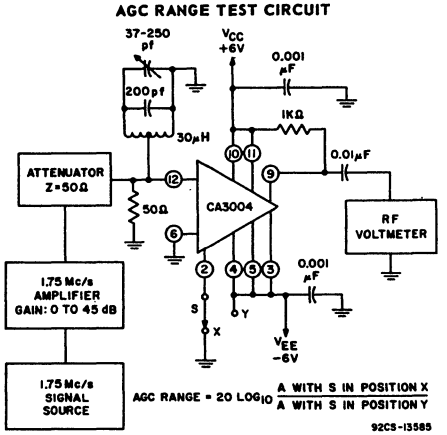


Fig. 14

RCA
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CA3005
CA3006

RF Amplifiers
Monolithic Silicon

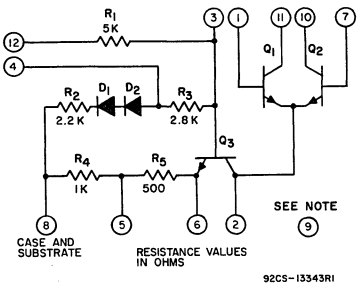
- Designed for use in Communications Equipment
- Balanced Differential Amplifier Configuration with Controlled Constant-Current Source to Provide Unexcelled Versatility

- Push-Pull Input and Output
- Wide and Narrow Band Amplifier
- AGC
- Detector
- RF, IF, and Video Frequency Capability
- Operation from DC to 100 MHz
- Mixer
- Limiter
- Modulator
- Cascode Amplifier



- Built-in Temperature Stability for Operation from -55° C to +125° C
- Companion Application Note, ICAN 5022 "Application of RCA CA3004, CA3005, and CA3006 Integrated Circuit RF Amplifiers", covers characteristics of different operating modes, noise performance, cross-modulation, mixer, AGC limiter, detector, and amplifier design considerations.

SCHEMATIC DIAGRAM FOR CA3005 AND CA3006



NOTE: Connect Terminal No.9 to most positive dc supply voltage used for circuit.

Fig.1

ABSOLUTE-MAXIMUM VOLTAGE LIMITS, at T_{FA} = 25°C

Voltage limits shown for each terminal can be applied under the indicated voltage conditions for other terminals.
All voltages are with respect to GROUND (common terminal of Positive and Negative DC Supplies)

TERMINAL	VOLTAGE LIMITS		CONDITIONS	
	NEGATIVE	POSITIVE	TERMINAL	VOLTAGE
1	-3.5	+3.5	7 8 9 10 11 12	0 -6 +6 +6 +6 0
2	TEST POINT: DO NOT APPLY VOLTAGE FROM EXTERNAL SOURCE			
3	-9.5	0	1 7 8 9 10 11 12	0 0 -9.5 +6 +6 +6 0
4	-6	0	1 7 8 9 10 11 12	0 0 -6 +6 +6 +6 0
5	-12	0	1 7 9 10 11 12	0 0 +6 +6 +6 0
6	-6	0	1 7 9 10 11 12	0 0 +6 +6 +6 -6
7	-3.5	+3.5	1 8 9 10 11 12	0 -6 +6 +6 +6 0
8	-12	0	1 7 9 10 11 12	0 0 +6 +6 +6 0
9	0	+12	1 7 8 10 11 12	0 0 -6 +6 +6 0
10	0	+12	1 7 8 9 11 12	0 0 -6 +6 +6 0
11	0	+12	1 7 8 9 10 12	0 0 -6 +6 +6 0
12	-9.5	0	8 9 10 11	-9.5 +6 +6 +6
CASE	Internally connected to Terminal No.8 (substrate) DO NOT GROUND			

OPERATING-TEMPERATURE RANGE . . . . . -55°C to +125°C

STORAGE-TEMPERATURE RANGE . . . . . -65°C to +150°C

LEAD TEMPERATURE (During Soldering)
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm)
from case for 10 seconds max. . . . . +265°C

MAXIMUM SINGLE-ENDED INPUT-
SIGNAL VOLTAGE . . . . . ±3.5 V

MAXIMUM COMMON-MODE INPUT-
SIGNAL VOLTAGE . . . . . -2.5 V, +3.5 V

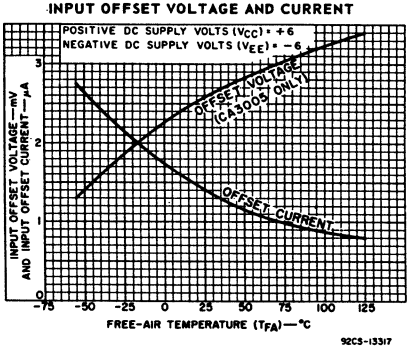
MAXIMUM DEVICE DISSIPATION . . . . . 300 mW

ELECTRICAL CHARACTERISTICS, at T_A = 25° C, V_{CC} = +6V, V_{EE} = -6V

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS Terminals No.3,4,5, and 6 Not Connected Except Where Noted	TEST CIRCUITS	LIMITS TYPE CA3005			LIMITS TYPE CA3006				TYPICAL CHARACTERISTICS CURVES
			Fig.	Min.	Typ.	Max.	Min.	Typ.	Max.		Fig.
STATIC CHARACTERISTICS											
Input Offset Voltage	V_{IO}		Fig.3	–	2.6	5	–	0.8	1	mV	Fig.2
Input Offset Current	I_{IO}		Fig.4	–	1.4	–	–	1.4	–	μA	Fig.2
Input Bias Current	I_{IB}		Fig.4	–	19	40	–	19	40	μA	Fig.5
Quiescent Operating Current	I_{10} or I_{11}	TERMINALS 4 / 5									
		NC / NC	Fig.8	–	1	–	–	1	–	mA	Fig.6
		NC / -V_{EE}	Fig.8	–	2.7	–	–	2.7	–	mA	NONE
		-V_{EE} / NC	Fig.8	–	0.45	–	–	0.45	–	mA	NONE
		-V_{EE} / -V_{EE}	Fig.8	–	1.25	–	–	1.25	–	mA	Fig.6
Quiescent Operating Current Ratio	$\frac{I_{10}}{I_{11}}$		Fig.8	–	1.05	–	–	1.05	–	–	Fig.7
Device Dissipation	P_T		Fig.8	–	26	–	–	26	–	mW	NONE
DYNAMIC CHARACTERISTICS											
Power Gain	G_p	f = 100 MHz, Cascode Configuration	Fig.10	16	20	–	16	20	–	dB	Fig.9
		f = 100 MHz, Differential-Ampl. Configuration	Fig.12	14	16	–	14	16	–	dB	Fig.11
Noise Figure	NF	f = 100 MHz, Cascode Configuration	Fig.10	–	7.8	9	–	7.8	9	dB	Fig.13
		f = 100 MHz, Differential Ampl. Configuration	Fig.12	–	7.8	9	–	7.8	9	dB	Fig.14
Common-Mode Rejection Ratio	CMR	f = 1 kHz	Fig.16	–	101	–	–	101	–	dB	Fig.15
AGC Range (Max. Voltage Gain to Complete Cutoff)	AGC	f = 1.75 MHz	Fig.17	-60	–	–	-60	–	–	dB	NONE

TYPICAL STATIC CHARACTERISTICS AND TEST CIRCUITS FOR TYPES CA3005 AND CA3006

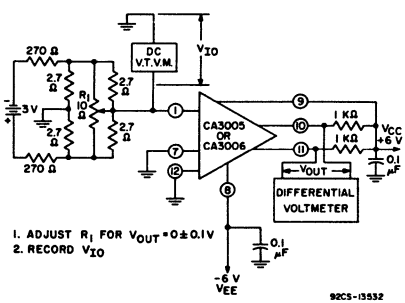
INPUT OFFSET VOLTAGE AND CURRENT



92CS-13317

Fig.2

INPUT OFFSET VOLTAGE TEST CIRCUIT



92CS-13532

Fig.3

TYPICAL STATIC CHARACTERISTICS AND TEST CIRCUITS FOR TYPES CA3005 AND CA3006

INPUT BIAS CURRENT

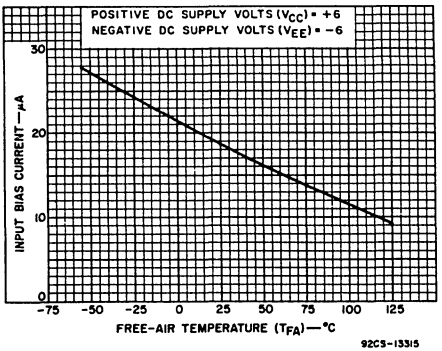


Fig.4

QUIESCENT OPERATING CURRENT

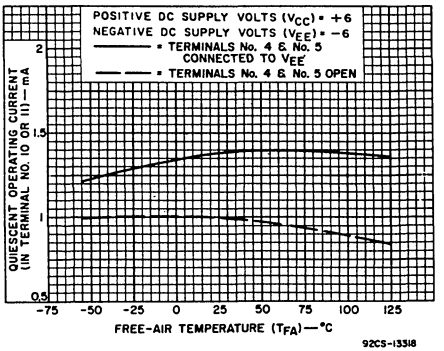


Fig.5

QUIESCENT OPERATING CURRENT RATIO

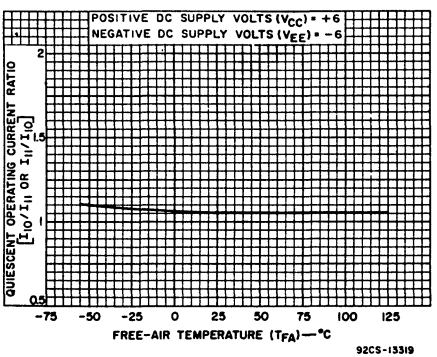


Fig.6

TYPICAL DYNAMIC CHARACTERISTICS AND TEST CIRCUITS FOR TYPES CA3005 AND CA3006

POWER-GAIN (CASCODE CONFIGURATION)

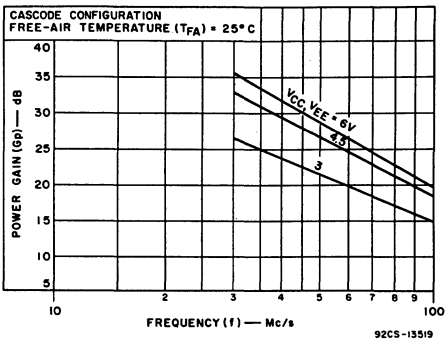


Fig.7

POWER-GAIN (DIFFERENTIAL-AMPLIFIER CONFIGURATION)

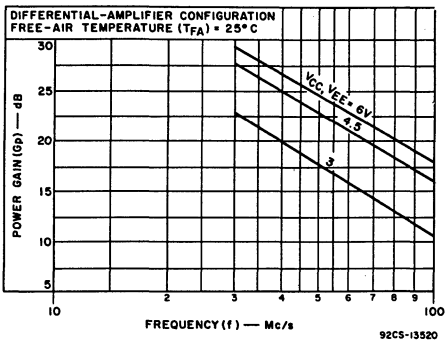
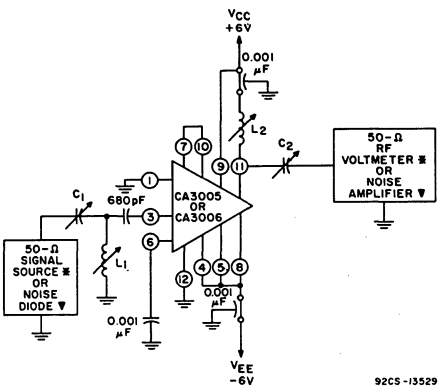


Fig.9

NOISE FIGURE AND POWER GAIN TEST CIRCUIT (CASCODE CONFIGURATION)



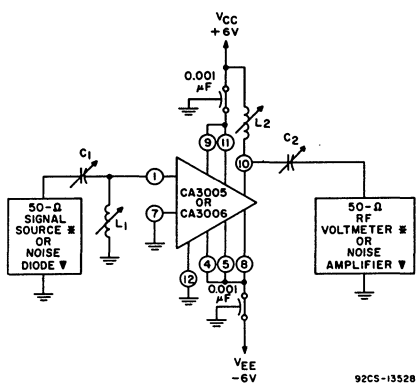
f Mc/s	C_1 pF	C_2 pF	L_1 μH	L_2 μH
30	14-150	5-40	0.3-0.6	0.8-1.4
100	5-40	5-40	0.07-0.12	0.15-0.3

* FOR POWER-GAIN TEST

▼ FOR NOISE-FIGURE TEST

Fig.8

NOISE FIGURE AND POWER-GAIN TEST CIRCUIT (DIFFERENTIAL AMPLIFIER CONFIGURATION)



f Mc/s	C_1 pF	C_2 pF	L_1 μH	L_2 μH
30	5-40	1.5-20	1.2-2	1.2-2
100	1-12	1-12	0.4-0.7	0.25-0.5

* FOR POWER-GAIN TEST

▼ FOR NOISE-FIGURE TEST

Fig.10

TYPICAL DYNAMIC CHARACTERISTICS FOR TYPES CA3005 AND CA3006

100-Mc/s NOISE FIGURE VS. V_{EE} (CASCODE CONFIGURATION)

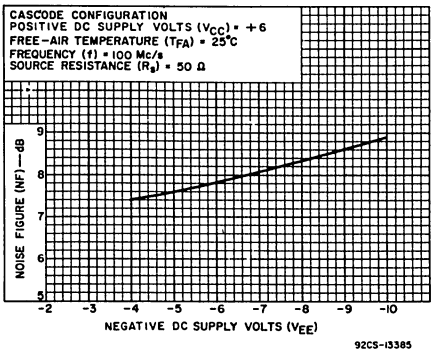


Fig. 11

100 Mc/s NOISE FIGURE VS. V_{EE} (DIFFERENTIAL AMPLIFIER CONFIGURATION)

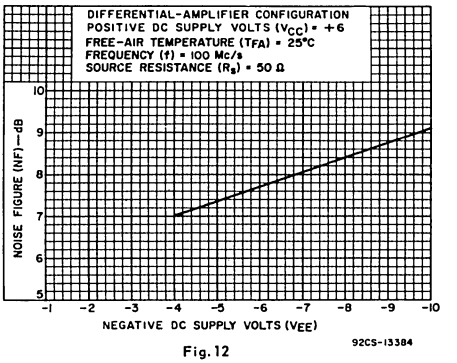


Fig. 12

COMMON-MODE-REJECTION RATIO

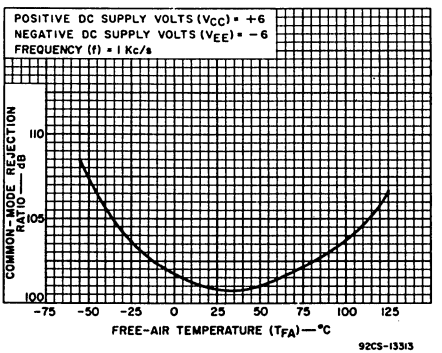


Fig. 13

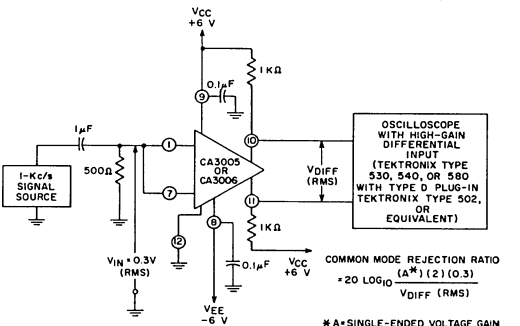


Fig. 14

AGC RANGE TEST CIRCUIT

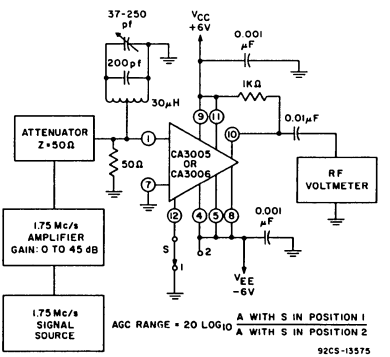


Fig. 15

RCA
Solid State
Division

Linear Integrated Circuits

CA3007

AF Amplifier



- Designed for use in Sound Systems and Communication Equipment
- Balanced differential-amplifier configuration with controlled constant-current source provides for both audio amplification and phase inversion
- Built-in temperature stability for operation from -55°C to +125°C
- Eliminates need for audio driver transformer
- Companion Application Note, ICAN 5037 "Application of the RCA-CA3007 Integrated Circuit Audio Driver" covers design of a dual supply audio driver in a direct-coupled audio amplifier, and a single supply audio driver in a capacitor-coupled audio amplifier

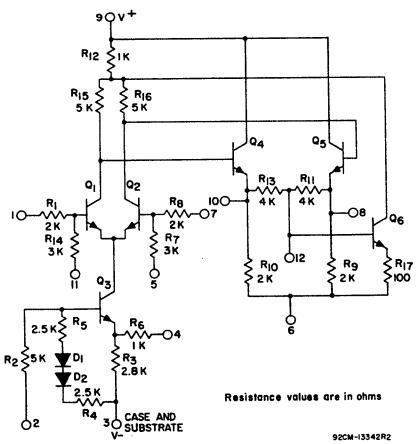
HIGHLIGHTS

- Input Impedance. . . . 4 kΩ typ.
- Output Impedance . . . 60 Ω typ.
- Power Gain . . . . . 22 dB typ.
- Push-Pull Input & Output
- Direct Coupling to Class B Audio Output Stage

APPLICATIONS

- Audio Amplifier
- Audio Driver

SCHEMATIC DIAGRAM



ABSOLUTE-MAXIMUM VOLTAGE LIMITS, at T_A = 25° C

Indicated voltage limits for each terminal can be applied under the specified operating conditions for other terminals. All voltages are with respect to ground ($-V_{CC}$, $+V_{EE}$, or common terminal of Positive and Negative DC supplies).

TERMINAL	VOLTAGE LIMITS		CONDITIONS	
	NEGATIVE	POSITIVE	TERMINAL	VOLTAGE
1	-2.5	+2.5	2	0
			3	-6
			6	0
			7	0
			9	+6
			11	0
2	-8	0	3	-8
			6	0
			7	0
			9	+6
			11	0
3	-10	0	6	0
			7	0
			9	+6
			11	0
4	-8.5	0	6	0
			7	0
			9	+6
			11	0
5	-2.5	+2.5	2	0
			3	-6
			6	0
			7	0
			9	+6
			11	0
6	-3	0	2	0
			3	-6
			7	0
			9	+6
			11	0
7	-2.5	+2.5	1	0
			2	0
			3	-6
			5	0
			6	0
			9	+6
8	-2	0	2	0
			3	-6
			6	0
			7	0
			9	+6
			11	0
9	0	+10	2	0
			3	-6
			6	0
			7	0
			11	0
10	-2	0	2	0
			3	-6
			6	0
			7	0
			9	+6
			11	0
11	-2.5	+2.5	1	0
			2	0
			3	-6
			6	0
			7	0
			9	+6
12	-2	0	2	0
			3	-6
			6	0
			7	0
			9	+6
			11	0
CASE	INTERNALLY CONNECTED TO TERMINAL No.3 (SUBSTRATE) DO NOT GROUND			

OPERATING-TEMPERATURE RANGE . . . . . -55°C to +125°C

STORAGE-TEMPERATURE RANGE . . . . . -65°C to +150°C

LEAD TEMPERATURE (During Soldering)
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm)
from case for 10 seconds max. . . . . +265°C

MAXIMUM SINGLE-ENDED INPUT-
SIGNAL VOLTAGE . . . . . ±2.5 V

MAXIMUM COMMON-MODE INPUT-
SIGNAL VOLTAGE . . . . . ±2.5 V

MAXIMUM DEVICE DISSIPATION . . . . . 300 mW

ELECTRICAL CHARACTERISTICS, at T_{FA} = 25°C, V_{CC} = +6 V, V_{EE} = -6 V,

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS Pin 4 Not Connected Unless Otherwise Noted	TEST CIRCUITS	LIMITS TYPE CA3007				TYPICAL CHARACTERISTICS CURVES
			Fig.	Min.	Typ.	Max.	Units	Fig.
STATIC CHARACTERISTICS								
Input Unbalance Voltage	V_{IU}		3	–	0.57	5	mV	2
Input Unbalance Current	I_{IU}		3	–	0.57	5	μA	2
Input Bias Current	I_I		3	–	11	34	μA	4
Quiescent Operating Voltage	V_8 or V_{10}		3	–	0.87	–	V	5
Device Dissipation	P_T		3	–	30	–	mW	NONE
DYNAMIC CHARACTERISTICS								
Power Gain	G_P	f = 1 Kc/s	6	20	22	–	dB	NONE
Total Harmonic Distortion	THD	f = 1 Kc/s	6	–	0.28	–	%	NONE
Input Impedance	Z_{IN}	f = 1 Kc/s	7	–	4K	–	Ω	NONE
Common-Mode Rejection Ratio	CMR	f = 1 Kc/s	9(A) 9(B)	–	77	–	dB	8

TYPICAL STATIC CHARACTERISTICS AND TEST CIRCUIT FOR CA3007

INPUT UNBALANCE VOLTAGE AND CURRENT vs TEMPERATURE

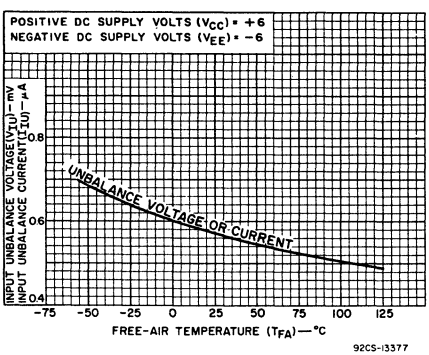
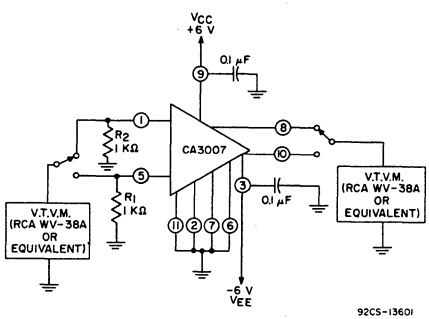


Fig.2

INPUT UNBALANCE VOLTAGE & CURRENT, INPUT BIAS CURRENT, QUIESCENT OPERATING VOLTAGE, AND DEVICE DISSIPATION TEST CIRCUIT



R_1 and R_2 matched to ±1%.

$P_T = V_{CC}I_9 + V_{EE}I_3$

I_9 = Direct Current into Terminal No.9

I_3 = Direct Current out of Terminal No.3

Fig.3

INPUT BIAS CURRENT vs TEMPERATURE

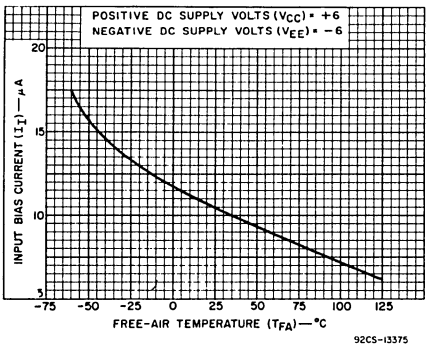


Fig.4

QUIESCENT OPERATING VOLTAGE vs TEMPERATURE

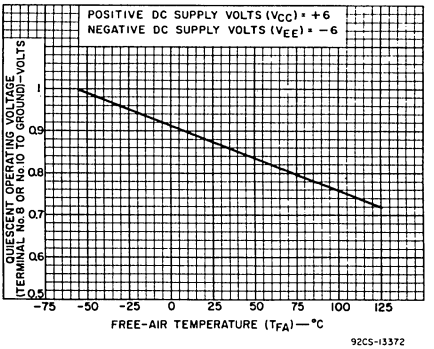
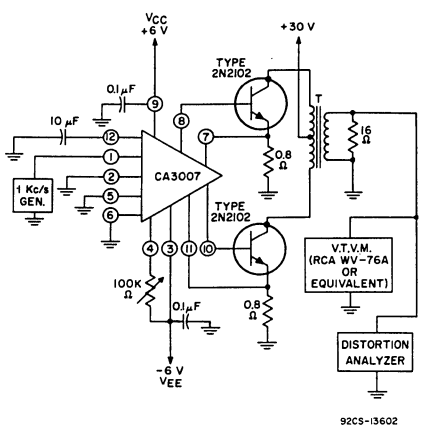


Fig.5

TYPICAL DYNAMIC TEST CIRCUITS FOR CA3007

POWER GAIN AND TOTAL HARMONIC DISTORTION TEST CIRCUIT



T (Output Transformer):
Primary Impedance = 2000 Ω C.T.
Secondary Impedance = 16 Ω
Efficiency = 45% approx.
(STANCOR TYPE TA-10 OR EQUIVALENT)

Fig.6

INPUT IMPEDANCE TEST CIRCUIT

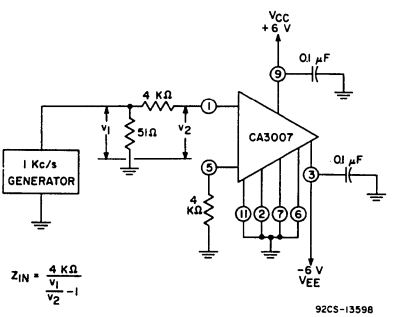


Fig.7

TYPICAL DYNAMIC CHARACTERISTIC AND TEST CIRCUITS FOR CA3007

COMMON-MODE REJECTION RATIO vs TEMPERATURE

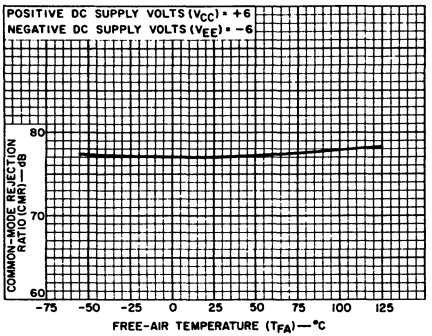
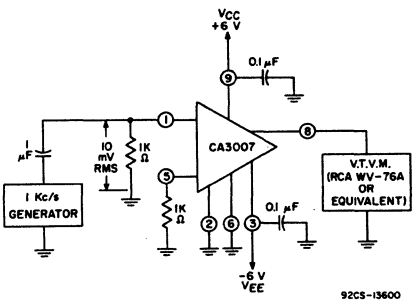
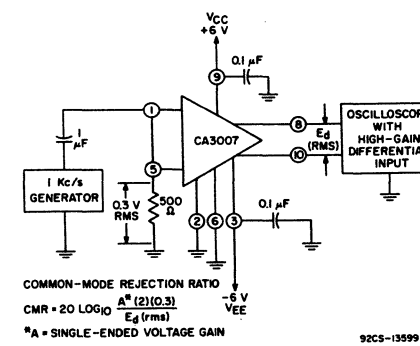


Fig.8

COMMON-MODE REJECTION-RATIO TEST CIRCUITS



(A) Single-Ended Differential Voltage Gain



(B) Common-Mode Voltage Gain

Fig.9

RCA
Solid State
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Linear Integrated Circuits

CA3008 CA3015 CA3030
CA3010 CA3016 CA3037
CA3029 CA3038

Operational Amplifiers

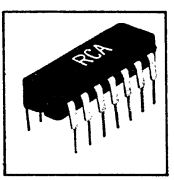
Monolithic Silicon

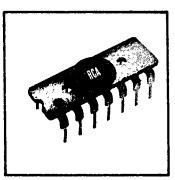
6-VOLT TYPES	*12-VOLT TYPES*	*PACKAGE*
CA3008	CA3016	14-Lead Flat Pack
CA3010	CA3015	12-Lead TO-5 Style
CA3029	CA3030	14-Lead Plastic Dual In-Line (TO-116)
CA3037	CA3038	14-Lead Ceramic Dual In-Line (TO-116)


CA3008
CA3016


CA3010
CA3015

- All types are electrically identical within their voltage groups
- Designed for use in Telemetry, Data-Processing, Instrumentation, and Communication Equipment
- Built-in temperature stability from -55°C to +125°C for flatpack, TO-5 style, and ceramic dual in-line packages; 0°C to +70°C for plastic dual in-line package
- Companion Application Notes ICAN-5290, "Integrated Circuit Operational Amplifiers"; ICAN-5213, "Application of the RCA-CA3015, CA3016 Integrated Circuit Operational Amplifiers"; and ICAN-5015, "Application of the RCA-CA3008, CA3010 Integrated Circuit Operational Amplifiers".


CA3029, CA3030


CA3037, CA3038

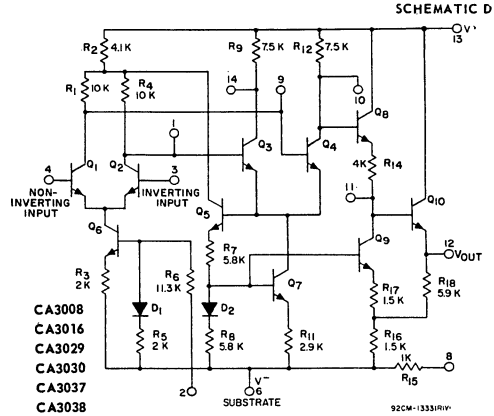
HIGHLIGHTS

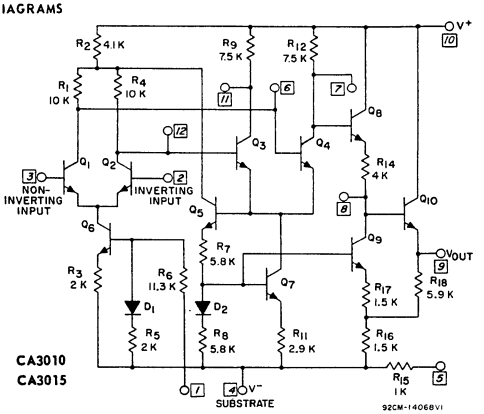
	6 V Types	12 V Types	
• Open-Loop Voltage Gain	60	70	dB typ.
• Common-Mode Rejection Ratio	94	103	dB typ.
• Output Impedance	200	92	Ω typ.
• Input Offset Voltage	1	1	mV typ.
• Static Power Drain at ± 12 V	-	175	mW typ.
± 6 V	30	30	mW typ.
± 3 V	7	7	mW typ.

APPLICATIONS

- Narrow-Band and Band-pass Amplifier
- Operational Functions
- Feedback Amplifier
- DC and Video Amplifier
- Multivibrator
- Oscillator
- Comparator
- Servo Driver
- Scaling Adder
- Balanced Modulator-Driver

SCHEMATIC DIAGRAMS





ABSOLUTE-MAXIMUM VOLTAGE AND CURRENT LIMITS, T_A = 25°C

Voltage or current limits shown for each terminal can be applied under the indicated voltage or other circuit conditions for other terminals

All voltages are with respect to ground (common terminal of Positive and Negative DC Supplies)

Terminal		Voltage or Current Limits		Circuit Conditions		
CA3010	CA3008 CA3029 CA3037	Negative	Positive	Terminal		Voltage
12	1	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
				CA3010	CA3008 CA3029 CA3037	
1	2	-8 V	0 V	4 10	6 13	-8 +6
2	3	-4 V	+1 V	1 3 4 10	2 4 6 13	0 0 -6 +6
3	4	-4 V	+1 V	1 2 4 10	2 3 6 13	0 0 -6 +6
	5	NO CONNECTION				
4	6	-10 V	0 V	1 10	2 13	0 +6
	7	NO CONNECTION				
5	8	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
6	9	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
7	10	0 V	+7 V	1 4 10	2 6 13	0 -6 +6
8	11	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
9	12	30 mA		4 10 200 Ω Between Terminals 6 & 12 (CA3008, CA3029, CA3037) 4 & 9 (CA3010)	6 13	-6 +6
10	13	0 V	+10 V	1 4	2 6	0 -6
11	14	0 V	+7 V	1 4 10	2 6 13	0 -6 +6
CASE		Internally connected to Terminal No.4, CA3010 (Substrate) DO NOT GROUND				

Terminal		Voltage or Current Limits		Circuit Conditions		
CA3015	CA3016 CA3030 CA3038	Negative	Positive	Terminal		Voltage
12	1	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
				CA3015	CA3016 CA3030 CA3038	
1	2	-16 V	0 V	4 10	6 13	-16 +12
2	3	-8 V	+1 V	1 3 4 10	2 4 6 13	0 0 -12 +12
3	4	-8 V	+1 V	1 2 4 10	2 3 6 13	0 0 -12 +12
	5	NO CONNECTION				
4	6	-20 V	0 V	1 10	2 13	0 +12
	7	NO CONNECTION				
5	8	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
6	9	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
7	10	0 V	+14 V	1 4 10	2 6 13	0 -12 +12
8	11	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
9	12	30 mA		4 10 400 Ω Between Terminals 6 & 12 (CA3016, CA3030, CA3038) 4 & 9 (CA3015)	6 13	-12 +12
10	13	0 V	+20 V	1 4	2 6	0 -12
11	14	0 V	+14 V	1 4 10	2 6 13	0 -12 +12
CASE		Internally connected to Terminal No.4, CA3015 (Substrate) DO NOT GROUND				

	CA3008 CA3016 CA3037	CA3010 CA3015 CA3038	CA3029 CA3030
OPERATING TEMPERATURE RANGE	-55°C to +125°C	-40°C to +85°C	
STORAGE TEMPERATURE RANGE	-65°C to +150°C	-65°C to +150°C	

	CA3016 CA3030 CA3038	CA3015	CA3008 CA3029	CA301 CA303
MAXIMUM SIGNAL VOLTAGE	-8 V to +1 V		-4 V to +1 V	
MAXIMUM DEVICE DISSIPATION	600 mW		300 mW	

ELECTRICAL CHARACTERISTICS at T_A = 25°C

Characteristics	Symbols	Special Test Conditions Terminal No.8 (CA3008, CA3016, CA3029, CA3030, CA3037, CA3038) Terminal No.5 (CA3010, CA3015) Not Connected Unless Otherwise Specified	Test Circuit Fig.	CA3008 CA3010 CA3029 CA3037 Min.	Typ.	Max.	CA3016 CA3015 CA3030 CA3038 Min.	Typ.	Max.	Units	Typical Characteristic Curves Fig.
STATIC CHARACTERISTICS:											
Input Offset Voltage	V_{IO}	V_{CC} = +6V, V_{EE} = -6V = +12V = -12V	4	- -	1.08 -	5 -	- -	- 1.37	- 5	mV	2
Input Offset Current	I_{IO}	= +6V = -6V = +12V = -12V	5	- -	0.54 -	5 -	- -	- 1.07	- 5	μA	2
Input Bias Current	I_{IB}	= +6V = -6V = +12V = -12V	5	- -	5.3 -	12 -	- -	- 9.6	- 24	μA	3
Input Offset Voltage Sensitivity: Positive	$\Delta V_{IO}/\Delta V_{CC}$	= +6V = -6V = +12V = -12V	4	- -	0.10 -	1 -	- -	- 0.096	- 0.5	mV/V	none
Negative	$\Delta V_{IO}/\Delta V_{EE}$	= +6V = -6V = +12V = -12V		- -	0.26 -	1 -	- -	- 0.156	- 0.5		
Device Dissipation	P_D	= +6 V = -6 V = +12V = -12V	4	- -	30 -	- -	- -	- 175	- -	mW	none
		[5] shorted to [9] V_{CC} = +6V, V_{EE} = -6V 8 shorted to 12 V_{CC} = +12V, V_{EE} = -12V		- -	102 -	- -	- -	- 500	- -		
DYNAMIC CHARACTERISTICS: All tests at f = 1 kHz except BW_{OL}											
Open-Loop Differential Voltage Gain	A_{OL}	V_{CC} = +6V, V_{EE} = -6V = +12V = -12V	8	57 -	60 -	- -	- 66	- 70	- -	dB	6 & 7
Open-Loop Bandwidth at -3 dB Point	BW_{OL}	= +6V = -6V = +12V = -12V	8	200 -	300 -	- -	- 200	- 320	- -	kHz	6 & 7
Common-Mode Rejection Ratio	CMRR	V_{CC} = +6V, V_{EE} = -6V = +12V = -12V	11	70 -	94 -	- -	- 80	- 103	- -	dB	12
Maximum Output-Voltage Swing	$V_{O(P-P)}$	= +6V = -6V = +12V = -12V	8	4 -	6.75 -	- -	- 12	- 14	- -	V_{P-P}	9 & 10
Input Impedance	Z_{IN}	= +6V = -6V = +12V = -12V	14	10 -	14 -	- -	- 5	- 7.8	- -	kΩ	13
Output Impedance	Z_{OUT}	= +6V = -6V = +12V = -12V	15	- -	200 -	- -	- -	- 92	- -	Ω	16
Common-Mode Input-Voltage Range	V_{ICR}	= +6V = -6V = +12V = -12V	11	0.5 to -4 -	- -	- -	- 0.65 to -8	- -	- -	V	none

LEAD TEMPERATURE (During Soldering):
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm)
from case for 10 seconds max. ..... +265°C

TYPICAL STATIC CHARACTERISTICS AND TEST CIRCUITS

Terminal Numbers in Circles are for CA3008, CA3016, CA3029, CA3030, CA3037, CA3038; Italic Numbers in Square Boxes are for CA3010, CA3015

INPUT OFFSET VOLTAGE AND CURRENT

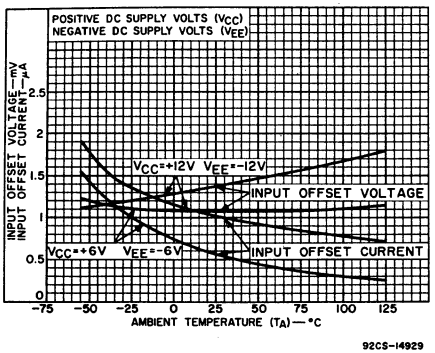


Fig.2

INPUT BIAS CURRENT

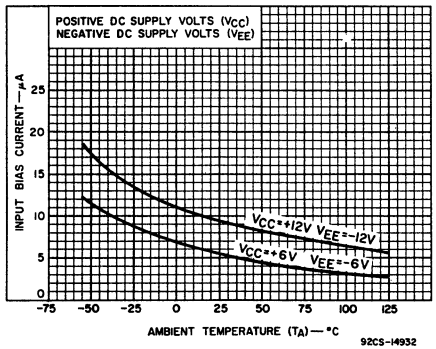


Fig.3

INPUT OFFSET VOLTAGE, INPUT OFFSET VOLTAGE SENSITIVITY, AND DEVICE DISSIPATION TEST CIRCUIT

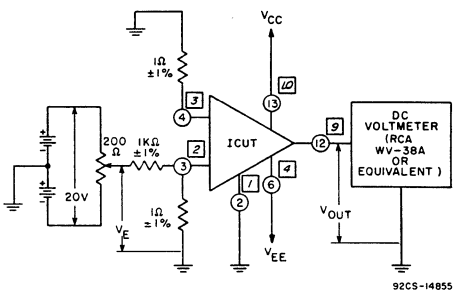


Fig.4

Procedure:

Input Offset Voltage

1. Adjust V_E for a DC Output Voltage (V_{OUT}) of 0 ± 0.1 volts.
2. Measure V_E and record Input Offset Voltage in millivolts as $V_E/1000$.

Input Offset Voltage Sensitivity

1. Adjust V_E for a DC Output Voltage (V_{OUT}) of 0 ± 0.1 volts.
2. Increase $|V_{CC}|$ by 1 volt and record output voltage (V_{OUT}).
3. Decrease $|V_{CC}|$ by 1 volt and record output voltage (V_{OUT}).
4. Divide the difference between V_{OUT} measured in steps 2 and 3 by the change in V_{CC} in steps 2 and 3.

$$\frac{V_{OUT}}{V_{CC}} = \frac{V_{OUT}\ (\text{Step 2}) - V_{OUT}\ (\text{Step 3})}{2\ \text{volts}}$$

5. Refer the reading to the input by dividing by Open Loop Voltage Gain (A_{OL}).

$$V_{IO}/V_{CC} = \frac{V_{OUT}/V_{CC}}{A_{OL}}$$

6. Repeat procedures 1 through 5 for the Negative Supply (V_{EE}).
7. Device Dissipation

$P_T = V_{CC}I_C + V_{EE}I_E$

I_C = Direct Current into Terminal (13) or *10*

I_E = Direct Current out of Terminal (6) or *4*

INPUT OFFSET CURRENT AND INPUT BIAS CURRENT TEST CIRCUIT

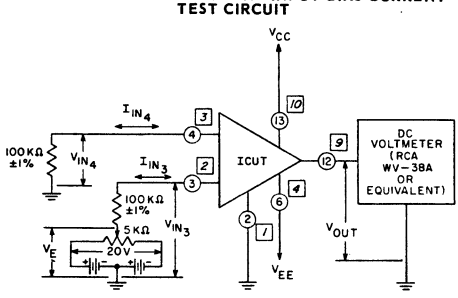


Fig.5

Procedure:

Input Bias Current and Input Offset Current

1. Adjust V_E for $|V_{OUT}| < 0.1$ V DC.
2. Measure and record V_E and V_{IN_4}.
3. Calculate the Input Bias Current using the following equation:

$$I_{I4} = \frac{V_{IN_4}}{100\ k\Omega}$$

4. Calculate the Input Offset Current using the following equation:

$$I_{IO} = V_E/100\ k\Omega$$

TYPICAL DYNAMIC CHARACTERISTICS AND TEST CIRCUITS

Terminal Numbers in Circles are for CA3008, CA3016, CA3029, CA3030, CA3037, CA3038; Italic Numbers in Square Boxes are for CA3010, CA3015

OPEN-LOOP VOLTAGE GAIN vs. FREQUENCY FOR CA3008, CA3010, CA3015, CA3016, CA3037, CA3038

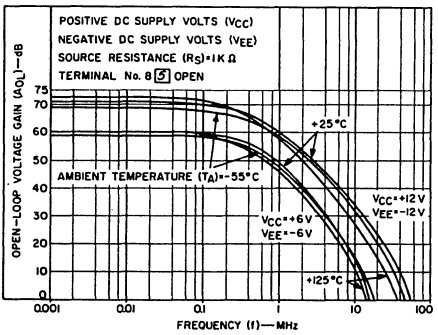


Fig.6

OPEN-LOOP VOLTAGE GAIN vs. FREQUENCY FOR CA3029 AND CA3030

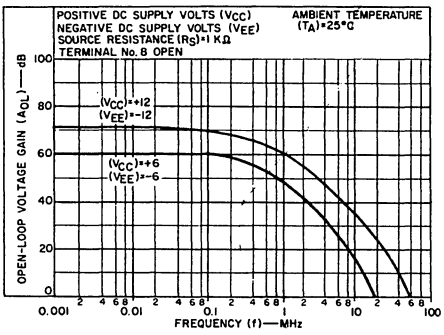
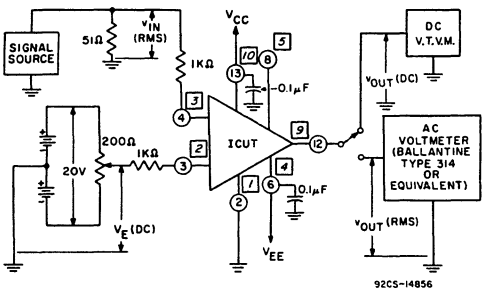


Fig.7

OPEN-LOOP DIFFERENTIAL VOLTAGE GAIN, MAXIMUM PEAK-TO-PEAK OUTPUT VOLTAGE, AND OPEN-LOOP BANDWIDTH AT -3 dB POINT TEST CIRCUIT



Procedure:

1. Adjust V_E for V_{OUT} = ±0.1 V DC.
2. Measure Open-Loop Differential Voltage Gain (A_{OL}) at f = 1 kHz.

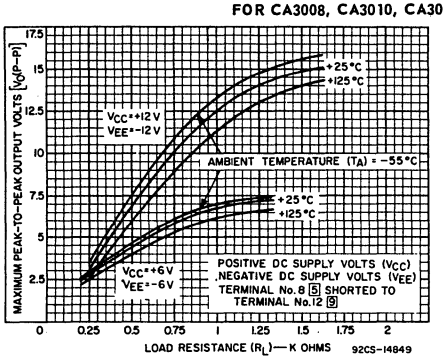
$$A_{OL} = 20 \log_{10} \frac{V_{OUT}}{V_{IN}}$$

3. Measure Maximum Peak-to-Peak Output Voltage at f = 1 kHz.
4. Measure Open-Loop Bandwidth at -3 dB Point.

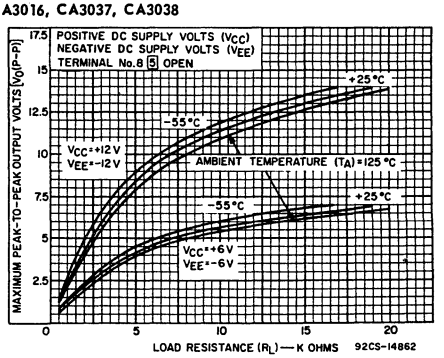
Reference Level = A_{OL} at 1 kHz.

Fig.8

MAXIMUM PEAK-TO-PEAK OUTPUT VOLTAGE vs. LOAD RESISTANCE FOR CA3008, CA3010, CA3015, CA3016, CA3037, CA3038



(a)



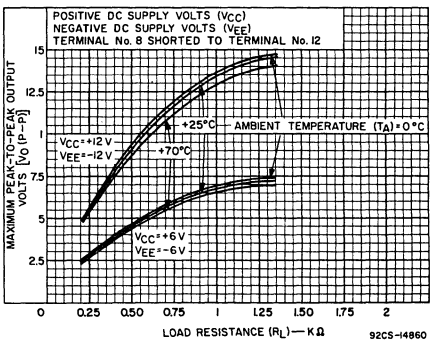
(b)

Fig.9

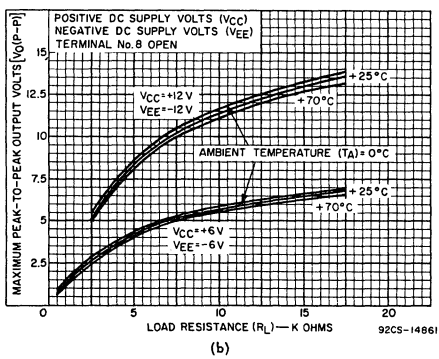
TYPICAL DYNAMIC CHARACTERISTICS AND TEST CIRCUITS

Terminal Numbers in Circles are for CA3008, CA3016, CA3029, CA3030, CA3037, CA3038; Italic Numbers in Square Boxes are for CA3010, CA3015

MAXIMUM PEAK-TO-PEAK OUTPUT VOLTAGE vs. LOAD RESISTANCE FOR CA3029 AND CA3030



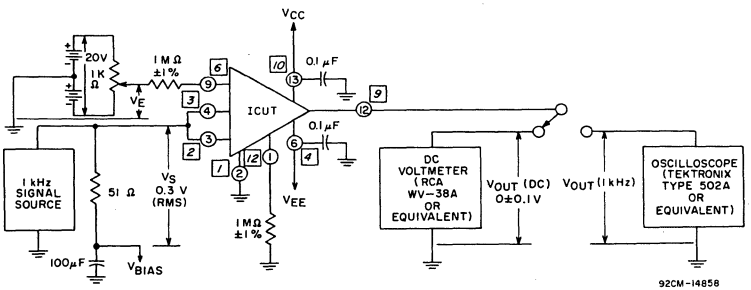
(a)



(b)

Fig.10

COMMON-MODE REJECTION RATIO AND COMMON-MODE INPUT-VOLTAGE-RANGE TEST CIRCUIT



Procedures:

Common-Mode Rejection Ratio:

1. Set V_{BIAS} = 0. Adjust V_E for V_{OUT}(DC) = 0 ± 0.1 V.
2. Apply 1-kHz sinusodial input signal and adjust for V_S = 0.3 V (RMS).
3. Measure and record the RMS value of V_{OUT}. An oscilloscope is used for this measurement so that the output signal may be visually separated.from noise output.
4. Calculate Common-Mode Voltage Gain:

$$A_{CM} = V_{OUT}/V_S$$

$$A_{CM} \text{ in dB} = -20 \log_{10} V_S/V_{OUT}$$

5. Calculate Common-Mode Rejection Ratio:

$$CMR \text{ in dB} = A_{DIFF} \text{ in dB} - A_{CM} \text{ in dB}.$$

Common-Mode Input-Voltage Range:

1. Calculate and record CMR for various positive and negative values of V_{BIAS} within the maximum limits shown on Page 2. The Common-Mode Input-Voltage Range limits are those values of V_{BIAS} at which CMR is 6 dB less than that calculated in Step 5 of the procedure given above.

Fig.11

COMMON-MODE REJECTION RATIO vs. FREQUENCY

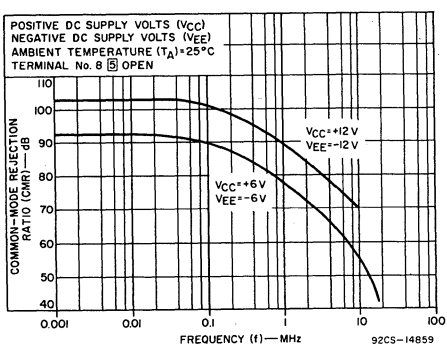


Fig.12

TYPICAL DYNAMIC CHARACTERISTICS AND TEST CIRCUITS

Terminal Numbers in Circles are for CA3008, CA3016, CA3029, CA3030, CA3037, CA3038;
Italic Numbers in Square Boxes are for CA3010, CA3015

SINGLE-ENDED INPUT IMPEDANCE vs. TEMPERATURE

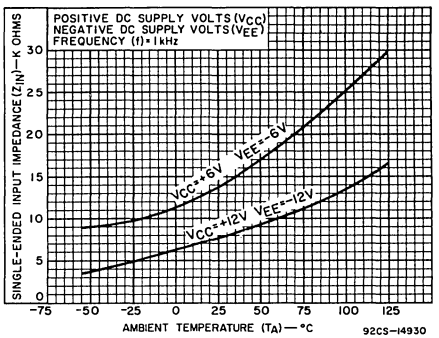


Fig.13

SINGLE-ENDED INPUT IMPEDANCE TEST CIRCUIT

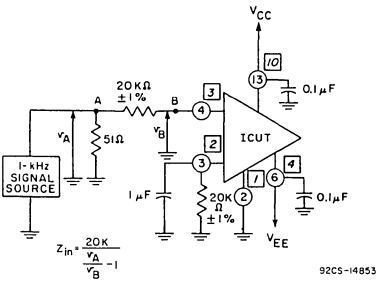


Fig.14

OUTPUT IMPEDANCE TEST CIRCUIT

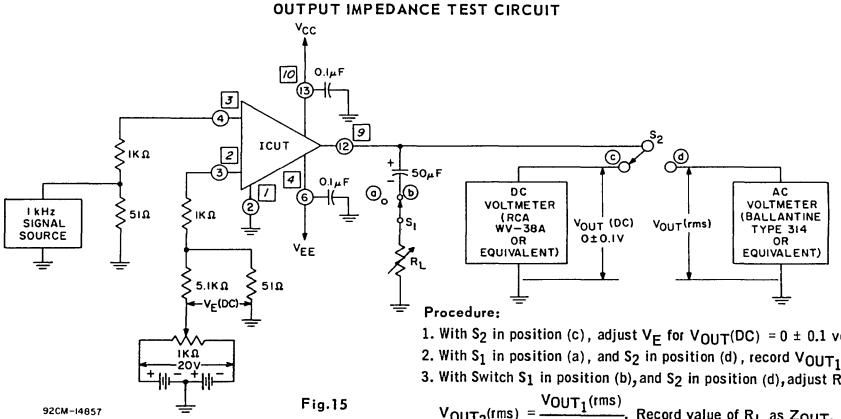
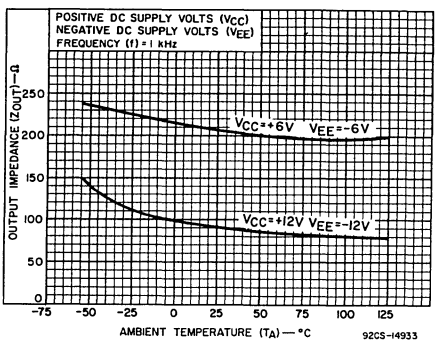


Fig.15

Procedure:

1. With S_2 in position (c), adjust V_E for $V_{OUT}(DC) = 0 \pm 0.1$ volt.
2. With S_1 in position (a), and S_2 in position (d), record $V_{OUT_1}(rms)$.
3. With Switch S_1 in position (b), and S_2 in position (d), adjust R_L until $V_{OUT_2}(rms) = \frac{V_{OUT_1}(rms)}{2}$. Record value of R_L as Z_{OUT}.



OUTPUT IMPEDANCE vs. TEMPERATURE

Fig.16

RCA
Solid State
Division

Linear Integrated Circuits

CA3008A CA3015A CA3030A
CA3010A CA3016A CA3037A
CA3029A CA3038A

Operational Amplifiers

Monolithic Silicon

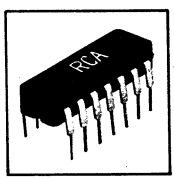
6-VOLT TYPES	12-VOLT TYPES	PACKAGE
CA3008A	CA3016A	14-Lead Flat Pack
CA3010A	CA3015A	12-Lead TO-5 Style
CA3029A	CA3030A	14-Lead Plastic Dual In-Line (TO-116)
CA3037A	CA3038A	14-Lead Ceramic Dual In-Line (TO-116)

- These new types have all the desirable features and characteristics of their prototypes plus lower noise figures and improved input characteristics for offset voltage, offset current, bias current, and impedance.
- All types are electrically identical within their voltage groups
- Designed for use in Telemetry, Data-Processing, Instrumentation, and Communication Equipment
- Built-in temperature stability from -55°C to +125°C for Flatpack, TO-5 style, and ceramic dual in-line packages; 0°C to +70°C for plastic dual in-line package
- Companion Application Notes ICAN-5290, "Integrated Circuit Operational Amplifiers"; ICAN-5213, "Application of the RCA-CA3015, CA3016 Integrated Circuit Operational Amplifiers"; and ICAN-5015, "Application of the RCA-CA3008, CA3010 Integrated Circuit Operational Amplifiers" cover Bode characteristics, phase compensation, frequency shaping, and amplifier design.

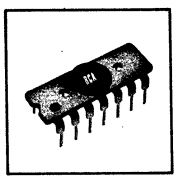


CA3008A, CA3016A


CA3010A, CA3015A



CA3029A, CA3030A


CA3037A, CA3038A

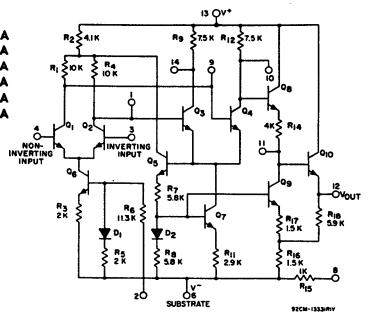
HIGHLIGHTS

	6 V Types	12 V Types	
• Open-Loop Voltage Gain	60	70	dB typ.
• Common-Mode Rejection Ratio	94	103	dB typ.
• Input Impedance	20	10	k Ω typ.
• Input Offset Voltage	0.9	1	mV typ.
• Input Offset Current	0.3	0.5	μA typ.
• Input Bias Current	2.5	4.7	μA typ.
• Static Power Drain at ±12 V	-	175	mW typ.
at ±6 V	30	30	mW typ.
at ±3 V	7	7	mW typ.

APPLICATIONS

- Narrow-Band and Band-pass Amplifier
- Operational Functions
- Feedback Amplifier
- DC and Video Amplifier
- Multivibrator
- Oscillator
- Comparator
- Servo Driver
- Scaling Adder
- Balanced Modulator-Driver

CA3008A
CA3016A
CA3029A
CA3030A
CA3037A
CA3038A



CA3010A
CA3015A

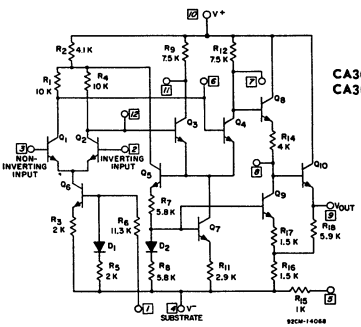


Fig. 1

SCHEMATIC DIAGRAMS

ABSOLUTE-MAXIMUM VOLTAGE AND CURRENT LIMITS, T_A = 25°C

Voltage or current limits shown for each terminal can be applied under the indicated voltage or other circuit conditions for other terminals

All voltages are with respect to ground (common terminal of Positive and Negative DC Supplies)

Terminal		Voltage or Current Limits		Circuit Conditions		
CA3010A	CA3008A, CA3029A, CA3037A	Negative	Positive	Terminal		Voltage
12	1	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
				CA3010A	CA3008A, CA3029A, CA3037A	
1	2	-8 V	0 V	4 10	6 13	-8 +6
2	3	-4 V	+1 V	1 3 4 10	2 4 6 13	0 0 -6 +6
3	4	-4 V	+1 V	1 2 4 10	2 3 6 13	0 0 -6 +6
-	5	NO CONNECTION				
4	6	-10 V	0 V	1 10	2 13	0 +6
-	7	NO CONNECTION				
5	8	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
6	9	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
7	10	0 V	+7 V	1 4 10	2 6 13	0 -6 +6
8	11	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
9	12	30 mA		4 10 200 Ω Between Terminals 6 & 12 (CA3008A, CA3029A, CA3037A) 4 & 9 (CA3010A)	6 13	-6 +6
10	13	0 V	+10 V	1 4	2 6	0 -6
11	14	0 V	+7 V	1 4 10	2 6 13	0 -6 +6
CASE		Internally connected to Terminal No.4, CA3010A (Substrate) DO NOT GROUND				

Terminal		Voltage or Current Limits		Circuit Conditions		
CA3015A	CA3016A, CA3030A, CA3038A	Negative	Positive	Terminal		Voltage
12	1	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
				CA3015A	CA3016A, CA3030A, CA3038A	
1	2	-16 V	0 V	4 10	6 13	-16 +12
2	3	-8 V	+1 V	1 3 4 10	2 4 6 13	0 0 -12 +12
3	4	-8 V	+1 V	1 2 4 10	2 3 6 13	0 0 -12 +12
	5	NO CONNECTION				
4	6	-20 V	0 V	1 10	2 13	0 +12
	7	NO CONNECTION				
5	8	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
6	9	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
7	10	0 V	+14 V	1 4 10	2 6 13	0 -12 +12
8	11	DO NOT APPLY VOLTAGE FROM AN EXTERNAL SOURCE TO THIS TERMINAL				
9	12	30 mA		4 10 400 Ω Between Terminals 6 & 12 (CA3016A, CA3030A, CA3038A) 4 & 9 (CA3015A)	6 13	-12 +12
10	13	0 V	+20 V	1 4	2 6	0 -12
11	14	0 V	+14 V	1 4 10	2 6 13	0 -12 +12
CASE		Internally connected to Terminal No.4, CA3015A (Substrate) DO NOT GROUND				

	CA3008A, CA3016A, CA3037A, CA3038A, CA3010A, CA3015A, CA3029A, CA3030A
OPERATING TEMPERATURE RANGE	-55°C to +125°C / -40°C to +85°C
STORAGE TEMPERATURE RANGE	-65°C to +200°C / -65°C to +150°C

	CA3016A, CA3030A, CA3038A, CA3015A, CA3008A, CA3029A, CA3010A, CA3037A
MAXIMUM SIGNAL VOLTAGE	-8 V to +1 V / -4 V to +1 V
MAXIMUM DEVICE DISSIPATION	600 mW / 300 mW

ELECTRICAL CHARACTERISTICS at T_A = 25°C

Characteristics	Symbols	Special Test Conditions Terminal No.8 (CA3008A, CA3016A, CA3029A, CA3030A, CA3037A, CA3038A), Terminal No.5 (CA3010A, CA3015A) Not Connected Unless Otherwise Specified	Test Circuit Fig.	CA3008A CA3010A CA3029A CA3037A Min.	Typ.	Max.	CA3016A CA3015A CA3030A CA3038A Min.	Typ.	Max.	Units	Typical Characteristic Curves Fig.
STATIC CHARACTERISTICS:											
Input Offset Voltage	V_{IO}	V_{CC} = +6V, V_{EE} = -6V = +12V = -12V	4	- -	0.9 -	2 -	- -	- 1	- 2	mV	2
Input Offset Current	I_{IO}	= +6V = -6V = +12V = -12V	5	- -	0.3 -	1.5 -	- -	- 0.5	- 1.6	μA	2
Input Bias Current	I_{IB}	= +6V = -6V = +12V = -12V	5	- -	2.5 -	4 -	- -	- 4.7	- 6	μA	3
Input Offset Voltage Sensitivity: Positive	$\Delta V_{IO}/\Delta V_{CC}$	= +6V = -6V = +12V = -12V	4	- -	0.10 -	1 -	- -	- 0.096	- 0.5	mV/V	none
Negative	$\Delta V_{IO}/\Delta V_{EE}$	= +6V = -6V = +12V = -12V		- -	0.26 -	1 -	- -	- 0.156	- 0.5		
Device Dissipation	P_D	= +6 V = -6 V = +12V = -12V	4	- -	40 -	- -	- -	- 175	- -	mW	none
		[5] shorted to [9] V_{CC} = +6V, V_{EE} = -6V		-	102	-	-	-	-		
		8 shorted to 12 V_{CC} = +12V, V_{EE} = -12V		-	-	-	-	500	-		
DYNAMIC CHARACTERISTICS: All tests at f = 1 kHz except BW_{OL}											
Open-Loop Differential Voltage Gain	A_{OL}	V_{CC} = +6V, V_{EE} = -6V = +12V = -12V	8	57 -	60 -	- -	- 66	- 70	- -	dB	6 & 7
Open-Loop Bandwidth at -3 dB Point	BW_{OL}	= +6V = -6V = +12V = -12V	8	200 -	300 -	- -	- 200	- 320	- -	kHz	6 & 7
Slew Rate	SR	V_{CC} = +6V V_{EE} = -6V R_S = 1 kΩ = +12V = -12V	none	- -	3 -	- -	- -	- 7	- -	V/μs	none
Common-Mode Rejection Ratio	CMR	V_{CC} = +6V, V_{EE} = -6V = +12V = -12V	11	70 -	94 -	- -	- 80	- 103	- -	dB	12
Maximum Output-Voltage Swing	$V_{O(P-P)}$	= +6V = -6V = +12V = -12V	8	4 -	6.75 -	- -	- 12	- 14	- -	V_{P-P}	9 & 10
Input Impedance	Z_{IN}	= +6V = -6V = +12V = -12V	14	15 -	20 -	- -	- 7.5	- 10	- -	kΩ	13
Output Impedance	Z_{OUT}	= +6V = -6V = +12V = -12V	15	- -	160 -	- -	- -	- 85	- -	Ω	16
Common-Mode Input-Voltage Range	V_{ICR}	= +6V = -6V = +12V = -12V	11	+0.5 -4 -	- - -	- - -	- +0.65 -8	- - -	- - -	V	none
Noise Figure	NF	V_{CC} = +3V , V_{EE} = -3V R_S = 1 kΩ = +6V = -6V = +9V = -9V = +12V = -12V	18	- - - -	6.3 8.3 - -	9 12 - -	- - - -	6.3 8.3 10 11	9 12 14 16	dB	17

LEAD TEMPERATURE (During Soldering): **ALL TYPES**

At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max. ..... +265°C

TYPICAL STATIC CHARACTERISTICS AND TEST CIRCUITS

Terminal Numbers in Circles are for CA3008A, CA3016A, CA3029A, CA3030A, CA3037A, CA3038A; Italic Numbers in Square Boxes are for CA3010A, CA3015A

INPUT OFFSET VOLTAGE AND CURRENT

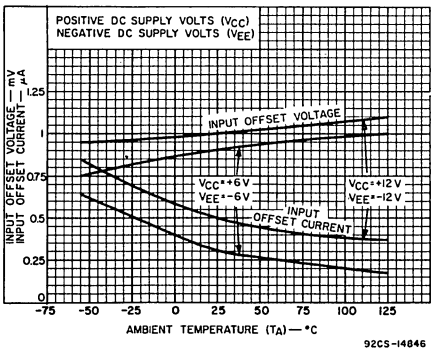


Fig.2

INPUT BIAS CURRENT

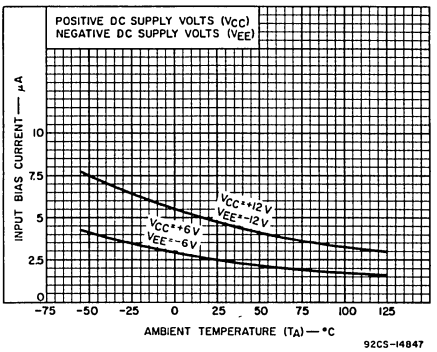


Fig.3

INPUT OFFSET VOLTAGE, INPUT OFFSET VOLTAGE SENSITIVITY, AND DEVICE DISSIPATION TEST CIRCUIT

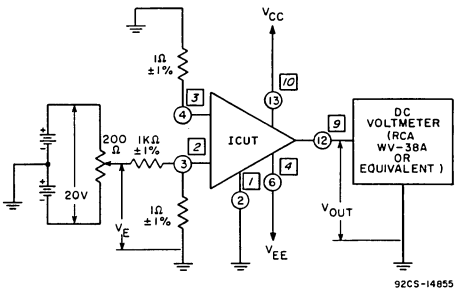


Fig.4

INPUT OFFSET CURRENT AND INPUT BIAS CURRENT TEST CIRCUIT

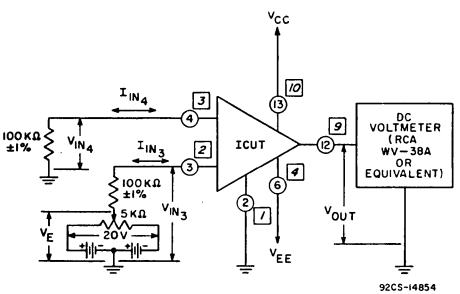


Fig.5

Procedure:

Input Offset Voltage

1. Adjust V_E for a DC Output Voltage (V_{OUT}) of 0 ± 0.1 volts.
2. Measure V_E and record Input Offset Voltage in millivolts as $V_E/1000$.

Input Offset Voltage Sensitivity

1. Adjust V_E for a DC Output Voltage (V_{OUT}) of 0 ± 0.1 volts.
2. Increase $|V_{CC}|$ by 1 volt and record output voltage (V_{OUT}).
3. Decrease $|V_{CC}|$ by 1 volt and record output voltage (V_{OUT}).
4. Divide the diference between V_{OUT} measured in steps 2 and 3 by the change in V_{CC} in steps 2 and 3.

$$\frac{V_{OUT}}{V_{CC}} = \frac{V_{OUT}\ (\text{Step 2}) - V_{OUT}\ (\text{Step 3})}{2\ \text{volts}}$$

5. Refer the reading to the input by dividing by Open Loop Voltage Gain (A_{OL}).

$$V_{IO}/V_{CC} = \frac{V_{OUT}/V_{CC}}{A_{OL}}$$

6. Repeat procedures 1 through 5 for the Negative Supply (V_{EE}).
7. Device Dissipation

$P_T = V_{CC}I_C + V_{EE}I_E$

I_C = Direct Current into Terminal 13 or *10*

I_E = Direct Current out of Terminal 6 or *4*

Procedure:

Input Bias Current and Input Offset Current

1. Adjust V_E for $|V_{OUT}| < 0.1$ V DC.
2. Measure and record V_E and V_{IN_4}
3. Calculate the Input Bias Current using the following equation:

$$I_{I4} = \frac{V_{IN_4}}{100\ k\Omega}$$

4. Calculate the Input Offset Current using the following equation:

$$I_{IO} = V_E/100\ k\Omega$$

TYPICAL DYNAMIC CHARACTERISTICS AND TEST CIRCUITS

Terminal Numbers in Circles are for CA3008A, CA3016A, CA3029A, CA3030A, CA3037A, CA3038A; Italic Numbers in Square Boxes are for CA3010A, CA3015A

OPEN LOOP VOLTAGE GAIN vs. FREQUENCY FOR CA3008A, CA3010A, CA3015A, CA3016A, CA3037A, CA3038A

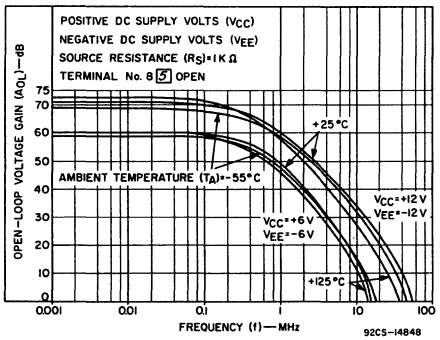


Fig.6

OPEN LOOP VOLTAGE GAIN vs. FREQUENCY FOR CA3029A AND CA3030A.

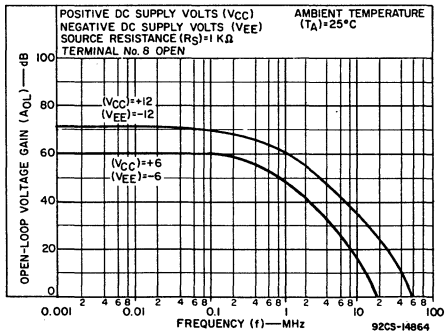
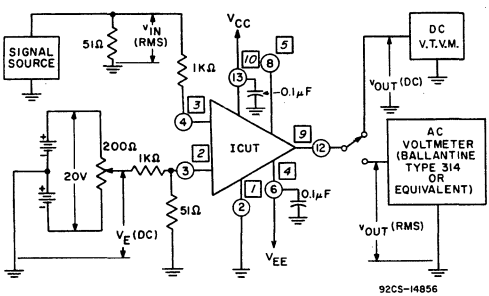


Fig.7

OPEN-LOOP DIFFERENTIAL VOLTAGE GAIN, MAXIMUM PEAK-TO-PEAK OUTPUT VOLTAGE, AND OPEN-LOOP BANDWIDTH AT -3 POINT TEST CIRCUIT



Procedure:

1. Adjust V_E for V_{OUT} = ±0.1 V DC.
2. Measure Open-Loop Differential Voltage Gain (A_{OL}) at f = 1 kHz

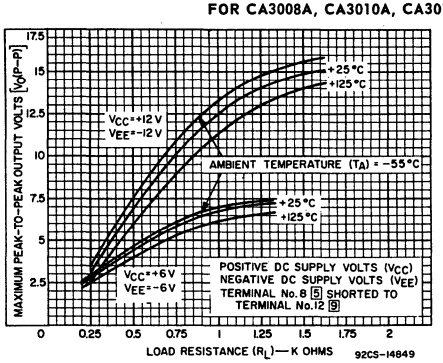
$$A_{OL} = 20 \log_{10} \frac{V_{OUT}}{V_{IN}}$$

3. Measure Maximum Peak-to-Peak Output Voltage at f = 1 kHz
4. Measure Open-Loop Bandwidth at -3 dB Point

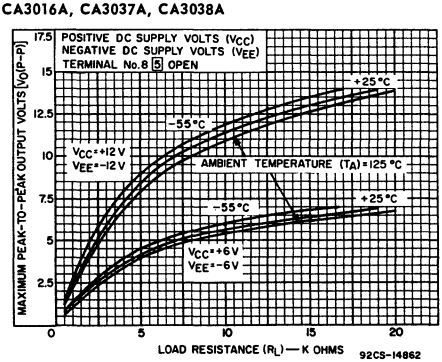
Reference Level = A_{OL} at 1 kHz

Fig.8

MAXIMUM PEAK-TO-PEAK OUTPUT VOLTAGE vs. LOAD RESISTANCE FOR CA3008A, CA3010A, CA3015A, CA3016A, CA3037A, CA3038A



(a)



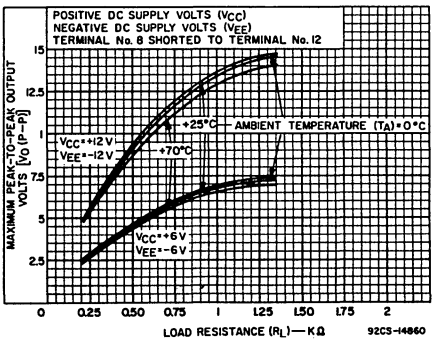
(b)

Fig.9

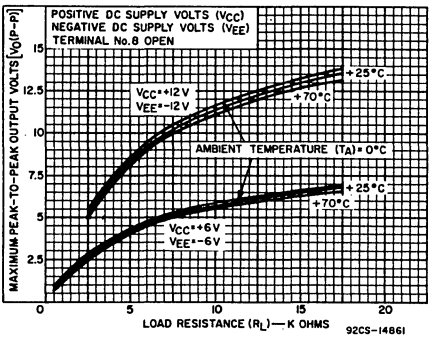
TYPICAL DYNAMIC CHARACTERISTICS AND TEST CIRCUITS

Terminal Numbers in Circles are for CA3008A, CA3016A, CA3029A, CA3030A, CA3037A, CA3038A; Italic Numbers in Square Boxes are for CA3010A, CA3015A

MAXIMUM PEAK-TO-PEAK OUTPUT VOLTAGE vs. LOAD RESISTANCE FOR CA3029A AND CA3030A



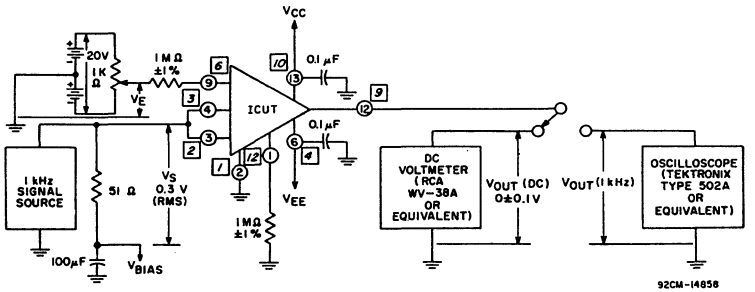
(a)



(b)

Fig.10

COMMON-MODE REJECTION RATIO AND COMMON-MODE INPUT-VOLTAGE-RANGE TEST CIRCUIT



Procedures:

Common-Mode Rejection Ratio:

1. Set V_{BIAS} = 0. Adjust V_E for $V_{OUT(DC)}$ = 0 ± 0.1 V.
2. Apply 1-kHz sinusoidal input signal and adjust for V_S = 0.3 V (RMS).
3. Measure and record the RMS value of V_{OUT}. An oscilloscope is used for this measurement so that the output signal may be visually separated from noise output.
4. Calculate Common-Mode Voltage Gain:

$$A_{CM} = V_{OUT}/V_S$$

$$A_{CM} \text{ in dB} = -20 \log_{10} V_S/V_{OUT}$$

5. Calculate Common-Mode Rejection Ratio:

CMR in dB = A_{DIFF} in dB - A_{CM} in dB.

Common-Mode Input-Voltage Range:

1. Calculate and record CMR for various positive and negative values of V_{BIAS} within the maximum limits shown on Page 2. The Common-Mode Input-Voltage Range limits are those values of V_{BIAS} at which CMR is 6 dB less than that calculated in Step 5 of the procedure given above.

Fig.11

COMMON-MODE REJECTION RATIO vs. FREQUENCY

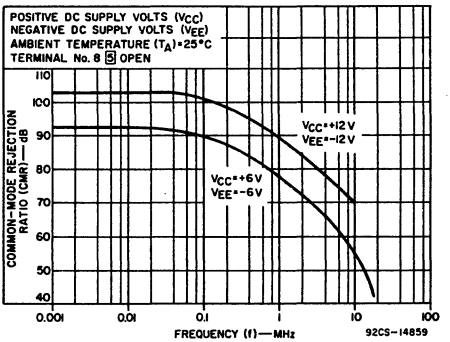


Fig.12

TYPICAL DYNAMIC CHARACTERISTICS AND TEST CIRCUITS

Terminal Numbers in Circles are for CA3008A, CA3016A, CA3029A, CA3030A, CA3037A, CA3038A; Italic Numbers in Square Boxes are for CA3010A, CA3015A

SINGLE-ENDED INPUT IMPEDANCE vs. TEMPERATURE

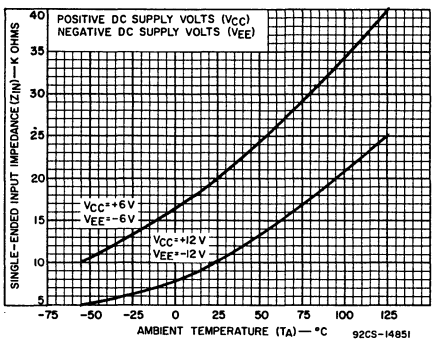


Fig.13

SINGLE-ENDED INPUT IMPEDANCE TEST CIRCUIT

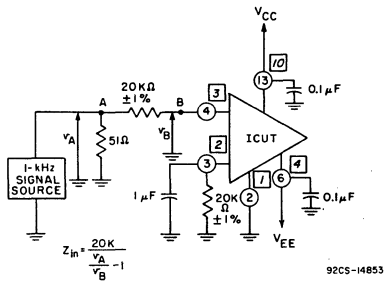


Fig.14

OUTPUT IMPEDANCE TEST CIRCUIT

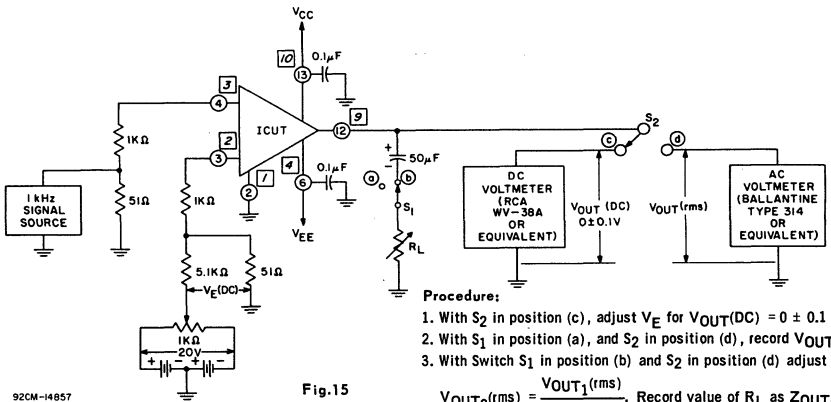


Fig.15

Procedure:

1. With S_2 in position (c), adjust V_E for $V_{OUT}(DC) = 0 \pm 0.1$ volt.
2. With S_1 in position (a), and S_2 in position (d), record $V_{OUT_1}(rms)$.
3. With Switch S_1 in position (b) and S_2 in position (d) adjust R_L until $V_{OUT_2}(rms) = \frac{V_{OUT_1}(rms)}{2}$. Record value of R_L as Z_{OUT}.

OUTPUT IMPEDANCE vs. TEMPERATURE

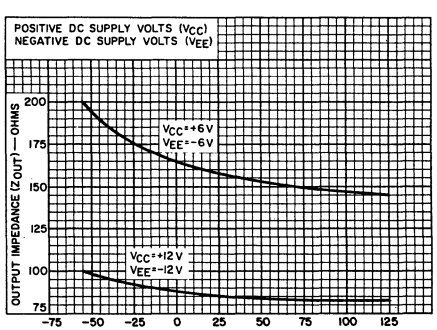


Fig.16

NOISE FIGURE vs. FREQUENCY

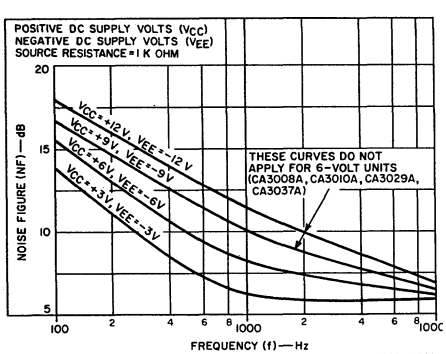


Fig.17

RCA
Solid State
Division

Linear Integrated Circuits

CA3011
CA3012

Wide-Band Amplifiers

Monolithic Silicon



FEATURES & APPLICATIONS

- exceptionally high amplifier gain: power gain at 4.5 MHz – 75 dB typ.
- excellent limiting characteristics – Input limiting voltage (knee) = 600 μV typ. at 10.7 MHz
- wide frequency capability – 100 kHz to > 20 MHz

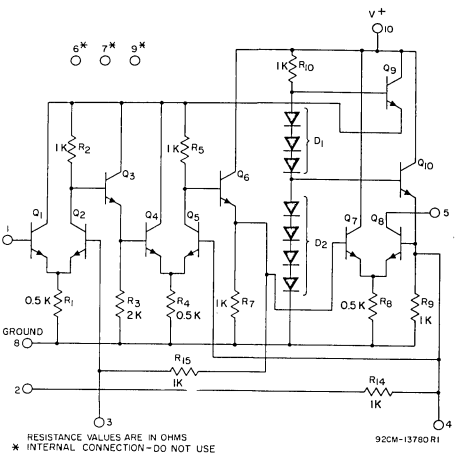


Fig. 1 SCHEMATIC DIAGRAM FOR CA3011 AND CA3012

BLOCK DIAGRAM OF TYPICAL FM RECEIVER USING RCA-CA3011 OR CA3012 INTEGRATED CIRCUIT WIDE-BAND AMPLIFIER

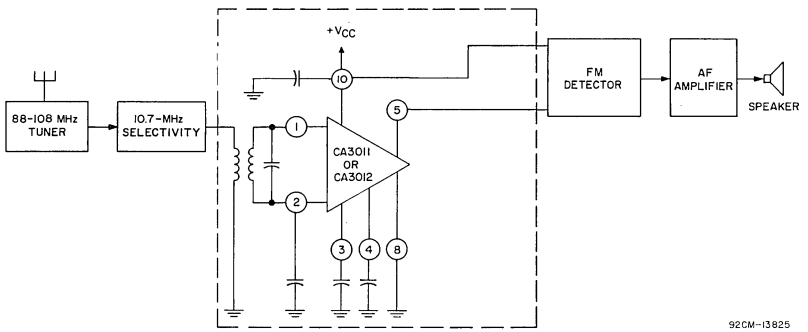


Fig. 2

ABSOLUTE-MAXIMUM VOLTAGE LIMITS AT T_A = 25° C

Indicated voltage limits for each terminal can be applied under the specified voltage conditions for other terminals. All voltages are with respect to ground (Terminal 8).

NOTE: TERMINALS 6, 7, AND 9 OF RCA-CA3011 AND CA3012 ARE USED FOR INTERNAL CONNECTIONS. DO NOT APPLY VOLTAGES OR MAKE EXTERNAL CONNECTIONS TO THESE TERMINALS.

CA3011

TERMINAL	VOLTAGE LIMITS		VOLTAGE CONDITIONS AT OTHER TERMINALS						
			1	2	3	4	5	8	10
1	-3	+3	–	Same as 1	Do Not Apply External Voltage	+2.5 to +7.5	+7.5	Ground	+7.5
2	-3	+3	Same as 2	–		+2.5 to +7.5	+7.5	Ground	+7.5
3	-3	+3	-3 to +3	Same as 1		+2.5 to +7.5	+7.5	Ground	+7.5
4	+2.5	+7.5	-3 to +3	Same as 1		–	+7.5	Ground	+7.5
5	0	+10	-3 to +3	Same as 1		+2.5 to +7.5	–	Ground	+7.5
8	-3	+7.5	-3 to +3	Same as 1		+2.5 to +7.5	+7.5	Ground	+7.5
10	0	+10	-3 to +3	Same as 1		+2.5 to +7.5	+7.5	Ground	–
CASE	INTERNALLY CONNECTED TO TERMINAL NO.8 (GROUND TERMINAL)								

CA3012

TERMINAL	VOLTAGE LIMITS		VOLTAGE CONDITIONS AT OTHER TERMINALS						
			1	2	3	4	5	8	10
1	-3	+3	–	Same as 1	Do Not Apply External Voltage	+2.5 to +10	+10	Ground	+10
2	-3	+3	Same as 2	–		+2.5 to +10	+10	Ground	+10
3	-3	+3	-3 to +3	Same as 1		+2.5 to +10	+10	Ground	+10
4	+2.5	+10	-3 to +3	Same as 1		–	+10	Ground	+10
5	0	+13	-3 to +3	Same as 1		+2.5 to +10	–	Ground	+10
8	-3	+10	-3 to +3	Same as 1		+2.5 to +10	+10	Ground	+10
10	0	+13	-3 to +3	Same as 1		+2.5 to +10	+10	Ground	–
CASE	INTERNALLY CONNECTED TO TERMINAL NO.8 (GROUND TERMINAL)								

OPERATING-TEMPERATURE RANGE ..... -55 to +125° C
STORAGE-TEMPERATURE RANGE ..... -65 to +150° C

LEAD TEMPERATURE (During Soldering):
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm)
from case for 10 seconds max. ..... +265°C

MAXIMUM INPUT-SIGNAL VOLTAGE:
Between Terminals 1 and 2 ..... ±3 V

MAXIMUM DEVICE DISSIPATION ..... 300 mW

RECOMMENDED MINIMUM DC SUPPLY VOLTAGE (V_{CC}) .. 5.5 V

Example of Use of LIMITS TABLE:

For RCA-3012, a maximum voltage of ±3 volts may be applied to Terminal 1 under the following conditions:

Terminal 2 is at the same dc potential as Terminal 1
Terminal 3: do not apply external voltage
Terminal 4 is at any dc potential between +2.5 and +10 volts
Terminal 5 is at a dc potential of +10 volts
Terminals 6, 7, and 9 are at 0 dc potential (NOT USED)
Terminal 8 is at dc ground potential
Terminal 10 is at a dc potential of +10 volts

ELECTRICAL CHARACTERISTICS

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS				LIMITS						UNITS	TYPICAL CHARACTERISTICS CURVES
		SETUP & PROCEDURE	FREQUENCY	DC SUPPLY VOLTAGE	AMBIENT TEMPERATURE	RCA CA3011			RCA CA3012				
		Fig.	f Mc/s	V_{CC} Volts	T_A °C	Min.	Typ.	Max.	Min.	Typ.	Max.		Fig.
Total Device Dissipation*	P_T	3	–	6	-55	–	80	–	66	80	135	mW	4
					+25	60	90	133	66	90	121	mW	
					+125	–	70	–	65	70	121	mW	
			–	7.5	-55	–	130	–	97	130	190	mW	4
					+25	95	120	187	97	120	167	mW	
					+125	–	100	–	95	100	167	mW	
			–	10	-55	–	–	–	150	210	275	mW	4
					+25	–	–	–	150	190	255	mW	
					+125	–	–	–	150	160	255	mW	
Voltage Gain**	A	5	1	6	-55	–	55	–	50	55	–	dB	6
					+25	60	66	–	60	66	–	dB	
					+125	–	61	–	50	61	–	dB	
		5	1	7.5	-55	–	59	–	55	59	–	dB	6
					+25	65	70	–	65	70	–	dB	
					+125	–	65	–	55	65	–	dB	
		5	1	10	-55	–	–	–	55	61	–	dB	6
					+25	–	–	–	65	71	–	dB	
					+125	–	–	–	55	66	–	dB	
		5	4.5	7.5	+25	60	67	–	60	67	–	dB	7
			10.7	7.5	+25	55	61	–	55	61	–	dB	
Input-Impedance Components: Parallel Input Resistance	R_{IN}	8	4.5	7.5	+25	–	3	–	–	3	–	kΩ	9
Parallel Input Capacitance	C_{IN}	8	4.5	7.5	+25	–	7	–	–	7	–	pF	9
Output Impedance Components: Parallel Output Resistance	R_{OUT}	10	4.5	7.5	+25	–	31.5	–	–	31.5	–	kΩ	11
Parallel Output Capacitance	C_{OUT}	10	4.5	7.5	+25	–	4.2	–	–	4.2	–	pF	11
Noise Figure	NF	12	4.5	7.5	+25	–	8.7	–	–	8.7	–	dB	13
Input Limiting Voltage (Knee)	$v_{i(lim)}$	5	4.5	7.5	+25	–	300	450	–	300	400	μV	6

* The total current drain may be determined by dividing P_T by V_{CC}.

** Recommended minimum dc supply voltage (V_{CC}) is 5.5 V. Nominal load current flowing into terminal 5 is 1.5 mA at 7.5 V.

TYPICAL CHARACTERISTICS AND TEST SETUPS

DISSIPATION TEST SETUP

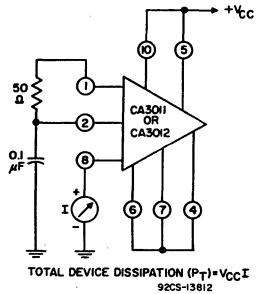


Fig.3

DISSIPATION VS TEMPERATURE

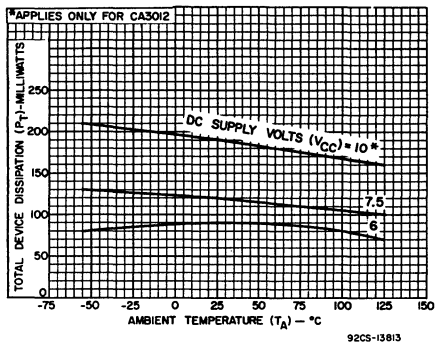
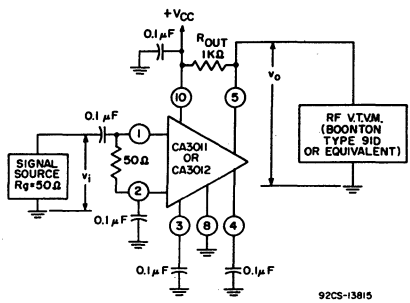


Fig.4

VOLTAGE-GAIN TEST SETUP



PROCEDURES

A - Voltage Gain:

1) Set input frequency at desired value, v_i = 100 μV rms.
2) Record v_o.
3) Calculate Voltage Gain A from $A = 20 \log_{10} v_o/v_i$
4) Repeat Steps 1, 2, and 3 for each frequency and/or for temperature desired.

B - Input Limiting Voltage (Knee):

1) Repeat Steps A1 and A2, using v_i = 100 mV
2) Decrease v_i to the level at which v_o is 3 dB below its value for v_i = 100 mV.
3) Record v_i as Input Limiting Voltage (Knee).

Fig.5

VOLTAGE GAIN & INPUT LIMITING VOLTAGE VS TEMPERATURE

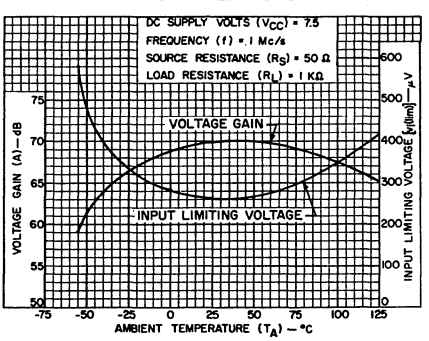


Fig.6

VOLTAGE GAIN AND INPUT LIMITING VOLTAGE VS FREQUENCY

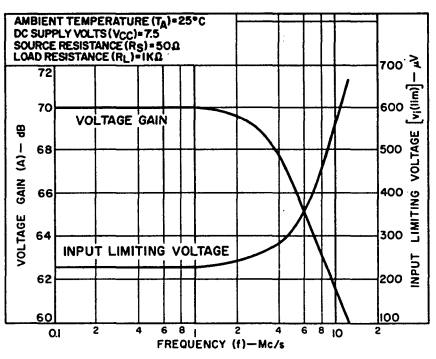


Fig.7

TYPICAL CHARACTERISTICS AND TEST SETUPS

INPUT-IMPEDANCE COMPONENTS TEST SETUP

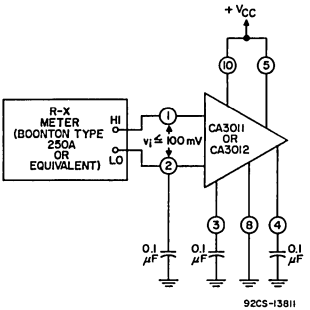


Fig.8

INPUT-IMPEDANCE COMPONENTS VS FREQUENCY

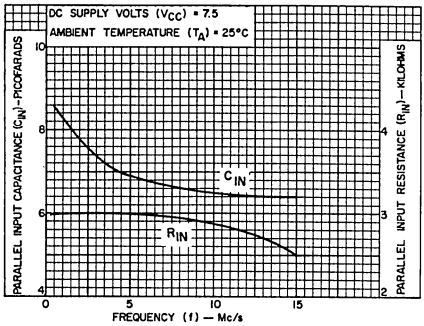


Fig.9

OUTPUT-IMPEDANCE COMPONENTS TEST SETUP

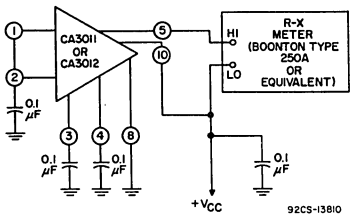


Fig.10

OUTPUT-IMPEDANCE COMPONENTS VS FREQUENCY

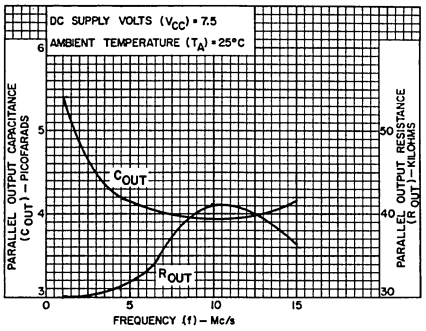
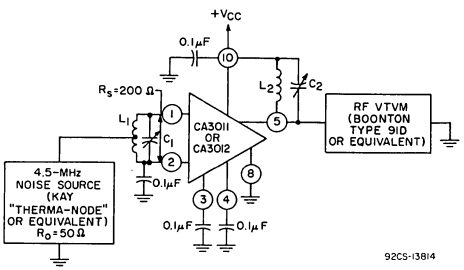


Fig.11

NOISE FIGURE TEST SETUP



L_1 = 82 μH, center-tapped

L_2 = 2.36 μH

C_1,C_2 = Arco Type 423 padder, or equivalent

Fig.12

NOISE FIGURE VS DC SUPPLY VOLTAGE

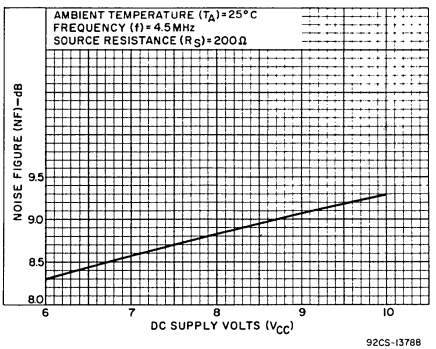


Fig.13

RCA
Solid State
Division

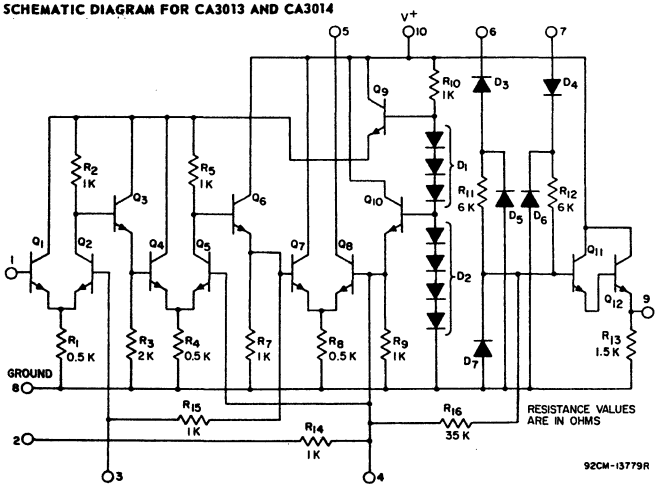
Linear Integrated Circuits

CA3013
CA3014

Wide-Band Amplifier-Discriminators

Monolithic Silicon

SCHEMATIC DIAGRAM FOR CA3013 AND CA3014

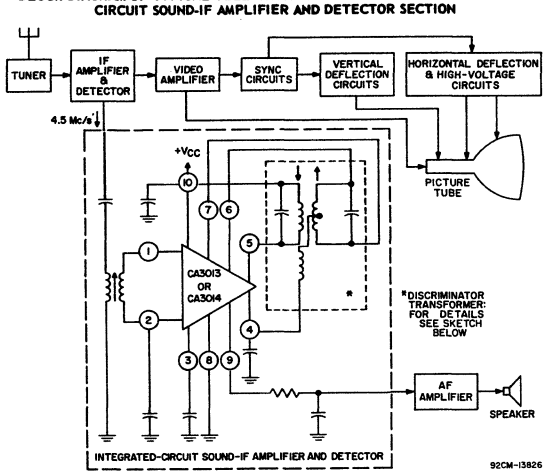




FEATURES & APPLICATIONS:

- exceptionally high gain:
  power gain at 4.5 MHz — 75 dB typ.
- excellent limiting characteristics — input limiting voltage (knee) = 300 μV typ. at 4.5 MHz
- excellent AM rejection: > 50 dB at 4.5 MHz
- high audio-voltage recovery — 220 mV typ. at 4.5 MHz 25 kHz deviation
- wide frequency capability — 100 kHz to > 20 MHz
- comprehensive circuit functions:
  if amplifier, AM and noise limiter, FM detector, audio preamplifier

BLOCK DIAGRAM OF TYPICAL TELEVISION RECEIVER USING RCA INTEGRATED-CIRCUIT SOUND-IF AMPLIFIER AND DETECTOR SECTION



ABSOLUTE-MAXIMUM VOLTAGE LIMITS AT T_A = 25° C

Indicated voltage limits for each terminal can be applied under the specified voltage conditions for other terminals. All voltages are with respect to ground (Terminal 8).

CA3013

TERMINAL	VOLTAGE LIMITS		VOLTAGE CONDITIONS AT OTHER TERMINALS									
			1	2	3	4	5	6	7	8	9	10
1	-3	+3	-	Same as 1	Do Not Apply External Voltage	+2.5 to +7.5	+7.5	Same as 4	Same as 4	Ground	AF Output	+7.5
2	-3	+3	Same as 2	-		+2.5 to +7.5	+7.5	Same as 4	Same as 4	Ground	AF Output	+7.5
3	-3	+3	-3 to +3	Same as 1		+2.5 to +7.5	+7.5	Same as 4	Same as 4	Ground	AF Output	+7.5
4	+2.5	+7.5	-3 to +3	Same as 1		-	+7.5	Same as 4	Same as 4	Ground	AF Output	+7.5
5	0	+10	-3 to +3	Same as 1		+2.5 to +7.5	-	Same as 4	Same as 4	Ground	AF Output	+7.5
6	+2.5	+7.5	-3 to +3	Same as 1		Same as 6	+7.5	-	Same as 4	Ground	AF Output	+7.5
7	+2.5	+7.5	-3 to +3	Same as 1		+2.5 to +7.5	+7.5	Same as 4	-	Ground	AF Output	+7.5
8	-3	+7.5	-3 to +3	Same as 1		+2.5 to +7.5	+7.5	Same as 4	Same as 4	Ground	AF Output	+7.5
9	0	+7.5	-3 to +3	Same as 1		+2.5 to +7.5	+7.5	Same as 4	Same as 4	Ground	-	+7.5
10	0	+10	-3 to +3	Same as 1		+2.5 to +7.5	+7.5	Same as 4	Same as 4	Ground	AF Output	-
CASE	INTERNALLY CONNECTED TO TERMINAL No.8 (GROUND TERMINAL)											

CA3014

TERMINAL	VOLTAGE LIMITS		VOLTAGE CONDITIONS AT OTHER TERMINALS									
			1	2	3	4	5	6	7	8	9	10
1	-3	+3	-	Same as 1	Do Not Apply External Voltage	+2.5 to +10	+10	Same as 4	Same as 4	Ground	AF Output	+10
2	-3	+3	Same as 2	-		+2.5 to +10	+10	Same as 4	Same as 4	Ground	AF Output	+10
3	-3	+3	-3 to +3	Same as 1		+2.5 to +10	+10	Same as 4	Same as 4	Ground	AF Output	+10
4	+2.5	+10	-3 to +3	Same as 1		-	+10	Same as 4	Same as 4	Ground	AF Output	+10
5	0	+13	-3 to +3	Same as 1		+2.5 to +10	-	Same as 4	Same as 4	Ground	AF Output	+10
6	+2.5	+10	-3 to +3	Same as 1		Same as 6	+10	-	Same as 4	Ground	AF Output	+10
7	+2.5	+10	-3 to +3	Same as 1		+2.5 to +10	+10	Same as 4	-	Ground	AF Output	+10
8	-3	+10	-3 to +3	Same as 1		+2.5 to +10	+10	Same as 4	Same as 4	Ground	AF Output	+10
9	0	+10	-3 to +3	Same as 1		+2.5 to +10	+10	Same as 4	Same as 4	Ground	-	+10
10	0	+13	-3 to +3	Same as 1		+2.5 to +10	+10	Same as 4	Same as 4	Ground	AF Output	-
CASE	INTERNALLY CONNECTED TO TERMINAL No.8 (GROUND TERMINAL)											

OPERATING-TEMPERATURE RANGE ..... -55 to +125°C

STORAGE-TEMPERATURE RANGE ..... -65 to +150°C

LEAD TEMPERATURE (During Soldering):
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm)
from case for 10 seconds max ..... +265°C

MAXIMUM INPUT-SIGNAL VOLTAGE:
Between Terminals 1 and 2 ..... ± 3 V

MAXIMUM DEVICE DISSIPATION ..... 300 mW

RECOMMENDED MINIMUM DC
SUPPLY VOLTAGE (V_{CC}) ..... 5.5 V

Example of use of LIMITS TABLE:

For RCA-CA3013, a maximum voltage of ±3 volts may be applied to Terminal 1 under the following conditions:

Terminal 2 is at the same dc potential as Terminal 1
Terminal 3: do not apply external voltage
Terminal 4 is at any dc potential between +2.5 and +7.5 volts
Terminal 5 is at a dc potential of +7.5 volts
Terminals 6 and 7 are at the same dc potential as Terminal 4
Terminal 8 is at dc ground potential
Terminal 9 is used as the af output terminal
Terminal 10 is at a dc potential of +7.5 volts

ELECTRICAL CHARACTERISTICS (See Page 8 for Definitions of Terms)	SYMBOLS	TEST CONDITIONS				LIMITS						UNITS	TYPICAL CHARACTERISTICS CURVES
		SETUP & PROCEDURE	FREQUENCY f	DC SUPPLY VOLTAGE V_{CC}	AMBIENT TEMPERATURE T_A	RCA CA3013			RCA CA3014				
		Fig.	Mc/s	volts	°C	Min.	Typ.	Max.	Min.	Typ.	Max.		Fig.
Total Device Dissipation*	P_T	3	–	6	-55	–	80	–	73	80	120	mW	4
					+25	60	90	133	73	90	110	mW	
					+125	–	70	–	60	70	110	mW	
		3	–	7.5	-55	–	130	–	106	130	170	mW	4
					+25	87	120	187	106	120	150	mW	
					+125	–	100	–	90	100	150	mW	
		3	–	10	-55	–	–	–	165	210	250	mW	4
					+25	–	–	–	165	190	230	mW	
					+125	–	–	–	150	160	230	mW	
Voltage Gain**	A	5	1	6	-55	–	55	–	50	55	–	dB	6
					+25	60	66	–	60	66	–	dB	
					+125	–	61	–	50	61	–	dB	
		5	1	7.5	-55	–	59	–	55	59	–	dB	6
					+25	65	70	–	65	70	–	dB	
					+125	–	65	–	55	65	–	dB	
		5	1	10	-55	–	–	–	55	61	–	dB	6
					+25	–	–	–	65	71	–	dB	
					+125	–	–	–	55	66	–	dB	
		5	4.5	7.5	+25	60	67	–	60	67	–	dB	7
			10.7	7.5	+25	55	60	–	55	60	–	dB	
Input-Impedance Components: Parallel Input Resistance	R_{IN}	8	4.5	7.5	+25	–	3	–	–	3	–	kΩ	9
Parallel Input Capacitance	C_{IN}	8	4.5	7.5	+25	–	7	–	–	7	–	pF	9
Output-Impedance Components: Parallel Output Resistance	R_{OUT}	10	4.5	7.5	+25	–	31.5	–	–	31.5	–	kΩ	11
Parallel Output Capacitance	C_{OUT}	10	4.5	7.5	+25	–	4.2	–	–	4.2	–	pF	11
Noise Figure	NF	12	4.5	7.5	+25	–	8.7	–	–	8.7	–	dB	13
Input Limiting Voltage (Knee)	v_i(lim)	14	4.5	7.5	+25	–	300	450	–	300	400	μV	15
Recovered AF Voltage	v_o(af)	14	4.5	6	+25	–	155	–	–	155	–	mV	15
				7.5	+25	128	188	–	135	188	–	mV	
				10	+25	–	–	–	–	220	–	mV	
Amplitude-Modulation Rejection	AMR	16	4.5	7.5	+25	–	50	–	–	50	–	dB	–
Discriminator Output Resistance	R_O(disc)	–	4.5	7.5	+25	–	60	–	–	60	–	Ω	–
Total Harmonic Distortion	THD	14	4.5	7.5	+25	–	1.8	–	–	1.8	–	%	17

* Total current drain may be determined by dividing P_T by V_{CC}.

** Recommended minimum dc supply voltage (V_{CC}) is 5.5 V. Nominal load current flowing into terminal 5 is 1.5 mA at 7.5 V.

TYPICAL CHARACTERISTICS AND TEST SETUPS

DISSIPATION TEST SETUP

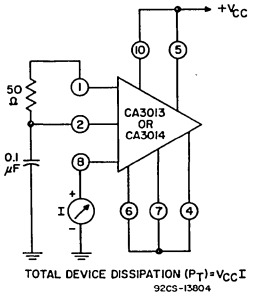


Fig.3

DISSIPATION vs. TEMPERATURE

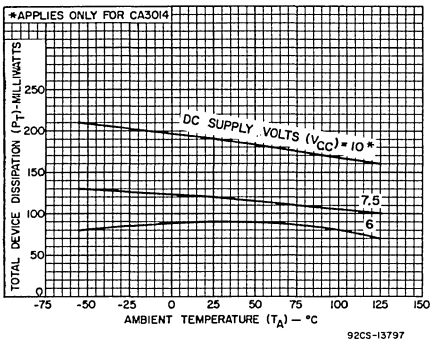
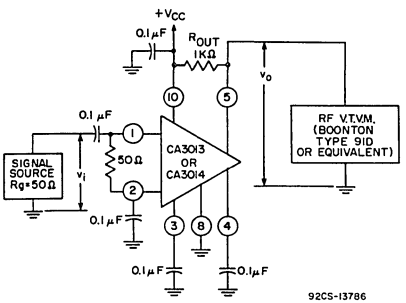


Fig.4

VOLTAGE-GAIN TEST SETUP



PROCEDURE:

1) Set input frequency at desired value, v_i = 100 μV rms.
2) Record v_o.
3) Calculate Voltage Gain A from A = 20 $\log_{10} v_o/v_i$.
4) Repeat Steps 1, 2, and 3 for each frequency and/or temperature desired.

Fig.5

1-Mc/s VOLTAGE GAIN vs. TEMPERATURE

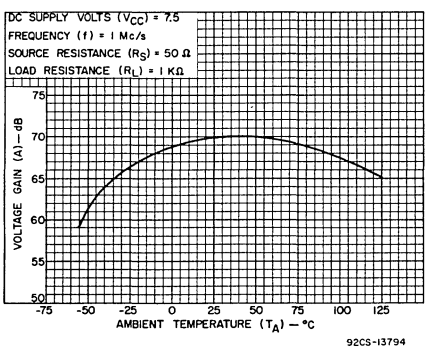


Fig.6

VOLTAGE GAIN vs. FREQUENCY

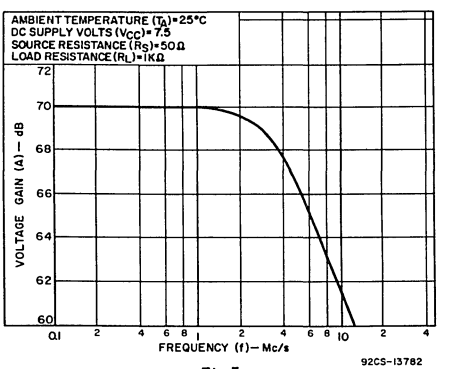


Fig.7

TYPICAL CHARACTERISTICS AND TEST SETUPS

INPUT-IMPEDANCE COMPONENTS TEST SETUP

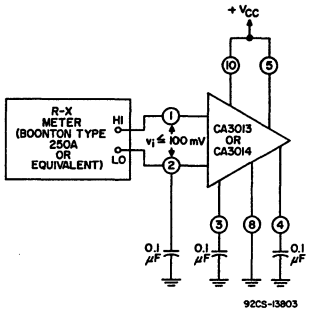


Fig.8

INPUT-IMPEDANCE COMPONENTS vs. FREQUENCY

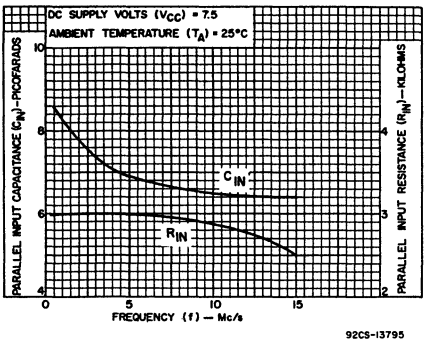


Fig.9

OUTPUT-IMPEDANCE COMPONENTS TEST SETUP

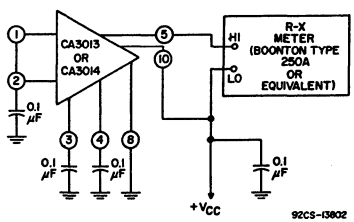


Fig.10

OUTPUT-IMPEDANCE COMPONENTS vs. FREQUENCY

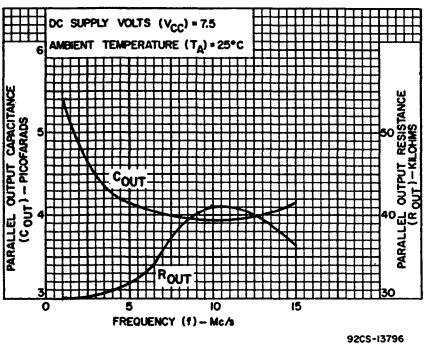
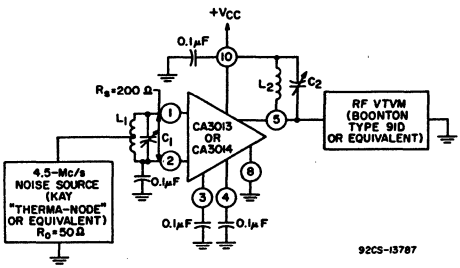


Fig.11

NOISE FIGURE TEST SETUP



L_1 = 82 μH, center-tapped

L_2 = 2.36 μH

C_1, C_2 = Arco Type 423 padder, or equivalent

Fig.12

NOISE FIGURE vs. DC SUPPLY VOLTAGE

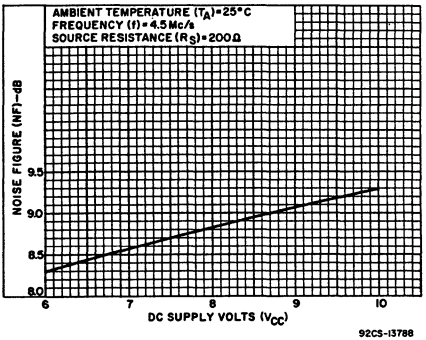
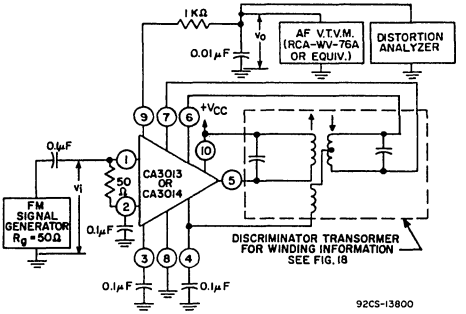


Fig.13

TYPICAL CHARACTERISTICS AND TEST SETUPS

INPUT LIMITING VOLTAGE, RECOVERED AF VOLTAGE, AND TOTAL HARMONIC DISTORTION TEST SETUP



PROCEDURE:

A - Recovered-AF Voltage Output:

1) Set input frequency = 4.5 Mc/s, v_i = 100 mV rms, modulating frequency = 1 kc/s, frequency deviation = ±25 kc/s.
2) Record v_o as Recovered-AF Voltage Output.

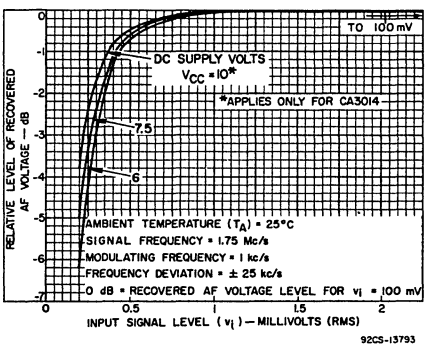
B - Input Limiting Voltage (Knee):

1) Repeat Steps A1 and A2, using v_i = 100 mV rms.
2) Decrease v_i to the level at which v_o is 3 dB below its value for v_i = 100 mV.
3) Record v_i as Input Limiting Voltage (Knee).

Fig. 14

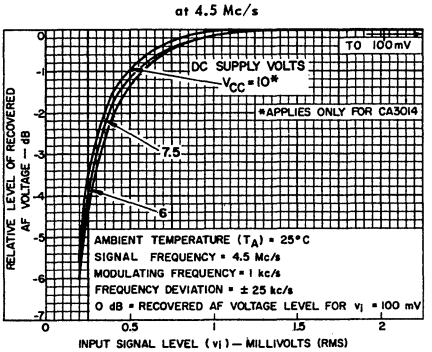
INPUT LIMITING VOLTAGE (KNEE) AND RECOVERED AF VOLTAGE

at 1.75 Mc/s



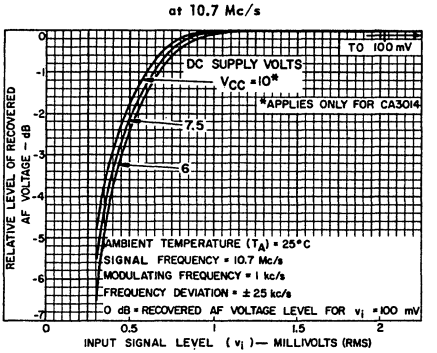
(a)

at 4.5 Mc/s



(b)

at 10.7 Mc/s



(c)

Fig. 15

TYPICAL CHARACTERISTICS AND TEST SETUPS

AM-REJECTION TEST SETUP

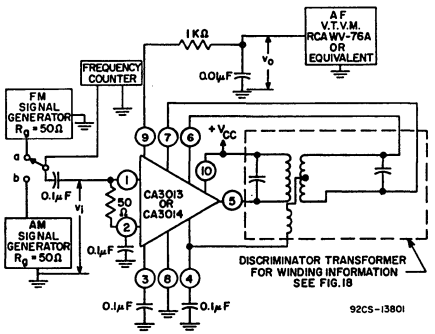


Fig. 16

PROCEDURE:

1) With Switch S in position "a", set input frequency = 4.5 Mc/s, v_i = 10 mV rms, modulating frequency = 1 kc/s, frequency deviation = ±25 kc/s.
2) Record v_o.
3) Place Switch S in position "b", and set input frequency = 4.5 Mc/s, v_i = 10 mV rms, modulating frequency = 1 kc/s, % modulation = 50.
4) Measure v_o, and record value in dB below value in Step 2 as AM Rejection.

TOTAL HARMONIC DISTORTION vs. DC SUPPLY VOLTAGE

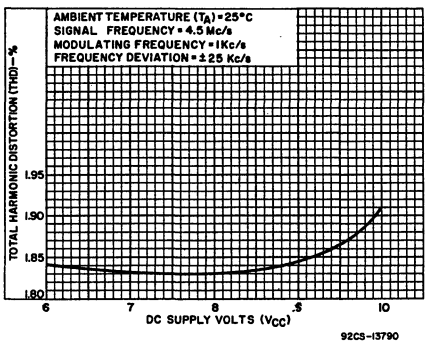
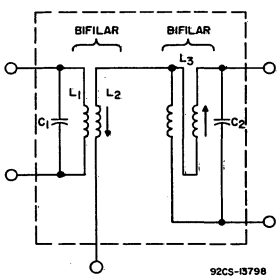


Fig. 17

DISCRIMINATOR TRANSFORMER SCHEMATIC



(a)

Fig. 18

CONSTRUCTION DETAILS OF DISCRIMINATOR TRANSFORMERS SHOWN IN FIGS. 2, 14 AND 16

Coil-Form Outside Diameter = 7/32 inch

Slugs: Radio Industries, Inc. Type "E" Material, or equivalent

Wire Type: "GRIPEZE"*, or equivalent

Operating Frequency Mc/s	Wire Size (AWG #)	Turns L_1▲	Turns L_2▲	Turns L_3	C_1 pF	C_2 pF
1.75	40	44	20	44 total (22 bifilar wound)	820	820
4.5	36	18	7	22 total (11 bifilar wound)	560	330
10.7	36	18	18	18 total (9 bifilar wound)	100	100

* Registered Trade Mark, Phelps-Dodge Copper Products.

▲ wound bifilar.

NOTE: The mutual coupling between L_1 and L_3 is adjusted for the desired degree of linearity.

(b)

RCA
Solid State
Division

Linear Integrated Circuits
CA3018
CA3018A

General-Purpose Transistor Arrays

Monolithic Silicon

The CA3018 and CA3018A consist of four general purpose silicon n-p-n transistors on a common monolithic substrate.

Two of the four transistors are connected in the Darlington configuration. The substrate is connected to a separate terminal for maximum flexibility.

The transistors of the CA3018 and the CA3018A are well suited to a wide variety of applications in low-power systems in the DC through VHF range. They may be used as discrete transistors in conventional circuits but in addition they provide the advantages of close electrical and thermal matching inherent in integrated circuit construction.

The CA3018A is similar to the CA3018 but features tighter control of current gain, leakage, and offset parameters making it suitable for more critical applications requiring premium performance.

APPLICATIONS

- General use in signal processing systems in DC through VHF range
- Custom designed differential amplifiers
- Temperature compensated amplifiers
- See RCA Application Note, ICAN-5296 "Application of the RCA CA3018 Integrated-Circuit Transistor Array" for suggested Applications.

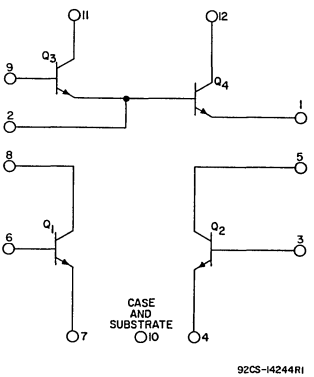


Fig. 1 - Schematic Diagram for CA3018 and CA3018A

TWO ISOLATED TRANSISTORS AND A DARLINGTON-CONNECTED TRANSISTOR PAIR

For Low-Power Applications at Frequencies from DC Through the VHF Range



12-Lead TO-5 Style

FEATURES

- Matched monolithic general purpose transistors
- H_{FE} matched ± 10%
- V_{BE} matched ± 2 mV CA3018A (± 5mV CA3018)
- Operation from DC to 120 MHz
- Wide operating current range
- CA3018A performance characteristics controlled from 10μ A to 10mA
- Low noise figure - - 3.2 dB typical at 1KHz
- Full military temperature range capability (-55 to + 125°C)

Maximum Ratings, Absolute-Maximum Values, at TA=25°C

	CA3018	CA3018A	
Power Dissipation, P:			
Any one transistor	300	300	mW
Total package	450	450	mW
Derate at 5 mW/°C for $T_A > 85°C$			
Temperature Range:			
Operating	-55 to + 125	-55 to + 125	°C
Storage.	-65 to + 150	-65 to + 150	°C
LEAD TEMPERATURE (During Soldering)			
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm)			
from case for 10 seconds max.	+265		°C

The following ratings apply for each transistor in the device:

	CA3018	CA3018A	
Collector-to-Emitter Voltage, V_{CEO} ·	15	15	V
Collector-to-Base Voltage, V_{CBO} · ·	20	30	V
Collector-to-Substrate Voltage, V_{CIO}*	20	40	V
Emitter-to-Base Voltage, V_{EBO} · · ·	5	5	V
Collector Current, I_C	50	50	mA

*The collector of each transistor of the CA3018 and CA3018A is isolated from the substrate by an integral diode. *The substrate (terminal 10) must be connected to the most negative point in the external circuit to maintain isolation between transistors and to provide for normal transistor action.*

Characteristics apply for each transistor in the CA3018 and CA3018A as specified.

ELECTRICAL CHARACTERISTICS at T_A = 25°C	SYMBOLS	SPECIAL TEST CONDITIONS	CA3018 LIMITS			CA3018A LIMITS			Units	CHARACTERISTICS CURVES
			Min.	Typ.	Max.	Min.	Typ.	Max.		Fig.
STATIC CHARACTERISTICS										
Collector-Cutoff Current	I_{CBO}	V_{CB}=10V, I_E=0	–	0.002	100	–	0.002	40	nA	2
Collector-Cutoff Current	I_{CEO}	V_{CE}=10V, I_B=0	–	See Curve	5	–	See Curve	0.5	μA	3
Collector-Cutoff Current Darlington Pair	I_{CEOD}	V_{CE}=10V, I_B=0	–	–	–	–	–	5	μA	–
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C=1mA, I_B=0	15	24	–	15	24	–	V	–
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C=10μA, I_E=0	20	60	–	30	60	–	V	–
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E=10μA, I_C=0	5	7	–	5	7	–	V	–
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_C=10μA, I_{CI}=0	20	60	–	40	60	–	V	–
Collector-to-Emitter Saturation Voltage	V_{CES}	I_B=1mA, I_C=10mA	–	0.23	–	–	0.23	0.5	V	–
Static Forward Current Transfer Ratio	h_{FE}	V_{CE}=3V, I_C=10mA I_C= 1mA I_C=10μA	– 30 –	100 100 54	– 200 –	50 60 30	100 100 54	– 200 –	– – –	4
Magnitude of Static-Beta Ratio (Isolated Transistors Q_1 and Q_2)		V_{CE}=3V, I_{C1}=I_{C2}=1mA	0.9	0.97	–	0.9	0.97	–	–	4
Static Forward Current Transfer Ratio Darlington Pair (Q_3 & Q_4)	h_{FED}	V_{CE}=3V I_C= 1mA I_C=100μA	1500 –	5400 –	– –	2000 1000	5400 2800	– –	–	5
Base-to-Emitter Voltage	V_{BE}	V_{CE}=3V I_E=1mA I_E=10mA	– –	0.715 0.800	– –	0.600 –	0.715 0.800	0.800 0.900	V	6
Input Offset Voltage	$\lvert V_{BE_1} - V_{BE_2} \rvert$	V_{CE}=3V, I_E=1mA	–	0.48	5	–	0.48	2	mV	6,8
Temperature Coefficient: Base-to-Emitter Voltage Q_1,Q_2	$\frac{\lvert \Delta V_{BE} \rvert}{\Delta T}$	V_{CE}=3V, I_E=1mA	–	-1.9	–	–	-1.9	–	$mV/^oC$	7
Base (Q_3)-to-Emitter (Q_4) Voltage-Darlington Pair	V_{BED} (V_{9-1})	V_{CE}=3V I_E=10mA I_E= 1mA	– –	1.46 1.32	– –	– 1.10	1.46 1.32	1.60 1.50	V	9
Temperature Coefficient: Base-to-Emitter Voltage Darlington Pair-Q_3,Q_4	$\frac{\lvert \Delta V_{BED} \rvert}{\Delta T}$	V_{CE}=3V, I_E=1mA	–	4.4	–	–	4.4	–	$mV/^oC$	10
Temperature Coefficient: Magnitude of Input-Offset Voltage	$\frac{\lvert V_{BE_1} - V_{BE_2} \rvert}{\Delta T}$	V_{CC}=+6V, V_{EE}=-6V, I_{C1}=I_{C2}=1mA	–	10	–	–	10	–	$\mu V/^oC$	–

ELECTRICAL CHARACTERISTICS, (CONT'D)

DYNAMIC CHARACTERISTICS			CA3018			CA3018A				
Low Frequency Noise Figure	NF	f=1 KHz, V_{CE}=3V, I_C=100μA Source resistance=1 KΩ	–	3.25	–	–	3.25	–	dB	11(b)
Low-Frequency, Small-Signal Equivalent-Circuit Characteristics:										
Forward Current-Transfer Ratio	h_{fe}	f=1kHz, V_{CE}=3V, I_C=1mA	–	110	–	–	110	–	–	12
Short-Circuit Input Impedance	h_{ie}		–	3.5	–	–	3.5	–	KΩ	12
Open-Circuit Output Impedance	h_{oe}		–	15.6	–	–	15.6	–	μmho	12
Open-Circuit Reverse Voltage-Transfer Ratio	h_{re}		–	$1.8x10^{-4}$	–	–	$1.8x10^{-4}$	–	–	12
Admittance Characteristics:										
Forward Transfer Admittance	Y_{fe}	f=1MHz, V_{CE}=3V, I_C=1mA	–	31-j1.5	–	–	31-j1.5	–	mmho	13
Input Admittance	Y_{ie}		–	0.3+j0.04	–	–	0.3+j0.04	–	mmho	14
Output Admittance	Y_{oe}		–	0.001+j0.03	–	–	0.001+j0.03	–	mmho	15
Reverse Transfer Admittance	Y_{re}		See Curve			See Curve			mmho	16
Gain-Bandwidth Product	f_T	V_{CE}=3V, I_C=3mA	300	500	–	300	500	–	MHz	17
Emitter-to-Base Capacitance	C_{EB}	V_{EB}=3V, I_E=0	–	0.6	–	–	0.6	–	pF	–
Collector-to-Base Capacitance	C_{CB}	V_{CB}=3V, I_C=0	–	0.58	–	–	0.58	–	pF	–
Collector-to-Substrate Capacitance	C_{CI}	V_{CI}=3V, I_C=0	–	2.8	–	–	2.8	–	pF	–

STATIC CHARACTERISTICS

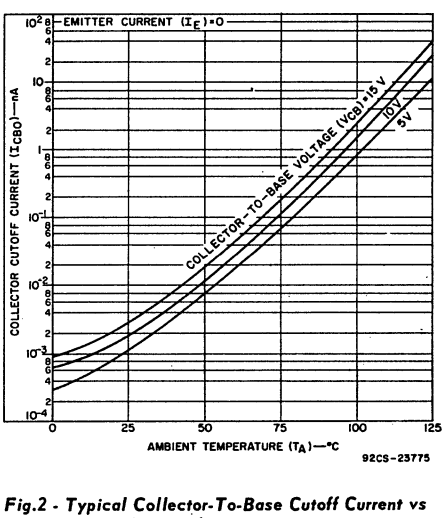


Fig.2 - Typical Collector-To-Base Cutoff Current vs Ambient Temperature for Each Transistor.

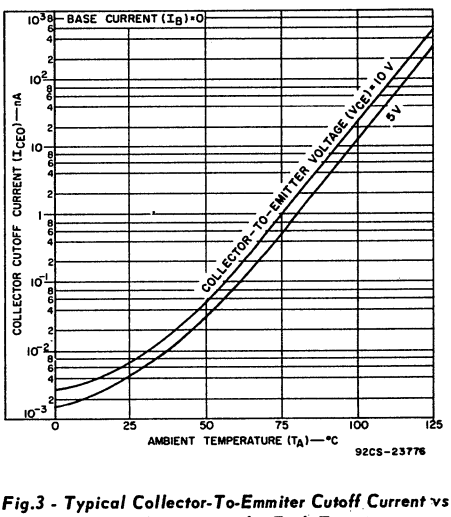


Fig.3 - Typical Collector-To-Emmiter Cutoff Current vs Ambient Temperature for Each Transistor.

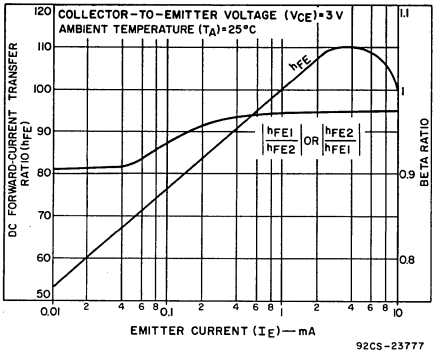


Fig.4 - Typical Static Forward Current-Transfer Ratio and Beta Ratio for Transistors Q, and Q_2 vs Emitter Current.

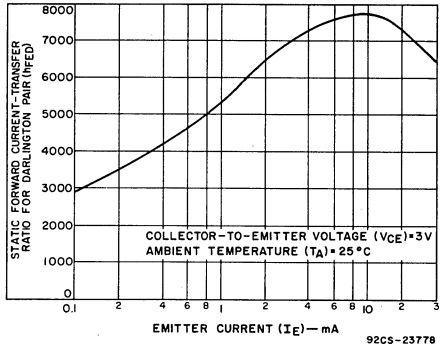


Fig.5 - Typical Static Forward Current - Transfer Ratio for Darlington-connected Transisters Q_3 and Q_4 vs Emitter Current.

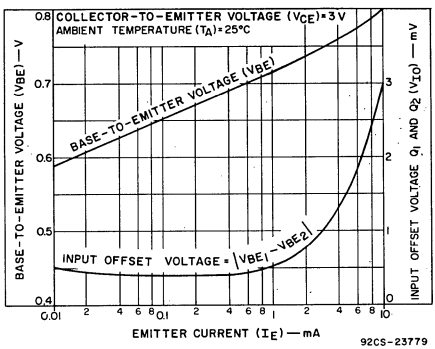


Fig.6 - Typical Static Base-to-Emitter Voltage Characteristic and Input Offset Voltage for Q_1 and Q_2 vs Emitter Current.

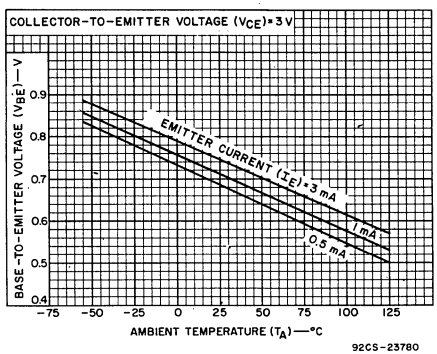


Fig.7 - Typical Base-To-Emitter Voltage Characteristic for Each Transistor vs Ambient Temperature

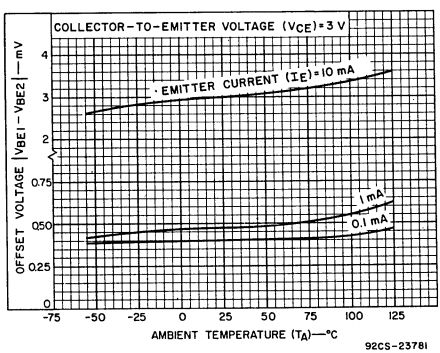


Fig.8 - Typical Offset Voltage Characteristic vs Ambient Temperature

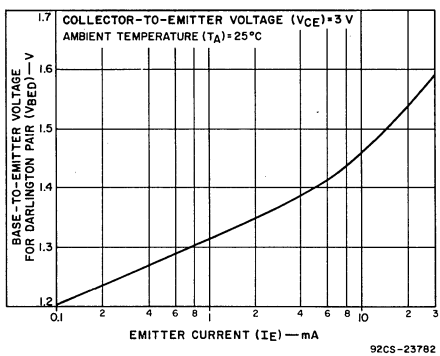


Fig.9 - Typical Static Input Voltage Characteristic for Darlington Pair (Q_3 and Q_4) vs Emitter Current

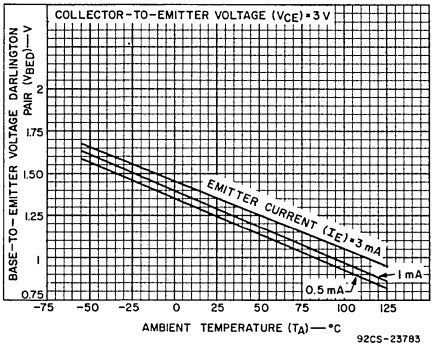


Fig.10 - Typical Static Input Voltage Characteristic for Darlington Pair (Q_3 and Q_4) vs Ambient Temperature.

TYPICAL DYNAMIC CHARACTERISTICS FOR EACH TRANSISTOR

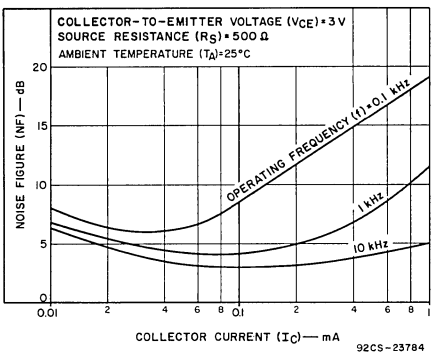


Fig.11(a) - Noise Figure vs Collector Current, R_S = 500 Ω.

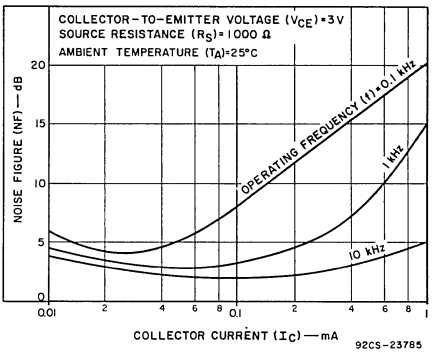


Fig.11(b) - Noise Figure vs Collector Current, R_S = 1 KΩ.

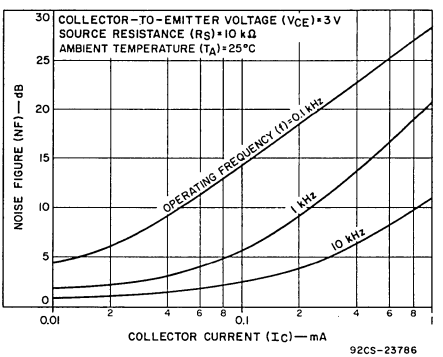


Fig.11(c) - Noise Figure vs Collector Current, R_S = 10 KΩ.

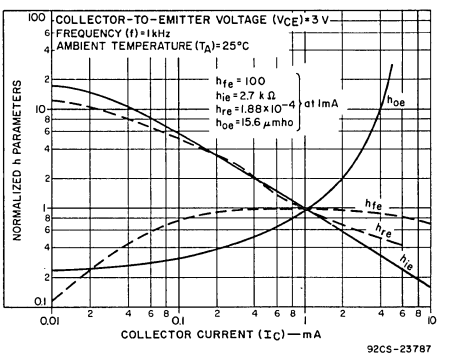


Fig.12 - Forward Current-Transfer Ratio (h_{fe}), Short-Circuit Input Impedance (h_{ie}), Open-Circuit Output Impedance (h_{oe}), and Open-Circuit Reverse Voltage-Transfer Ratio (h_{re}) vs Collector Current

TYPICAL DYNAMIC CHARACTERISTICS FOR EACH TRANSISTOR

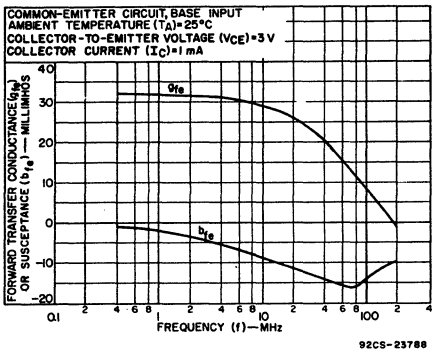


Fig. 13 - Forward Transfer Admittance (Y_{fe})

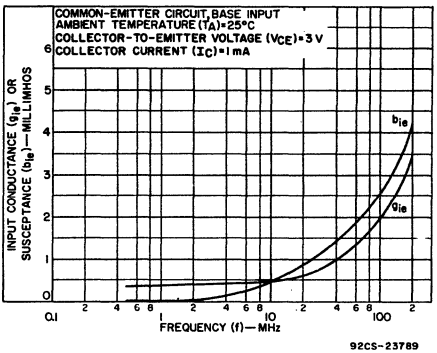


Fig. 14 - Input Admittance (Y_{ie})

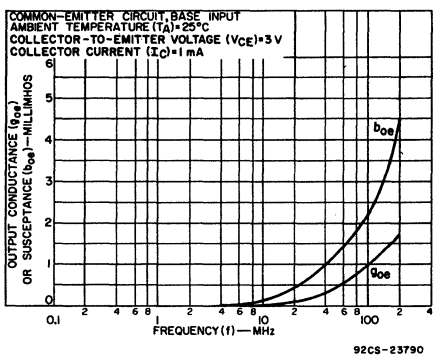


Fig. 15 - Output Admittance (Y_{oe})

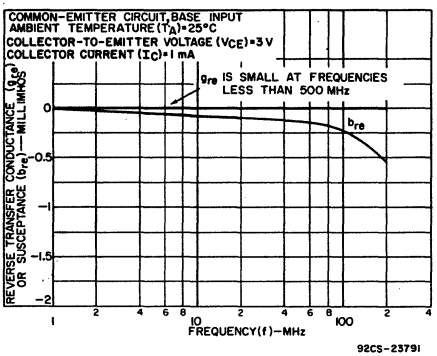


Fig. 16 - Reverse Transfer Admittance (Y_{re})

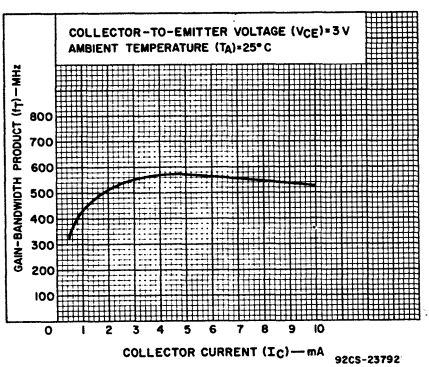


Fig. 17 - Typical Gain-Bandwidth Product (f_T) vs Collector Current

RCA
Solid State
Division

Linear Integrated Circuits

CA3019

DIODE ARRAY

Monolithic Silicon

The CA3019 consists of one Diode "Quad" and two Isolated Diodes on a Common Substrate.



10-Pin TO-5

- **Designed for use in Telemetry, Data-Processing, Instrumentation, and Communication Equipment**
- **Built-in Temperature Stability for Operation from -55°C to +125°C**
- **10-Terminal TO-5 Package**
- **Hermetically Sealed**
- **Companion Application Note, ICAN-5299 "Application of the RCA CA3019 Integrated-Circuit Diode Array"**

HIGHLIGHTS

- **Excellent Diode Match**
- **Low Leakage Current**
- **Low Pedestal Voltage when Gating**

APPLICATIONS

- **Modulator**
- **Mixer**
- **Balanced Modulator**
- **Analog Switch**
- **Diode Gate for Chopper-Modulator Applications**

ABSOLUTE-MAXIMUM RATINGS:

DISSIPATION:
Any one diode unit . . . . . 20 max. mW
Total for device . . . . . 120 max. mW

TEMPERATURE RANGE:
Storage . . . . . -65 to +150 °C
Operating . . . . . -55 to +125 °C

LEAD TEMPERATURE (During Soldering)
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm)
from case for 10 seconds max. . . . . +265°C

VOLTAGE: See Table

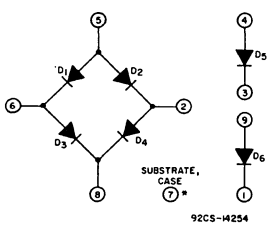


Fig. 1 - Schematic Diagram for CA3019.

Absolute-Maximum Voltage Limits at $T_A = 25°C$

TERMINAL	VOLTAGE LIMITS		CONDITIONS	
	NEGATIVE	POSITIVE	TERMINAL	VOLTAGE
1	-3	+12	7	-6
2	-3	+12	7	-6
3	-3	+12	7	-6
4	-3	+12	7	-6
5	-3	+12	7	-6
6	-3	+12	7	-6
7	-18	0	1, 2, 3, 6, 8	0
8	-3	+12	7	-6
9	-3	+12	7	-6
10	NO CONNECTION			
CASE	INTERNALLY CONNECTED TO TERMINAL 7 **DO NOT GROUND**			

ELECTRICAL CHARACTERISTICS, at an Ambient Temperature, T_A, of 25°C
CHARACTERISTICS APPLY FOR EACH DIODE UNIT, UNLESS OTHERWISE SPECIFIED.

CHARACTERISTICS	SYMBOLS	TEST CIRCUITS	SPECIAL TEST CONDITIONS	LIMITS TYPE CA3019				TYPICAL CHARACTERISTICS CURVES
		Fig.		Min.	Typ.	Max.	Units	Fig.
DC Forward Voltage Drop	V_F	-	DC Forward Current (I_F) = 1 mA	-	0.73	0.78	V	2
DC Reverse Breakdown Voltage	$V_{(BR)R}$	-	DC Reverse Current (I_R) = -10 μA	4	6	-	V	-
DC Reverse Breakdown Voltage Between any Diode Unit and Substrate	$V_{(BR)R}$	-	DC Reverse Current (I_R) = -10 μA	25	80	-	V	-
DC Reverse (Leakage) Current	I_R	-	DC Reverse Voltage (V_R) = -4 V	-	0.0055	10	μA	3
DC Reverse (Leakage) Current Between any Diode Unit and Substrate	I_R	-	DC Reverse Voltage (V_R) = -4 V	-	0.010	10	μA	-
Magnitude of Diode Offset Voltage (Difference in DC Forward Voltage Drops of any Two Diode Units)	$\lvert V_{F1} - V_{F2} \rvert$	-	DC Forward Current (I_F) = 1 mA	-	1	5	mV	-
Single Diode Capacitance	C_D	-	Frequency (f) = 1 MHz DC Reverse Voltage (V_R) = -2 V	-	1.8	-	pF	4
Diode Quad-to-Substrate Capacitance	C_{DQ-I}		Frequency (f) = 1 MHz DC Reverse Voltage (V_R) between Terminal 2,5,6, or 8 of Diode Quad and Terminal 7 (Substrate) = -2 V					
			Terminal 2 or 6 to Terminal 7	-	4.4	-	pF	5
			Terminal 5 or 8 to Terminal 7	-	2.7	-	pF	6
Series Gate Switching Pedestal Voltage	V_S	7		-	10	-	mV	-

TYPICAL CHARACTERISTICS

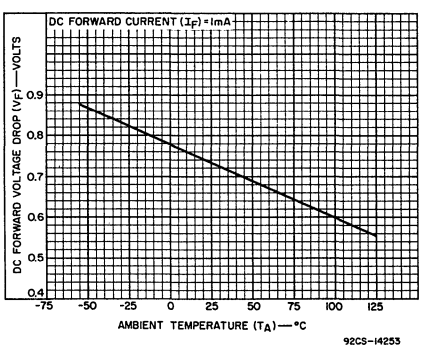


Fig.2 - DC Forward Voltage Drop (any Diode) vs Temperature for CA3019.

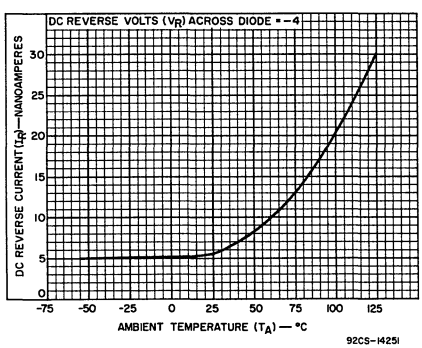


Fig.3 - Reverse (Leakage) Current (any Diode) vs Temperature for CA3019.

TYPICAL CHARACTERISTICS

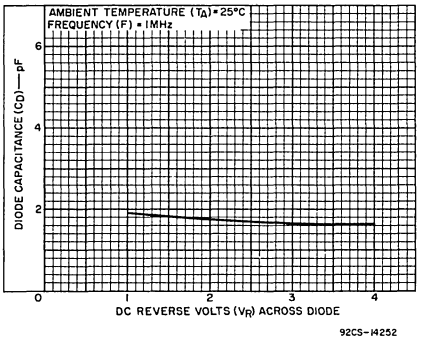


Fig.4 - Diode Capacitance (any Diode) vs Reverse Voltage for CA3019.

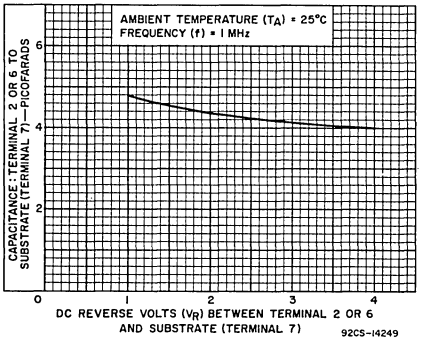


Fig.5 - Diode Quad-to-Substrate Capacitance vs Reverse Voltage for CA3019.

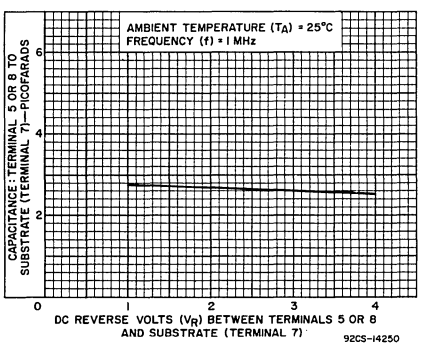


Fig.6 - Diode Quad-to-Substrate Capacitance vs Reverse Voltage for CA3019.

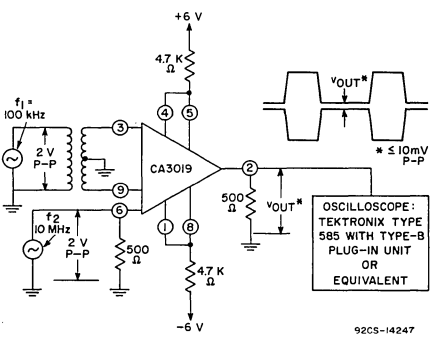


Fig.7 - Series Gate Switching Test Setup for CA3019.

RCA
Solid State
Division

Linear Integrated Circuits
CA3020
CA3020A

Multipurpose Wide-Band Power Amplifiers

Monolithic Silicon

The RCA-CA3020 and CA3020A are Integrated-Circuit, Multistage, Multipurpose, Wide-Band Power Amplifiers on a single monolithic silicon chip. They employ a highly versatile and stable direct-coupled circuit configuration featuring wide frequency range, high voltage and power gain, and high power output. These features plus inherent stability over a wide temperature range make the CA3020 and CA3020A extremely useful for a wide variety of applications in military, industrial, and commercial equipment.

The CA3020 and CA3020A are particularly suited for service as Class B power amplifiers. The CA3020A can provide a maximum power output of 1 watt from a 12-volt DC supply with a typical power gain of 75 dB. The CA3020 provides 0.5 watt power output from a 9-volt supply with the same power gain.

These types are supplied in hermetically sealed, TO-5 style 12-lead packages.

SCHEMATIC DIAGRAM FOR CA3020 AND CA3020A

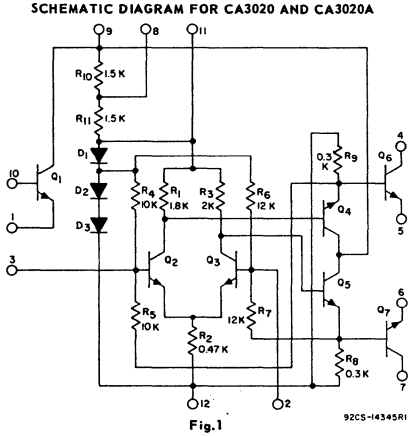


Fig.1

The resistance values included on the schematic diagram have been supplied as a convenience to assist Equipment Manufacturers in optimizing the selection of "outboard" components of equipment designs. The values shown may vary as much as ± 30%.

RCA reserves the right to make any changes in the Resistance Values provided such changes do not adversely affect the published performance characteristics of the device.

MULTIPURPOSE WIDE-BAND POWER AMPLIFIERS

For Military, Industrial, and Commercial Equipment at Frequencies up to 8 MHz



12-Lead TO-5

FEATURES

- High power output - class B amplifier —
  CA3020 .... 0.5 watt typ. at V_{CC} = +9V
  CA3020A ... 1.0 watt typ. at V_{CC} = +12V
- Wide frequency range —
  Up to 8 MHZ with resistive loads
- High power gain ..... 75db typ.
- Single power supply for class B operation with transformer —
  CA3020 ..... 3 to 9V
  CA3020A ..... 3 to 12V
- Built-in temperature-tracking voltage regulator provides stable operation over -55°C to +125°C temperature range

APPLICATIONS

- AF power amplifiers for portable and fixed sound and communications systems
- Servo-control amplifiers
- Wide-band linear mixers
- Video power amplifiers
- Transmission-line driver amplifiers (balanced and unbalanced)
- Fan-in and fan-out amplifiers for computer logic circuits
- Lamp-control amplifiers
- Motor-control amplifiers
- Power multivibrator
- Power switches
- Companion Application Note, ICAN 5766 "Application of CA3020 and CA3020A Integrated Circuit Multipurpose Wide-Band Power Amplifiers."

ABSOLUTE-MAXIMUM RATINGS:

DISSIPATION:

WITHOUT HEAT SINK	WITH HEAT SINK
At T_A = 25°C . . . . . 1 W	At T_C = 25°C . . . . . 2 W
Above T_A = 25°C . . . . . derate linearly 6.7 mW/°C	At T_C = 25°C to T_C = 55°C . . . . . 2 W
	Above T_C = 55°C . . derate linearly 16.7 mW/°C

TEMPERATURE RANGE:

Operating . . . . . -55°C to +125°C
Storage . . . . . -65°C to +150°C

LEAD TEMPERATURE (During Soldering):

At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm)
from case for 10 seconds max. . . . . +265°C

MAXIMUM VOLTAGE RATINGS at T_A = 25°C

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range of the vertical terminal 1 with respect to terminal 12 is 0 to +10 volts.

TERMINAL No.	1	2	3	4	5	6	7	8	9	10	11	12
1		*	*	*	*	*	*	*	▲ 0 -10/-12	+3 Note 1	*	+10 0
2			*	*	*	*	*	*	*	*	*	+2 -2
3				*	*	*	*	*	*	*	*	+2 -2
4					▲ +18/+25 0	*	*	*	*	*	*	▲ +18/+25 0
5						*	*	*	*	*	*	+3 Note 2
6							▲ 0 -18/-25	*	*	*	*	+3 Note 2
7								*	*	*	*	▲ +18/+25 0
8									Note 3	*	*	Note 3 0
9										+10 0	Note 1 0	+10/+12 0
10											*	+10 0
11												*
12												REF. SUB- STRATE

MAXIMUM CURRENT RATINGS

TERMINAL No.	I_{IN} mA	I_{OUT} mA
1	-	20
2	-	-
3	-	-
4	300	-
5	-	300
6	-	300
7	300	-
8	-	-
9	20	-
10	1	-
11	20	-
12	-	-

Note 1: This voltage is established by the maximum current rating.

Note 2: The emitters of Q_6 and Q_7 may be returned to a negative voltage supply through emitter resistors. Current into terminal No.9 should not be exceeded and the total device dissipation should not be exceeded.

Note 3: Terminal No.8 may be connected to terminals Nos.9, 11, or 12.

* Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe if the specified limits between all other terminals are not exceeded.

▲ Higher value is for CA3020A.

ELECTRICAL CHARACTERISTICS AT T_A = 25°C

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS			LIMITS CA3020			LIMITS CA3020A			UNITS
		CIRCUIT AND PROCEDURE	DC SUPPLY VOLTAGE								
		FIG.	V_{CC1}	V_{CC2}	MIN.	TYP.	MAX.	MIN.	TYP.	MAX.	
Collector-to-Emitter Breakdown Voltage, Q_6 & Q_7 at 10 mA	$V_{(BR)CER}$	2a	-	-	18	-	-	25	-	-	V
Collector-to-Emitter Breakdown Voltage, Q_1 at 0.1 mA	$V_{(BR)CEO}$	-	-	-	10	-	-	10	-	-	V
Idle Currents, Q_6 & Q_7	I_4 IDLE I_7 IDLE	8	9.0	2.0	-	5.5	-	-	5.5	-	mA
Peak Output Currents, Q_6 & Q_7	I_4PK I_7PK	8	9.0	2.0	140	-	-	180	-	-	mA
Cutoff Currents, Q_6 & Q_7	I_4 CUTOFF I_7 CUTOFF	8	9.0	2.0	-	-	1.0	-	-	1.0	mA
Differetial Amplifier Current Drain	I_{CC1}	8	9.0	9.0	6.3	9.4	12.5	6.3	9.4	12.5	mA
Total Current Drain	I_{CC1} + I_{CC2}	8	9.0	9.0	8.0	21.5	35.0	14.0	21.5	30.0	mA
Differential Amplifier Input Terminal Voltages	V_2 V_3	8	9.0	2.0	-	1.11	-	-	1.11	-	V
Regulator Terminal Voltage	V_{11}	8	9.0	2.0	-	2.35	-	-	2.35	-	V
Q, Cutoff (Leakage) Currents: Collector-to-Emitter	I_{CEO}	-	10.0	-	-	-	100	-	-	100	μA
Emitter-to-Base	I_{EBO}		3.0	-	-	-	0.1	-	-	0.1	
Collector-to-Base	I_{CBO}		3.0	-	-	-	0.1	-	-	0.1	
Forward Current Transfer Ratio, Q_1 at 3 mA	h_{FE1}	-	6.0	-	30	75	-	30	75	-	
Bandwidth at -3 dB Point	BW	9	6.0	6.0	-	8	-	-	8	-	MHz
Maximum Power Output	$P_{O(MAX)}$	10	6.0	6.0	200	300[a]	-	200	300[a]	-	mW
			9.0	9.0	400	550[a]	-	400	550[a]	-	
			9.0	12.0	-	-	-	800	1000[b]	-	
Sensitivity for P_{OUT} = 400 mW	e_{IN}	10	9.0	9.0	-	35[a]	55	-	-	-	mV
Sensitivity for P_{OUT} = 800 mW	e_{IN}	10	9.0	12.0	-	-	-	-	50[b]	100	mV
Input Resistance---- Terminal 3 to Ground	R_{IN3}	11	6.0	6.0	-	1000	-	-	1000	-	Ω
Junction-to-Case Thermal Resistance	θ_{J-C}	-	-	-	-	-	60	-	-	60	°C/W

a R_{CC} = 130 Ω

b R_{CC} = 200 Ω

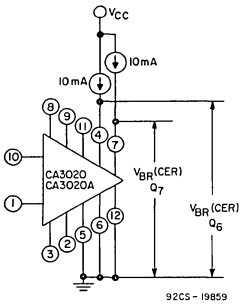


Fig.2

a. Collector-to-emitter breakdown voltage (Q_6 & Q_7) circuit

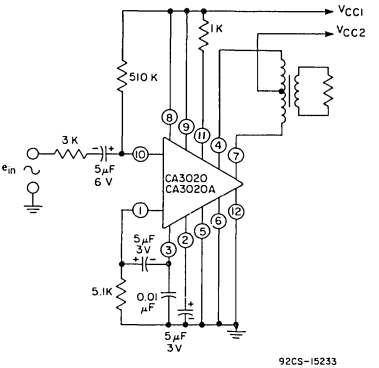


Fig.2

b. Typical audio amplifier circuit utilizing the CA3020 or CA3020A as an audio preamplifier and class B power amplifier

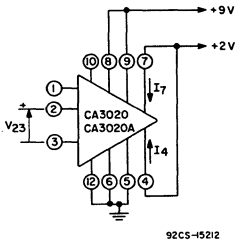
TYPICAL PERFORMANCE DATA*

An External Radiator is Recommended for High Ambient Temperature Operation

CHARACTERISTICS	SYMBOLS	CA3020	CA3020A	UNITS
Power Supply Voltage	V_{CC_1}	9.0	9.0	V
	V_{CC_2}	9.0	12.0	
Zero Signal Current — Diff. Ampl.	I_{CC_1}	15	15	mA
Zero Signal Current — Output Ampl.	I_{CC_2}	24	24	
Maximum Signal Current — Diff. Ampl.	I_{CC_1}	16	16.6	mA
Maximum Signal Current — Output Ampl.	I_{CC_2}	125	140	
Maximum Power Output at THD = 10%	Po	550	1000	mW
Sensitivity	e_{IN}	35	45	mV
Power Gain	G_P	75	75	dB
Input Resistance	R_{IN}	55	55	kΩ
Efficiency	η	45	55	%
Signal-to-Noise Ratio	S/N	70	66	dB
THD at 150 mW level		3.1	3.3	%
Test Signal Frequency from 600Ω Generator		1000	1000	Hz
Equivalent Collector-to-Collector Load Resistance	R_{CC}	130	200	Ω

* Refer to Figs.8 through 12 for Measurement and Symbol Information.

TYPICAL TRANSFER CHARACTERISTICS



a. Test Setup

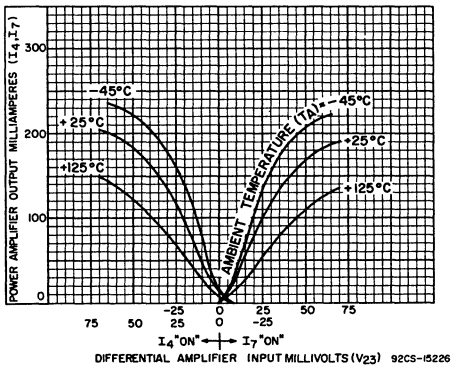
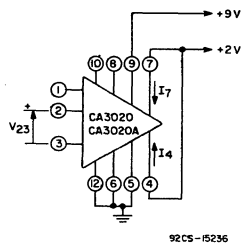


Fig.3 b. Characteristics with R_{10} shorted out



a. Test Setup

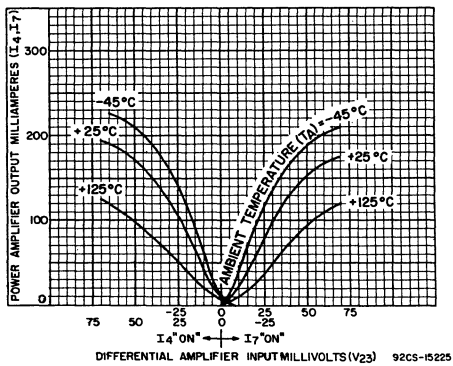
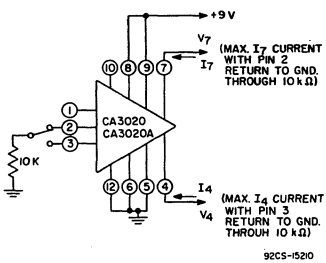


Fig.4 b. Characteristics with R_{10} in circuit

"MINIMUM DRIVE" TYPICAL CURRENT-VOLTAGE SATURATION CURVE



a. Test Setup

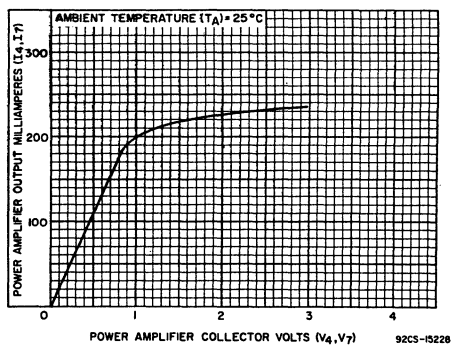
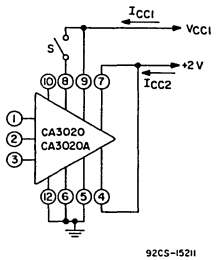
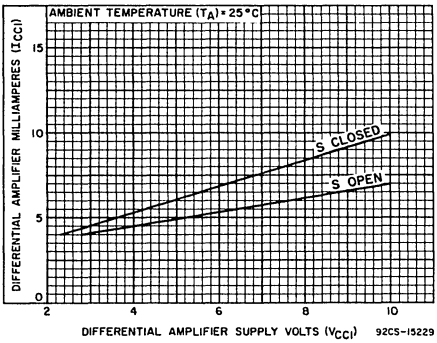


Fig.5 b. Characteristic

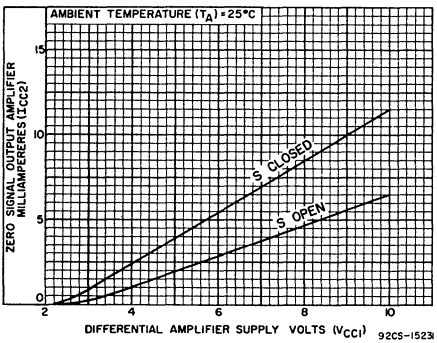
ZERO SIGNAL AMPLIFIER CURRENT vs DIFFERENTIAL AMPLIFIER SUPPLY VOLTAGE



a. Test Setup



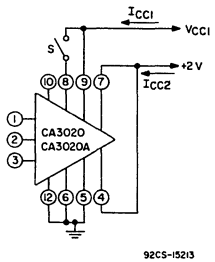
b. Differential Amplifier Characteristics



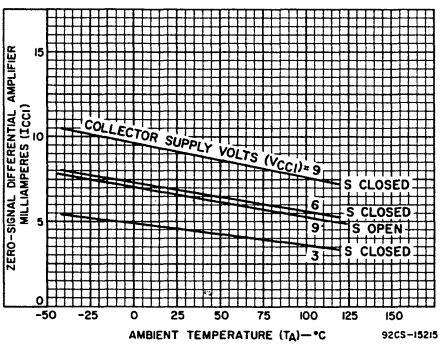
c. Output Amplifier Characteristics

Fig.6

ZERO SIGNAL AMPLIFIER CURRENT vs AMBIENT TEMPERATURE

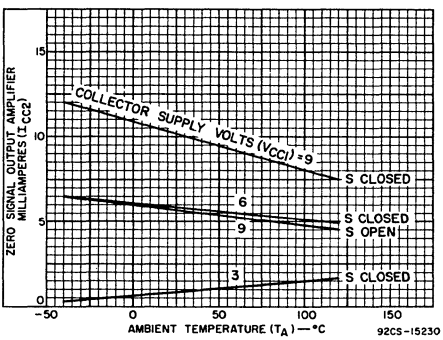


a. Test Setup



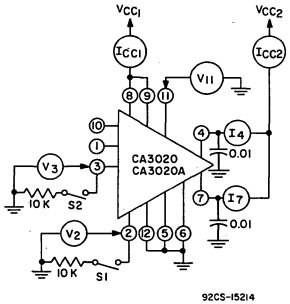
b. Differential Amplifier Characteristics

Fig.7



c. Output Amplifier Characteristics

STATIC CURRENT AND VOLTAGE TEST CIRCUIT



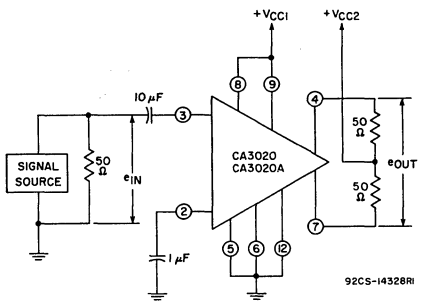
92CS-15214

CURRENTS OR VOLTAGES	S1	S2
$I_{4\text{-IDLE}}$	open	open
$I_{7\text{-IDLE}}$	open	open
$I_{4\text{-PEAK}}$	open	close
$I_{7\text{-PEAK}}$	close	open
$I_{4\text{-CUTOFF}}$	close	open
$I_{7\text{-CUTOFF}}$	open	close

CURRENTS OR VOLTAGES	S1	S2
I_{CC1}	open	open
I_{CC2}	open	open
V_2	open	open
V_3	open	open
V_{11}	open	open

Fig.8

MEASUREMENT OF BANDWIDTH AT -3 dB POINTS



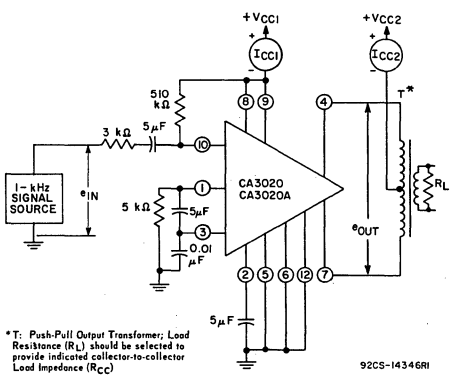
92CS-14328RI

Fig.9

PROCEDURES:

1. Apply desired value of V_{CC_1} and V_{CC_2}
2. Apply 1 kHz input signal and adjust for e_{IN} = 5 mV (rms)
3. Record the resulting value of e_{OUT} in dB (reference value)
4. Vary input-signal frequency, keeping e_{IN} constant at 5 mV, and record frequencies above and below 1 kHz at which e_{OUT} decreases 3 dB below reference value.
5. Record bandwidth as frequency range between -3 dB points.

MEASUREMENTS OF ZERO-SIGNAL DC CURRENT DRAIN, MAXIMUM-SIGNAL DC CURRENT DRAIN, MAXIMUM POWER OUTPUT, CIRCUIT EFFICIENCY, SENSITIVITY, AND TRANSDUCER POWER GAIN



*T: Push-Pull Output Transformer; Load Resistance (R_L) should be selected to provide indicated collector-to-collector Load Impedance (R_{CC})

92CS-14346RI

Fig.10

PROCEDURES:

Zero-Signal DC Current Drain

1. Apply desired Value of V_{CC_1} and V_{CC_2} and reduce e_{IN} to 0V
2. Record resulting values of I_{CC_1} and I_{CC_2} in mA as Zero-Signal DC Current Drain.

Maximum-Signal DC Current Drain, Maximum Power Output, Circuit Efficiency, Sensitivity, and Transducer Power Gain

1. Apply desired value of V_{CC_1} and V_{CC_2} and adjust e_{IN} to the value at which the Total Harmonic Distortion in the output of the amplifier = 10%
2. Record resulting value of I_{CC_1} and I_{CC_2} in mA as Maximum-Signal DC Current Drain
3. Determine resulting amplifier power output in watts and record as Maximum Power Output (P_{OUT})
4. Calculate Circuit Efficiency (η) in % as follows:

$$\eta = 100\,\frac{P_{OUT}}{V_{CC_1}I_{CC_1}+V_{CC_2}I_{CC_2}}$$

where P_{OUT} is in watts, V_{CC_1} and V_{CC_2} are in volts, and I_{CC_1} and I_{CC_2} are in amperes.

5. Record value of e_{IN} in mV (rms) required in Step 1 as Sensitivity (e_{IN})
6. Calculate Transducer Power Gain (G_p) in dB as follows:

$$G_p = 10\log_{10}\frac{P_{OUT}}{P_{IN}}$$

where P_{IN} (in mW) $= \dfrac{e_{IN}^2}{3000 + R_{IN(10)}}$

MEASUREMENT OF INPUT RESISTANCE

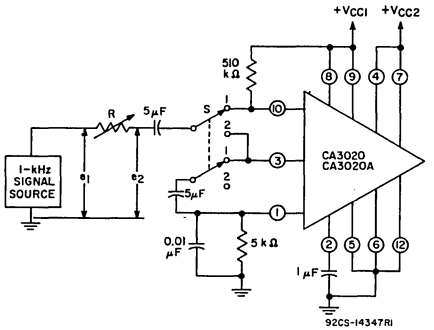


Fig.11

PROCEDURES:

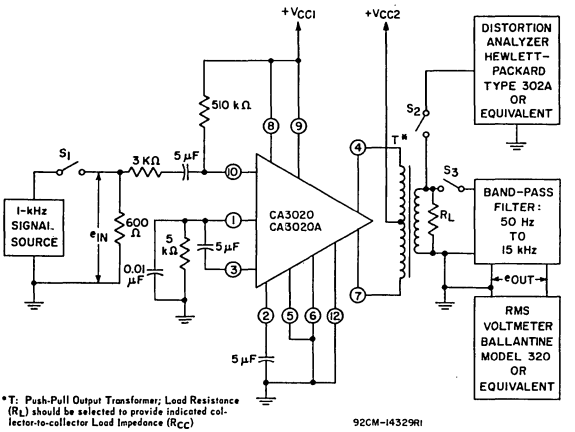
Input Resistance Terminal 10 to Ground ($R_{IN_{10}}$)

1. Apply desired value of V_{CC_1} and V_{CC_2} and set S in Position 1
2. Adjust 1-kHz input for desired signal level of measurement
3. Adjust R for $e_2 = e_1/2$
4. Record resulting value of R as $R_{IN_{10}}$

Input Resistance Terminal 3 to Ground (R_{IN_3})

1. Apply desired value of V_{CC_1} and V_{CC_2} set S in Position 2
2. Adjust 1-kHz input for desired signal level of measurement
3. Adjust R for $e_2 = e_1/2$
4. Record resulting value of R as R_{IN_3}

MEASUREMENT OF SIGNAL-TO-NOISE RATIO AND TOTAL HARMONIC DISTORTION



PROCEDURES:

Signal-to-Noise Ratio

1. Close S_1 and S_3; open S_2
2. Apply desired values of V_{CC_1} and V_{CC_2}
3. Adjust e_{IN} for an amplifier output of 150mW and record resulting value of E_{OUT} in dB as e_{OUT_1} (reference value)
4. Open S_1 and record resulting value of e_{OUT} in dB as e_{OUT_2}
5. Signal-to-Noise Ratio (S/N) = $20\log_{10} \frac{e_{OUT_1}}{e_{OUT_2}}$.

Total Harmonic Distortion

1. Close S_1 and S_2; open S_3
2. Apply desired values of V_{CC_1} and V_{CC_2}
3. Adjust e_{IN} for desired level amplifier output power
4. Record Total Harmonic Distortion (THD) in %

Fig.12

RCA
Solid State
Division

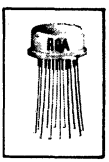
Linear Integrated Circuits

CA3021
CA3022
CA3023

Low-Power Video and Wideband Amplifiers

Monolithic Silicon

RCA-CA3021, CA3022, and CA3023 are low-power integrated-circuit wideband amplifiers with a wide range of applications in industrial, military, and commercial communications equipment. Each consists of a multistage amplifier circuit and unconnected diodes on a single chip, hermetically sealed in a 12-lead TO-5 style package. The diodes may be connected to provide limiting in FM applications.



The CA3021, CA3022, and CA3023 have the same maximum ratings, and differ principally in dissipation (dc power requirements) and bandwidth capability. All three devices are designed for operation over the temperature range from -55° C to +125° C.

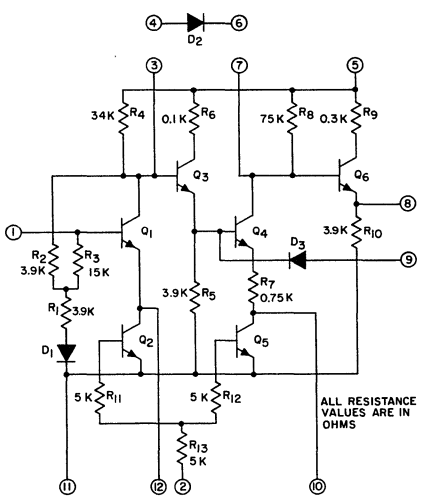
HIGHLIGHTS

- Low DC Power Drain:
  P_D { CA3021 = 4 mW typ. / CA3022 = 12.5 mW typ. / CA3023 = 35 mW typ. } at V_{CC} = 6 V
- Excellent frequency response:
  -3 dB BW { CA3021 = 2.4 MHz typ. / CA3022 = 7.5 MHz typ. / CA3023 = 16 MHz typ.
- High Voltage Gain:
  A { CA3021 = 56 dB typ. at 0.5 MHz / CA3022 = 57 dB typ. at 2.5 MHz / CA3023 = 53 dB typ. at 5 MHz
- Wide AGC Range: 33 dB typ.
- Only one power supply (4.5 to 12 V) required
- Hermetically Sealed 12-Lead TO-5-style package
- Operation from -55° C to +125° C

APPLICATIONS

- Gain-Controlled Linear Amplifiers
- AM/FM IF Amplifiers • Video Amplifiers • Limiters

SCHEMATIC DIAGRAM FOR CA3021, CA3022, AND CA3023



ABSOLUTE-MAXIMUM RATINGS:

OPERATING-TEMPERATURE RANGE	-55°C to +125°C	
STORAGE-TEMPERATURE RANGE	-65°C to +150°C	
LEAD TEMPERATURE (During Soldering): At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max.	+265°C	
DEVICE DISSIPATION, P_T	120 max.	mW
INPUT-SIGNAL VOLTAGE	-3, +3 max.	V
DC VOLTAGES AND CURRENTS	See Table Below	

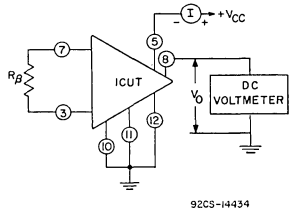
	VOLTAGE OR CURRENT LIMITS		CIRCUIT CONDITIONS	
TERMINAL	NEGATIVE	POSITIVE	TERMINAL	CONDITIONS
1	-3V	+3V	1	Connected to Voltage Source through 100Ω Resistor
			5	+12V
			10, 11, 12	Ground
2	-3V	+12V	5	+12V
			10, 11, 12	Ground
3	0V	+12V	5	+12V
			10, 11, 12	Ground
4	-12V	+12V	6, 11	Ground
	10 max. mA			
5	0V	+18V	10, 11, 12	Ground
6	-12V	+12V	5, 11	Ground
	10 max. mA			

	VOLTAGE OR CURRENT LIMITS		CIRCUIT CONDITIONS	
TERMINAL	NEGATIVE	POSITIVE	TERMINAL	CONDITIONS
7	0V	+12V	5	+12V
			10, 11, 12	Ground
8	20 max. mA		5	+12V
			10, 11, 12	Ground
9	-0.5V	+3V	5	+12V
			10, 11, 12	Ground
10	0V	+4V	2,5	+12V
			11	Ground
11	-6V	+12V	2	Ground
			5	+12V
12	0V	+4V	2,5	+12V
			11	Ground

ELECTRICAL CHARACTERISTICS, at $T_A = 25^oC$, $V_{CC} = +6V$, unless otherwise specified

CHARACTERISTIC		SYMBOL	TEST CONDITIONS			LIMITS									UNITS	TYPICAL CHARACTERISTIC CURVE
			TEST SETUP AND PROCEDURE	FEEDBACK RESISTANCE (R_β) BETWEEN TERMINALS 3 AND 7	FREQUENCY f	CA3021 (TA5219)			CA3022 (TA5236)			CA3023 (TA5218)				
			Fig.	Ω	MHz	Min.	Typ.	Max.	Min.	Typ.	Max.	Min.	Typ.	Max.	Units	Fig.
Device Dissipation		P_T	2	∞	-	1	4	8	–	–	–	–	–	–	mW	3a,d
				∞	–	–	–	-	5	12.5	24	–	–	–	mW	3b,d
				∞	–	–	–	–	–	–	–	24	35	48	mW	3c,d
Quiescent Output Voltage		V_O	2	39k	–	–	2.2	–	–	–	–	–	–	–	V	–
				10k	–	–	–	–	–	1.9	–	–	–	–	V	
				4.7k	–	–	–	–	–	–	–	–	1.3	–	V	
AGC Source Current		I_{AGC}	4	V_{AGC} =+6V		–	0.8	–	–	0.8	–	–	0.8	–	mA	–
Voltage Gain		A	5	560k	0.5	50	56	–	–	–	–	–	–	–	dB	6a
				39k	0.8	40	46	–	–	–	–	–	–	–	dB	6a,d
				39k	2.5	–	–	–	50	57	–	–	–	–	dB	6b
				10k	3	–	–	–	40	44	–	–	–	–	dB	6b,d
				18k	5	–	–	–	–	–	–	50	53	–	dB	6c
				4.7k	10	–	–	–	–	–	–	40	44	–	dB	6c,d
Bandwidth at -3 dB Point		BW	5	39k	–	0.8	2.4	–	–	–	–	–	–	–	MHz	6a
				10k	–	–	–	–	3	7.5	–	–	–	–	MHz	6b
				4.7k	–	–	–	–	–	–	–	10	16	–	MHz	6c
Input-Impedance Components	Input Resistance	R_{IN}	7	39k	1	–	4000	–	–	–	–	–	–	–	Ω	–
				10k	5	–	–	–	–	1300	–	–	–	–	Ω	
				4.7k	10	–	–	–	–	–	–	–	300	–	Ω	
	Input Capacitance	C_{IN}	7	39k	1	–	11	–	–	–	–	–	–	–	pF	–
				10k	5	–	–	–	–	18	–	–	–	–	pF	
				4.7k	10	–	–	–	–	–	–	–	13	–	pF	
Output Resistance		R_{OUT}	8	39k	1	–	300	–	–	–	–	–	–	–	Ω	–
				10k	5	–	–	–	–	120	–	–	–	–	Ω	
				4.7k	10	–	–	–	–	–	–	–	100	–	Ω	
Noise Figure		NF	9	39k	1	–	4.2	8.5	–	–	–	–	–	–	dB	–
				10k	1	–	–	–	–	4.4	8.5	–	–	–	dB	
				4.7k	1	–	–	–	–	–	–	–	6.5	8.5	dB	
AGC Range		AGC	10	–	1	–	33	–	–	–	–	–	–	–	dB	–
				–	5	–	–	–	–	33	–	–	–	–	dB	
				–	10	–	–	–	–	–	–	–	33	–	dB	
Maximum Output Voltage (RMS Value)		v_{out}	5	39k	1	–	0.6	–	–	–	–	–	–	–	V(rms)	–
				10k	5	–	–	–	–	0.7	–	–	–	–	V(rms)	
				4.7k	10	–	–	–	–	–	–	–	0.5	–	V(rms)	

TEST SETUP FOR MEASUREMENT OF DEVICE DISSIPATION AND QUIESCENT OUTPUT VOLTAGE



$P_T = V_{CC}\ (I)$

Fig.2

DEVICE DISSIPATION VS DC SUPPLY VOLTAGE FOR CA3022

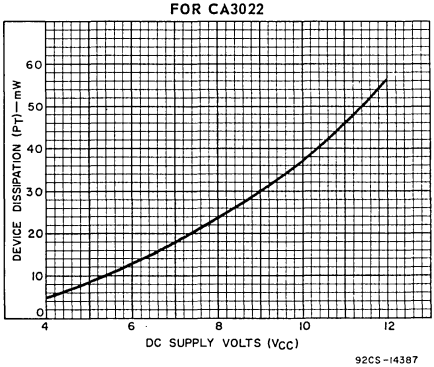


Fig.3(b)

DEVICE DISSIPATION VS TEMPERATURE FOR CA3021, CA3022, AND CA3023

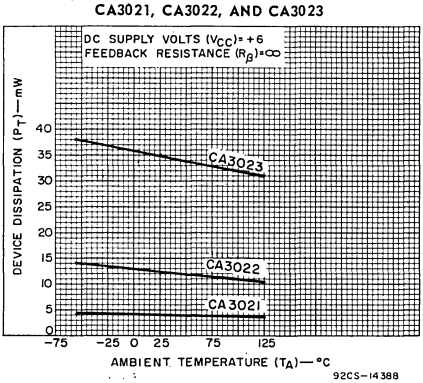


Fig.3(d)

DEVICE DISSIPATION VS DC SUPPLY VOLTAGE FOR CA3021

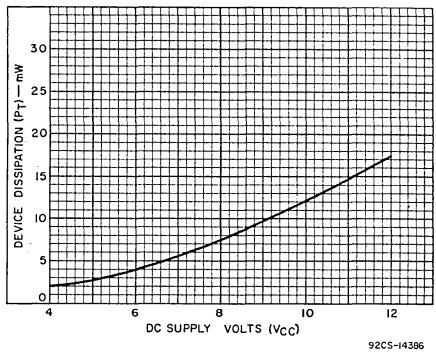


Fig.3(a)

DEVICE DISSIPATION VS DC SUPPLY VOLTAGE FOR CA3023

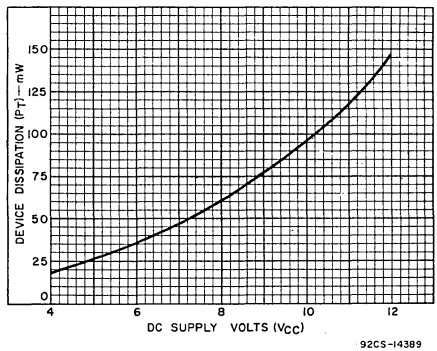
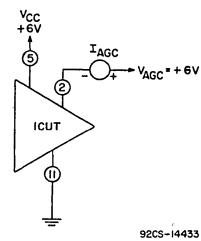


Fig.3(c)

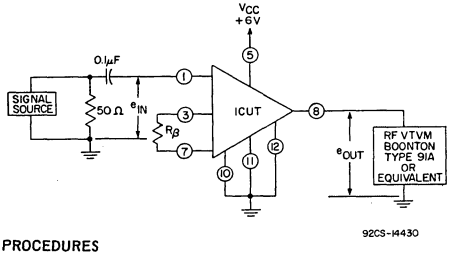
TEST SETUP FOR MEASUREMENT OF AGC SOURCE CURRENT



I_{AGC} IS THE CURRENT FLOWING INTO TERMINAL 2.

Fig.4

TEST SETUP FOR MEASUREMENTS OF VOLTAGE-GAIN, -3dB BANDWIDTH, AND MAXIMUM OUTPUT VOLTAGE



PROCEDURES

Voltage Gain:

(a) Set e_{in} = 0.5 mVat frequency specified, read e_{out} Voltage Gain (A) = $20 \log_{10} \frac{e_{out}}{e_{in}}$

Bandwidth:

(a) Set e_{out} to a convenient reference voltage at f = 100 kHz and record corresponding value of e_{in}.

(b) Increase the frequency, keeping e_{in} constant until e_{out} drops 3-dB. Record Bandwidth.

Fig.5

VOLTAGE GAIN VS FREQUENCY FOR CA3021

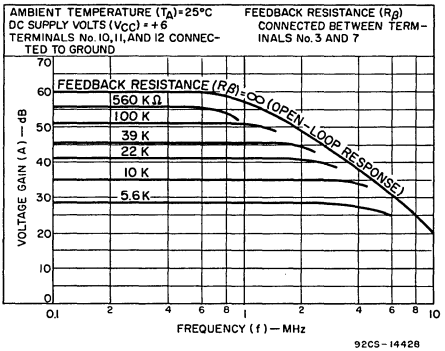


Fig.6(a)

VOLTAGE GAIN VS FREQUENCY FOR CA3022

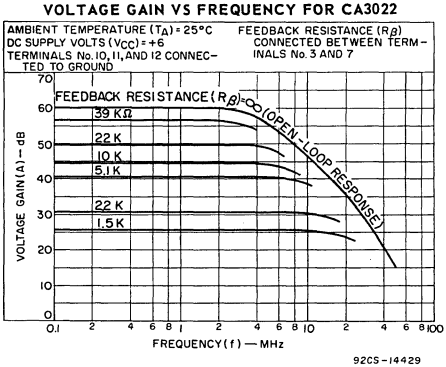


Fig.6(b)

VOLTAGE GAIN VS FREQUENCY FOR CA3023

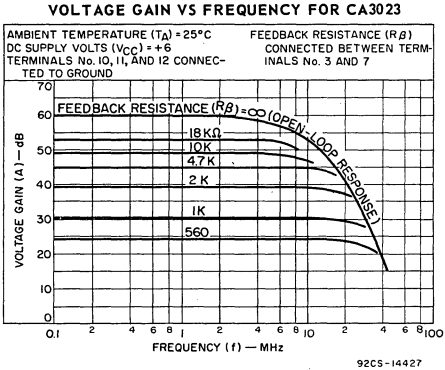


Fig.6(c)

DC SUPPLY VOLTS (V_{CC}) = +6 TERMINALS No. 10, 11, AND 12 CONNECTED TO GROUND FEEDBACK RESISTANCE (R_B) CONNECTED BETWEEN TERMINALS No. 3 AND No. 7	TYPE	R_B KΩ	f MHz
	CA3021	39	1
	CA3022	10	5
	CA3023	4.7	10

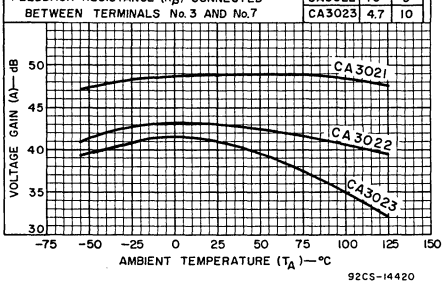
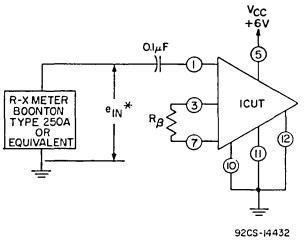


Fig.6(d)

VOLTAGE GAIN VS TEMPERATURE FOR CA3021, CA3022, AND CA3023

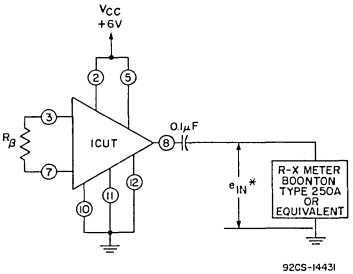
TEST SETUP FOR MEASUREMENT OF INPUT-IMPEDANCE COMPONENTS



* $e_{in} \leq 10$ mV

Fig.7

TEST SETUP FOR MEASUREMENT OF OUTPUT RESISTANCE

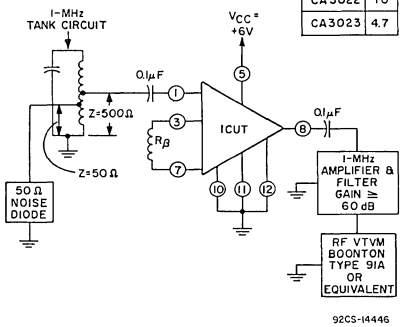


* $e_{in} \leq 10$ mV

Fig.8

TEST SETUP FOR MEASUREMENT OF NOISE FIGURE

ICUT	Rβ KΩ
CA3021	39
CA3022	10
CA3023	4.7

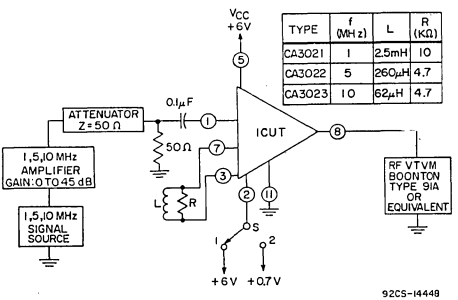


CA3021 - R_β = 39 kΩ
CA3022 - R_β = 10 kΩ
CA3023 - R_β = 4.7 kΩ

Fig.9

TEST SETUP FOR MEASUREMENT OF AGC RANGE

TYPE	f (MHz)	L	R (KΩ)
CA3021	1	2.5mH	10
CA3022	5	260μH	4.7
CA3023	10	62μH	4.7



$$\text{AGC RANGE} = 20 \log_{10} \frac{\text{A WITH S IN POSITION 1}}{\text{A WITH S IN POSITION 2}}$$

(A = VOLTAGE GAIN)

	f MHz
CA3021	1
CA3022	5
CA3023	10

Fig. 10

RCA
Solid State
Division

Linear Integrated Circuits

CA3026
CA3054

Transistor Array

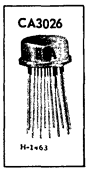
Monolithic Silicon

The CA3026 and CA3054 each consists of two independent differential amplifiers with associated constant-current transistors on a common monolithic substrate. The six n-p-n transistors which comprise the amplifiers are general purpose devices which exhibit low 1/f noise and a value of f_T in excess of 300 MHz. These features make the CA3026 and CA3054 useful from dc to 120 MHz. Bias and load resistors have been omitted to provide maximum application flexibility.

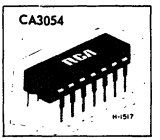
The monolithic construction of the CA3026 and CA3054 provides close electrical and thermal matching of the amplifiers. This feature makes these devices particularly useful in dual channel applications where matched performance of the two channels is required.

DUAL INDEPENDENT DIFFERENTIAL AMPLIFIERS

For Low-Power Applications at Frequencies from DC to 120 MHz



12-Lead TO-5



14-Lead
Dual-In-Line
Plastic Package

APPLICATIONS

- Dual sense amplifiers
- Dual Schmitt triggers
- Multifunction combinations -- RF/Mixer/Oscillator; Converter/IF
- IF amplifiers (differential and/or cascode)
- Product detectors
- Doubly balanced modulators and demodulators
- Balanced quadrature detectors
- Cascade limiters
- Synchronous detectors
- Pairs of balanced mixers
- Synthesizer mixers
- Balanced (push-pull) cascode amplifiers

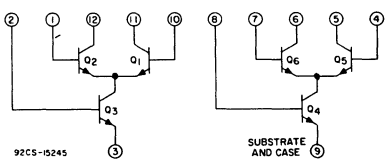


Fig.1a - Schematic Diagram for CA3026.

FEATURES

- Two differential amplifiers on a common substrate
- Independently accessible inputs and outputs
- Maximum input offset voltage -- ± 5 mV
- Full military temperature range capability -- -55°C to +125°C
- Limited temperature range -- 0°C to 85°C for CA3054

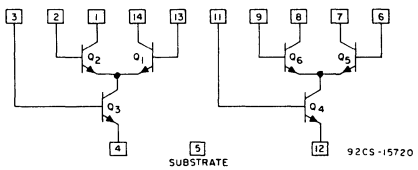


Fig.1b - Schematic Diagram for CA3054.

CAUTION: Substrate MUST be maintained negative with respect to all collector terminals of this device. See Maximum Voltage Ratings chart.

MAXIMUM RATINGS, ABSOLUTE-MAXIMUM VALUES, AT T_A = 25°C

Power Dissipation, P:	CA3026	CA3054	
Any one transistor	300	300	mW
Total package	600	750	mW
For T_A > 55°C	Derate at 5	6.67	mW/°C
Temperature Range:			
Operating	-55 to + 125		°C
Storage	-65 to + 150		°C
Lead Temperature (During Soldering):			
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max.	+265		°C

The following ratings apply for each transistor in the device:

Collector-to-Emitter Voltage, V_{CEO}	15	V
Collector-to-Base Voltage, V_{CBO}	20	V
Collector-to-Substrate Voltage, V_{CIO}*	20	V
Emitter-to-Base Voltage, V_{EBO}	5	V
Collector Current, I_C	50	mA

* The collector of each transistor of the CA3026 and CA3054 is isolated from the substrate by an integral diode. *The substrate must be connected to a voltage which is more negative than any collector voltage in order to maintain isolation between transistors and provide for normal transistor action. The substrate should be maintained at signal (AC) ground by means of a suitable grounding capacitor, to avoid undesired coupling between transistors.*

Maximum Voltage Ratings

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range between vertical terminal 1† and horizontal terminal 3† is +15 to -5 volts.

† For CA3026; corresponding terminals for CA3054 are vertical terminal 2 and horizontal terminal 4.

CA3054 TERMINAL No. →		13	14	1	2	3	4	6	7	8	9	11	12	5
↓	CA3026 TERMINAL No. →	10	11	12	1	2	3	4	5	6	7	8	Note 1 9	Note 1 9
13	10		0 -20	*	+5 -5	*	+15 -5	*	*	*	*	*	*	*
14	11			*	*	*	+20 0	*	*	*	*	*	*	+20 0
1	12				+20 0	*	+20 0	*	*	*	*	*	*	+20 0
2	1					*	+15 -5	*	*	*	*	*	*	*
3	2						+1 -5	*	*	*	*	*	*	*
4	3							*	*	*	*	*	*	*
6	4								0 -20	*	+5 -5	*	+15 -5	*
7	5									*	*	*	*	+20 0
8	6										+20 0	*	*	+20 0
9	7											*	+15 -5	*
11	8												+1 -5	*
12	9													*
5	9													Ref Sub-strate

* Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe if the specified limits between all other terminals are not exceeded.

Note 1: In the CA3026 terminal No.9 is connected to the emitter of Q_4, the reference substrate, and the case; therefore, the case should not be grounded. Two terminal 9 columns (CA3026) appear in the voltage rating chart because it is a composite chart for both the CA3026 and the CA3054. Wherever an asterisk is shown in one column 9 and a rating is shown in the other column 9, the asterisk should be ignored.

Maximum Current Ratings

CA3054 TERMINAL No.o	CA3026 TERMINAL No.	I_{IN} mA	I_{OUT} mA
13	10	5	0.1
14	11	50	0.1
1	12	50	0.1
2	1	5	0.1
3	2	5	0.1
4	3	0.1	-50
6	4	5	0.1
7	5	50	0.1
8	6	50	0.1
9	7	5	0.1
11	8	5	0.1
12	9	0.1	50

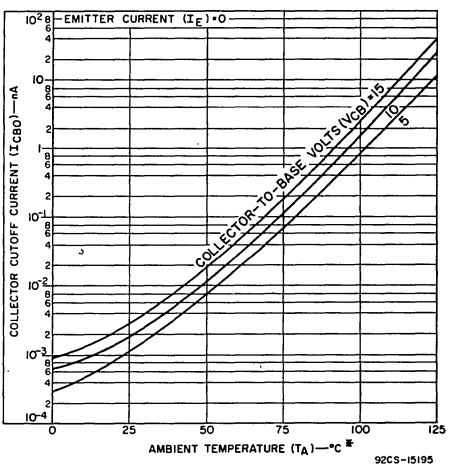
o Terminal No.10 of CA3054 is not used

ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	TEST CIR-CUIT	CA3026 CA3054 LIMITS			UNITS	TYPICAL CHARAC-TERISTICS CURVES
			FIG.	MIN.	TYP.	MAX.		FIG.
STATIC CHARACTERISTICS								
For Each Differential Amplifier								
Input Offset Voltage	V_{IO}	V_{CB} = 3 V $I_{E(Q3)} = I_{E(Q4)}$ = 2 mA	-	-	0.45	5	mV	6
Input Offset Current	I_{IO}		-	-	0.3	2	μA	7
Input Bias Current	I_I		-	-	10	24	μA	3
Quiescent Operating Current Ratio	$\frac{I_{C(Q1)}}{I_{C(Q2)}}$ or $\frac{I_{C(Q5)}}{I_{C(Q6)}}$		-	-	0.98 to 1.02	-	-	3
Temperature Coefficient Magnitude of Input-Offset Voltage	$\frac{\lvert\Delta V_{IO}\rvert}{\Delta T}$		-	-	1.1	-	μV/°C	5
For Each Transistor								
DC Forward Base-to-Emitter Voltage	V_{BE}	V_{CB} = 3 V, I_C = 50 μA	-	-	0.630	0.700	V	6
		1 mA	-	-	0.715	0.800		
		3 mA	-	-	0.750	0.850		
		10 mA	-	-	0.800	0.900		
Temperature Coefficient of Base-to-Emitter Voltage	$\frac{\Delta V_{BE}}{\Delta T}$	V_{CB} = 3 V, I_C = 1 mA	-	-	-1.9	-	μV/°C	4
Collector-Cutoff Current	I_{CBO}	V_{CB} = 10 V, I_E = 0	-	-	0.002	100	nA	2
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1 mA, I_B = 0	-	15	24	-	V	-
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 10 μA, I_E = 0	-	20	60	-	V	-
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_C = 10 μA, I_{CI} = 0	-	20	60	-	V	-
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 10 μA, I_C = 0	-	5	7	-	V	-
DYNAMIC CHARACTERISTICS								
Common-Mode Rejection Ratio For Each Amplifier	CMR	V_{CC} = 12 V V_{EE} = -6 V V_X = -3.3 V f = 1 kHz	8a	-	100	-	dB	8b
AGC Range, One Stage	AGC		9a	-	75	-	dB	9b
Voltage Gain, Single Stage Double-Ended Output	A		9a	-	32	-	dB	9b
AGC Range, Two Stage	AGC		10a	-	105	-	dB	10b
Voltage Gain, Two Stage Double-Ended Output	A		10a	-	60	-	dB	10b
Low-Frequency, Small-Signal Equivalent-Circuit Characteristics: (For Single Transistor)								
Forward Current-Transfer Ratio	h_{fe}	f = 1 kHz, V_{CE} = 3 V, I_C = 1 mA	-	-	110	-	-	11
Short-Circuit Input Impedance	h_{ie}		-	-	3.5	-	kΩ	11
Open-Circuit Output Impedance	h_{oe}		-	-	15.6	-	μmho	11
Open-Circuit Reverse Voltage-Transfer Ratio	h_{re}		-	-	1.8×10^{-4}	-	-	11

DYNAMIC CHARACTERISTICS CONT'D.								
1/f Noise Figure (For Single Transistor)	NF	f = 1 kHz, V_{CE} = 3 V	-	-	3.25	-	dB	-
Gain-Bandwidth Product (For Single Transistor)	f_T	V_{CE} = 3 V, I_C = 3 mA	-	-	550	-	MHz	12
Admittance Characteristics; Differential Circuit Configuration: (For Each Amplifier)								
Forward Transfer Admittance	y_{21}	V_{CB} = 3 V Each Collector $I_C \approx$ 1.25 mA f = 1 MHz	-	-	-20+j0	-	mmho	13a
Input Admittance	y_{11}		-	-	0.22+j0.1	-	mmho	13b
Output Admittance	y_{22}		-	-	0.01+j0	-	mmho	13c
Reverse Transfer Admittance	y_{12}		-	-	-0.003 +j0	-	mmho	13d
Admittance Characteristics; Cascode Circuit Configuration: (For Each Amplifier)								
Forward Transfer Admittance	y_{21}	V_{CB} = 3 V Total Stage $I_C \approx$ 2.5 mA f = 1 MHz	-	-	68-j0	-	mmho	14a
Input Admittance	y_{11}		-	-	0.55+j0	-	mmho	14b
Output Admittance	y_{22}		-	-	0+j0.02	-	mmho	14c
Reverse Transfer Admittance	y_{12}		-	-	0.004-j0.005	-	μmho	14d
Noise Figure	NF	f = 100 MHz	-	-	8	-	dB	-

TYPICAL STATIC CHARACTERISTICS



* For CA3054: use data from 0°C to 85°C only

Fig.2 - Collector-to-base cutoff current vs ambient temperature for each transistor.

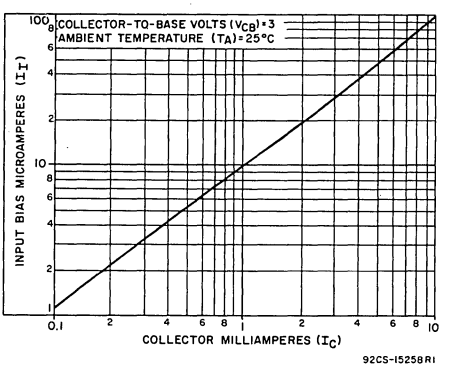


Fig.3 - Input bias current characteristic vs collector current for each transistor.

TYPICAL STATIC CHARACTERISTICS

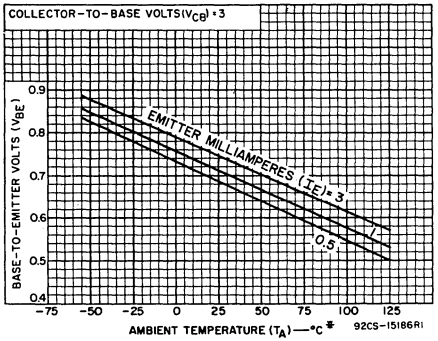


Fig.4 - Base-to-emitter voltage characteristic for each transistor vs ambient temperature.

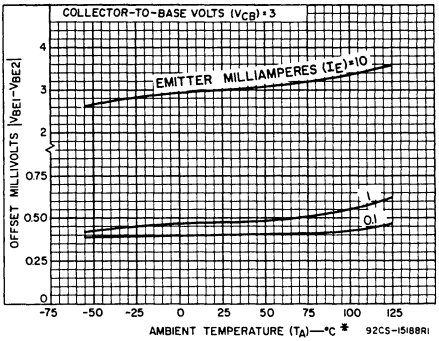


Fig.5 - Offset voltage characteristic vs ambient temperature for differential pairs.

* For CA3054: use data from 0°C to 85°C only

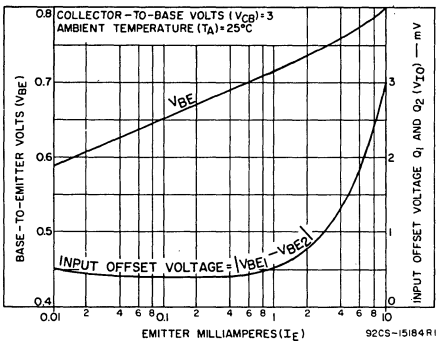


Fig.6 - Static base-to-emitter voltage characteristic and input offset voltage for differential pairs vs emitter current.

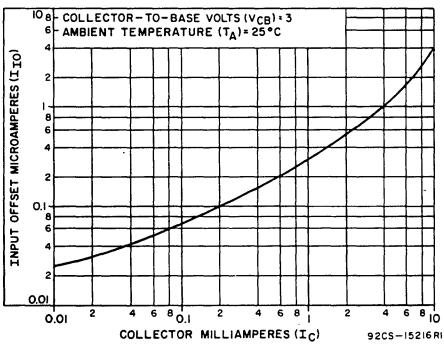
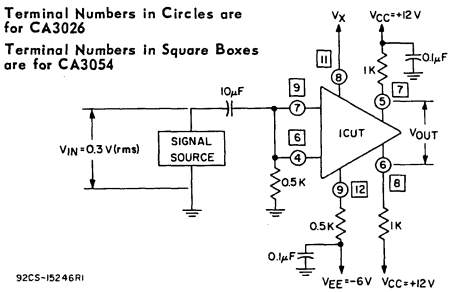


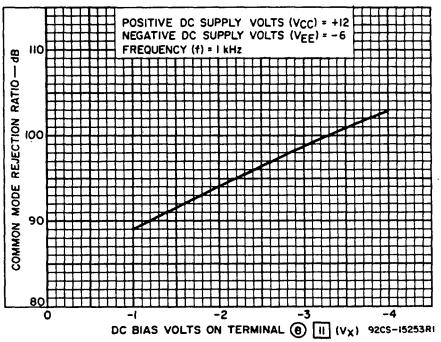
Fig.7 - Input offset current for matched differential pairs vs collector current.

TYPICAL DYNAMIC CHARACTERISTICS

COMMON MODE REJECTION RATIO



(a) Test setup

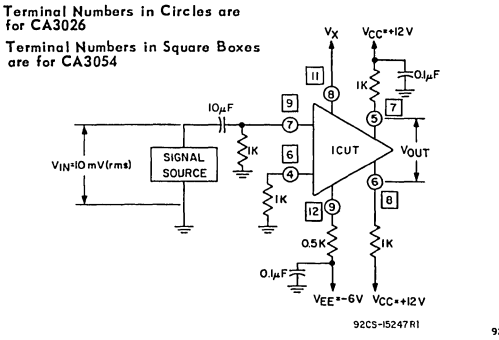


(b) Characteristic

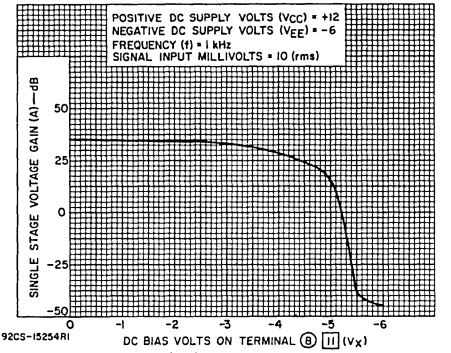
Fig.8

TYPICAL DYNAMIC CHARACTERISTICS (cont'd)

SINGLE-STAGE VOLTAGE GAIN



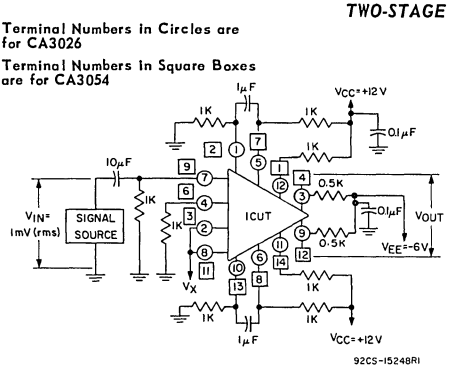
(a) Test setup



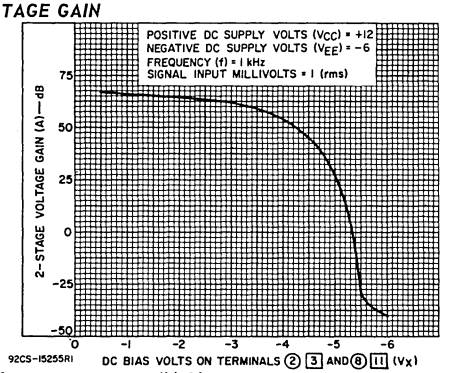
(b) Characteristic

Fig.9

TWO-STAGE VOLTAGE GAIN



(a) Test setup



(b) Characteristic

Fig.10

TYPICAL DYNAMIC CHARACTERISTICS FOR EACH TRANSISTOR

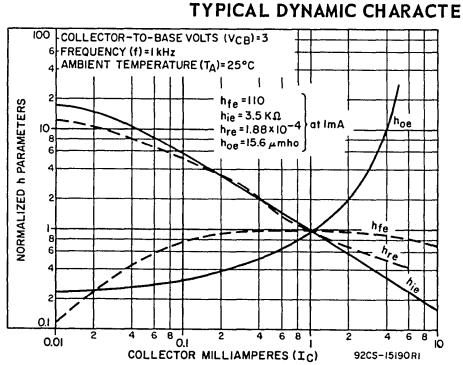


Fig.11 - Forward current-transfer ratio (h_{fe}), short-circuit input impedance (h_{ie}), open-circuit output impedance (h_{oe}), and open-circuit reverse voltage-transfer ratio (h_{re}) vs collector current for each transistor.

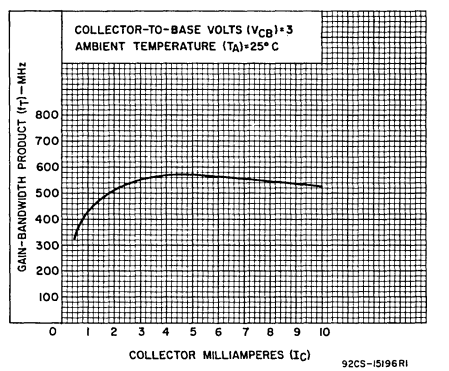


Fig.12 - Gain-bandwidth product (f_T) vs collector current.

TYPICAL DYNAMIC CHARACTERISTICS FOR EACH DIFFERENTIAL AMPLIFIER

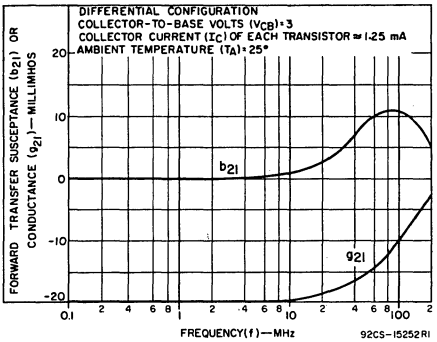


Fig.13(a) - Forward transfer admittance (Y_{21}) vs frequency.

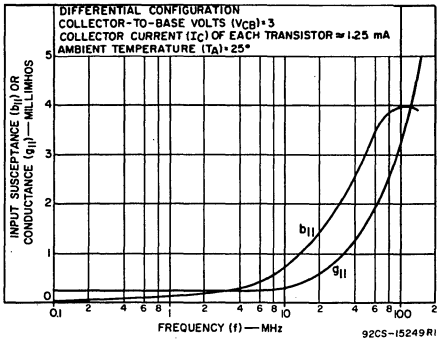


Fig.13(b) - Input admittance (Y_{11}).

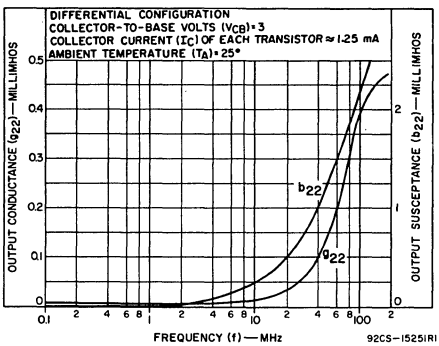


Fig.13(c) - Output admittance (Y_{22}) vs frequency.

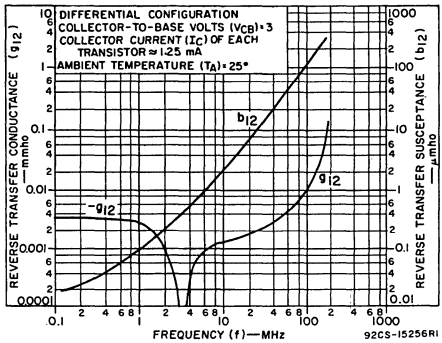


Fig.13(d) - Reverse transfer admittance (Y_{12}) vs frequency.

TYPICAL DYNAMIC CHARACTERISTICS FOR EACH CASCODE AMPLIFIER

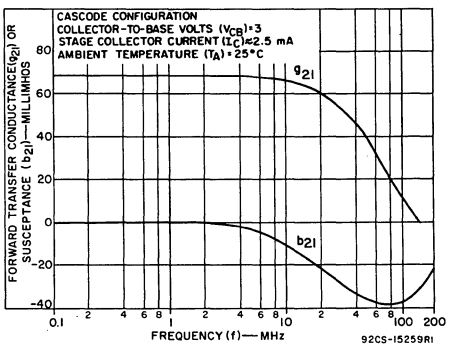


Fig.14(a) - Forward transfer admittance (Y_{21}) vs frequency.

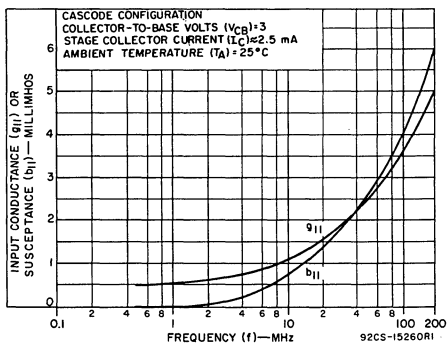


Fig.14(b) - Input admittance (Y_{11}) vs frequency.

TYPICAL CHARACTERISTICS FOR EACH CASCODE AMPLIFIER (cont'd)

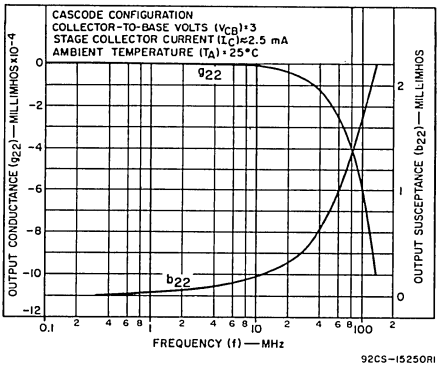


Fig.14(c) - Output admittance (Y_{22}) vs frequency.

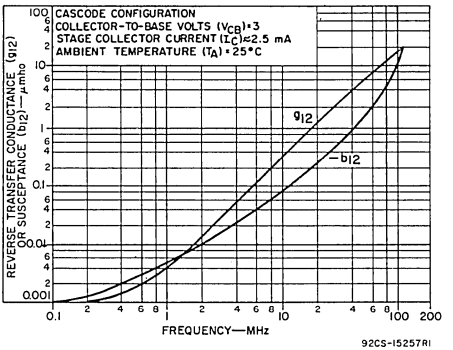


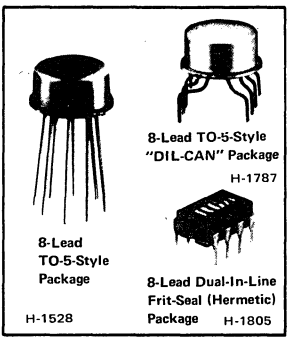
Fig.14(d) - Reverse transfer admittance (Y_{12}) vs frequency.

RCA
Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA3028A, CA3028AF, CA3028AS
CA3028B, CA3028BF, CA3028BS
CA3053, CA3053F, CA3053S



DIFFERENTIAL/CASCODE AMPLIFIERS

For Communications and Industrial Equipment at Frequencies from DC to 120 MHz

FEATURES

- Controlled for Input Offset Voltage, Input Offset Current, and Input Bias Current (CA3028B)
- Balanced Differential Amplifier Configuration with Controlled Constant-Current Source to Provide Unexcelled Versatility
- Single- and Dual-Ended Operation
- Operation from DC to 120 MHz
- Balanced-AGC Capability
- Wide Operating-Current Range

The CA3028A and CA3028B are differential/cascode amplifiers designed for use in communications and industrial equipment operating at frequencies from dc to 120 MHz.

The CA3028B is like the CA3028A but is capable of premium performance particularly in critical dc and differential amplifier applications requiring tight controls for input offset voltage, input offset current, and input bias current.

The CA3053 is similar to the CA3028A and CA3028B but is recommended for IF amplifier applications.

The CA3028A, CA3028B, and CA3053 are supplied in a hermetic 8-lead TO-5-style package. The "F" versions are supplied in a frit-seal package and the "S" versions in formed-lead (DIL-CAN) packages.

APPLICATIONS

- RF and IF Amplifiers (Differential or Cascode)
- DC, Audio, and Sense Amplifiers
- Converter in the Commercial FM Band
- Oscillator • Mixer • Limiter
- Companion Application Note, ICAN 5337 "Application of the RCA CA3028 Integrated Circuit Amplifier in the HF and VHF Ranges." This note covers characteristics of different operating modes, noise performance, mixer, limiter, and amplifier design considerations.

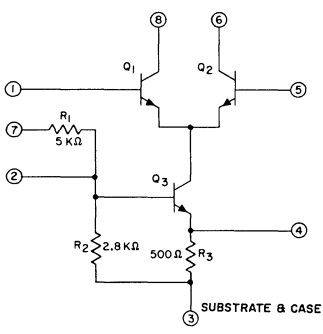


Fig.1 - Schematic diagram for CA3028A, CA3028B and CA3053.

ABSOLUTE MAXIMUM RATINGS AT $T_A = 25^oC$

DISSIPATION:
- At T_A up to 55°C (CA3028AF, CA3028BF, CA3053F) ..... 750 mW
- At $T_A > 55^oC$ (CA3028AF, CA3028BF, CA3053F) ..... Derate linearly 6.67 mW/°C
- At T_A up to 85°C (CA3028A, CA3028B, CA3053) ..... 450 mW
- At $T_A > 85^oC$ (CA3028A, CA3028B, CA3053) Derate linearly 5 mW/°C

AMBIENT-TEMPERATURE RANGE:
- Operating ..... −55°C to +125°C
- Storage ..... −65°C to +150°C

LEAD TEMPERATURE (During Soldering):
- At distance 1/16 ± 1/32" (1.59 ± 0.79 mm) from case for 10 seconds max. ..... +265°C

MAXIMUM VOLTAGE RATINGS at T_A = 25°C

TERMINAL No.	1	2	3	4	5	6	7	8
1		0 to -15▲	0 to -15▲	0 to -15▲	+5 to -5	*	*	+20⊕ to 0
2			+5 to -11	+5 to -1	+15◆ to 0	*	+15◆ to 0	*
3‡				+10 to 0	+15◆ to 0	+30● to 0	+15◆ to 0	+30● to 0
4					+15◆ to 0	*	*	*
5						+20⊕ to 0	*	*
6							*	*
7								*
8								

This chart gives the range of voltages which can be applied to the terminals listed horizontally with respect to the terminals listed vertically. For example, the voltage range of the horizontal terminal 4 with respect to terminal 2 is -1 to +5 volts.

‡ Terminal #3 is connected to the substrate and case.

* Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe, if the specified voltage limits between all other terminals are not exceeded.

▲ Limit is -12V for CA3053

⊕ Limit is +15V for CA3053

◆ Limit is +12V for CA3053

● Limit is +24V for CA3028A and +18V for CA3053

MAXIMUM CURRENT RATINGS

TERMINAL No.	I_{IN} mA	I_{OUT} mA
1	0.6	0.1
2	4	0.1
3	0.1	23
4	20	0.1
5	0.6	0.1
6	20	0.1
7	4	0.1
8	20	0.1

ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTIC	SYMBOL	TEST CIRCUIT Fig.	SPECIAL TEST CONDITIONS		LIMITS TYPE CA3028A			LIMITS TYPE CA3028B			LIMITS TYPE CA3053			UNITS	TYPICAL CHARACTERISTICS CURVES Fig.
					Min.	Typ.	Max.	Min.	Typ.	Max.	Min.	Typ.	Max.		
STATIC CHARACTERISTICS															
			$+V_{CC}$	$-V_{EE}$											
Input Offset Voltage	V_{IO}	2	6V 12V	6V 12V	- -	- -	- -	- -	0.98 0.89	5 5	- -	- -	- -	mV	4
Input Offset Current	I_{IO}	3a	6V 12V	6V 12V	- -	- -	- -	- -	0.56 1.06	5 6	- -	- -	- -	μA	4
Input Bias Current	I_I	3a	6V 12V	6V 12V	- -	16.6 36	70 106	- -	16.6 36	40 80	- -	- -	- -	μA	5a
		3b	9V 12V	- -	- -	- -	- -	- -	- -	- -	- -	29 36	85 125		5b
Quiescent Operating Current	I_6 or I_8	3a	6V 12V	6V 12V	0.8 2	1.25 3.3	2 5	1 2.5	1.25 3.3	1.5 4	- -	- -	- -	mA	6a 7
		3b	9V 12V	- -	- -	- -	- -	- -	- -	- -	1.2 2.0	2.2 3.3	3.5 5.0		6b
AGC Bias Current (Into Constant-Current Source Terminal No.7)	I_7	8a	12V 12V	V_{AGC} =+9 V_{AGC} = +12	- -	1.28 1.65	- -	- -	1.28 1.65	- -	- -	- -	- -	mA	8b
		-	9V 12V	- -	- -	- -	- -	- -	- -	- -	- -	1.15 1.55	- -		-
Input Current (Terminal No.7)	I_7	-	6V 12V	6V 12V	0.5 1	0.85 1.65	1 2.1	0.5 1	0.85 1.65	1 2.1	- -	- -	- -	mA	-
Device Dissipation	P_T	3a	6V 12V	6V 12V	24 120	36 175	54 260	24 120	36 175	42 220	- -	- -	- -	mW	9
		3b	9V 12V	- -	- -	- -	- -	- -	- -	- -	- -	50 100	80 150		-

ELECTRICAL CHARACTERISTICS at T_A = 25°C (cont'd)

CHARACTERISTIC	SYMBOL	TEST CIRCUIT Fig.	SPECIAL TEST CONDITIONS		LIMITS TYPE CA3028A Min.	Typ.	Max.	LIMITS TYPE CA3028B Min.	Typ.	Max.	UNITS	TYPICAL CHARACTERISTICS CURVE Fig.
DYNAMIC CHARACTERISTICS												
Power Gain	G_P	10a	f = 100 MHz	Cascode	16	20	-	16	20	-	dB	10b
		11a,d	V_{CC} = +9V	Diff.-Ampl.	14	17	-	14	17	-		11b,e
		10a	f = 10.7 MHz	Cascode	35	39	-	35	39	-	dB	10b
		11a	V_{CC} = +9V	Diff.-Ampl.	28	32	-	28	32	-		11b
Noise Figure	NF	10a	f = 100 MHz	Cascode	-	7.2	9	-	7.2	9	dB	10c
		11a,d	V_{CC} = +9V	Diff.-Ampl.	-	6.7	9	-	6.7	9		11c,e
Input Admittance	Y_{11}	-	f = 10.7 MHz V_{CC} = +9V	Cascode				-	0.6 + j 1.6	-	mmho	12
		-		Diff.-Ampl.				-	0.5 + j 0.5	-		13
Reverse Transfer Admittance	Y_{12}	-		Cascode				-	0.0003 - j0	-	mmho	14
		-		Diff.-Ampl.				-	0.01 - j0.0002	-		15
Forward Transfer Admittance	Y_{21}	-		Cascode				-	99 - j18	-	mmho	16
		-		Diff.-Ampl.				-	-37 + j0.5	-		17
Output Admittance	Y_{22}	-		Cascode				-	0. + j0.08	-	mmho	18
		-		Diff.-Ampl.				-	0.04 + j0.23	-		19
Power Output (Untuned)	P_o	20a	f = 10.7 MHz	Diff.-Ampl. 50 Ω Input-Output	-	5.7	-	-	5.7	-	μW	20b
AGC Range (Max. Power Gain to Full Cutoff)	AGC	21a	V_{CC} = +9V	Diff.-Ampl.	-	62	-	-	62	-	dB	21b
Voltage Gain at f = 10.7 MHz	A	22a	f = 10.7 MHz V_{CC} = +0V R_L = 1 kΩ	Cascode	-	40	-	-	40	-	dB	22b
		22c		Diff.-Ampl.	-	30	-	-	30	-		22d
Voltage Gain Differential at f = 1 kHz		23	V_{CC} = +6V, V_{EE} = -6V, R_L = 2 kΩ		-	-	-	35	38	42	dB	-
			V_{CC} = +12V, V_{EE} = -12V R_L = 1.6 kΩ		-	-	-	40	42.5	45		
Max. Peak-to-Peak Output Voltage at f = 1 kHz	V_o(P-P)	23	V_{CC} = +6V, V_{EE} = -6V, R_L = 2 kΩ		-	-	-	7	11.5	-	V_{P-P}	-
			V_{CC} = +12V, V_{EE} = -12V R_L = 1.6 kΩ		-	-	-	15	23	-		
Bandwidth at -3 dB point	BW	23	V_{CC} = +6V, V_{EE} = -6V, R_L = 2 kΩ		-	-	-	-	7.3	-	MHz	-
			V_{CC} = +12V, V_{EE} = -12V R_L = 1.6 kΩ		-	-	-	-	8	-		
Common-Mode Input-Voltage Range	V_{CMR}	24	V_{CC} = +6V, V_{EE} = -6V V_{CC} = +12V, V_{EE} = -12V		- -	- -	- -	-2.5 -5	(-3.2 - 4.5) (-7 - 9)	4 7	V	-
Common-Mode Rejection Ratio	CMR	24	V_{CC} = +6V, V_{EE} = -6V V_{CC} = +12V, V_{EE} = -12V		- -	- -	- -	60 60	110 90	- -	dB	-
Input Impedance at f = 1 kHz	Z_{IN}		V_{CC} = +6V, V_{EE} = -6V V_{CC} = +12V, V_{EE} = -12V		- -	- -	- -	- -	5.5 3	- -	kΩ	-
Peak-to-Peak Output Current	I_{P-P}		V_{CC} = +9V V_{CC} = +12V	f = 10.7 MHz e_{in} = 400 mV Diff.-Ampl.	2 3.5	4 6	7 10	2.5 4.5	4 6	6 8	mA	-

ELECTRICAL CHARACTERISTICS at T_A = 25°C (cont'd)

CHARACTERISTIC		SYMBOL	TEST CIRCUIT Fig.	SPECIAL TEST CONDITIONS		LIMITS TYPE CA3053 Min.	Typ.	Max.	UNITS	TYPICAL CHARACTERISTICS CURVE Fig.
DYNAMIC CHARACTERISTICS										
Power Gain		G_P	10a	f = 10.7 MHz	Cascode	35	39	-	dB	
			11a	V_{CC} = +9V	Diff.-Ampl.	28	32			
Input Admittance		Y_{11}	-	f = 10.7 MHz V_{CC} = +9V	Cascode	-	0.6 + j 1.6	-	mmho	12
			-		Diff.-Ampl.	-	0.5 + j 0.5	-		13
Reverse Transfer Admittance		Y_{12}	-		Cascode	-	0.0003 - j0	-	mmho	14
			-		Diff.-Ampl.	-	0.01 - j0.0002	-		15
Forward Transfer Admittance		Y_{21}	-		Cascode	-	99 - j18	-	mmho	16
			-		Diff.-Ampl.	-	-37 + j0.5	-		17
Output Admittance		Y_{22}	-		Cascode	-	0. + j0.08	-	mmho	18
			-		Diff.-Ampl.	-	0.04 + j0.23	-		19
Voltage Gain	at f = 10.7 MHz	A	22a	f = 10.7 MHz V_{CC} = +0V R_L = 1 kΩ	Cascode	-	40	-	dB	22b
			22c		Diff.-Ampl.	-	30	-		22d
Peak-to-Peak Output Current		P-P		V_{CC} = +9V V_{CC} = +12V	f = 10.7 MHz e_{in} = 400 mV Diff.-Ampl.	2	4	7	mA	
						3.5	6	10		

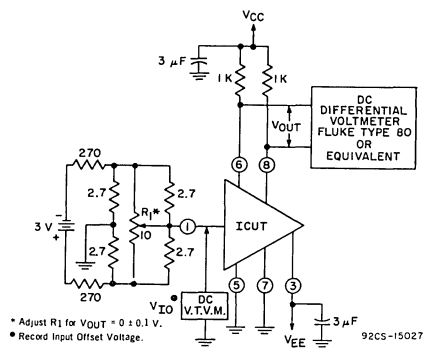


Fig.2 - Input offset voltage test circuit for CA3028B.

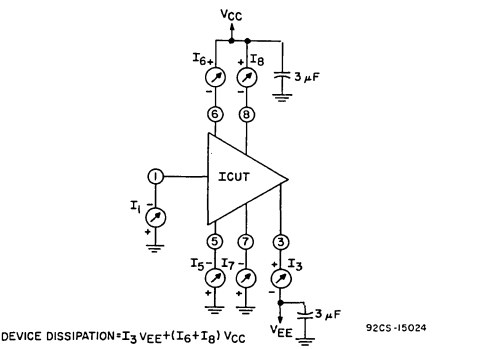


Fig.3a - Input offset current, input bias current, device dissipation, and quiescent operating current test circuit for CA3028A and CA3028B.

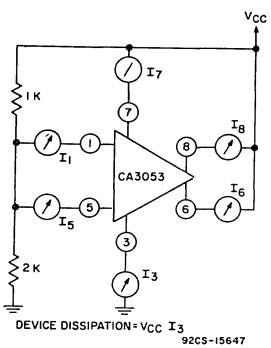


Fig.3b - Input bias current, device dissipation, and quiescent operating current test circuit for CA3053.

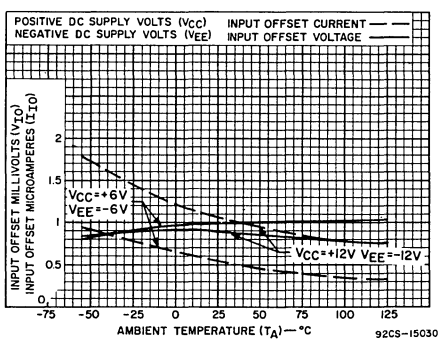


Fig.4 - Input offset voltage and input offset current for CA3028B.

TYPICAL CHARACTERISTICS

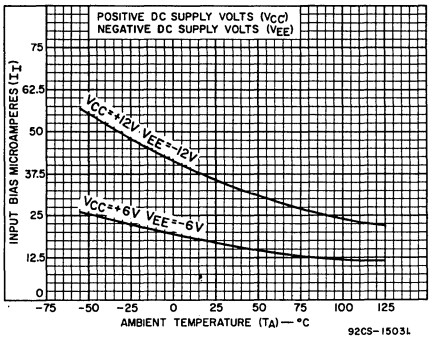


Fig.5a - Input bias current vs. ambient temperature for CA3028A and CA3028B.

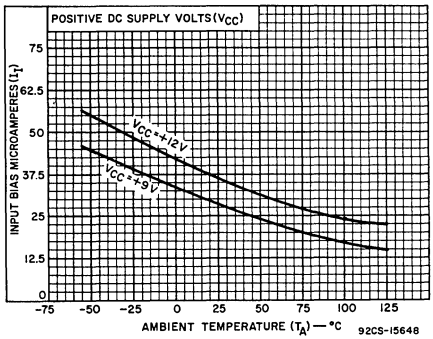


Fig.5b - Input bias current vs. ambient temperature for CA3053.

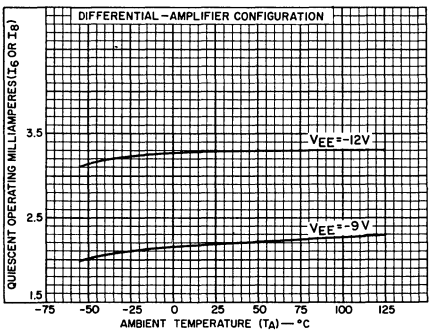


Fig.6a - Quiescent operating current vs. ambient temperature for CA3028A and CA3028B.

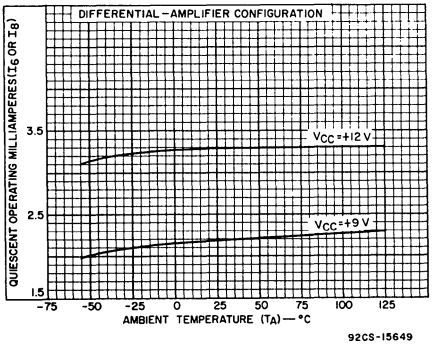


Fig.6b - Quiescent operating current vs. ambient temperature for CA3053.

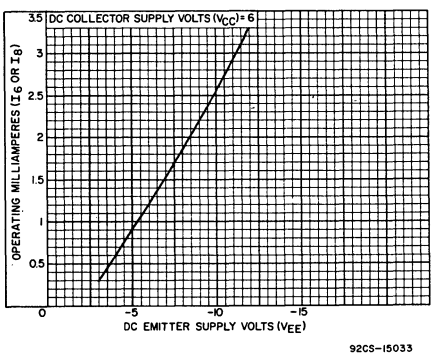


Fig.7 - Operating current vs. V_{EE} voltage for CA3028A and CA3028B.

TYPICAL CHARACTERISTICS AND TEST CIRCUITS

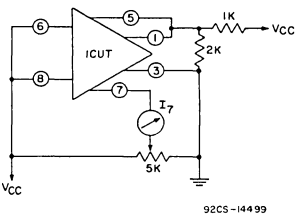


Fig.8a - AGC bias current test circuit (differential-amplifier configuration) for CA3028A and CA3028B.

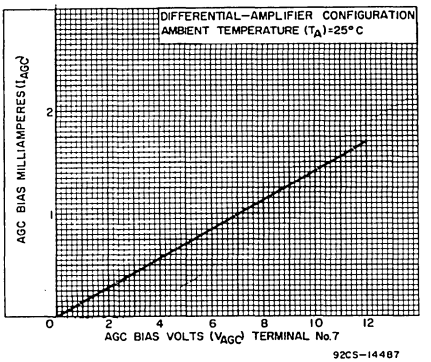


Fig.8b - AGC bias current vs. bias volts (terminal No.7) for CA3028A and CA3028B.

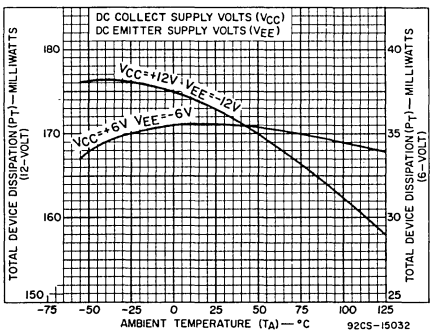
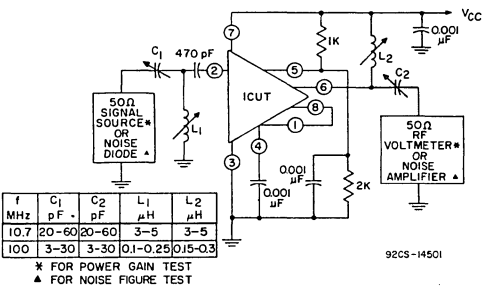


Fig.9 - Device dissipation vs. temperature for CA3028A and CA3028B.



f MHz	C1 pF	C2 pF	L1 μH	L2 μH
10.7	20-60	20-60	3-5	3-5
100	3-30	3-30	0.1-0.25	0.15-0.3

* FOR POWER GAIN TEST
▲ FOR NOISE FIGURE TEST

Fig.10a - Power gain and noise figure test circuit (cascode configuration) for CA3028A, CA3028B and CA3053.*

* 10.7 MHz Power Gain Test Only.

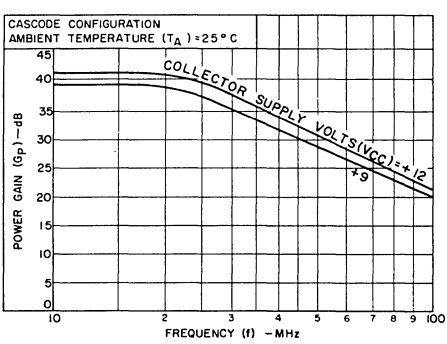


Fig.10b - Power gain vs. frequency (cascode configuration) for CA3028A and CA3028B.

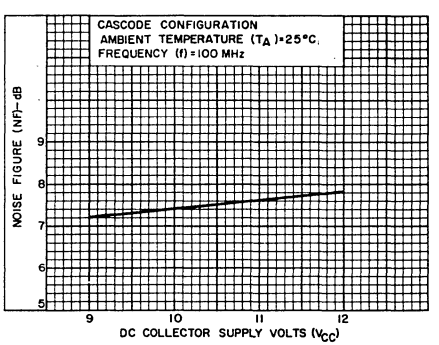
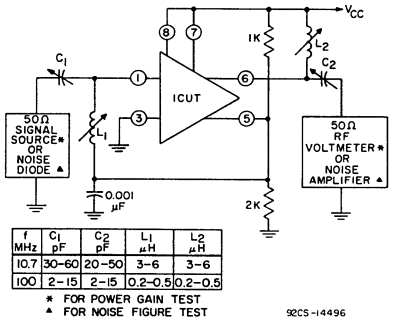


Fig.10c - 100 MHz noise figure vs. collector supply volts (cascode configuration) for CA3028A and CA3028B.

TYPICAL NOISE FIGURE AND POWER GAIN TEST CIRCUITS AND CHARACTERISTICS



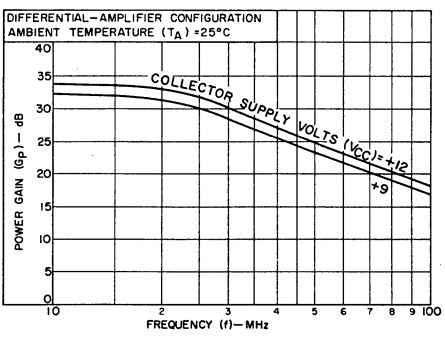
f MHz	C_1 pF	C_2 pF	L_1 μH	L_2 μH
10.7	30-60	20-50	3-6	3-6
100	2-15	2-15	0.2-0.5	0.2-0.5

* FOR POWER GAIN TEST
▲ FOR NOISE FIGURE TEST

92CS-14496

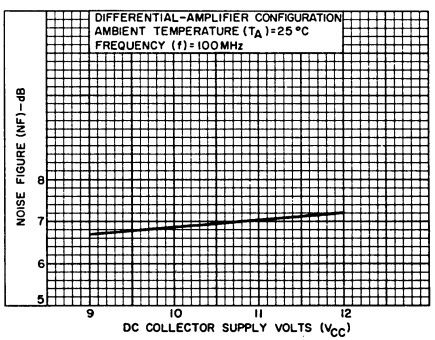
Fig.11a - Power gain and noise figure test circuit (differential-amplifier configuration and terminal No.7 connected to V_{CC}) for CA3028A, CA3028B and CA3053.*

* 10.7 MHz Power Gain Test Only.



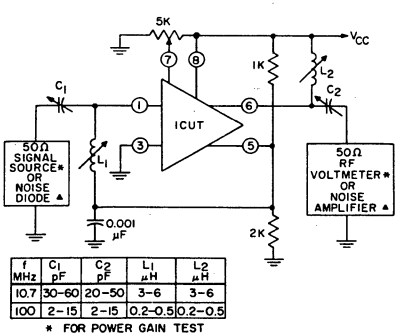
92CS-14495

Fig.11b - Power gain vs. frequency (differential-amplifier configuration) for CA3028A and CA3028B.



92CS-14485

Fig.11c - 100 MHz noise figure vs. collector supply voltage (differential-amplifier configuration) for CA3028A and CA3028B.

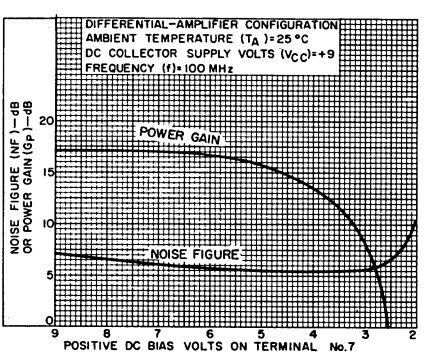


f MHz	C_1 pF	C_2 pF	L_1 μH	L_2 μH
10.7	30-60	20-50	3-6	3-6
100	2-15	2-15	0.2-0.5	0.2-0.5

* FOR POWER GAIN TEST
▲ FOR NOISE FIGURE TEST

92CS-14516

Fig.11d - Power gain and noise figure test circuit (differential-amplifier configuration for CA3028A and CA3028B.



92CS-14484

Fig.11e - 100 MHz noise figure and power gain vs. base-to-emitter bias (terminal No.7) for CA3028A and CA3028B.

TYPICAL ADMITTANCE PARAMETERS

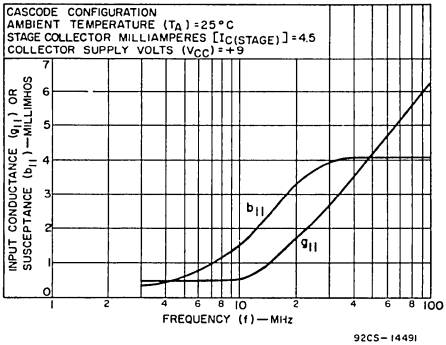


Fig.12 - Input admittance (Y_{11}) vs. frequency (cascode configuration) for CA3028A, CA3028B and CA3053.

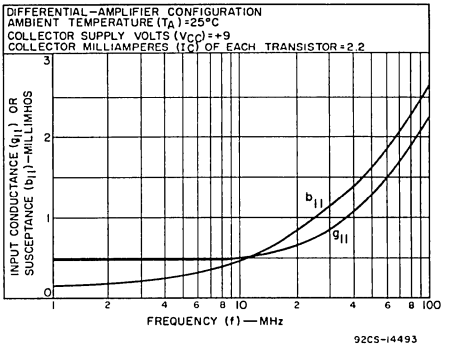


Fig.13 - Input admittance (Y_{11}) vs. frequency (differential-amplifier configuration) for CA3028A, CA3028B and CA3053.

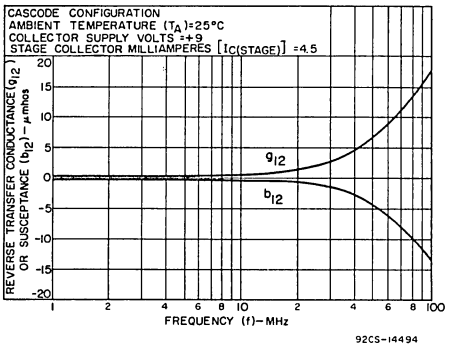


Fig.14 - Reverse transadmittance (Y_{12}) vs. frequency (cascode configuration) for CA3028A, CA3028B and CA3053.

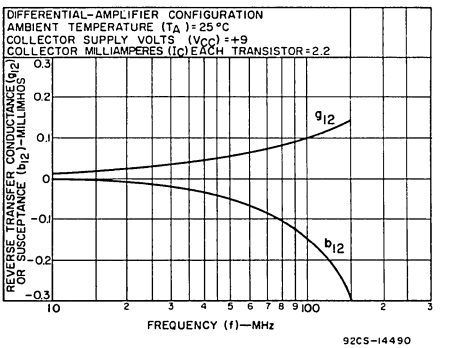


Fig.15 - Reverse transadmittance (Y_{12}) vs. frequency (differential-amplifier configuration) for CA3028A, CA3028B and CA3053.

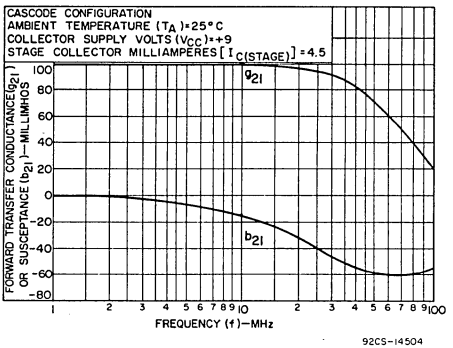


Fig.16 - Forward transadmittance (Y_{21}) vs. frequency (cascode configuration) for CA3028A, CA3028B and CA3053.

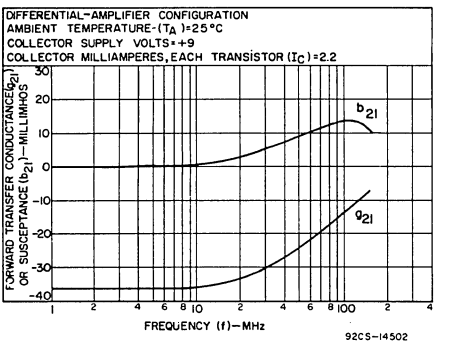


Fig.17 - Forward transadmittance (Y_{21}) vs. frequency (differential-amplifier configuration) for CA3028A, CA3028B and CA3053.

TYPICAL ADMITTANCE PARAMETERS

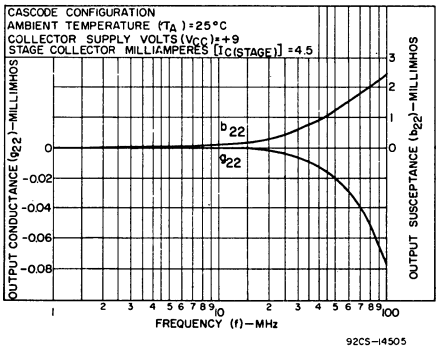


Fig.18 - Output admittance (Y_{22}) vs. frequency (cascode configuration) for CA3028A, CA3028B and CA3053.

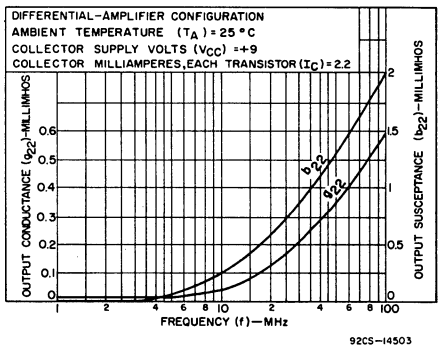


Fig.19 - Output admittance (Y_{22}) vs. frequency (differential-amplifier configuration) for CA3028A, CA3028B and CA3053.

TYPICAL TEST CIRCUITS AND CHARACTERISTICS

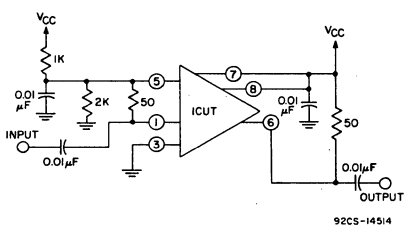


Fig.20a - Output power test circuit for CA3028A and CA3028B.

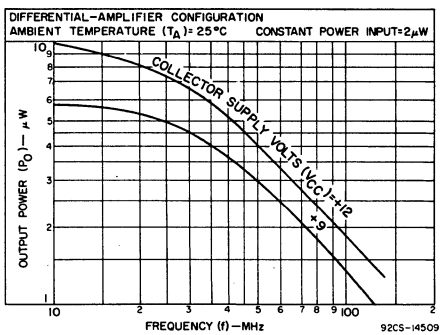
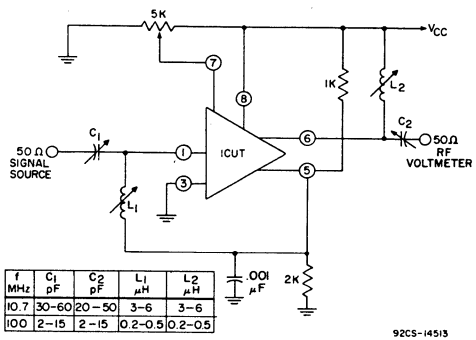


Fig.20b - Output power vs. frequency — 50 Ω input and 50 Ω output (differential-amplifier configuration) for CA3028A and CA3028B.



f MHz	C_1 pF	C_2 pF	L_1 μH	L_2 μH
10.7	30-60	20-50	3-6	3-6
100	2-15	2-15	0.2-0.5	0.2-0.5

Fig.21a - AGC range test circuit (differential amplifier) for CA3028A and CA3028B.

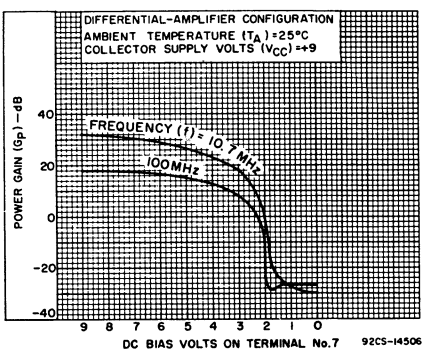


Fig.21b - AGC characteristics for CA3028A and CA3028B.

TEST CIRCUITS AND TYPICAL CHARACTERISTICS

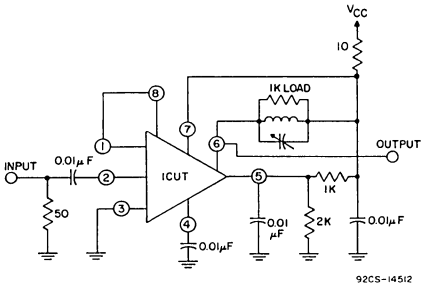


Fig.22a - Transfer characteristic (voltage gain) test circuit (10.7 MHz) cascode configuration for CA3028A, CA3028B and CA3053.

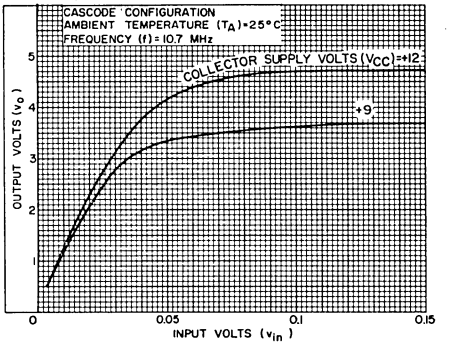


Fig.22b - Transfer characteristics (cascode configuration) for CA3028A, CA3028B and CA3053.

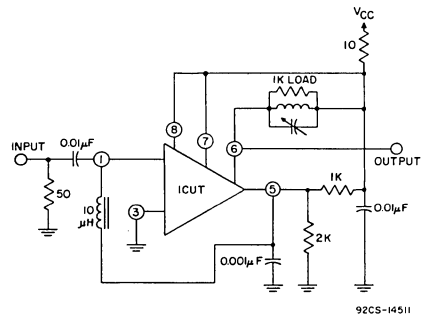


Fig.22c - Transfer characteristic (voltage gain) test circuit (10.7 MHz) differential-amplifier configuration for CA3028A, CA3028B and CA3053.

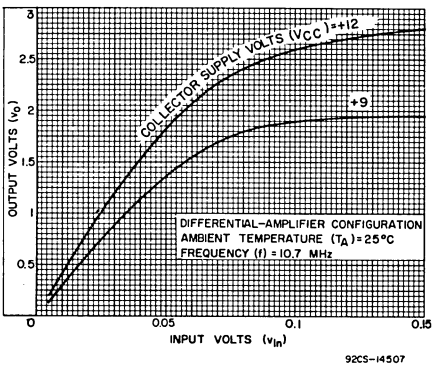
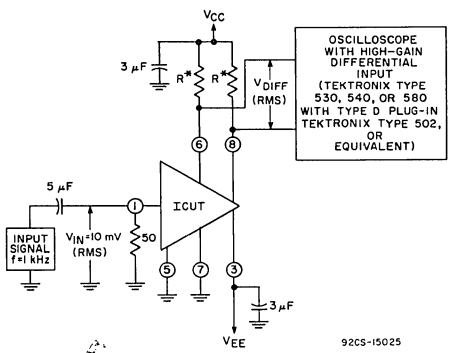
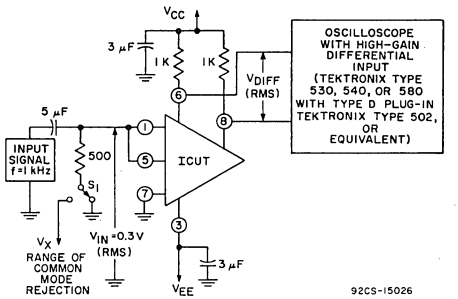


Fig.22d - Transfer characteristics (differential-amplifier configuration) for CA3028A, CA3028B and CA3053.



* For R = 1.6 kΩ – (V_{CC} = 12V, V_{EE} = -12V)
For R = 2 kΩ – (V_{CC} = 6V, V_{EE} = -6V)

Fig.23 - Differential voltage gain, maximum peak-to-peak output voltage, and bandwidth test circuit for CA3028B.



For CMR test: S_1 to ground
For input common-mode voltage range test: S_1 to V_X

$$\text{Common mode rejection ratio} = 20 \log_{10} \frac{(A^*)\ (2)\ (0.3)}{V_{DIFF}\ (RMS)}$$

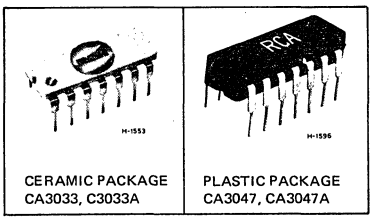
* A = Single-ended voltage gain.

Fig.24 - Common-mode rejection ratio and common-mode input-voltage range test circuit for CA3028B.

RCA Solid State Division

Linear Integrated Circuits

CA3033 CA3033A CA3047 CA3047A



CERAMIC PACKAGE
CA3033, C3033A

PLASTIC PACKAGE
CA3047, CA3047A

Operational Amplifiers

For High-Output-Current Applications

APPLICATIONS

- Comparator
- Integrator
- Differentiator
- Audio Amplifier
- Summing Amplifier
- Servo Driver
- DC Amplifier
- Multivibrator
- Narrow Band and Band Pass Amplifier

FEATURES

	CA3033 CA3047 $V^+ = +12$ V $V^- = -12$ V	CA3033A CA3047A $V^+ = 15$ V $V^- = -15$ V	
Output Current	36	76	mA min.
Input Offset Current	35	25	nA max.
Open Loop Differential Gain	84	87	dB min.
Output Voltage Swing.	18	23	V_{p-p} min.
Input Bias Current	350	180	nA max.
Power Output	80	220	mW min.
Common Mode Rejection Ratio	84	93	dB min.

RCA-CA3033 is a high-performance integrated circuit operational amplifier featuring high input impedance, high gain, high power output, and low input-offset voltage and current. The device consists of two differential amplifiers in cascade and a single-ended class-B power output stage on a single monolithic silicon chip.

RCA-CA3033A has all the superior features and characteristics of the CA3033 but, in addition, can be operated at higher supply voltages to provide higher gain, higher common mode rejection, greater maximum output voltage swing, and more than double the power output.

RCA-CA3033 and CA3033A are hermetically sealed in 14-lead "dual-in-line" ceramic packages and are designed for operation over the full military temperature range of −55 °C to +125 °C.

The RCA-CA3047 and CA3047A are electrically identical to the CA3033 and CA3033A, respectively, but are limited in operating and storage temperature range.

The RCA-CA3047 and CA3047A are supplied in 14-lead, "dual-in-line" plastic packages and are designed to operate over the temperature range of 0 °C to +70 °C, ambient.

Companion Application Note, ICAN-5641 "Application of RCA CA3033 and CA3033A High Performance Integrated-Circuit Operational Amplifiers."

The resistance values included on the schematic diagram have been supplied as a convenience to assist Equipment Manufacturers in optimizing the selection of "outboard" components of equipment designs. The values shown may vary as much as ±30%.

RCA reserves the right to make any changes in the Resistance Values provided such changes do not adversely affect the published performance characteristics of the device.

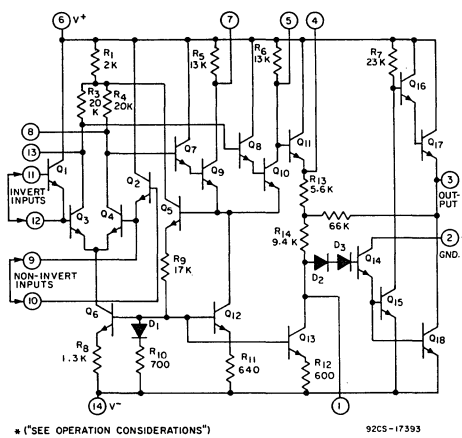


Fig. 1 - Schematic diagram of operational amplifiers, CA3033, CA3033A, CA3047, CA3047A.

ABSOLUTE-MAXIMUM RATINGS

	CA3033	CA3033A	CA3047	CA3047A
INPUT SIGNAL VOLTAGE	±10 V	−13 V, +10 V	±10 V	−13V, +10 V
DEVICE DISSIPATION:				
Up to T_A = 25 °C	1.2 W	1.2 W	750 mW	750 mW
Above T_A = 25 °C	Derate at 8 mW/°C		Derate at 6.67 mW/°C	
TEMPERATURE RANGE:				
Operating	−55 °C to +125 °C		0 °C to +70 °C	
Storage	−65 °C to +150 °C		−65 °C to +150 °C	
LEAD TEMPERATURE (During Soldering):				
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max.	+265 °C			

MAXIMUM VOLTAGE RATINGS at T_A = 25° C — CA3033, CA3047

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range of the vertical terminal 1 with respect to the horizontal terminal 14 is 0 to +4 volts.

TERMINAL No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1		*	*	*	*	*	*	*	*	*	*	*	*	+4 0
2			*	*	*	*	*	*	*	*	*	*	*	+26 0
3				*	*	0 -26	*	*	*	*	*	*	*	+26 0
4					+5 -1	0 -15	*	*	*	*	*	*	*	+26 0
5						0 -26	*	+20 -1 Note 1	*	*	*	*	+20 -1 Note 1	*
6							+26 0	+26 0	+26 0	+26 0	+26 0	+26 0	+26 0	+26 0
7								+20 -2 Note 1	*	*	*	*	+20 -2 Note 1	+26 0
8									+20 -1 Note 2	+20 -2 Note 3	+20 -2 Note 3	+20 -1 Note 2	*	+26 0
9										+1 -5	*	+5 -5	+1 -20 Note 2	+26 -5
10											+10 -10	*	+2 -20 Note 3	+26 -10
11												+1 -5	+2 -20 Note 3	+26 -10
12													+1 -20 Note 2	+26 -5
13														*
14														Sub- strate

MAXIMUM CURRENT RATINGS

CA3033 CA3047
CA3033A CA3047A

TERMINAL No.	I_{IN} mA	I_{OUT} mA
1	5	5
2	20	-
3	50	50
4	10	10
5	5	5
6	-	-
7	5	5
8	1	1
9	1	0.1
10	1	0.1
11	1	0.1
12	1	0.1
13	1	1
14	-	-

Notes: 1. This rating applies to the more positive terminal of terminals 8 and 13.
2. This rating applies to the more positive terminal of terminals 9 and 12.
3. This rating applies to the more positive terminal of terminals 10 and 11.

*Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe if the specified limits between all other terminals are not exceeded.

MAXIMUM VOLTAGE RATINGS at T_A = 25° C — **CA3033A, CA3047A**

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range of the vertical terminal 1 with respect to the horizontal terminal **14** is **0** to **+4** volts.

TERMINAL No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1		*	*	*	*	*	*	*	*	*	*	*	*	+4 0
2			*	*	*	*	*	*	*	*	*	*	*	+38 0
3				*	*	0 -38	*	*	*	*	*	*	*	+38 0
4					+5 -1	0 -22	*	*	*	*	*	*	*	+38 0
5						0 -38	*	+30 -1 Note 1	*	*	*	*	+30 -2 Note 1	*
6							+38 0	+38 0	+38 0	+38 0	+38 0	+38 0	+38 0	+38 0
7								+30 -2 Note 1	*	*	*	*	+20 -2 Note 1	+38 0
8									+30 -1 Note 2	+30 -2 Note 3	+30 -2 Note 3	+30 -1 Note 2	*	+38 0
9										+1 -5	*	+5 -5	+1 -30 Note 2	+38 -5
10											+10 -10	*	+2 -20 Note 3	+38 -10
11												+1 -5	+2 -30 Note 3	+38 -10
12													+1 -30 Note 2	+38 -5
13														*
14														Sub-strate

Notes: See CA3033, CA3047 Rating Chart Notes.

MAXIMUM CURRENT RATINGS are identical for all four types (See CA3033, CA3047 chart)

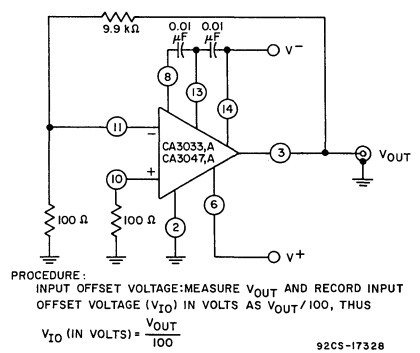


Fig. 2a - Input offset voltage, input offset voltage sensitivity, and device dissipation test circuit.

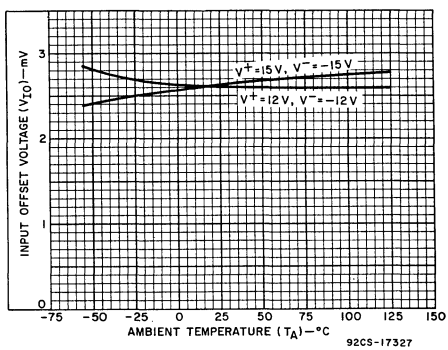


Fig. 2b - Typical input offset voltage vs. ambient temperature.

ELECTRICAL CHARACTERISTICS
For Equipment Design

Characteristics	Symbols	Test Conditions			Typical Characteristics Curves	LIMITS						Units
						CA3033 CA3047			CA3033A CA3047A			
		Circuit	T_A = 25° C			DC Supply Voltage V^+ = 12 V V^- = −12 V			V^+ = 15 V V^- = −15 V			
		Fig.			Fig.	Min.	Typ.	Max.	Min.	Typ.	Max.	
Input Offset Voltage	V_{IO}	2a			2b	–	2.6	5	–	2.9	5	mV
Input Offset Current	I_{IO}	3a			3b	–	5	35	–	9	25	nA
Input Bias Current	I_I	3a			3c	–	70	350	–	100	180	nA
Input Offset Voltage Sensitivity:												
Positive	$\triangle V_{IO}/\triangle V^+$	2a			–	–	0.3	0.5	–	0.2	0.5	mV/V
Negative	$\triangle V_{IO}/\triangle V^-$	2a			–	–	0.3	0.5	–	0.2	0.5	mV/V
Device Dissipation	P_T	2a			–	60	120	180	80	170	300	mW
Open-Loop Differential Voltage Gain	A_{OL}	–	f = 1 kHz		4	84	90	–	87	93	–	dB
Common-Mode Rejection Ratio	CMRR	–			5	84	100	–	93	105	–	dB
Common-Mode Input-Voltage Range	V_{ICR}	–			–	−7.5	+5,−9	+3.5	−9.7	6,−11	4.7	V
Maximum Output-Voltage Swing	V_O(P-P)	–	f = 1 kHz	R_L = 500 Ω	–	18	22	–	–	–	–	V_{P-P}
				R_L = 300 Ω		–	–	–	23	25	–	
Input Impedance	Z_I	–			–	0.25	1.5	–	0.6	1	–	MΩ
Output Current	I_O	–		R_L = 500 Ω	6	35	44	–	–	–	–	mA-(P-P)
				R_L = 300 Ω		–	–	–	76	83	–	
Power Output THD <5%	P_o	–		R_L = 500 Ω	7	80	122	–	–	–	–	mW
				R_L = 300 Ω		–	–	–	220	255	–	

ELECTRICAL CHARACTERISTICS
Typical Values Intended Only for Design Guidance

Input Offset Voltage Drift −55° C to 125° C	$V_{IO}/\triangle T$	2a			2b	–	6.6	–	–	6.6	–	μV/°C
Input Offset Current Drift												
−55° C to 25° C	$I_{IO}/\triangle T$	3a			3b	–	1	–	–	1	–	nA/°C
25° C to 125° C						–	0.08	–	–	0.08	–	
60-dB Amplifier Bandwidth	BW	8a	C_X, C_Y = 0.001 μF		8b,c	–	230	–	–	350	–	kHz
Slew Rate	SR	9	(amplifier circuit only)		–	–	2.7	–	–	3	–	V/μs

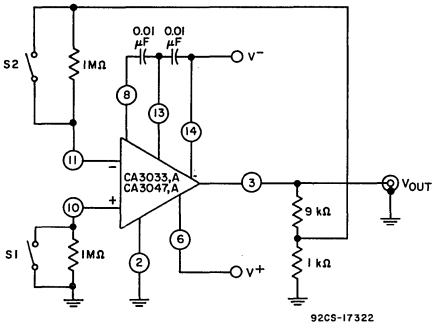


Fig. 3a - Input offset current and input bias current test circuit.

PROCEDURES:

A. Inverting Input Current

Set switch, S_1 in closed position and set switch, S_2 in open position.

Measure output voltage and convert this reading to inverting input current using the following relation:

$$I_I \text{ inverting (in } \mu A) = \frac{V_{OUT} \text{ (in volts)}}{10}$$

B. Non-inverting Input Current

Set switch, S_1 in open position and set switch, S_2 in closed position.

Measure output voltage and convert this reading to non-inverting input current using the following relation:

$$I_I \text{ non-inverting (in } \mu A) = \frac{-V_{OUT} \text{ (in volts)}}{10}$$

C. Input Offset Current

Set switches, S_1 and S_2 in open positions.

Measure output voltage and convert this reading to input offset current using the following relation:

$$I_{IO} \text{ (in } \mu A) = \frac{V_{OUT} \text{ (in volts)}}{10}$$

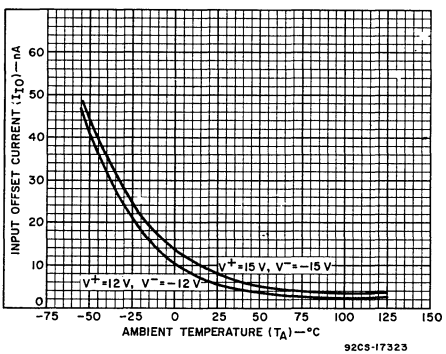


Fig. 3b - Typical input offset current vs. ambient temperature.

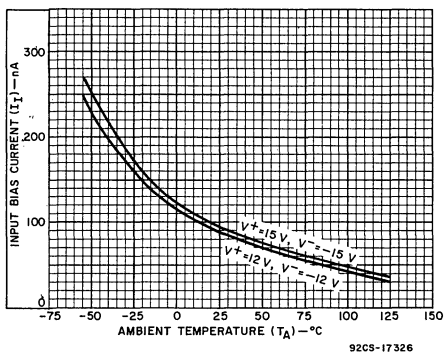


Fig. 3c - Typical input bias current vs. ambient temperature.

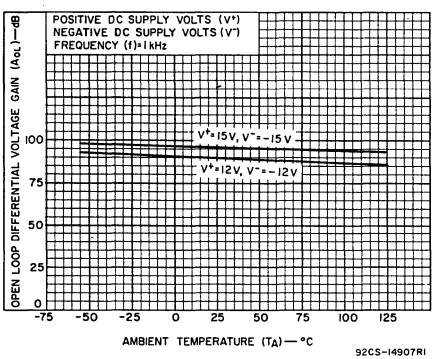


Fig. 4 - Typical open-loop differential voltage gain vs. ambient temperature.

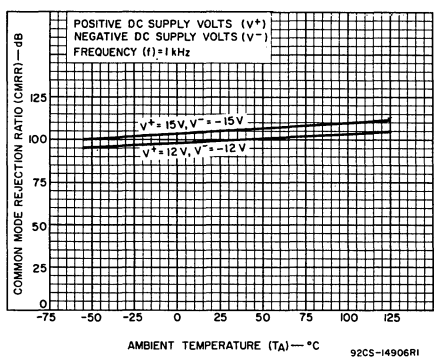


Fig. 5 - Typical common mode rejection ratio vs. ambient temperature.

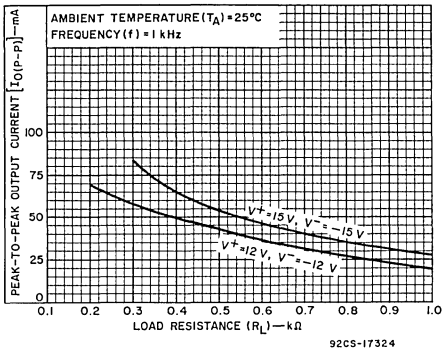


Fig. 6 - Typical peak-to-peak output current vs. load resistance.

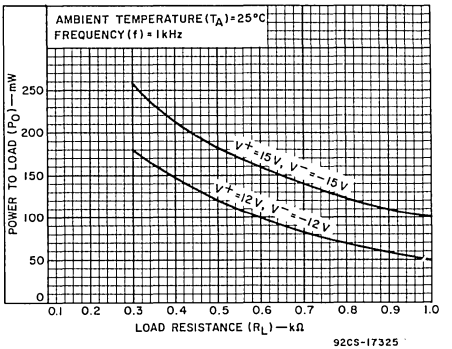


Fig. 7 - Typical power output vs. load resistance.

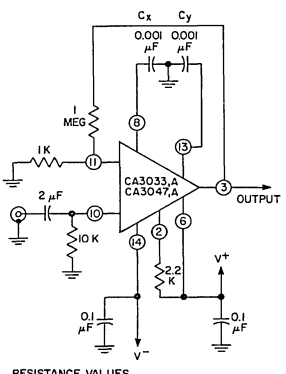


Fig. 8a - Typical 60-dB amplifier.

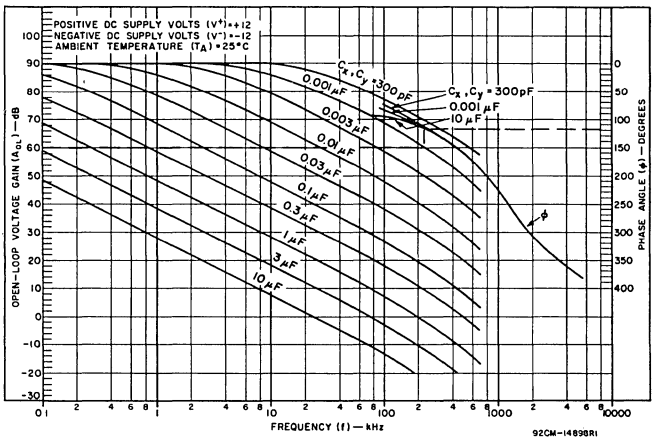


Fig. 8b - Typical phase compensation characteristics for CA3033, CA3047 (V^+ = +12 V, V^- = −12 V)

For any desired closed loop gain (in decibels), read horizontally along the gain line to the attenuation curve which provides the desired closed loop bandwidth. The required values for the compensation capacitors is shown on the curve. Move vertically from the intersection of the gain and attenuation lines until the phase angle curve (ϕ) is reached and read the phase angle between the input and output on the right-hand scale. The difference between the indicated phase angle and 180° is the typical phase margin. (A minimum phase margin of 45° is recommended to allow for component variations and differences among amplifiers.) If the phase margin is smaller than required, the desired bandwidth can be stably achieved through the use of a more complex feedback network. As the closed loop gain approaches unity, the compensating capacitors required (0.3 μF to 1.0 μF) are bulky and costly. A capacitor one-half the value shown on the chart, connected between terminals 8 and 13, and a 0.001 μF capacitor from either terminals 8 or 13 to ground or V^- is an acceptable alternative method. This arrangement provides the same gain-phase roll-off shown on the curves and permits the use of more readily available, lower-voltage disc capacitors which are smaller and cost less. For linear operation, the maximum expected difference voltage between the two collectors is less than 1 volt.

Figure 8a shows the phase compensating capacitors (C_x, C_y) returned to ground. In some systems with large parasitic impedances in the power supply system, returning these capacitors to the negative (V^-) supply may result in more stable operation.

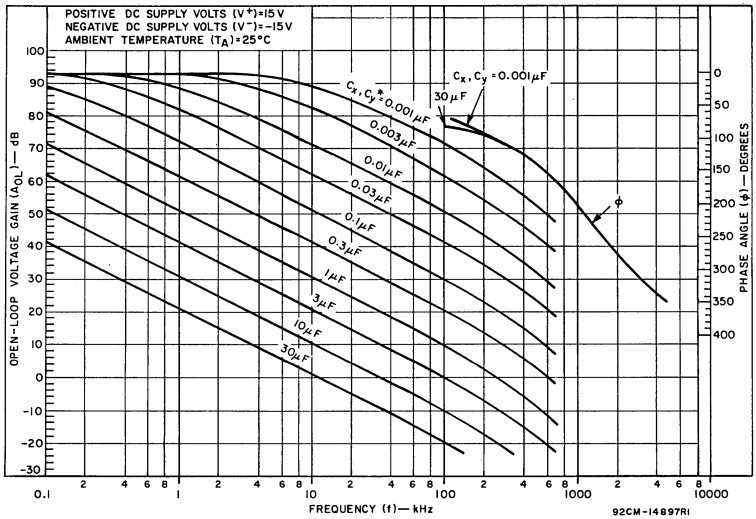
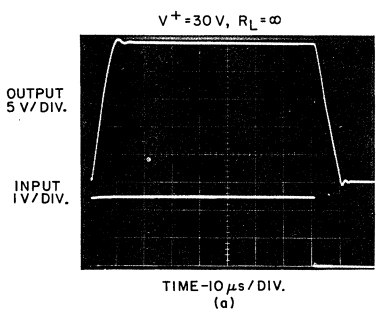
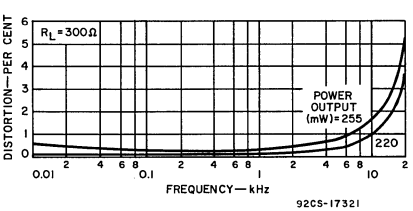
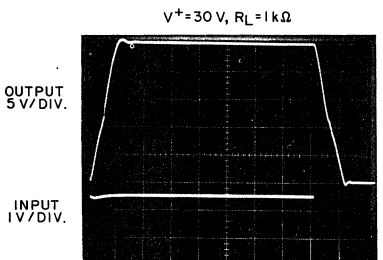


Fig. 8c - Typical phase compensation characteristics for CA3033A, CA3047A (V^+ = 15 V, V^- = −15 V).







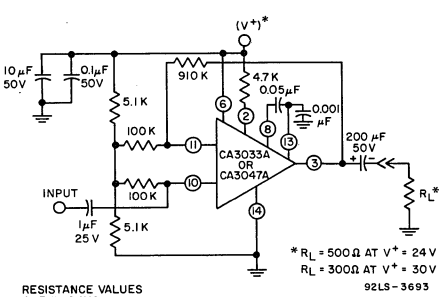


Fig. 9 - Amplifier with single voltage supply and associated pulse response waveforms and distortion curves.

OPERATING CONSIDERATIONS

The CA3033, CA3033A, CA3047, and CA3047A operational amplifiers have very high peak-pulse current capability. The open-loop output impedance is typically less than 30 ohms at 10 kHz and the peak short-circuit output current may exceed 100 milliamperes. To prevent possible damage to the chip because of excessive dissipation it is important that the output stage is not subjected to sustained high peak currents. To minimize the possibility of damage from accidental shorts, it is recommended that a 51-ohm resistor be placed in series with the output circuit.

When high peak output currents are required of the amplifier, it is desirable to provide a current-limiting resistor of about 2200 ohms in series with the collector of transistor Q_{14}. This resistor may be returned to ground, or, if its value is increased to 4700 ohms; it may be returned to the V^+ terminal.

RCA
Solid State
Division

Linear Integrated Circuits

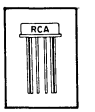
CA3035
CA3035V1

Ultra-High-Gain
Wide-Band Amplifier Array

Monolithic Silicon

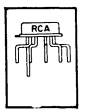
- Three Individual General-Purpose Amplifiers
- Ideal for service in Remote-Control Amplifiers — — e.g., TV Receivers
- Available in two electrically identical versions: CA3035 with straight leads; CA3035V1 with formed leads

CA3035



10-LEAD
TO-5

CA3035V1



FORMED-LEAD
10-LEAD TO-5

HIGHLIGHTS

- Three separate amplifiers — gain and bandwidth for each amplifier can be adjusted with suitable external circuitry
- Amplifiers operable independently or in cascade
- Exceptionally high cascade voltage gain — 129 dB typ. at 40 kHz
- Low noise performance
- Wide-band response
- All amplifiers single-ended — only one power supply required
- Wide operating temperature range — -55°C to +125°C
- Built-in temperature compensation
- Hermetically sealed, all-welded 10-lead TO-5-style metal package with straight or formed leads

SCHEMATIC DIAGRAM FOR CA3035 AND CA3035V1

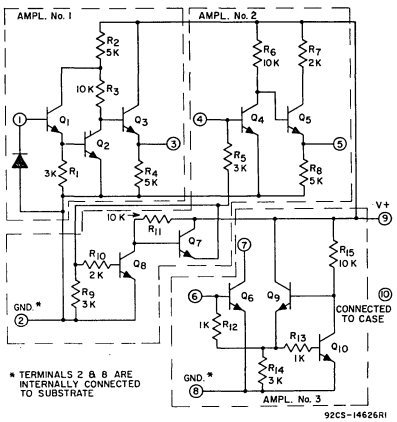


Fig. 1

TYPICAL REMOTE CONTROL SYSTEM

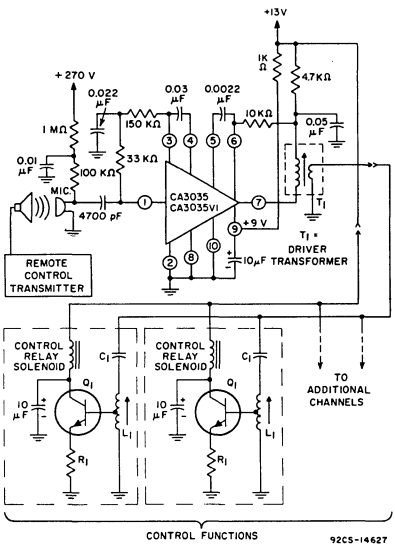


Fig. 2

ABSOLUTE-MAXIMUM RATINGS:

Operating Temperature Range ..... -55°C to +125°C
Storage Temperature Range ..... -65°C to +150°C
Device Dissipation ..... 300 mW
Input Voltage ..... 1 V p-p
Supply Voltage ..... +15V
Lead Temperature (During Soldering):
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm)
from case for 10 seconds max. ..... +265°C

ELECTRICAL CHARACTERISTICS AT T_A = 25°C

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS	TEST CIRCUITS AND CHARACTERISTICS CURVES	LIMITS CA3035, CA3035V1 Min.	Typ.	Max.	UNITS
STATIC CHARACTERISTICS							
Quiescent Operating Voltage	V3 V5 V7	V_{CC} = +9V	Fig.3	– – –	2 1.9 4.9	– – –	V V V
Total Current Drain	I_d	V_{CC} = +9V, R_{L3} = 5KΩ	Fig.3	3.5	5	7.5	mA
DYNAMIC CHARACTERISTICS							
Voltage Gain: Amplifier No.1 Amplifier No.2 Amplifier No.3	 A_1 A_2 A_3	 f = 40 kHz, V_{CC} = +9V		 40 40 38	 44 46 42	 – – –	 dB dB dB
Output Voltage Swing	V_{out} V_{1out} V_{2out} V_{3out}	 R_{L1} = 10KΩ R_{L2} = 10KΩ R_{L3} = 5KΩ Sinusoidal Output, V_{CC} = +9V		 – – –	 2 2.6 8	 – – –	 Vp-p Vp-p Vp-p
Input Resistance: Amplifier No.1 Amplifier No.2 Amplifier No.3	 R_{1in} R_{2in} R_{3in}	 f = 40 kHz		 – – –	 50K 2K 670	 – – –	 Ω Ω Ω
Output Resistance	R_{1out} R_{2out} R_{3out}	f = 40 kHz		– – –	270 170 100K	– – –	Ω Ω Ω
Bandwidth at -3dB point: Amplifier No.1 Amplifier No.2 Amplifier No.3	 BW_1 BW_2 BW_3	 V_{CC} = +9V	 Fig.5 Fig.6 Fig.7	 – – –	 500 2.5 2.5	 – – –	 kHz MHz MHz
Noise Figure Amplifier No.1	 NF_1	 f = 1 kHz, R_S = 1 KΩ	 Fig.4	 –	 6	 7	 dB
Sensitivity		V_{CC} = +13 V Relay (K_1) Current = 7.5 mA	Fig.2	–	100	150	μV

STATIC CHARACTERISTICS

TEST CIRCUIT

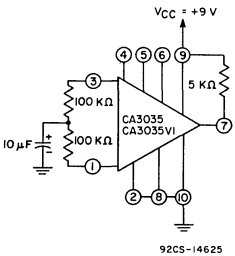
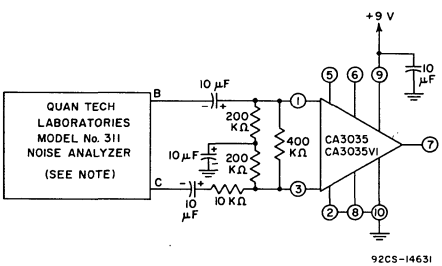


Fig.3

NOISE FIGURE TEST CIRCUIT



NOTE: SET ALL INTERNAL POWER SUPPLIES ON QUAN TECH NOISE ANALYZER TO ZERO VOLTS.

Fig.4

TYPICAL 1st-AMPLIFIER RESPONSE

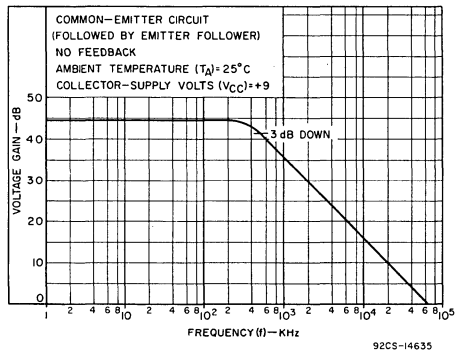


Fig.5

TYPICAL 2nd-AMPLIFIER RESPONSE

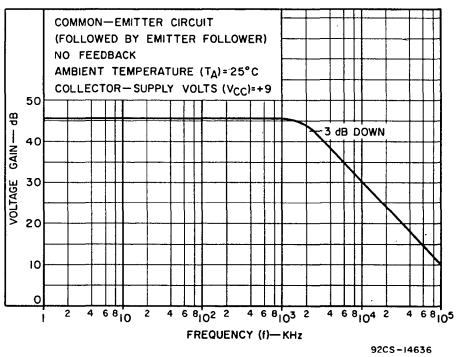


Fig.6

TYPICAL 3rd-AMPLIFIER RESPONSE

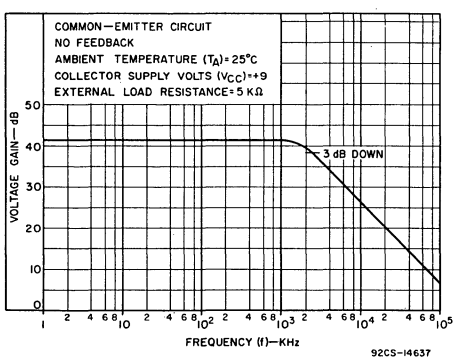


Fig.7

RCA
Solid State
Division

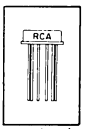
Linear Integrated Circuits

CA3036

DUAL DARLINGTON ARRAY

Monolithic Silicon

- Two independent low-noise wide-band amplifier channels
- Particularly useful for preamplifier and low-level amplifier applications in single-channel and stereo systems
- Wide application in low-noise industrial instrumentation amplifiers



10-Lead
TO-5

HIGHLIGHTS

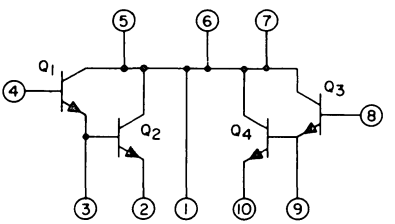
- Matched transistors with emitter-follower outputs
- Low-noise performance
- 200-MHz gain-bandwidth product
- Operation from -55°C to +125°C
- Hermetically sealed, all-welded 10-lead TO-5-style metal package

APPLICATIONS

- Stereo phonograph preamplifiers
- Low-level stereo and single channel ⑧ amplifier stages
- Low-noise, emitter-follower differential amplifiers
- Operational amplifier drivers

MAXIMUM RATINGS, Absolute-Maximum Values:

POWER DISSIPATION, P:		
Any one transistor	300 max.	mW
Total for array	600 max.	mW
TEMPERATURE RANGE:		
Operating	−55 to +125	°C
Storage	−65 to +150	°C
LEAD TEMPERATURE (During Soldering):		
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max.	+265	°C
The following ratings apply for each transistor in the array:		
Collector-to-Emitter Voltage, V_{CEO}	15 max.	V
Collector-to-Base Voltage, V_{CBO}	30 max.	V
Emitter-to-Base Voltage, V_{EBO}	5 max.	V
Collector Current, I_C	50 max.	mA



92CS-14624

Fig.1 — Schematic Diagram for CA3036.

ELECTRICAL CHARACTERISTICS, at $T_A = 25°C$

CHARACTERISTICS		SYMBOLS	TEST CONDITIONS	LIMITS TYPE CA3036 Min.	Typ.	Max.	UNITS
For Each Transistor (Q_1, Q_2, Q_3, Q_4)	Collector-Cutoff Current	I_{CBO}	$V_{CB}=5V$, $I_E=0$	--	--	0.5	μA
	Collector-Cutoff Current	I_{CEO}	$V_{CE}=10$ V, $I_B=0$	--	--	5	μA
	Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	$I_C=1$ mA, $I_B=0$	15	20	--	V
	Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	$I_C=10\,\mu A$, $I_E=0$	30	44	--	V
	Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	$I_E=10\,\mu A$, $I_C=0$	5	6	--	V
For Either Input Transistor (Q_1 or Q_3)	Static Forward Current-Transfer Ratio	h_{FE}	I_{C1} or $I_{C3}=1$ mA	30	82	--	--
For Either Darlington Pair (Q_1, Q_2 or Q_3, Q_4)	Emitter-to-Base Breadkown Voltage	$V_{(BR)EBO(D)}$	I_{E2} or $I_{E4}=10\,\mu A$	10	12.6	--	V
	Static Forward Current-Transfer Ratio	$h_{FE(D)}$	$I_{C1}+I_{C2}$ or $I_{C3}+I_{C4}$ $=1$ mA	1000	4540	--	--
For Each Input Transistor (Q_1 or Q_3)	Short-Circuit Forward Current-Transfer Ratio	h_{fe}	$f=1$ kHz, I_{C1} or $I_{C3}=1$ mA	--	82	--	--
	Short-Circuit Input Impedance	h_{ie}		--	2.6K	--	Ω
	Open-Circuit Output Admittance	h_{oe}		--	7	--	μmho
	Open-Circuit Reverse Voltage-Transfer Ratio	h_{re}		--	9.8×10^{-5}	--	--
For Either Darlington Pair (Q_1, Q_2 or Q_3, Q_4)	Short-Circuit Forward Current-Transfer Ratio	$h_{fe(D)}$	$f=1$ kHz, $I_{C1}+I_{C2}$ or $I_{C3}+I_{C4}$ $=1$ mA	--	1300	--	--
	Short-Circuit Input Impedance	$h_{ie(D)}$		--	82K	--	Ω
	Open-Circuit Output Admittance	$h_{oe(D)}$		--	108	--	μmho
	Open-Circuit Reverse Voltage-Transfer Ratio	$h_{re(D)}$		--	2.7×10^{-3}	--	--
	Voltage Gain	$A_{(D)}$		--	26	--	dB
	Power Gain	$G_{p(D)}$		--	47	--	dB
	Noise Voltage See Fig.3 for Test Circuit	E_N	$f=100$ Hz	--	0.2	3	$\mu V(rms)/\sqrt{f(Hz)}$
			$f=1$ kHz	--	0.05	0.3	
			$f=10$ kHz	--	0.012	0.1	
For Either Input Transistor (Q_1 or Q_3)	Forward Transfer Admittance	y_{fe}	$f=50$ MHz, I_{C1} or $I_{C3}=2$ mA	--	0.68 + j 7.9	--	mmho
	Input Admittance (Output Short-Circuited)	y_{ie}		--	4.14 + j 5.95	--	mmho
	Output Admittance (Input Short-Circuited)	y_{oe}		--	1.94 + j 2.64	--	mmho
	Reverse Transfer Admittance (Input Short-Circuited)	y_{re}		--	Negligible	--	mmho
For either Darlington Pair (Q_1, Q_2 or Q_3, Q_4)	Input Admittance (Output Short-Circuited)	$y_{ie(D)}$	$f=50$ MHz, $I_{C1}+I_{C2}$ or $I_{C3}+I_{C4}$ $=2$ mA	--	1.71 + j 2.8	--	mmho
	Output Admittance (Input Short-Circuited)	$y_{oe(D)}$		--	3.96 + j 2.6	--	mmho
	Gain-Bandwidth Product	$f_T(D)$		150	200	--	MHz

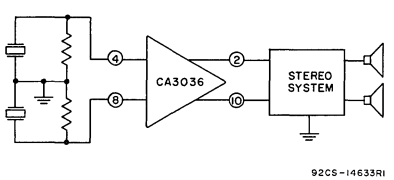


Fig.2 - Block Diagram of Stereo System using CA3036 as Phono Preamplifier.

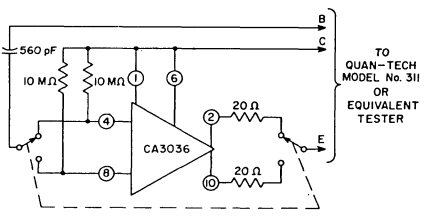


Fig.3 - Noise Voltage Test Circuit for CA3036.

RCA
Solid State
Division

Linear Integrated Circuits

CA3039

Diode Array

Six Matched Diodes on a Common Substrate
Monolithic Silicon

The RCA-CA3039 consists of six ultra-fast, low capacitance diodes on a common monolithic substrate. Integrated circuit construction assures excellent static and dynamic matching of the diodes, making the array extremely useful for a wide variety of applications in communication and switching systems.

Five of the diodes are independently accessible, the sixth shares a common terminal with the substrate.

For applications such as balanced modulators or ring modulators where capacitive balance is important, the substrate should be returned to a DC potential which is significantly more negative (with respect to the active diodes) than the peak signal applied.

ULTRA-FAST LOW-CAPACITANCE MATCHED DIODES

For Applications in Communications and Switching Systems



12-Lead TO-5

APPLICATIONS

- Balanced modulators or demodulators
- Ring modulators
- High speed diode gates
- Analog switches

FEATURES

- Excellent reverse recovery time – 1 ns typ.
- Matched monolithic construction – V_F matched within 5 mV
- Low diode capacitance – C_D = 0.65 pF typical at $V_R = -2$ V

ABSOLUTE MAXIMUM RATINGS AT $T_A = 25^oC$

DISSIPATION:

Any one diode unit	100 mW
Total for device	600 mW
For $T_A > 55^oC$	derate linearly 5.7 mW/°C

TEMPERATURE RANGE:

Operating	–55 to +125°C
Storage	–65 to +150°C

LEAD TEMPERATURE (During Soldering):

At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max.	+ 265°C
PEAK INVERSE VOLTAGE, PIV for: D_1–D_5	5 V
D_6	0.5 V
PEAK DIODE-TO-SUBSTRATE VOLTAGE, V_{DI} for D_1–D_5 (term. 1,4,5,8 or 12 to term. 10)	+ 20, –1 V
DC FORWARD CURRENT, I_F	25 mA
PEAK RECURRENT FORWARD CURRENT, I_f	100 mA
PEAK FORWARD SURGE CURRENT, $I_{f(surge)}$	100 mA

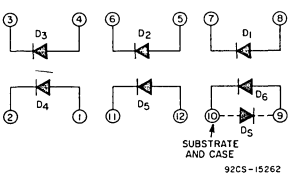


Fig. 1 - Schematic Diagram for CA3039

ELECTRICAL CHARACTERISTICS, at T_A = 25° C

Characteristics apply for each diode unit, unless otherwise specified.

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS	LIMITS			UNITS	CHARACTERISTIC CURVES
			MIN.	TYP.	MAX.		FIG.
DC Forward Voltage Drop	V_F	I_F = 50 μA	–	0.65	0.69	V	2
		1 mA	–	0.73	0.78	V	
		3 mA	–	0.76	0.80	V	
		10 mA	–	0.81	0.90	V	
DC Reverse Breakdown Voltage	$V_{(BR)R}$	I_R = −10 μA	5	7	–	V	–
DC Reverse Breakdown Voltage Between any Diode Unit and Substrate	$V_{(BR)R}$	I_R = −10 μA	20	–	–	V	–
DC Reverse (Leakage) Current	I_R	V_R = −4 V	–	0.016	100	nA	3
DC Reverse (Leakage) Current Between any Diode Unit and Substrate	I_R	V_R = −10 V	–	0.022	100	nA	4
Magnitude of Diode Offset Voltage (Difference in DC Forward Voltage Drops of any Two Diode Units)	$\lvert V_{F_1} - V_{F_2} \rvert$	I_F = 1 mA	–	0.5	5	mV	2
Temperature Coefficient of $\lvert V_{F_1} - V_{F_2} \rvert$	$\frac{\Delta \lvert V_{F_1} - V_{F_2} \rvert}{\Delta T}$	I_F = 1 mA	–	1	–	μV/°C	5
Temperature Coefficient of Forward Drop	$\frac{\Delta V_F}{\Delta T}$	I_F = 1 mA	–	−1.9	–	mV/°C	6
DC Forward Voltage Drop for Anode-to-Substrate Diode (D_S)	V_F	I_F = 1 mA	–	0.65	–	V	–
Reverse Recovery Time	t_{rr}	I_F = 10 mA, I_R = 10 mA	–	1	–	ns	–
Diode Resistance	R_D	f = 1 kHz, I_F = 1 mA	25	30	45	Ω	7
Diode Capacitance	C_D	V_R = −2 V, I_F = 0	–	0.65	–	pF	8
Diode-to-Substrate Capacitance	C_{DI}	V_{DI} = +4 V, I_F = 0	–	3.2	–	pF	9

TYPICAL CHARACTERISTICS

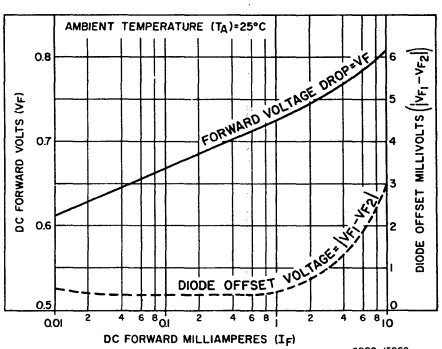


Fig. 2 - DC forward voltage drop (any diode) and diode offset voltage vs DC forward current

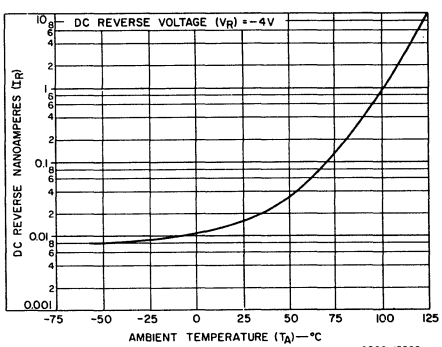


Fig. 3 - DC reverse (leakage) current (diodes 1,2,3,4,5) vs temperature

TYPICAL CHARACTERISTICS

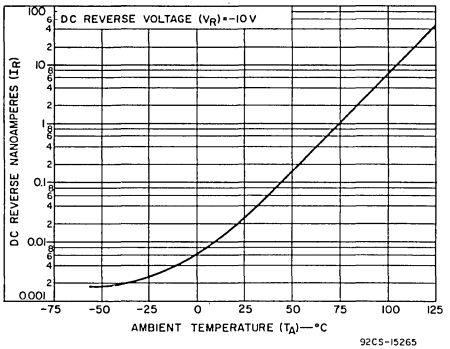


Fig. 4 - DC reverse (leakage) current between diodes (1,2,3,4,5) and substrate vs temperature

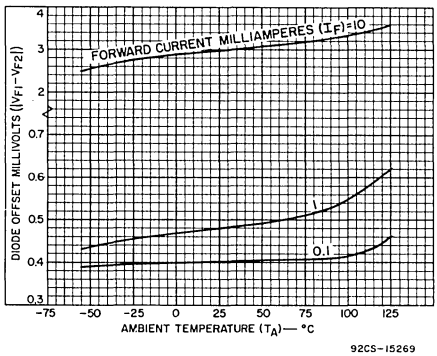


Fig. 5 - Diode offset voltage (any diode) vs temperature

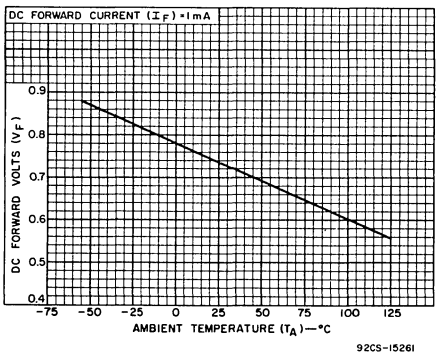


Fig. 6 - DC forward voltage drop (any diode) vs temperature

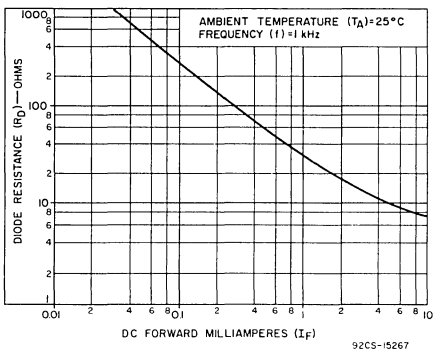


Fig. 7 - Diode resistance (any diode) vs DC forward current

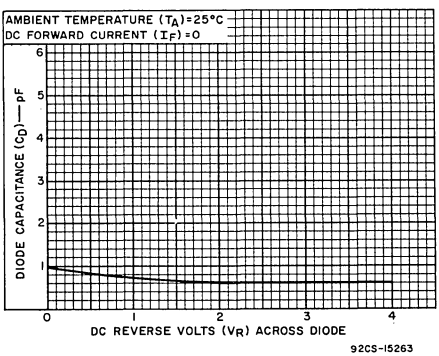


Fig. 8 - Diode capacitance (diodes 1,2,3,4,5) vs reverse voltage

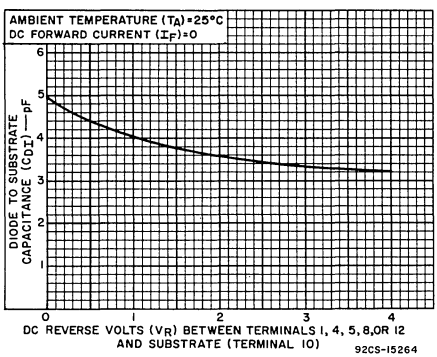


Fig. 9 - Diode-to-substrate capacitance vs reverse voltage

RCA
Solid State
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Linear Integrated Circuits

CA3040

The RCA CA3040 is a monolithic silicon integrated circuit designed to meet the requirements of a wide variety of applications requiring high gain and wide bandwidth. The cascode-connected differential amplifier achieves a double-ended gain of 37 dB with a typical 3 dB bandwidth of 55 MHz. Emitter-Follower input and output stages provide the desirable high input impedance and low output impedance for coupling to other circuits.

The CA3040 includes two biasing options, allowing the user to optimize his design over the entire military temperature range of -55 to +125°C. **Bias Mode A** yields a substantially constant voltage at the output terminals for applications using DC coupling to succeeding stages or requiring maximum dynamic range over the temperature range. DC output voltage varies less than 0.1 volt (typically) over the entire temperature range while gain varies ±2 dB. **Bias Mode B** provides extremely stable gain over the temperature range. Gain variation is 0 dB (typically) in this Bias Mode. DC variation is ±0.8 volt.

Provisions are also made for stabilizing the operating point for either single or split power supplies.

VIDEO and WIDE-BAND AMPLIFIER

For Industrial and Commercial Equipment at Frequencies up to 200 MHz



12-Lead TO-5

FEATURES

- High Differential Push-Pull Voltage Gain ..... 37 dB typ.
  Single-Ended Voltage Gain..... 31 dB typ.
- Wide (3dB) Bandwidth..... 55 MHz typ.
- Balanced Input and Output
- High Input Resistance ..... 150 kΩ typ.
- Low Output Resistance..... 125 Ω typ.
- Bias Options for Temperature Compensation:
  Bias Mode A: "Constant" Voltage
  Bias Mode B: "Constant" Gain

APPLICATIONS

- Video Amplifier
- Schmitt Trigger
- Modulator
- IF Amplifier
- Sense Amplifier
- Mixer
- DC Amplifier

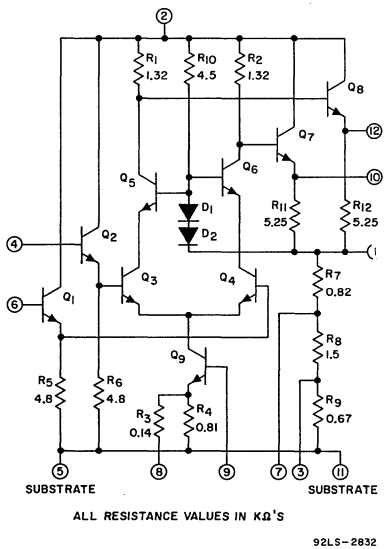


Fig.1 - Schematic Diagram for CA3040

ABSOLUTE-MAXIMUM RATINGS

DISSIPATION * . . . . . 450 mW
Derating factor for T_A > 85°C . . . . . 5 mW/°C
TEMPERATURE RANGE:
Operating . . . . . -55°C to +125°C
Storage . . . . . -65°C to +150°C

LEAD TEMPERATURE (During Soldering):

At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm)
from case for 10 seconds max. . . . . +265°C

* Limitation imposed by the thermal resistance of package.

MAXIMUM VOLTAGE RATINGS at T_A = 25°C

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range of the vertical terminal 2 with respect to terminal 11 is 0 to +14 volts.

TERMINAL No.	1	2	3	4	5▲	6	7	8	9	10	11▲	12
1		0 -14	*	*	+14 0	*	+10 -10	*	*	*	+14 0	*
2			*	+14 0	+14 0	+14 0	*	*	*	+14 0	+14 0	+14 0
3				*	+5 -3	*	*	*	*	*	+5 -3	*
4					*	+3 -3	*	*	*	*	*	*
5▲					▲	*	+10 -3	*	+3 -7	*	0 - Note 1	*
6							*	*	*	*	*	*
7								*	*	*	+10 -3	*
8									+3 -3	*	*	*
9										*	+7 -3	*
10											*	*
11▲											▲	*
12												

MAXIMUM CURRENT RATINGS

TERMINAL No.	I_{IN} mA	I_{OUT} mA
1	5	5
2	-	-
3	5	5
4	1	0.1
5	-	-
6	1	0.1
7	5	5
8	5	5
9	1	0.1
10	-	10
11	-	-
12	-	10

▲ Reference Substrate

Note 1: External connection required for proper operation.

* Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe if the specified limits between all other terminals are not exceeded.

ELECTRICAL CHARACTERISTICS AT T_A = 25°C Unless Otherwise Specified

Characteristics	Symbols	Test Circuits Fig.	Special Test Conditions	Limits Min.	Limits Typ.	Limits Max.	Units
STATIC CHARACTERISTICS	V_{CC} = +6V, V_{EE} = -6V						
Output Voltage	V_{10} or V_{12}	2(a) 2(b)	Bias Mode Switch A or B: Closed	1.4	2.7	3.7	V
Base Bias Voltage	V_9	2(a)	Bias Mode A Switch Closed	-	-1.7	-	V
		2(b)	Bias Mode B Switch Closed	-	-1.7	-	V
Input Bias Reference Voltage	V_1	2(a) 2(b)	Bias Mode Switch A or B: Open	-1	-	+1	V
Input Bias Current	I_4, I_6	2(a) 2(b)	Bias Mode Switch A or B: Closed	-	15	45	μA
Input Unbalance Current	$\lvert I_6 - I_4 \rvert$	2(a) 2(b)	Bias Mode Switch A or B: Closed	-	-	6	μA
Power Supply Current Drain	I_2 or $I_5 + I_{11}$	2(a)	Mode A Switch open or closed	4.7	8.5	15.5	mA
	I_2 or $I_5 + I_8 + I_{11}$	2(b)	Mode B Switch open or closed				
DYNAMIC CHARACTERISTICS	V_{CC} = +12V, V_{EE} = 0, Split Voltage Supply (Optional) = +6V						
Differential Voltage Gain							
Single-Ended Input Differential Output	$A_{DIFF(DE)}$	3(a)	f = 1 MHz R_S = 50 Ω	34	37	-	dB
Single-Ended Input and Output	$A_{DIFF(SE)}$	3(a)	f = 1 MHz R_S = 50 Ω	28	31	-	dB
-3 dB Bandwidth	BW	3(a)	R_S = 50 Ω	40	55	-	MHz
Differential Voltage Gain Balance	$A_{DIFF(SE)10} - A_{DIFF(SE)12}$	3(a)	f = 1 MHz	-1	0	+1	dB
Output Voltage Swing	V_8 or V_{10} RMS	3(a)	f = 1 MHz R_S = 50 Ω	-	0.5	-	V_{RMS}
Noise Figure	NF	3(a)	(Note 1) f = 30 MHz R_S = 400 Ω	-	7.5	9	dB
Parallel Input Resistance	R_I	3(a)	f = 1 MHz	-	150	-	kΩ
Parallel Input Capacitance	C_I	3(a)		-	2.2	-	pF
Output Resistance	R_O	3(a)		-	125	-	Ω
TEMPERATURE DEPENDENT CHARACTERISTICS Temperature coefficients for ambient temperature: -55°C ≤ T_A ≤ + 125°C							
Output Voltage	$\frac{\Delta V_{10} \text{ or } \Delta V_{12}}{^oC}$	3(a)	Bias Mode A	-	0	-	mV/°C
		3(b)	Bias Mode B	-	6.4	-	mV/°C
Power Supply Current Drain	$\Delta I_2/^oC$	3(a)	Bias Mode A	-	5	-	μA/°C
Differential Voltage Gain	$A_{DIFF}/^oC$	3(a)	Bias Mode A	-	0.0166	-	dB/°C
		3(b)	Bias Mode B	-	0	-	

Note 1: Replace 1-kΩ resistors between Term. 1 and 4 and Term. 1 and 6 with suitable chokes so that reactance at 30 MHz exceeds 5kΩ

STATIC CHARACTERISTICS TEST CIRCUITS FOR CA3040

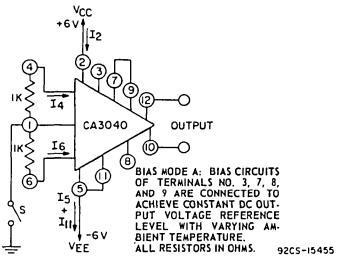


Fig.2(a) - Bias Mode A

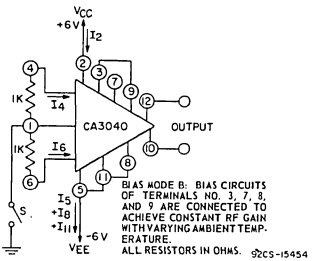
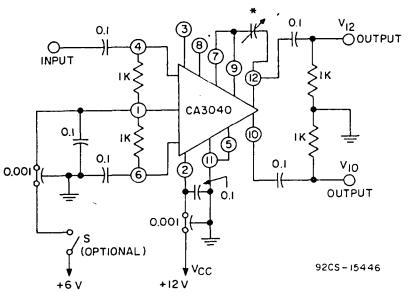


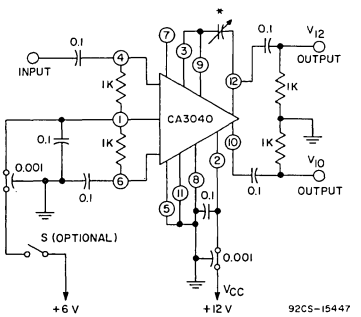
Fig.2(b) - Bias Mode B

DYNAMIC CHARACTERISTICS TEST CIRCUITS FOR CA3040



*VARIABLE CAPACITANCE (0.5-1.0 μF) ADJUSTMENT FOR EQUAL 3dB BANDWIDTH AT AMPLIFIER OUTPUTS, TERMINALS 10 AND 12.
ALL RESISTORS IN OHMS.
ALL CAPACITORS IN MICROFARADS (UNLESS OTHERWISE INDICATED).
BIAS MODE A IS AS DEFINED IN FIG. 2 (a)

Fig.3(a) - Bias Mode A



*SEE FIG 3(a)
BIAS MODE B IS AS DEFINED IN FIG 2(b)
ALL RESISTORS IN OHMS.
ALL CAPACITORS IN MICROFARADS (UNLESS OTHERWISE INDICATED).

Fig.3(b) - Bias Mode B

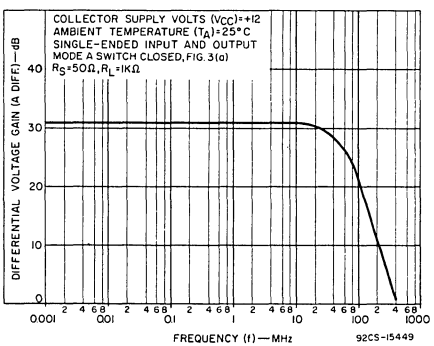


Fig.4 - Differential Voltage Gain vs Frequency

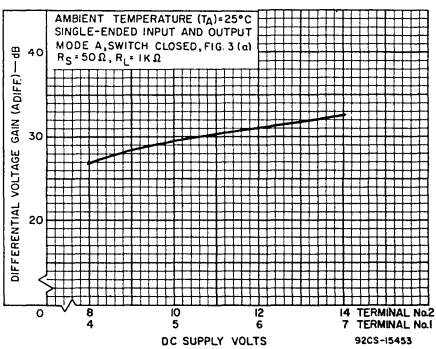


Fig.5 - Differential Voltage Gain vs DC Supply Voltages

OPERATING CONSIDERATIONS

General

The CA3040 is designed to provide flexibility in the selection of power supply configurations and to provide the circuit designer the choice between two modes of temperature-compensated performance. Mode A, which provides constant DC output voltage, is recommended for most applications. The control of the operating point provided by this mode maintains the dynamic range of the device while gain variation over most of the range is less than ±1 dB. Mode B provides constant gain for applications where this consideration is critical, but will exhibit a reduction of dynamic range at the temperature extremes.

Power Supply Considerations

Figures 2 and 3 illustrate the use of the CA3040 with balanced dual supplies and single power supplies, respectively. Both figures demonstrate that the inputs may be directly referenced to the center point of the supply (ground in Fig.2) by closing the included switch. This is the natural connection in Fig.2. This connection is optional, however, and need not be made. Use of this connection in Fig.3 implies the presence of another DC supply or a "stiff" bleeder. If such a source is present its use is suggested in order to maintain maximum common mode range. Dynamic performance and dynamic range of the output circuit are unaffected by the choice of biasing scheme used so that in most cases direct connection of Terminal No.1 to the center point of the supply is not required. Where direct connection is not used, Terminals No.4 and No.6 must be biased from Terminal No.1 for proper operation.

High-Frequency Considerations

Stable high-frequency operation requires that proper high-frequency construction techniques be followed. The photograph of Fig.6 illustrates the precautions taken in the construction of the test circuit of Fig.3.

Extreme caution is required because of the extended gain bandwidth capability of the device. Oscillations have been observed in the 400-to-800 MHz range when precautions were not taken. In addition to normal considerations of shielding, parts layout, and isolation, the following specific suggestions are made:

1. Use sockets only when necessary. Sockets, when used, must provide shielding within the pin circle. The socket shown in the chassis of Fig.6 is a Barnes MG-1201, or equivalent, modified by drilling a 1/8" hole in the center and inserting a grounded brass pin.
2. Do not bypass Terminal No.9 in normal operation. Fig.3 shows the use of neutralization between Terminal No.9 and one output to balance the amplifier at high frequencies. Experience shows that stable operation, while possible, is difficult to achieve if Terminal No.9 is bypassed to ground.
3. In DC testing, 1 kΩ, 1/4 W carbon resistors should be soldered directly to the socket Terminals No.4 and No.6 to suppress parasitic oscillations. All current carrying connections are made at the other end of the resistors. Direct sensing of Terminal No.4 or No.6 voltage should not be attempted.

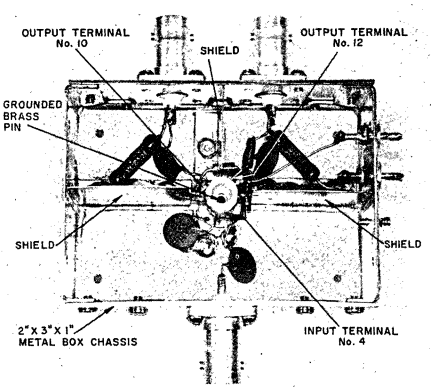


Fig.6 - Test Circuit Layout

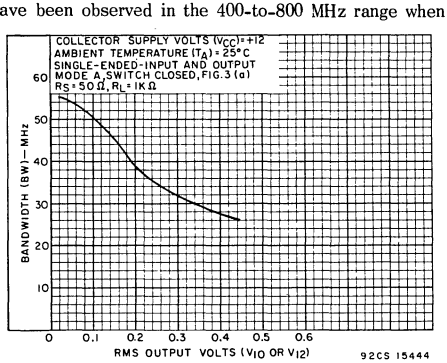


Fig.7 -3dB Bandwidth vs Single-Ended Output Voltage

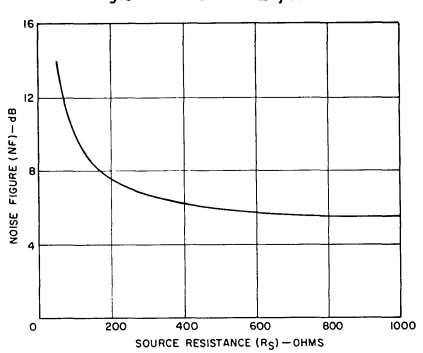


Fig.8 -Noise Figure (NF) vs Source Impedance

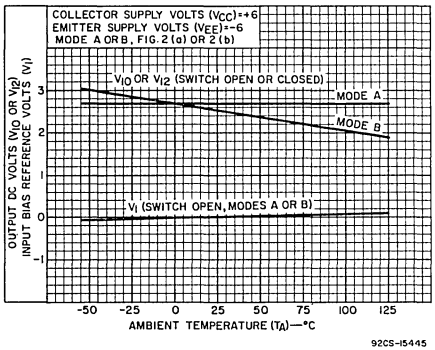


Fig.9 - Output Volts or Input Bias Reference Volts vs Ambient Temperature

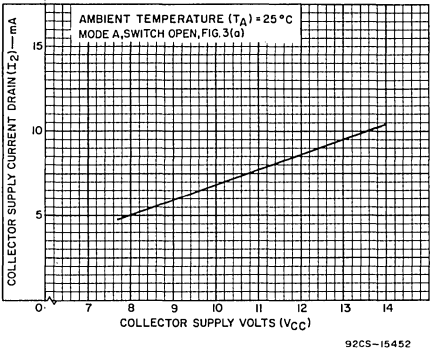


Fig.10 - Collector Supply Current Drain (I_2) vs Collector Supply Voltage (V_{CC})

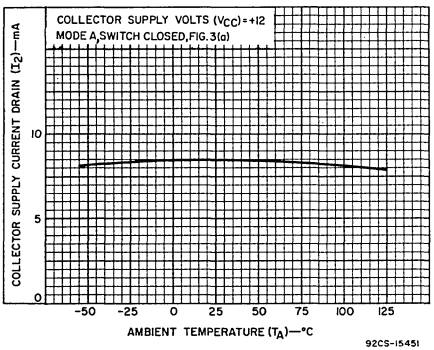


Fig.11 - Collector Supply Current Drain (I_2) vs Ambient Temperature

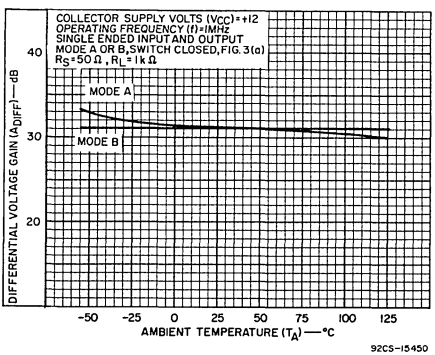


Fig.12 - Single-Ended Differential Voltage Gain vs Ambient Temperature

RCA Solid State Division

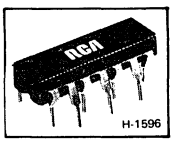
Linear Integrated Circuits

CA3041

WIDE-BAND AMPLIFIER, FM DETECTOR AF PREAMPLIFIER/DRIVER

Monolithic Silicon

For Sound Sections of TV Receivers Using Tube-Type AF Output Amplifiers



RCA Integrated Circuit Type CA3041 provides, in a single monolithic silicon chip, a major subsystem for the sound sections of TV receivers. As shown in the Schematic Diagram (Fig. 1) and the TV Receiver Block Diagrams (Fig. 2) the CA3041 contains a multistage wide-band if-amplifier/limiter section, an FM-detector stage, a Zener-diode-regulated power-supply section, and an af-amplifier section specifically designed to drive directly a 6AQ5 beam power tube or other audio output tube of similar characteristics.

In FM receivers, the CA3041 can be used to provide if amplification and limiting, FM detection, and af preamplification.

The CA3041 provides exceptional versatility of circuit design because the if-amplifier/limiter section, FM detector section, and af-preamplifier/driver section can be used independently of each other.

The CA3041 utilizes a 14-lead dual-in-line plastic package with leads specially formed to facilitate automatic insertion of the device in suitably punched printed-circuit boards. Templates showing recommended layout of printed-circuit boards for the CA3041 are provided in this bulletin (Figs. 13, 14 and 15).

FEATURES

- high-sensitivity – input limiting voltage (knee) = 150 μV typ. at 4.5 MHz
- large audio drive voltage capability
- excellent AM rejection – 58 dB typ. at 4.5 MHz
- inherent high stability – internally shielded
- internal Zener-diode-regulated voltage supply
- low harmonic radiation
- wide frequency capability – <100 kHz to > 20 MHz
- low harmonic distortion

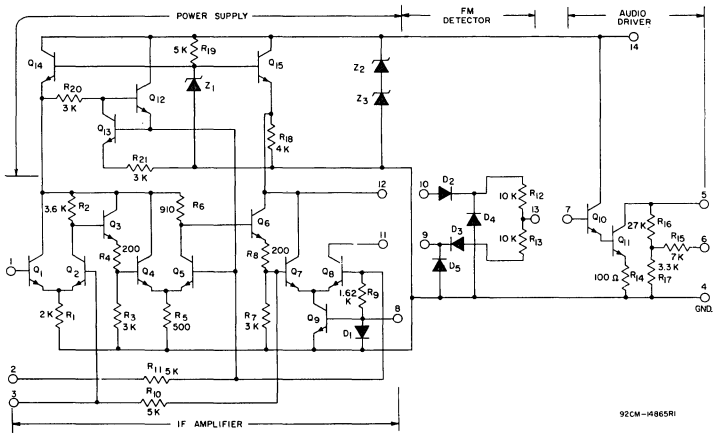


Fig. 1 - Schematic diagram.

ABSOLUTE-MAXIMUM VOLTAGE AND CURRENT LIMITS AT T_A = 25°C

Indicated voltage or current limits for each terminal may be applied under the specified voltage conditions for other terminals. All voltages are with respect to ground (Terminal 4).

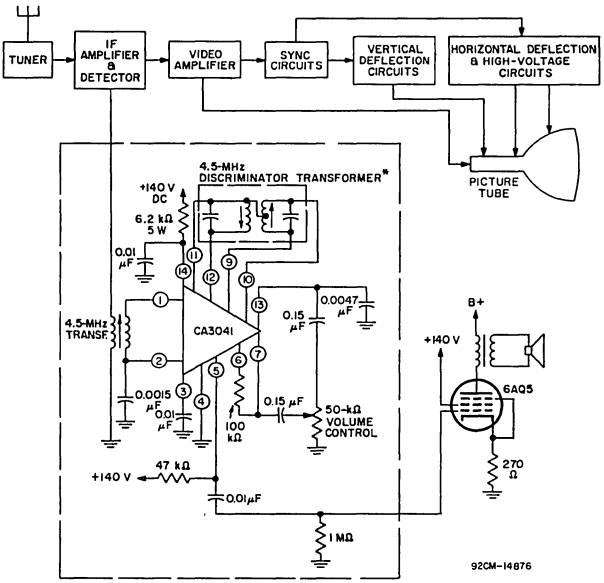
TERMINAL	VOLTAGE OR CURRENT LIMITS		VOLTAGE CONDITIONS AT OTHER TERMINALS													
			1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	-3 V	+3 V	–	AT SAME DC VOLTAGE AS TERMINAL 1	EXTERNAL DC VOLTAGE IS NOT NORMALLY APPLIED TO THIS TERMINAL	GROUND (VOLTAGE REFERENCE TERMINAL)	CONNECTED TO +140 V THROUGH 47 kΩ RESISTOR*	CONNECTED TO TERMINAL 7 THROUGH 100 kΩ RESISTOR*	AF-INPUT TERMINAL (EXTERNAL DC VOLTAGE IS NOT NORMALLY APPLIED TO THIS TERMINAL)	DC VOLTAGE IS NOT NORMALLY APPLIED TO THIS TERMINAL	AT SAME DC VOLTAGE AS TERMINAL 12 (EXCEPT TERMINAL 9)	AT SAME DC VOLTAGE AS TERMINAL 12 (EXCEPT TERMINAL 10)	AT SAME DC VOLTAGE AS TERMINAL 12 (EXCEPT TERMINAL 11)	EXTERNAL DC VOLTAGE IS NOT NORMALLY APPLIED TO THIS TERMINAL	AT SAME DC VOLTAGE AT TERMINAL 12 (EXCEPT TERMINAL 13)	CONNECTED TO +140 V DC THROUGH 6.2-kΩ RESISTOR*
2	-3 V	+3 V	-3 to +3													
3	-3 V	+3 V	-3 to +3													
4	GROUND (VOLTAGE REFERENCE TERMINAL)		-3 to +3													
5	20 mA		-3 to +3													
6	0 V	+10 V	-3 to +3													
7	10 mA		-3 to +3													
8	10 mA		-3 to +3													
9	10 mA		-3 to +3													
10	10 mA		-3 to +3													
11	+2.5 V	+5 V	-3 to +3													
12	+2.5 V	+5 V	-3 to +3													
13	+2.5 V	+5 V	-3 to +3													
14	50 mA		-3 to +3													

* Any other combination of DC Supply Voltage and Series Resistance which will not cause the Maximum Device Dissipation Limit or any of the Maximum Voltage or Current Limits for the CA3041 to be exceeded may be used.

OPERATING-TEMPERATURE RANGE ..... −40° to +85°C
STORAGE-TEMPERATURE RANGE ..... −65° to +150°C
LEAD TEMPERATURE (During Soldering):
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm)
from case for 10 seconds max. ..... +265°C
MAXIMUM INPUT-SIGNAL VOLTAGE:
Between Terminals 1 and 3 ..... ±3 V
MAXIMUM DEVICE DISSIPATION:
At Ambient Temperatures up to +25°C ..... 950 mW
above +25°C ..... Derate at 10.8 mW/°C

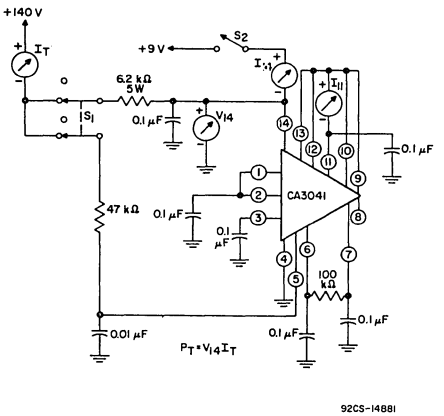
ELECTRICAL CHARACTERISTICS, *at an Ambient Temperature, T_A, of 25°C, and a DC Supply Voltage, V_{CC}, of +140 Volts applied to Terminal 14 through a resistance of 6.2 kΩ, unless otherwise indicated. Any other combination of DC Supply Voltage and Series Resistance which will not cause the Maximum Dissipation Limit or any of the Maximum Voltage or Current Limits for the CA3041 to be exceeded may be used.*

CHARACTERISTICS (See Page 7 for Definitions of Terms)	SYMBOLS	TEST CONDITIONS: SETUP AND PROCEDURE	TEST CONDITIONS: SPECIAL CONDITIONS	LIMITS: TYPE CA3041 Min.	Typ.	Max.	Units	TYPICAL CHARACTERISTICS CURVES
		Fig.						Fig.
Total Device Dissipation	P_T	3	T_A = 0°C	220	245	270	mW	4
			T_A = +25°C	225	250	275	mW	
			T_A = +85°C	230	255	280	mW	
Zener Regulating Voltage (DC Supply Voltage at Terminal 14)	V_{14}	–		10.5	11.2	12.3	V	–
Quiescent Operating Current (into Terminal 11)	I_{11}	3		0.25	0.63	1	mA	–
9-Volt Current Drain (Quiescent Operating Current into Terminal 14)	I_{14}	3	V_{CC} = +9 V applied directly to Terminal 14	7	11	16	mA	–
Input-Impedance Components: Parallel Input Resistance	R_i	5	f = 4.5 MHz	–	11	–	kΩ	–
Parallel Input Capacitance	C_i	5		–	5	–	pF	–
Output-Impedance Components: Parallel Output Resistance	R_o	–		–	100	–	kΩ	–
Parallel Output Capacitance	C_o	–		–	4	–	pF	–
Input Limiting Voltage (Knee)	$V_{i(lim)}$	6		–	150	200	μV (rms)	10
Amplitude-Modulation Rejection	AMR	7		45	58	–	dB	8
IF-Amplifier Voltage Gain	$A_{(IF)}$	9		–	67	–	dB	10
Recovered AF Voltage: 1. At FM-Detector Output	$V_{o(af)}$	–	R_L = 50 kΩ, Δf = ±25 kHz, THD = 0.7% (typ.)	–	250	–	mV (rms)	–
2. At AF-Driver Output in Test Setup		–	THD < 5%	8	9	–	V (rms)	–
Total Harmonic Distortion	THD	6	$V_{o(af)}$ = 8 V(rms)	–	1.5	5	%	–
Discriminator Output Resistance	$R_{o(dis)}$	–	f = 1 kHz	–	10	–	kΩ	–
AF-Amplifier Input Resistance	$R_{i(af)}$	–		–	100	–	kΩ	–
AF-Amplifier Output Resistance	$R_{o(af)}$	–		–	30	–	kΩ	–
AF-Driver Voltage Gain	A_{af}	11		–	41	–	dB	12



* TRW Electronics, Des Plaines, Illinois. Part No. EO23874, or equivalent.

Fig.2 - Block diagram of typical TV receiver using CA3041.



PROCEDURES:

Total Device Dissipation:

1. Close S_1, open S_2.
2. Measure and record V_{14} and I_T.
3. Determine Total Device Dissipation from $P_T = V_{14} I_T$.

Quiescent Operating Current into Terminal 11:

1. Close S_1, open S_2.
2. Measure I_{11} and record as Quiescent Operating Current into Terminal 11.

9-Volt Current Drain:

1. Open S_1, close S_2.
2. Measure I_{14} and record as 9-Volt Current Drain.

Fig.3 - Test setup for total dissipation, quiescent operating current into terminal No.11, and 9-volt current drain.

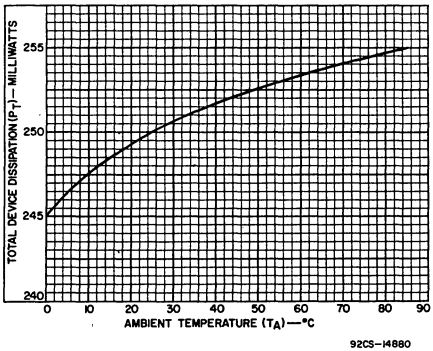


Fig.4 - Typical dissipation characteristic for CA3041.

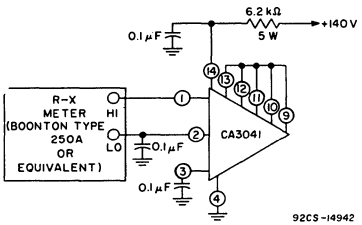
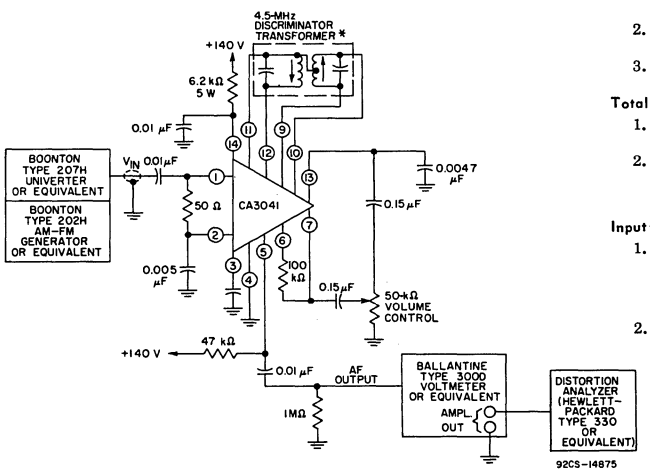


Fig.5 - Test setup for measurement of input-impedance components.



PROCEDURES:

Recovered AF Voltage:

1. Set Input Signal Generator as follows:
   Output frequency = 4.5 MHz
   Modulating frequency = 1 kHz
   Deviation = ± 25 kHz
   Output level for V_{in} = 100 mV rms
2. Set volume control for maximum af output.
3. Measure af output voltage and record as Recovered AF Voltage.

Total Harmonic Distortion:

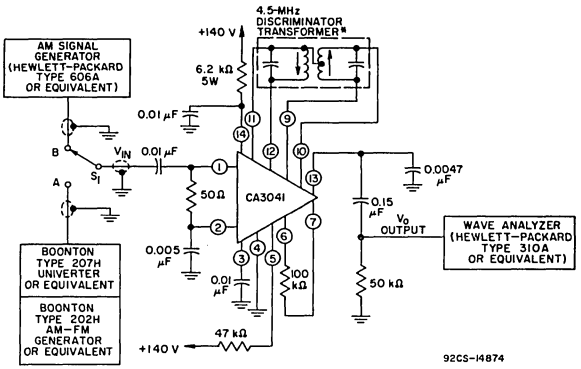
1. Adjust volume control for an af output voltage of 300 mV rms.
2. Measure Total Harmonic Distortion of the output signal in accordance with the Operating Instructions for the Distortion Analyzer.

Input Limiting Voltage (Knee):

1. Decrease V_{in} until the af output voltage is 3 dB less than the value set in Step 1 of the procedure for measurement of Total Harmonic Distortion (300 mV - 3 dB = 210 mV)
2. Measure resulting value of V_{in} and record as Input Limiting Voltage (Knee).

* TRW Electronics, Des Plaines, Illinois. Part No. EO23874, or equivalent.

Fig.6 - Test setup for measurement of input limiting voltage (Knee), recovered AF voltage, and total harmonic distortion.



* TRW Electronics, Des Plaines, Illinois. Part No. EO23874, or equivalent.

PROCEDURES:

1. Set FM Signal Generator as follows:
   Output frequency = 4.5 MHz
   Modulating frequency = 1000 Hz
   Deviation = ± 25 kHz
   Output level for V_{in} = 100 mV rms
2. Set AM Signal Generator as follows:
   Output frequency = 4.5 MHz
   Modulating frequency = 1000 Hz
   Per cent modulation = 30
   Output level for V_{in} = 10 mV rms
3. With S_1 in Position A measure AF Output Voltage and record as $V_{o(FM)}$.
4. With S_1 in Position B measure AF Output Voltage and record as $V_{o(AM)}$.
5. Determine AM Rejection from AMR = $V_{o(FM)}/V_{o(AM)}$

Fig.7 - Test setup for measurement of AM rejection.

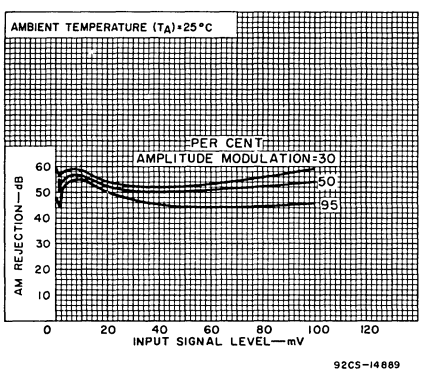
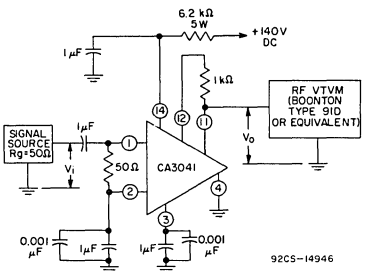


Fig.8 - Typical AM rejection characteristics for CA3041.



PROCEDURE:

A - Voltage Gain:

1) Set input frequency at desired value, v_i = 100 μV rms.
2) Record v_o.
3) Calculate Voltage Gain A from A = 20 $\log_{10}$ v_o/v_i
4) Repeat Steps 1, 2, and 3 for each frequency and/or for temperature desired.

Fig.9 - Test setup for measurement of IF-amplifier voltage gain.

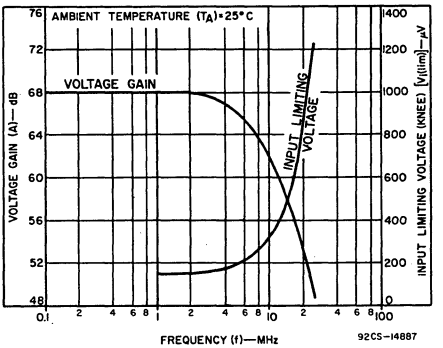


Fig.10 - Typical IF-amplifier voltage gain and input-limiting voltage (knee) characteristics.

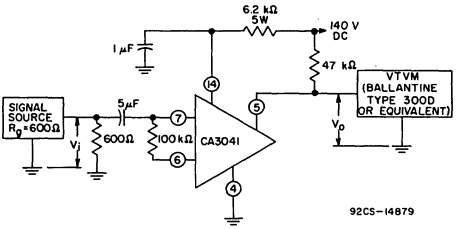


Fig.11 - Test setup for measurement of AF-amplifier voltage gain.

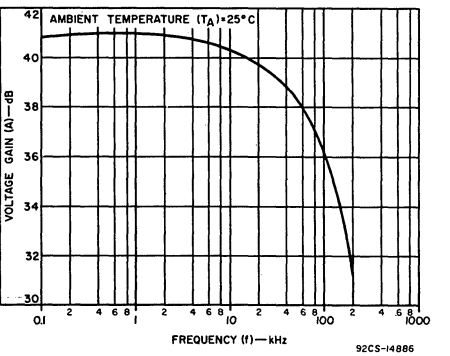


Fig.12 - Typical AF-driver voltage-gain characteristic.

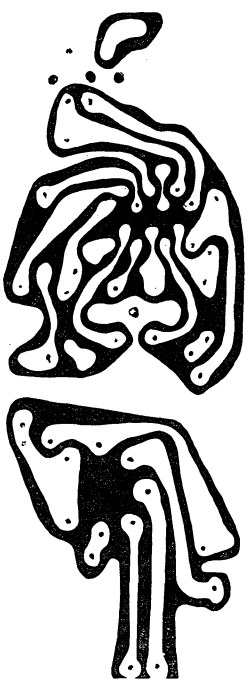


Fig.13 - Recommended layout of printed-circuit board for complete TV-receiver sound strip utilizing RCA-CA3041 (Top View).

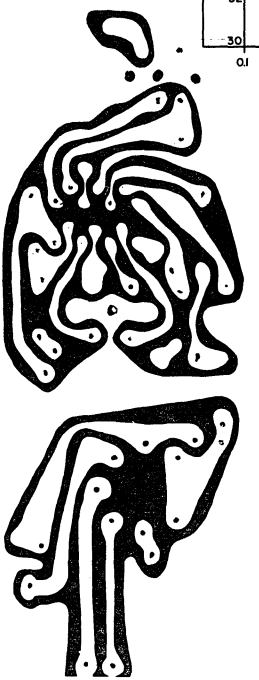


Fig.14 - Recommended layout of printed-circuit board for complete TV-receiver sound strip utilizing RCA-CA3041 (Bottom View).

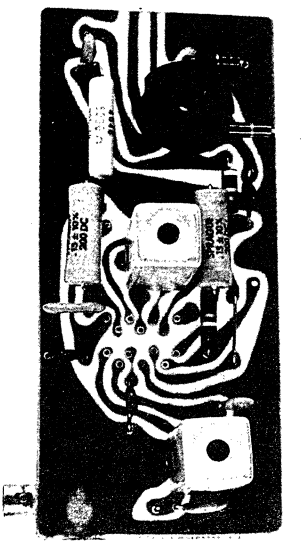


Fig.15 - Recommended parts layout for TV-receiver sound strip utilizing RCA-CA3041. (Top View)

RCA Solid State Division

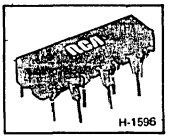
Linear Integrated Circuits

CA3042

WIDE-BAND AMPLIFIER, FM DETECTOR AF PREAMPLIFIER/DRIVER

Monolithic Silicon

For Sound Sections of TV Receivers Using Transistor-Type AF Output Amplifiers



RCA Integrated Circuit Type CA3042 provides, in a single monolithic silicon chip, a major sub-system for the sound sections of TV receivers. As shown in the Schematic Diagram (Fig.1) and the TV Receiver Block Diagrams (Figs.2A and 2B) the CA3042 contains a multistage wide-band if-amplifier section, an FM-detector stage, a Zener-diode-regulated power-supply section, and an af-amplifier section specifically designed to drive directly an n-p-n audio output transistor or a high-gain audio output pentode tube.

In FM receivers, the CA3042 can be used to provide if amplification and limiting, FM detection, and af preamplification.

The CA3042 provides exceptional versatility of circuit design because the if-amplifier/limiter section, FM detector section, and af-preamplifier/driver section can be used independently of each other.

The CA3042 utilizes a 14-lead dual-in-line plastic package with leads specially formed to facilitate automatic insertion of the device in suitably punched printed-circuit boards. Templates showing recommended layout of printed-circuit boards for the CA3042 are provided in this bulletin (Figs.13 & 14).

FEATURES

- high sensitivity — input limiting voltage (knee) = 150 μV typ. at 4.5 MHz
- 6-mA audio drive capability
- excellent AM rejection — 58 dB typ. at 4.5 MHz
- inherent high stability — internally shielded
- internally Zener-diode-regulated voltage supply
- low harmonic radiation
- wide frequency capability — <100 kHz to >20 MHz
- low harmonic distortion

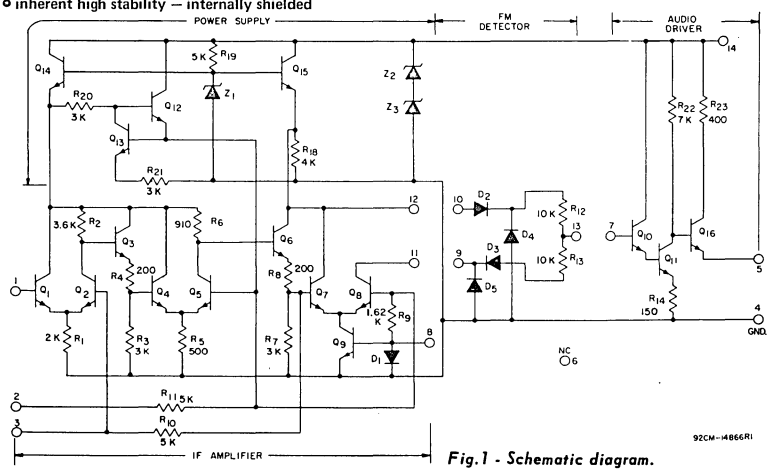


Fig.1 - Schematic diagram.

ABSOLUTE-MAXIMUM VOLTAGE AND CURRENT LIMITS AT T_A = 25°C

Indicated voltage or current limits for each terminal may be applied under the specified voltage conditions for other terminals. All voltages are with respect to ground (Terminal 4).

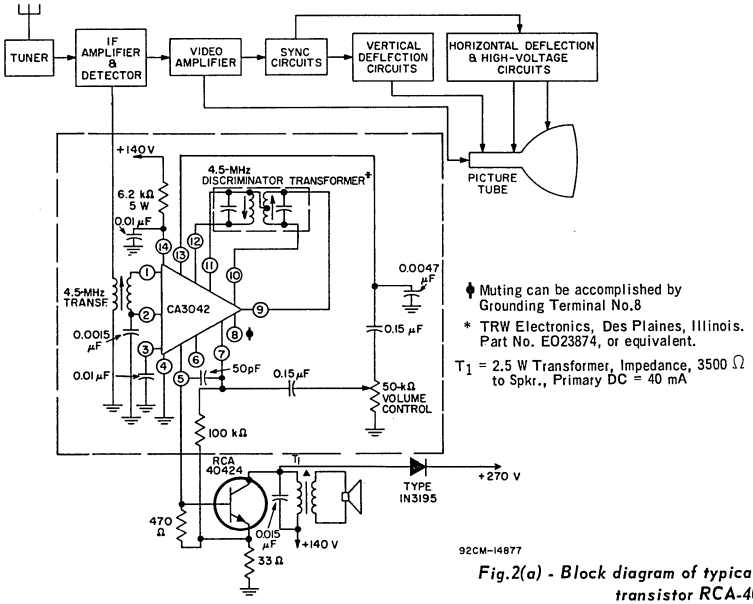
TERMINAL	VOLTAGE OR CURRENT LIMITS		VOLTAGE CONDITIONS AT OTHER TERMINALS													
			1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	-3 V	+3 V	–	AT SAME DC VOLTAGE AS TERMINAL 1	EXTERNAL DC VOLTAGE IS NOT NORMALLY APPLIED TO THIS TERMINAL	GROUND (VOLTAGE REFERENCE TERMINAL)	AF-DRIVER OUTPUT TERMINAL (EXTERNAL DC VOLTAGE IS NOT NORMALLY APPLIED TO THIS TERMINAL)	NO CONNECTION	AF-INPUT TERMINAL (EXTERNAL DC VOLTAGE IS NOT NORMALLY APPLIED TO THIS TERMINAL)	MUTING TERMINAL (EXTERNAL DC VOLTAGE IS NOT NORMALLY APPLIED TO THIS TERMINAL EXCEPT THAT TERMINAL MAY BE GROUNDED TO OBTAIN MUTING ACTION)	AT SAME DC VOLTAGE AS TERMINAL 12 (EXCEPT TERMINAL 9)	AT SAME DC VOLTAGE AS TERMINAL 12 (EXCEPT TERMINAL 10)	AT SAME DC VOLTAGE AS TERMINAL 12 (EXCEPT TERMINAL 11)	EXTERNAL DC VOLTAGE IS NOT NORMALLY APPLIED TO THIS TERMINAL	AT SAME DC VOLTAGE AT TERMINAL 12 (EXCEPT TERMINAL 13)	CONNECTED TO +140 V DC THROUGH 6.2-kΩ RESISTOR*
2	-3 V	+3 V	-3 to +3													
3	-3 V	+3 V	-3 to +3													
4	GROUND (VOLTAGE REFERENCE TERMINAL)		-3 to +3													
5	20 mA		-3 to +3													
6	NO CONNECTION		-3 to +3													
7	10 mA		-3 to +3													
8	10 mA		-3 to +3													
9	10 mA		-3 to +3													
10	10 mA		-3 to +3													
11	+2 V	+10 V	-3 to +3													
12	+2.5 V	+10 V	-3 to +3													
13	0 V	+10 V	-3 to +3													
14	50 mA		-3 to +3													

* Any other combination of DC Supply Voltage and Series Resistance which will not cause the Maximum Device Dissipation Limit or any of the Maximum Voltage or Current Limits for the CA3042 to be exceeded may be used.

OPERATING-TEMPERATURE RANGE ..... –40° to +85°C
STORAGE-TEMPERATURE RANGE ..... –65° to +150°C
LEAD TEMPERATURE (During Soldering):
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm)
from case for 10 seconds max. ..... +265°C
MAXIMUM INPUT-SIGNAL VOLTAGE:
Between Terminals 1 and 3 ..... ±3 V
MAXIMUM DEVICE DISSIPATION:
At Ambient Temperatures up to +25°C ..... 950 mW
above +25°C ..... Derate at 10.8 mW/°C

ELECTRICAL CHARACTERISTICS, *at an Ambient Temperature, T_A, of 25°C, and a DC Supply Voltage, V_{CC}, of +140 Volts applied to Terminal 14 through a resistance of 6.2 kΩ, unless otherwise indicated. Any other combination of DC Supply Voltage and Series Resistance which will not cause the Maximum Dissipation Limit or any of the Maximum Voltage or Current Limits for the CA3042 to be exceeded may be used.*

CHARACTERISTICS (See Page 7 for Definitions of Terms)	SYMBOLS	TEST CONDITIONS: SETUP AND PROCEDURE	TEST CONDITIONS: SPECIAL CONDITIONS	LIMITS: TYPE CA3042 Min.	Typ.	Max.	Units	TYPICAL CHARACTERISTICS CURVES
		Fig.						Fig.
Total Device Dissipation	P_T	3	T_A = 0°C	200	230	260	mW	4
			T_A = +25°C	210	240	270	mW	
			T_A = +85°C	220	250	280	mW	
Zener Regulating Voltage (DC Supply Voltage at Terminal 14)	V_{14}	–		10.5	11.2	12.3	V	–
Quiescent Operating Current (into Terminal 11)	I_{11}	3		0.25	0.63	1	mA	–
9-Volt Current Drain (Quiescent Operating Current into Terminal 14)	I_{14}	3	V_{CC} = +9 V applied directly to Terminal 14	8	12	18	mA	–
Input-Impedance Components: Parallel Input Resistance	R_I	5	f = 4.5 MHz	–	11	–	kΩ	–
Parallel Input Capacitance	C_I	5	f = 4.5 MHz	–	5	–	pF	–
Output-Impedance Components: Parallel Output Resistance	R_o	–	f = 4.5 MHz	–	100	–	kΩ	–
Parallel Output Capacitance	C_o	–	f = 4.5 MHz	–	4	–	pF	–
Input Limiting Voltage (Knee)	$V_{i(lim)}$	12	f = 4.5 MHz	–	150	200	μV (rms)	9
Amplitude-Modulation Rejection	AMR	6	f = 4.5 MHz	45	58	–	dB	7
IF-Amplifier Voltage Gain	$A_{(IF)}$	8	f = 4.5 MHz	–	67	–	dB	9
Recovered AF Voltage: 1. At FM-Detector Output	$V_{o(af)}$	12	f = 4.5 MHz, Δf = ±25 kHz, R_L = 50 kΩ, THD = 0.7% (typ.)	–	250	–	mV (rms)	–
2. At AF-Driver Output in Test Setup		12	f = 4.5 MHz, Δf = ±25 kHz, R_L = 322 Ω, THD < 5%	500	800	–	mV (rms)	–
3. At AF-Driver Output in TV-Receiver Sound System		2A or 2B	f = 4.5 MHz, Δf = ±25 kHz, R_L = 150 kΩ, THD = 1.5% (typ.)	–	3	–	V (rms)	–
Total Harmonic Distortion: 1. In Test Setup	THD	12	f = 4.5 MHz, Δf = ±25 kHz, $V_{o(af)}$ = 500 mV (rms)	–	1.5	5	%	–
2. In TV Receiver Sound System		2A or 2B	f = 4.5 MHz, Δf = ±25 kHz, $V_{o(af)}$ = 1.3 V (rms)	–	1	–	%	–
FM-Detector Output Resistance	$R_{o(det)}$	–	f = 1 kHz	–	10	–	kΩ	–
AF-Driver Input Resistance	$R_{I(af)}$	–	f = 1 kHz	–	100	–	kΩ	–
AF-Driver Output Resistance	$R_{o(af)}$	–	f = 1 kHz	–	250	–	Ω	–
AF-Driver Voltage Gain	A_{af}	10	f = 1 kHz, R_S = 50 Ω, C_1 = 0	–	30	–	dB	11

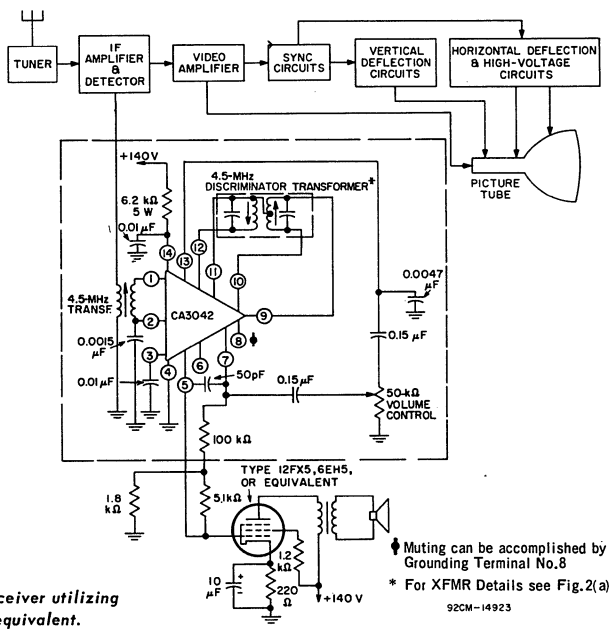


♦ Muting can be accomplished by Grounding Terminal No.8

* TRW Electronics, Des Plaines, Illinois. Part No. EO23874, or equivalent.

T_1 = 2.5 W Transformer, Impedance, 3500 Ω to Spkr., Primary DC = 40 mA

Fig.2(a) - Block diagram of typical TV receiver utilizing transistor RCA-40424.



♦ Muting can be accomplished by Grounding Terminal No.8

* For XFMR Details see Fig.2(a)

Fig.2(b) - Block diagram of typical TV receiver utilizing the CA3042 and a 12FX5, 6EH5, or equivalent.